A FINITE VOLUME METHOD FOR THE PREDICTION OF THREE-DIMENSIONAL FLUID FLOW IN COMPLEX DUCTS

by

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ABSTRACT

This thesis presents the development and application of a method for the prediction of steady, subsonic, two- or three-dimensional fluid flow in ducts of complex geometry.

Various mathematical formulations of the problem were examined, and a strong conservation form of the governing equations employing Cartesian vector and tensor components was adopted. The differential equations were discretised in a finite volume fashion, using a non-staggered arrangement of variables. The computational grid is completely characterized by the coordinates of the cell vertices, which are assumed to be joined by straight lines.

An efficient solution procedure, which facilitates the use of the non-staggered variable arrangement, was developed. It represents significant improvements on the SIMPLE algorithm for velocity-pressure coupling, and an existing method of its implementation when a non-staggered variable arrangement is used.

Several differencing schemes were evaluated, and a new flux blending technique for eliminating numerical oscillations associated with the higher-order ones was developed.

The method was first assessed by applying it to some flows for which "exact" solutions were known. It was then applied to a number of complex two- and three-dimensional, laminar and turbulent recirculating flows in a range of duct geometries. These calculations demonstrated the good robustness, accuracy and generality of the solution method presented.

The agreement between the predictions and the experimental data, where available, is found to be satisfactory, within the limitations of the turbulence model used (the standard k-ε model) and the possibility of achieving grid-independent solutions.

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Dedicated to my wife Kerstin and our parents, as a tribute to their sacrifices.

CONTENTS

NOMENCLATURE

Latin letters

a	general vector; coefficients in the discretised equations
A	general tensor; area; coefficient matrix
b	parts of discretised transformation factors β , defined in Eq. (3.24)
B	coefficients in transformed differential equations, defined in Eq. (3.13)
c	specific heat; coefficients in SCHEME 2, defined in Eq. (4.25)
C	coefficients in the turbulence model; source term in the discretised equations
d	diameter
D	deformation tensor; coefficients in the discretised equations defined in Eq. (3.36); diameter
e	base vectors
E	coefficients in the cross-derivative diffusion terms, defined in Eq. (4.21); coefficients in the pressure-correction equation, defined in Eq. (5.28); constant in wall functions; errors, defined in Eq. (8.1)
f	spatially constant base vectors; general vector function; linear interpolation factors
F	mass fluxes through a cell face
g	determinant of a metric tensor; magnitude of a base vector; quadratic interpolation factors
G	rate of production of turbulent kinetic energy
H	operator in discretised momentum equations, defined in Eq. (5.15)
i	Cartesian base vectors; node index in x^1 coordinate direction
I	unit tensor; flux through a cell face
j	node index in x^2 coordinate direction
J	Jacobian of coordinate transformation
k	kinetic energy of turbulence; node index in x^3 coordinate direction
l	arc length
L	length; large number
M	total number of computational nodes (cells)
n	normal (perpendicular) coordinate; a unit normal vector

N	number of nodes in a particular direction; normal vector
p	pressure
P	function in Eq. (6.36); forcing function in Eqs. (7.3) and (7.9)
Pe	Peclet number
q	flux vector
Q	coefficients in pressure terms, defined in Eq. (5.15); forcing function in Eqs. (7.3) and (7.9)
r	radial coordinate; aspect and expansion ratios, defined in Eqs. (7.1) and (7.2)
R	residual source, defined in Eq. (6.38); forcing function in Eq. (7.9)
Re	Reynolds number
s	volumetric source
S	integrated volumetric source (over cell volume)
SR	sum of residual sources
SN	normalization quantities, given in Table 6.1
t	time
T	stress tensor; wall shear force
u	Cartesian velocity components
U	covariant velocity components (normal to cell faces)
v	velocity vector
V	volume
x	arbitrary coordinates
y	Cartesian coordinates
z	axial coordinate

Greek letters

α	Cartesian components of a dual natural base vector; under-relaxation parameter
β	coordinate transformation factors, defined in Eq. (3.2)
γ	blending factor, defined in Eq. (4.31)
Γ	diffusivity coefficient
δ	difference; Kronecker's delta
ε	rate of dissipation of turbulent kinetic energy
κ	Von Karman's constant
λ	conductivity; coefficients in Eqs. (6.21), (6.23) and (6.37); convergence limit for normalized residual source sums
μ	dynamic viscosity
ω	differential operator in transformed momentum equations, defined in Eq. (3.20)

τ	anisotropic part of the stress tensor; shear stress
Ω	discretised ω operator
ρ	density
σ	Prandtl/Schmidt numbers
θ	circumferential coordinate; angle between grid lines
ψ	angle between coordinate lines of the arbitrary and Cartesian coordinate systems

Subscripts

B	lower-order scheme (bounded)
c	general cell face location
corr	correction
e,w,n,s,t,b	at the centre of cell face: e,w,n,s,t,b
E,W,N,S,T,B	at the neighbour node: E,W,N,S,T,B
eff	effective
i,j,k	coordinate directions; tensor indices
in	inlet
norm	normalization quantity
o	outlet
p	pressure-related
P	at the central node P
s	sum
u	velocity-rèlated
U	higher-order scheme (unbounded)
w	wall-related
ϕ	related to a scalar quantity ϕ
1,2,3	coordinate directions

Superscripts

C	convection
D	diffusion
DC	cross-derivative diffusion term
DN	normal-derivative diffusion term
i,j,k	coordinate directions; tensor indices
S	source
t	turbulent
1,2,3	coordinate directions
*	predictor stage; normalized coordinate
**	corrector stage

***	second corrector stage
'	fluctuating component; first correction
"	second correction
$\bar{}$	mean; average; linearly interpolated
$\vec{}$	vector
+	normalized quantities in wall functions
$\hat{}$	pseudo-quantities in the SIMPLER method
$\tilde{}$	pseudo-quantities in the PISO and SIMPLEC methods

CHAPTER 1

INTRODUCTION

1.1 Background

During the past three decades some new branches in the engineering sciences tree have emerged. Due to developments in computer technology, it has become possible to obtain (relatively) accurate solutions to many problems for which , although the mathematical formulations were laid down long ago, solutions were impossible to obtain analytically except for a limited number of special, simple cases. One of these new branches is computational fluid dynamics, which is attracting more and more interest as its achievements are becoming apparent.

Fluid flow, often coupled with heat transfer, and sometimes also with chemical reaction and mixing, finds many applications in engineering. The differential equations describing the motion of a fluid and associated phenomena have been known for more than a century; however, only for laminar flows in simple geometries (pipes, channels etc.) could analytic solutions be found. For more complex flows engineers have had to rely solely on the experimental information. The gathering of that information is, however, not only difficult and time consuming, it is also very expensive, since for every new experiment new test facilities have to be built. On the other hand, as computers have become more powerful, the possibility of numerical prediction of fluid flows has emerged. In this, the differential transport equations are discretised essentially by replacing differentials by differences, calculated from a finite number of values associated with the computational nodes, which are distributed on a suitable grid over the solution domain. The system of ordinary algebraic equations results, which can be efficiently solved by means of a digital computer. This is aided by the fact that computers are becoming ever faster in performing "number-crunching" (at current rates in excess of 200 million of floating point operations per second), and increasing in capacity to store variables (millions of values can be kept in internal memory and a virtually unlimited additional capacity is available in the form of external, disc and tape memory). Also important is the fact that the size of the computer is reducing: machines used to occupy large rooms and required powerful air conditioning system 10 years ago, whereas nowadays a computer of similar capacity is often a desk-top machine. All this adds to the attractiveness of numerical solution methods, for it is enabling them to become faster and cheaper than experiments.

Practical fluid flow problems, however, still pose substantial difficulties for numerical analysts. The three main obstacles to be overcome are the following:

1. The physical complexity of the flow;
2. The geometrical complexity of flow domains;
3. The limited accuracy, stability and economy of numerical solution methods.

Each of the above problems will now be discussed in turn.

Most real-life flows are unsteady and turbulent; even when the average flow is steady, the fluctuations induced by turbulence require the three-dimensional time-dependent form of equations to be solved. Even more problematic is the fact that the length scales of the turbulent motions can be very small in high Reynolds number flows, and to resolve these by numerical methods requires grids so fine, that the capacity of even the largest of todays supercomputers is strained * (although this approach is currently being used for fundamental studies of simple flows, eg. by Moser, 1984).

The only feasible practical approach is therefore to "model" the turbulance. Many such models exist; a comprehensive survey can be found in Rodi (1980). All of them rely on experimental data to determine empirical parameters featuring in the equations, which are often problem dependent. Thus none of these models can be said to be universal. The degree of accuracy they provide varies from case to case. The problem of modelling of turbulence is even more pronounced by the lack of detailed experimental studies of complex turbulent flows with benchmark accuracy, which could be used for the determination of model parameters. Recent developments in Laser Anemometry are already bringing progress in this direction, and the selection of detailed reliable data has already been accomplished for some flows (Johnston, 1981). This remains, however, an area in which substantial improvements are needed to enable accurate predictions of complex turbulent flows.

The differential transport equations are easily discretised if they

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* Leslie (1985) suggests that the storage and computing time requirements for the "full simulation" of turbulent flow at $Re=6000$ would be about 20 Mwords of fast store, and about 300 hours on CRAY XMP computer. With the increase in Re, the demands are likely to encrease with the power of Re: for storage, as $Re^{2.6}$ and for computational time as $Re^{3.5}$. If the current trend in computer technology developments continues, he expects the full simulations of engineering flows to be performed by the year 2060.

are in the Cartesian or polar cylindrical form. However, the range of practical flow configurations into which these coordinates fit is rather limited, and is already extensively covered by numerical and experimental studies. Most practical geometries are of irregular shape, and thus a different approach is needed. The present study is principally concerned with this subject, the aim being to develop a numerical solution method which is applicable to a wider range of geometries. More details about possible approaches, previous related work and present contributions will be given in the next two sections.

The demands on a numerical solution method from the user point of view are (i) accuracy, (ii) stability and (iii) economy. All three are difficult to meet simultaneously, as will be discussed later. However, some gerneral observations will be made here.

The accuracy* of numerical solutions is obviously of vital importance. It can generally be increased by: (i) using more computational nodes in the numerical grid, (ii) using higher-order interpolation formulae and (iii) controlling certain grid properties (eg. degree of alignment of grid and stream lines). Some of the measures aimed at increasing accuracy (eg. the use of higher-order interpolation schemes) may impair other important properties, namely stability and economy, of the numerical solution process.

Stability also depends on the type of numerical solution method adopted (ie., whether explicit, implicit or semi-implicit). A method which is not stable over a wide range of problems is of little practical value.

The economy of a numerical solution method is also important. Although computer resources are growing, they are by no means unexhaustable, and a method which requires less storage and computing time for the same level of accuracy than its competitors is to be preferred. Special computer hardware architecture (eg. vector or parallel processing) can sometimes be utilized to improve the efficiency, as discussed in Gentzsch (1984).

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* It is necessary to distinguish here between the "numerical" accuracy, ie. the degree to which the solution of discretised algebraic equations satisfies the original differential equation, and the "physical" accuracy, ie. the degree to which such solution represents the true solution of the physical problem under consideration. A numerically "exact" solution will result if the number of computational nodes tends to infinity. On the other hand, the "physical" accuracy depends on the correctness of the differential equations. These are usually not exact, but involve certain assumptions (eg. turbulence model etc.), which can greatly influence the physical accuracy of otherwise correct numerical solutions.

The present study is, as already noted, mostly concerned with overcoming the second-named diffilcuty, although due attention will also be given to the questions of accuracy and economy. Before, however, the present contributions are outlined, the need for the development of the present method, and some related work which provided both the background and inspiration for it, will briefly be discussed.

1.2 Previous Related Studies

As already noted, most fluid flows of engineering interest occur in domains of irregular geometry. Many such examples can be found in industry (turbine blade passages, heat exchangers, combustion chambers, internal combustion engines, aircrafts, vehicles, mixing vessels and other equipment), as well as in everyday life (air flow around buildings and in air-conditioning systems, water flow in rivers and lakes etc.). The ability to predict those flows is of vital importance for a number of reasons, including the reduction of energy losses and environment pollution.

Irregular shapes of flow domains can be treated in numerical solution methods in one of the following ways:

(i) the use of regular coordinates with stepwise approximation of curved boundaries (Fig. 1.1), or with a composite of two or more grids fitting to the boundaries and partly overlapping (Fig. 1.2);

(ii) the use of orthogonal curvilinear coordinates (Fig. 1.3);

(iii) the use of non-orthogonal curvilinear coordinates (Fig. 1.4).

A brief description of these approaches will now be given by considering a typical irregular flow domain, as found, for example, in heat exchangers. A more detailed discussion will appear in later chapters, in conjunction with the description of the approach developed in the present study.

The first approach uses the simplest equations (eg. Cartesian) whose discretisation is easy and straightforward (see, for example, Gosman and Ideriah, 1976 and Patankar, 1980). However, the treatment of irregular boundaries is extremely difficult, and has proved to be not only inconvenient, but also inaccurate (Koutmos and McGuirk, 1983). Another problem with this approach is the distribution of the grid lines. The introduction of fine grids in regions of steep gradients (eg. near walls) results in unnecessary refinement elsewhere, thus extending the computational storage and time requirements (see Fig. 1.1)

In some cases boundaries of the flow domain consist of different types of "regular" geometry, eg. circular and straight; it is then possible to, for example, construct polar grid around a circular boundary, and rectilinear elsewhere, with some overlapping (see Fig. 1.2). This approach has been used by eg. Launder and Massey (1978) and Fujii et al (1984) for study of flow in tube bank assemblies. It possesses several shortcomings, namely the necessity for separate solutions on each subgrid, the need for the extensive interpolation to define the internal boundary values for each

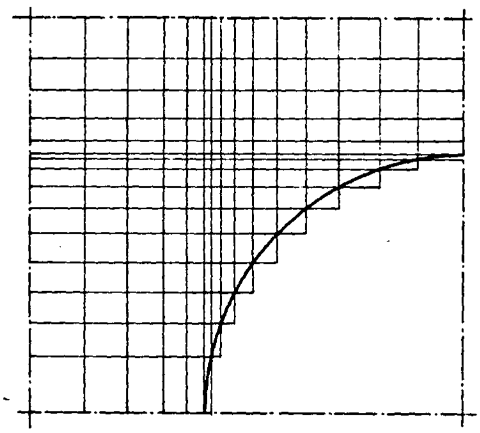


Fig. 1.1 Stepwise approximation of curved boundaries

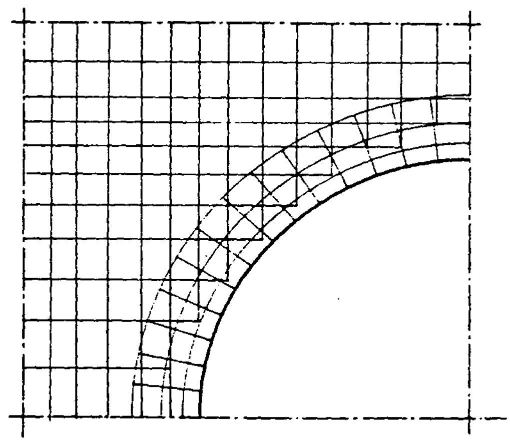


Fig. 1.2 A composite grid

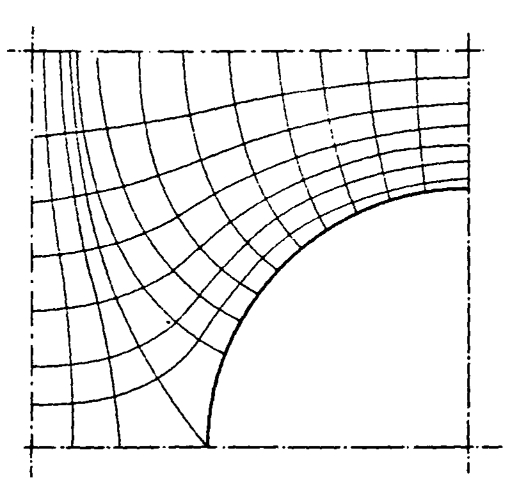


Fig. 1.3 Curvilinear orthogonal grid

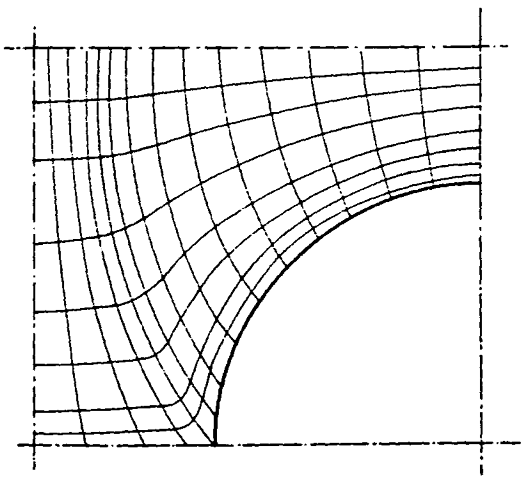


Fig. 1.4 Non-orthogonal grid

subgrid where they overlap, and the fact that different computer programs need be written for different problems.

The second mentioned approach utilizes general curvilinear orthogonal coordinates. The transport equations become somewhat more complex, but their discretised form is still very similar to that resulting from the use of Cartesian coordinates. The main shortcomings of such methods are: (i) the need to generate the orthogonal grid, which is sometimes difficult and time consuming, especially in three dimensions, and (ii) the limited control allowable on the distribution of the grid lines: in particular, in order to avoid intersection of neighbouring grid lines in one coordinate direction, excessive spacing between them may be required in some parts of the solution domain, which may result in poor resolution of the flow there (eg. near boundary "discontinuities", or corners, as illustrated in Fig. 1.3). Although not general, this approach extended the applicability of numerical solution methods to a wider class of problems. More details can be found in works of, eg. Pope (1978) and Antonopuolos (1979).

Methods which employ non-orthogonal grids do not suffer from the drawbacks mentioned earlier. Quite arbitrary shapes of solution domains can be tackled, and the grid properties can be controlled to much greater extent (see Fig. 1.4). Here a distinction can be made between methods which employ grid-oriented velocity components, and methods which use Cartesian velocity components. In the first-named group (eg. methods developed by Demirdžić et al, 1980 and Demirdžić, 1982) the curvature of grid lines plays an important role, in that smoothness and moderate rate of change of curvature from node to node is essential for the accuracy of the flow calculations, which is an important drawback. Another inconvenience is that the differential equations become considerably more complex than in the two previous approaches, especially in three dimensions. The second-named group has the advantage that, although the differential equations have an increased number of terms as compared to the Cartesian ones due to the coordinate transformation, they do not contain the undesirable curvature terms and thus are easier to discretise and less sensitive to grid properties. Various numerical solution methods based on this type of grid and working variables have been developed so far (eg., Hirt et al, 1974; Rhie, 1981; Hah, 1983, 1984; Vanka et al, 1980; Nakayama, 1985). The present method also falls within this group. More details about this class of solution methods will be given in Chapter 5.

It should be also noted here that finite element methods, which dominate the scene in solid mechanics, are being more and more used for fluid flow predictions. Their main feature, the ability to handle arbitrarily shaped solution domains, has long been held to be a unique advantage over finite volume (difference) methods. However, the recent development of versions of the latter methods employing non-orthogonal grids has shown this advantage not to be real. On the other hand, some serious problems exist with Galerkin finite element methods, which led to the development of a control volume version of a finite element method for two-dimensional fluid flow (Baliga and Patankar, 1983). In general, finite element methods place higher demands on computer resources, especially in three-dimensional cases. More details about this class of solution methods can be found in Schneider et al (1978), Chung (1978) and other references cited in Baliga and Patankar (1983).

1.3 Present Contributions

The present study contributes to the computational fluid dynamics field in the following areas:

(i) Further evaluation of the various forms of the differential equations which describe viscous fluid flow in general coordinates, along the lines of the earlier study of Demirdžić (1982). It is shown that the strong conservation form of governing equations, with vector and tensor components expressed through the arbitrary, but spatially constant base vectors are to be preferred.

(ii) Derivation and discretisation of the governing transport equations, written in terms of Cartesian vector and tensor components and arbitrary non-orthogonal coordinates. The finite volume approach is used for the discretisation which has been performed in physical space, in a novel way which minimizes the requirement for geometrical information about the grid, and the sensitivity to same.

(iii) Evaluation of various differencing schemes with respect to flow calculations on arbitrary three-dimensional grids. Five common differencing schemes for convective terms, and two for diffusion terms, have been adapted for use in the present method and their properties have been examined. It has been found that the unconditional boundedness is diffcult to guarantee when non-orthogonal grids are used, due to the presence of cross-derivative diffusion terms. Also the use of higher-order schemes for convective terms is found to violate the boundedness requirement under certain conditions, due to the negative coefficients in the final discretised equations.

(iv) Development of a new flux-blending method to remedy the unboundedness of the higher-order scheme solutions. Suitable boundedness criteria for convection and diffusion contributions are defined, and in regions where these criteria are not met the blending of lower- and higher-order fluxes is performed in proportions which guarantee boundedness of the solution.

(v) Evaluation of various alternative dependent variable arrangements on non-orthogonal grids. It is shown that the non-staggered arrangement is the simplest and most desirable, but all of arrangements studied may result in pressure-velocity decoupling when Cartesian velocity components are used, unless special care is taken.

(vi) Development of an interpolation practice which enables the use of the non-staggered variable arrangement. This is based on an idea first used by Rhie (1981), and ensures that no decoupling between the pressures and velocity occurs.

(vii) Evaluation of various pressure-derivation algorithms. Four available algorithms for the derivation of pressure have been adapted for use in conjuction with the non-staggered variable arrangement, and their relative merits have been examined. It is shown that one commonly-employed algorithm can be adapted for calculations on non-orthogonal grids in a way which makes it economical when the non-orthogonality of grid lines is not too severe.

(viii) Derivation of a relation between the optimum under-relaxation factors for velocity and pressure. It is shown that in the chosen algorithm the under-relaxation factor for pressure can be related to the under-relaxation factor for velocity components to assist in optimizing convergence rates. This has been demonstrated on predictions of several recirculating two-dimensional flows, where up to five-fold savings on computational effort have been obtained by using the optimized factors as compared to the use of standard ones found in the literature.

(ix) Prediction of several two- and three-dimensional complex flows and comparisions with experiment. The present method has been embodied in two computer programs: for two-dimensional, and for three-dimensional problems respectively. The performance of these is first assessed by predicting some simple flows for which solutions are known. Predictions are then made of some complex, laminar and turbulent recirculating flows of practical engineering interest, for which experimental results exist. Some examples of other applications are also given.

1.4 Contents of the Thesis

In Chapter 2 the discussion of various forms of governing differential equations of fluid flow in general coordinates is carried out, and the relative merits of the strong, semi-strong and weak conservation form of the differential equations are pointed out. Based on this discussion the choice of vector and tensor components and the form of governing equations to be used in the present study is then made. Next, the time-averaged equations of turbulent flow, and the k-ε turbulence model adopted for use in the present study, are described.

In Chapter 3 the governing differential equations are presented one by one in an expanded form which is suitable for discretisation in the finite volume fashion, and then so discretised. The evaluation of the key components of these equations, namely the convection, diffusion and source terms is then outlined, in a general way which allows for different differencing schemes (interpolation practices) to be later introduced and evaluated. Next, the form of the discretised (algebraic) equations to be solved, the general structure of the coefficient matrix resulting from various possible interpolation practices, and the ways of solving these equations are discussed. The notion of under-relaxation as a means of stabilizing the iterative solution process and improving the rate of convergence is also introduced there.

Chapter 4 deals with various interpolation (differencing) schemes which may be used to express the cell fluxes in terms of the neighbouring nodal values. First the desired properties of the differencing schemes are outlined. This is followed by the evaluation of five common schemes for the convective terms, where it is found that all so called "higher-order" schemes are prone to generate unbounded solutions under certain conditions. Next, two possible interpolation practices for diffusion terms are outlined, and their properties are also discussed. Special attention is paid to the cross-derivative terms and the effects they have on the numerical solution procedure. This is followed by a brief review of existing methods for overcoming unboundedness problems associated with the higher-order differencing schemes, and the presentation of a new flux-blending method, developed here for that purpose.

In Chapter 5 various possible grid and variable arrangements are dicussed. A non-staggered arrangement is adopted, and an interpolation practice for the discretisation of the continuity equation which avoids otherwise possible decoupling problems is introduced, which is inspired by the work of Rhie (1981) and Rhie and Chow (1982). Next some of the available methods for pressure derivation are adapted to the non-staggered variable arrangement and non-orthogonal grids, and then examined from the point of view of efficiency and economy of the overall solution process.

In Chapter 6 the final choices are made about the form of the discretised equations to be solved, the differencing schemes to be embodied and the pressure-derivation algorithm to be used. This is followed by a summary of the overall solution algorithm and associated convergence criteria.

Chapter 7 deals with the grid and associated geometrical parameters featuring in the discretised equations. The desired properties of a computational grid are first described, followed by a description of some simple grid generation methods which in many cases produce grids with satisfactory properties. Also described are the calculation of cell volumes, and other geometrical properties required for general two- and three-dimensional cases.

In Chapter 8 the present method is assessed by applying it to some simple inviscid, laminar and turbulent flows for which either analytic or well-known empirical solutions exist. The sensitivity of the method to the grid properties, the effects of under-relaxation factors on the rate of convergence, and the performance of various differencing schemes and the new flux blending method on some model problems, are also presented in this chapter.

In Chapter 9 applications of the method to several physically and geometrically complex flows are presented. These include flows in a circular pipe with a sudden contraction, across staggered banks of tubes, around a bluff body confined in a pipe, and in S- and C-shaped diffusers. For these cases some experimental data exists and it is used to further assess the performance of the method. In addition predictions are given of some complex flows for which no experimental data exist: these are shown in order to demonstrate the range of applicability of the present method.

Finally, conclusions and suggestions for future developments are given in Chapter 10.

Additionally, in an appendix the special case of two-dimensional axisymmetric flow is discussed. It is shown that the final form of the discretised equations is similar to that for a plain two-dimensional case, and the evaluation of the extra terms is described.

CHAPTER 2

DIFFERENTIAL EQUATIONS OF VISCOUS FLOW IN GENERAL COORDINATES

2.1 Introduction

When a simple coordinate system is used to express the differential transport equations (eg. Cartesian, polar cylindrical), the choice of vector and tensor components, and therefore the form of governing equations, is obvious. However, if a more complex coordinate system is used, there are different possibilities to choose from, and therefore various forms of governing equations can be obtained. The difference in the form of equations will have a great influence on the process of their discretisation, suitability of the discretised equations for numerical solution methods, computational effort needed to obtain solutions, and at the end on the accuracy of the solutions themselves. Before deciding which form to use, one should therefore first have a look at the relative merits of these different possibilities. This will be the subject of this chapter.

First, in section 2.2, the various forms of governing differential equations that can be obtained in general coordinate systems will be briefly discussed, and references for more detailed studies will be given.

In section 2.3 the choice of vector and tensor components to bo used in this study will be made, based on conclusions derived in the preceding section.

In section 2.4 the final form of governing differential equations adopted for use in this study will be given.

In section 2.5 time-averaged equations of turbulent flow and turbulence closure model will be presented.

Finally, the conclusions will be summarized in section 2.6.

Tensor notation will be used to present equations in their compact form, with superscripts denoting contravariant, and subscripts denoting covariant components. The Einstein summation convention is assumed, ie. matching upper and lower indices are to be summed over their range (1 to 3 for three-dimensional space).

2.2 Discussion of Various Forms of Governing Differential Equations in General Coordinates

2.2.1 Coordinate-free form of governing equations

The basic equations which describe conservation of mass, momentum and scalar quantities can be expressed in the following vector form, which is independent of coordinate system used:

$$\frac{\partial \rho}{\partial t} + \mathrm{div}(\rho\vec{v}) = s_m$$

$$\frac{\partial}{\partial t}(\rho\vec{v}) + \mathrm{div}(\rho\vec{v}\otimes\vec{v} - \vec{\vec{T}}) = \vec{s}_v \tag{2.1}$$

$$\frac{\partial}{\partial t}(\rho\phi) + \mathrm{div}(\rho\phi\vec{v} - \vec{q}) = s_\phi$$

Here the density ρ, velocity vector $\vec{v}$ and scalar quantity ϕ appear as the basic dependent variables, $\vec{\vec{T}}$ is the stress tensor and $\vec{q}$ is the flux vector. The latter two can be expressed in terms of basic dependent variables. The stress tensor is defined, for a Newtonian fluid, as:

$$\vec{\vec{T}} = -(p + \frac{2}{3}\mu\mathrm{div}\vec{v})\vec{\vec{I}} + 2\mu\vec{\vec{D}} \tag{2.2}$$

where p is the pressure, μ is dynamic viscosity of the transporting fluid, $\vec{\vec{I}}$ is the unit tensor of second order, and $\vec{\vec{D}}$ is the deformation (or rate of strain) tensor. The flux vector is usually given by the Fourier-type law:

$$\vec{q} = \lambda \mathrm{grad}\phi \tag{2.3}$$

where proportionality coefficient λ is sometimes called "diffusivity" or "conductivity", and is usually a property of the transporting fluid.

Demirdžić (1982) has conducted an extensive study of the possible forms of governing differential equations when a general, non-orthogonal coordinate system is used. Therefore, there is no need for a repetition of this exercise here; only the most important factors that will influence the choice of the form of equations to be used further in this study will be mentioned here. Readers interested in more details are directed to Demirdžić (1982) and other references (Vinokur, 1974; Anderson et al, 1968; Walkden, 1966; Tuesdell, 1977; Eiseman and Stone, 1980; Sedov, 1971; Sokolnikoff, 1964, etc.)

2.2.2 General coordinate system

Figure 2.1 shows a general, non-orthogonal coordinate system (x^1, x^2, x^3) and the reference Cartesian coordinate system (y^1, y^2, y^3). The latter is uniquely defined by three, mutually orthogonal unit vectors $\vec{i}_i$, directed along coordinate lines y^i. On the other hand, in general coordinate system x^i, several base vectors can be defined,with reference to which other vectors and tensors can be expressed (any set of three linearly independent vectors can form a basis).

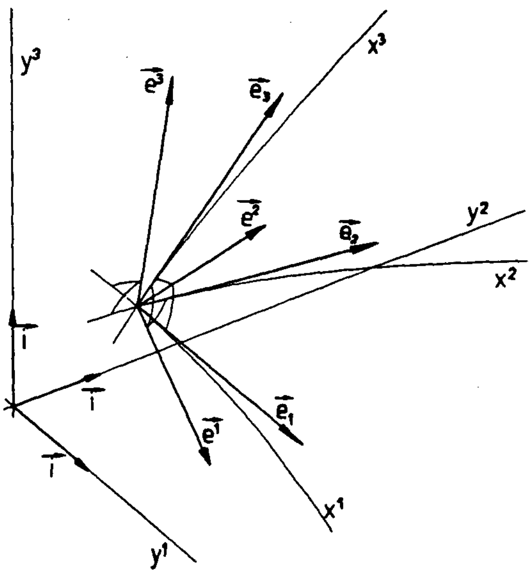


Fig. 2.1: General coordinate system (x^1, x^2, x^3), reference Cartesian frame (y^1, y^2, y^3), and their base vectors

The "natural" base vectors $\vec{e}_i$ are tangential to their coordinate line x^i, and are defined as:

$$\vec{e}_i = \frac{\partial y^m}{\partial x^i}\vec{i}_m \tag{2.4}$$

The "dual natural" base vectors $\vec{e}^{\,i}$ are normal to the coordinate surface of constant x^i, and are defined as:

$$\vec{e}^{\,i} = \frac{\partial x^i}{\partial y^m}\vec{i}^{\,m} \tag{2.5}$$

Both bases are shown in figure 3.1.

These base vectors are normally neither nondimensional, nor unit vectors; a set of nondimensional and unit base vectors in a general coordinate system represent physical natural base vectors, which are obtained by normalizing the natural base vectors, thus:

$$\vec{e}_{(i)} = \frac{\vec{e}_i}{\sqrt{g_{ii}}} \tag{2.6}$$

where $\sqrt{g_{ii}}$ is the magnitude of $\vec{e}_i$, ie. $g_{ii} = \vec{e}_i \cdot \vec{e}_i$. The reciprocal base vectors, defined as

$$\vec{e}^{\,(i)} = \sqrt{g_{ii}}\,\vec{e}^{\,i} \tag{2.7}$$

are also nondimensional, but not unit vectors.

Nondimensional unit base vectors can also be obtained by normalizing dual base vectors, in which case one gets dual physical base vectors:

$$\vec{e}^{\,[i]} = \frac{\vec{e}^{\,i}}{\sqrt{g^{ii}}} \tag{2.8}$$

The reciprocal base can also be defined as:

$$\vec{e}_{[i]} = \sqrt{g^{ii}}\;\vec{e}_i \tag{2.9}$$

These vectors are also nondimensional, but not unit vectors.

Depending on the type of basis used, vectors and tensors can accordingly be expressed in terms of their covariant (a_i and A_{ij}), contravariant (a^i and A^{ij}), physical ($a_{(i)}$, $A_{(ij)}$, $a^{(i)}$, $A^{(ij)}$) or dual physical ($a_{[i]}$, $A_{[ij]}$, $a^{[i]}$, $A^{[ij]}$) components. The essentials of tensor calculus in terms of the covariant and contravariant components can be found in most textbooks about tensors (Sokolnikoff, 1964; Aris, 1962; Sedov, 1971) while tensor calculus in terms of physical components has been presented by Demirdžić (1982).

2.2.3 Strong, semi-strong and weak conservation form of equations

Depending on the way the divergence operator is expressed when a general coordinate system is used, two basic forms of governing equations can result: a "weak conservation" form, in which some terms are not under a differential operator, and a "strong conservation" form, in which all terms arising from the divergence operator are under differential operators. A special case is the momentum equation, which describes the conservation of a vector quantity. To be solved, it has to be resolved into specific directions, which would result in three equations in terms of vector components. However, only if the direction of resolution is spatially invariant, does conservation of a vector mean conservation of its components too. When a strong conservation form of momentum equation is resolved into directions which vary in space (eg., in the direction aligned or normal to curvilinear coordinates), then in the equations for the velocity components undifferentiated terms will appear (centrifugal and Corriolis forces and stress-tensor counterparts $\left\{ {i \atop mj} \right\} \cdot (\rho v^m v^j - T^{mj})$, see Demirdžić, 1982). This form of equation is then said to be a "semi-strong" conservation form.

While all forms of differential equations describe the same laws and from mathematical point of view any would do, it is not irrelevant which form is used when numerical solution is attempted. When the differential equations are integrated over a finite number of control volumes and properly discretised, the resulting fluxes from the strong conservation form would cancel in pairs at all interior control volume (cell) faces when summed, so that only boundary fluxes remain. This guarantees overall conservation of the transported quantity (Roache, 1976), and for that reason such a form of equations is called a "strong conservation form".

On the other hand, when a weak conservation form of equations is integrated over control volumes, there remain undifferentiated terms which do not go into cell face fluxes , but are associated with cell center, and there is no obvious discretisation procedure which would guarantee overall conservation; hence the "weak conservation" name for such equations.

It is not only the fact that the weak conservation form of equations has some "non-conservative" terms, that causes worries to a numerical analyst; the nature of those terms is also of considerable concern. Specifically, these terms are all related to Christoffel symbols, which include what is commonly called "curvature terms", ie. the second derivatives of coordinates. These are very sensitive to grid smoothness, and difficult to calculate accurately by a numerical procedure; if strong curvature along any coordinate line is present, many nodes are needed on the numerical grid to assure reasonable accuracy of numerically calculated curvature terms.

From this brief discussion one can conceive that the strong conservation form of governing differential equations is to be preferred. It can be achieved if the base vectors at any point are expressed in terms of arbitrary, but spatially constant base vectors, eg.:

$$\vec{e}_i = \gamma_i^m \vec{f}_m \tag{2.9}$$

It can also be achieved by expressing vector and tensor components in terms of such constant base vectors, eg.:

$$\vec{v} = v^i \vec{f}_i \quad , \quad \vec{\vec{T}} = T^{ij} \vec{f}_i \otimes \vec{f}_j \tag{2.10}$$

The first method has a drawback in that the resulting individual momentum equations do not contain a dominant velocity component, which would make numerical solution difficult and expensive (instead of solving weakly coupled equations for velocity components by iterating for each in turn, one would have to solve the whole set simultaneously). The alternative way does not suffer from this problem, and has another advantage that stress tensor terms are simpler, because their components are expressed in terms of vectors which have fixed direction, and therefore contain no curvature terms.

The second approach seems to be superior, and the obvious choice would be to use Cartesian base vectors $\vec{i}_i$ as a fixed basis. That this approach, in spite of its advantages, has not been used very much in the past is due to some numerical problems. In such a case the velocity components do not follow the local coordinate directions , and decoupling problems and oscillations have usually been encountered when standard numerical solution techniques were used (Hirt et al, 1974; Rhie and Chow, 1982; Demirdžić, 1982). The arrangement of the variables on a numerical grid and the way of deriving a pressure equation are crucial for effective numerical solution if this form of equations is used. These matters will be addressed in more detail in Chapter 5.

Another approach to obtain (an apparent) strong conservation form of governing equations would be to express base vectors (eg. $\vec{e}_i$) in terms of locally fixed base (eg. $\vec{e}_{iP}$), in the centre of each computational cell, as done by Vinokur (1974), or to express vector and tensor components in the terms of such a locally fixed basis. Although this approach results in relatively simple form of equations, their strong conservation form is illusory, as now a discontinuity problem on subdomain boundaries arises, since the basis on one side is not the same as on the other. To account for this, one would have to introduce some extra terms, which would represent nothing else but the apparently avoided curvature terms!

2.3 Choice of Vector and Tensor Components for Use in this Study

As was mentioned in the previous section, vectors and tensors in a general coordinate system can be expressed in terms of as many types of components as there are types of base vectors that could be used. Only when nondimensional base vectors are used (physical or dual physical) will all components bare the same dimensions and physical significance as vectors and tensors themselves. For example, in plane polar coordinates the contravariant velocity components are $v^r = \frac{dr}{dt}$, $v^\theta = \frac{d\theta}{dt}$, and covariant ones are $v_r = \frac{dr}{dt}$, $v_\theta = r^2 \frac{d\theta}{dt}$; both v^θ and v_θ have dimensions which do not correspond to what is normally understood as a "velocity component". This makes understanding of results obtained by using such components more difficult, as they do not behave in an "expected" manner. For example, the above mentioned v_r and v_θ change (from point to point) in space even if a physically constant, uniform parallel flow field is being considered. Moreover, v^θ is, for given θ , equal to constant/r , and it tends to infinity as r tends to zero. It is not only difficult to appreciate what is the actual behaviour of the flow by looking at results in terms of such non-physical components, but they can also introduce additional inaccuracies in the numerical solution (eg., in the above example near $r = 0$).

The physical contravariant components ($a^{(i)}$ and $A^{(ij)}$), which are related to nondimensional unit base vectors $\vec{e}_{(i)}$, do not suffer from such problems, neither do the covariant dual physical components ($a_{[i]}$ and $A_{[ij]}$), which are related to normalized dual base vectors, $\vec{e}^{[i]}$. The first-mentioned ones were successfully used by Demirdžić (1982) in numerical predictions of two-dimensional flows in complex geometries.

However, the simplest equations result when vector and tensor components are referred to a fixed base vectors, and this is the only way of achieving the desired strong conservation form of the governing equations. The obvious choice is to use orthogonal, Cartesian base vectors $\vec{i}_i$, in which case there is no difference between covariant and contravariant, physical and non-physical components. Such representation of vectors is easy to understand; the magnitudes of vector and its components are related by the simple Pythagora's rule, and since most often measurements are made of Cartesian components, comparison of numerical results and experimental data is made easier. Implementation of boundary conditions is also easier in the case of Cartesian components.

Having all this in mind, it has been decided to express vector and tensor components with reference to the Cartesian base vectors for the purpose of this study. The resulting equations are far simpler for numerical integration in three dimensions than any other form, and they contain the minimum necessary information about the coordinate frame. This will aviod problems of sensivity of solutions to grid properties, possessed by methods which employ curvilinear (grid oriented) vector components, such as that of Demirdžić (1982). Numerical problems, associated with particular solution algorithms when Cartesian components are used on general grids, which have prevented their wider use so far, have been successfully overcome in this study. Details about this will be given in Chapter 5.

2.4 The Form of Governing Differential Equations Adopted for Use in this Study

Since Cartesian components of vectors and tensors will be used from now on, there is, as already noted, no difference between contravariant and covariant ones. The equations would be different only in form, if one strictly used covariant or contravariant notations; however, since in essence the difference does not exist, strictness can be omitted, and hereafter indices will be placed conveniently.

The compact form of differential equations, which describe transport of mass, momentum and an arbitrary scalar quantity over general coordinate system can, in case of steady, incompressible, subsonic flow (the only type of flows considered here), be written as (Demirdžić, 1982):

$$\frac{\Delta}{\Delta x^j}(\rho u_m \alpha^{mj}) = s_m \tag{2.11}$$

$$\frac{\Delta}{\Delta x^j}\left[(\rho u_m u_i - T_{mi})\alpha^{mj}\right] = s_i^u \tag{2.12}$$

$$\frac{\Delta}{\Delta x^j}\left[(\rho u_m \phi - q_m)\alpha^{mj}\right] = s_\phi \tag{2.13}$$

Here, the differential operator $\Delta/_{\Delta x}{}^j$ is defined as (see Demirdžić, 1982):

$$\frac{\Delta}{\Delta x^j}(\ldots) = \frac{1}{\sqrt{g}}\frac{\partial}{\partial x^j}[\sqrt{g}(\ldots)] \tag{2.14}$$

where g is the determinant of metric tensor, $g=J^2$, and J is the Jacobian of coordinate transformation $y^i=y^i(x^j)$ (see Fig. 2.1). In above expressions the α's represent the projections of the dual base vectors of the general coordinate frame $\vec{e}^{\,j}$ onto the Cartesian base vectors $\vec{i}_i$, ie. their Cartesian components:

$$\alpha_i^j = \alpha^{ij} = \vec{i}_i \cdot \vec{e}^{\,j} = \vec{i}^{\,i} \cdot \vec{e}^{\,j} = \frac{\partial x^j}{\partial y^i} \tag{2.15}$$

The stress tensor and the flux vector are defined as:

$$T_{mi} = -p\delta_{mi} + \tau_{mi} = -p\delta_{mi} + \mu\left(\frac{\partial u_i}{\partial x^n}\alpha_m^n + \frac{\partial u_m}{\partial x^l}\alpha_i^l\right) \tag{2.16}$$

$$q_m = \Gamma_\phi \frac{\partial \phi}{\partial x^n}\alpha_m^n \tag{2.17}$$

where δ_{mi} is the Kronecker's delta (1 if m=i, and 0 otherwise), and τ_{mi} is the anisotropic part of the stress tensor. All terms on the left-hand side of Eqs. (2.11)-(2.13) are contained under the differential operator, and therefore they represent a desired strong conservation form of governing equations, which, as already noted, is important for finite volume method of numerical solution adopted here. These equations do not contain undesirable "curvature terms"; the momentum equations do contain three pressure terms, but as will be seen later, this does not pose any difficulty to the present solution algorithm. These terms are far simpler for numerical calculation than extra stress terms in semi-strong and weak conservation form of governing equations when vectors and tensors are related to variable base vectors (see Walkden, 1966; Demirdžić et al, 1981; Demirdžić, 1982).

2.5 Time-Averaged Equations of Turbulent Flow and Turbulence Closure Model

It has been said earlier that the subject of present investigation is the steady flow of a fluid at low Mach numbers. However, it is well known that only laminar flows (at low Reynolds numbers) can be literally "steady"; high Reynolds number flows can be said to be steady on an average basis only, since small scale high frequency fluctuations are always present (Launder & Spalding, 1974). Therefore, unless a special treatment is applied, the full three-dimensional time-dependent form of the equations of motion would have to be solved, and a very fine numerical grid (with spacing smaller than the length scale of smallest turbulent eddies) would have to be employed. This is not a very practical approach, although it is being employed for fundamental studies of relatively simple flows.

The alternative is to use some sort of averaging to obtain equations for average quantities (eg. time-averaging, see Schlichting, 1979). The time-averaging process consists of expressing each variable through its "mean" value $\overline{\phi}$ and "fluctuating" component ϕ' , thus:

$$\phi = \overline{\phi} + \phi' \tag{2.18}$$

It is justifiable, for non-reacting incompressible turbulent flows, to neglect fluctuations of fluid viscosity and density. In this case, after introducing substitutions of the type of (2.18) for the dependent variables of equations (2.11), (2.12) and (2.13) and time-averaging, there results the following set of equations:

$$\frac{\Delta}{\Delta x^j} (\rho \overline{u}_m \alpha^{mj}) = \overline{s}_m \tag{2.19}$$

$$\frac{\Delta}{\Delta x^j} \{ [\rho \overline{u}_m \overline{u}_i - (\overline{T}_{mi} - \rho \overline{u'_m u'_i})] \alpha^{mj} \} = \overline{s}^u_i \tag{2.20}$$

$$\frac{\Delta}{\Delta x^j}\{[\rho\bar{u}_m\bar{\phi} - (\bar{q}_m - \rho\overline{u'_m\phi'})]\alpha^{mj}\} = \bar{s}_\phi \tag{2.21}$$

These are similar to their instantaneous counterparts (equations (2.11), (2.12) and (2.13), with the addition of time-derivative), if the instantaneous quantities in the latter are replaced by the mean ones, with the exception of extra terms appearing in scalar and momentum equations, which contain mean products of fluctuating components. These are usually called the Reynolds stresses:

$$T^t_{mi} = -\rho\overline{u'_m u'_i} \tag{2.22}$$

and turbulent scalar fluxes:

$$q^t_m = -\rho\overline{u'_m\phi'} \tag{2.23}$$

To close the system of equations (2.19)-(2.21) it is necessary either to relate these extra terms directly to the mean quantities, or to supply additional equations from which they can be derived. A wide variety of "turbulence models" exists for the latter purpose. It is beyond the scope of this study to review all of them; such reviews can be found in the literature, and readers interested in more details are directed to references (Launder and Spalding, 1972; Hanjalić and Launder, 1972; Jones and Launder, 1972; Rodi, 1982, and others).

For this study the most widely used model is adopted, namely the so called "k-ε" model of turbulence (Hanjalić, 1970; Jones and Launder, 1972). This is based on the mixing lenght theory (Bousinesq, 1877; Prandtl, 1925) and provides the following relations for the Reynolds stresses and turbulent scalar fluxes:

$$-\rho\overline{u'_m u'_i} = \mu_t\left(\frac{\partial\bar{u}_i}{\partial x^n}\alpha^n_m + \frac{\partial\bar{u}_m}{\partial x^l}\alpha^l_i\right) - \frac{2}{3}k\,\delta_{mi} = \tau^t_{mi} - P_k\delta_{mi} \tag{2.24}$$

$$-\rho\overline{u'_m\phi'} = \Gamma^t_\phi\,\frac{\partial\bar{\phi}}{\partial x^n}\alpha^n_m \tag{2.25}$$

where the turbulent viscosity μ_t and diffusivity coefficient Γ^t_ϕ are defined as:

$$\mu_t = C_\mu\rho\frac{k^2}{\varepsilon} \tag{2.26}$$

$$\Gamma^t_\phi = \frac{\mu_t}{\sigma^t_\phi} \qquad (2.27)$$

and the parts of turbulent normal stresses $\frac{2}{3}k\delta_{ij}$ are normally incorporated into pressure. Here, k stands for the kinetic energy of turbulence, and ε stands for the rate of dissipation of that energy. They are defined as:

$$k = \frac{1}{2}\overline{u'_i u'_i} \qquad (2.28)$$

$$\varepsilon = \frac{\mu}{\rho}\overline{\left(\frac{\partial u'_i}{\partial x^n}\alpha^n_j\right)\cdot\left(\frac{\partial u'_i}{\partial x^n}\alpha^n_j\right)} \qquad (2.29)$$

The quantities C_μ and σ^t_ϕ are empirical constants, derived by reference to detailed measurements of well-established turbulent flows (see Launder et al, 1972; Launder and Spalding, 1972).

The values of k and ε appearing in the definition (2.26) of μ_t are derived from the momentum equations with the aid of assumptions and empiricism (see Hanjalić, 1970). The left-hand sides of equations for k and ε (see Launder and Spalding, 1974) are the same as that for a general scalar quantity (ie. equation (2.13)), when k and ε are substituted for ϕ, and $\frac{\mu_t}{\sigma_k}$ and $\frac{\mu_t}{\sigma_\varepsilon}$ for Γ_ϕ. The respective source terms s_ϕ take the following form:

$$s_k = G - \rho\varepsilon \qquad (2.30)$$

and

$$s_\varepsilon = C_1\frac{\varepsilon}{k}G - C_2\rho\frac{\varepsilon^2}{k} \qquad (2.31)$$

where C_1 and C_2 are further empirical coefficients, and G is the rate of production of turbulent kinetic energy, defined as:

$$G = \tau^t_{ij}\frac{\partial \bar{u}_i}{\partial y^j} = \mu_t\left(\frac{\partial \bar{u}_i}{\partial x^n}\alpha^n_j + \frac{\partial \bar{u}_j}{\partial x^m}\alpha^m_i\right)\left(\frac{\partial \bar{u}_i}{\partial x^n}\alpha^n_j\right) \qquad (2.32)$$

The values of empirical coefficients (C_μ, C_1, C_2, σ_k and σ_ε) used in this study were taken from Launder and Spalding (1974) and are given in Table 2.1.

C_μ	C_1	C_2	σ_k	σ_ε
0.09	1.44	1.92	1.0	1.3

Table 2.1: The values of empirical coefficients in the k-ε model of turbulence

When (2.24) and (2.25) are introduced into momentum and scalar equations, (2.20) and (2.21) respectively, they reduce to the same form as that of equations (2.12) and (2.13), where now μ and Γ_ϕ are replaced by

$$\mu_{eff} = \mu + \mu_t \tag{2.33}$$

and

$$\Gamma_{\phi,eff} = \Gamma_\phi + \Gamma_\phi^t \tag{2.34}$$

This is one of the attractive features of the k-ε model of turbulence - the equations of laminar and turbulent flow have the same form; only the diffusivity coefficients Γ_ϕ are differently calculated.

Therefore, equations (2.11) - (2.13) will be used to describe transport of mass, momentum and arbitrary scalar quantity in both laminar and turbulent flows. The overbar denoting mean quantities is no longer needed, and hereafter it will be omitted, as well as the suffix "$_{eff}$" in (2.33) and (2.34).

2.6 Closure

In this chapter a brief discussion of the various possible forms of transport equations in general coordinates has been conducted. It has been recognized that, from the point of view of a numerical solution method, a strong conservation form of governing equations is superior to other alternatives (semi-strong and weak conservation form). It has also been recognized that the best way of achieving a strong conservation form of equations is to express vectors and tensors through spatially invariant, nondimensional unit base vectors. The choice has fallen onto Cartesian base vectors, since in that case the simplest form of governing equations results, and all problems of choice of the type of components (ie. covariant, contravariant, physical or non-physical) become redundant. Discretisation of this form of equations is also the easiest possible, as it requires the minimum information about the coordinate transformation (no curvature terms etc.), and thus the necessity of having a smooth grid is relaxed.

For the case of turbulent flow, time-averaging and the k-ε turbulence model have been used to derive transport equations in terms of mean quantities. These have the same form as those for the instantaneous quantities (with due modification to the diffusion coefficients as in Eqs. (2.33) and (2.34)), but of course entail the solution of two additional differential equations for k and ε.

The discretisation procedure and the numerical solution method for these equations will be presented in the next four chapters.

CHAPTER 3

DISCRETISATION OF GOVERNING DIFFERENTIAL EQUATIONS

3.1 Introduction

In the previous chapter a choice has been made about the vector and tensor components to be used in this study, and the final form of governing differential equations resulting from that choice has been given. This chapter deals with discretisation of those equations using the finite volume technique, in order to obtain a form suitable for numerical solution.

First in section 2, the expanded form of governing differential equations for three-dimensional problems is given. This will be the starting point for discretisation procedure, which is outlined in section 3.

Section 4 deals with the treatment of the various terms, that can be distinguished in the discretised equations, namely those describing convection, diffusion and sources.

In section 5 the general form of algebraic equations, to which discretised and linearized transport equations reduce, will be shown and discussed. Special attention will be paid to the structure of the resulting coefficient matrix, and the influence it has on the choice of solution method.

Section 6 gives the closing remarks on the topics of this chapter.

3.2 Expanded Form of Governing Differential Equations

Firstly, for the sake of easier subsequent manipulation, the coefficients α , featuring in Eqs. (2.11) to (2.13), will hereafter be expressed as:

$$\alpha^{ji} = \alpha^{i}_{j} = \frac{1}{J}\beta^{i}_{j} = \frac{1}{J}\beta^{ji} \qquad (3.1)$$

where the β's are defined as:

$$
\begin{aligned}
\beta_1^1 &= \left(\frac{\partial y^2}{\partial x^2}\frac{\partial y^3}{\partial x^3} - \frac{\partial y^2}{\partial x^3}\frac{\partial y^3}{\partial x^2}\right)\\
\beta_2^1 &= -\left(\frac{\partial y^1}{\partial x^2}\frac{\partial y^3}{\partial x^3} - \frac{\partial y^1}{\partial x^3}\frac{\partial y^3}{\partial x^2}\right)\\
\beta_3^1 &= \left(\frac{\partial y^1}{\partial x^2}\frac{\partial y^2}{\partial x^3} - \frac{\partial y^1}{\partial x^3}\frac{\partial y^2}{\partial x^2}\right)\\
\beta_1^2 &= -\left(\frac{\partial y^2}{\partial x^1}\frac{\partial y^3}{\partial x^3} - \frac{\partial y^3}{\partial x^1}\frac{\partial y^2}{\partial x^3}\right)\\
\beta_2^2 &= \left(\frac{\partial y^1}{\partial x^1}\frac{\partial y^3}{\partial x^3} - \frac{\partial y^3}{\partial x^1}\frac{\partial y^1}{\partial x^3}\right)\\
\beta_3^2 &= -\left(\frac{\partial y^1}{\partial x^1}\frac{\partial y^2}{\partial x^3} - \frac{\partial y^2}{\partial x^1}\frac{\partial y^1}{\partial x^3}\right)\\
\beta_1^3 &= \left(\frac{\partial y^2}{\partial x^1}\frac{\partial y^3}{\partial x^2} - \frac{\partial y^3}{\partial x^1}\frac{\partial y^2}{\partial x^2}\right)\\
\beta_2^3 &= -\left(\frac{\partial y^1}{\partial x^1}\frac{\partial y^3}{\partial x^2} - \frac{\partial y^3}{\partial x^1}\frac{\partial y^1}{\partial x^2}\right)\\
\beta_3^3 &= \left(\frac{\partial y^1}{\partial x^1}\frac{\partial y^2}{\partial x^2} - \frac{\partial y^2}{\partial x^1}\frac{\partial y^1}{\partial x^2}\right)
\end{aligned} \tag{3.2}
$$

Obviously, the β_i^j represent the cofactor of $\frac{\partial y^i}{\partial x^j}$ in the Jacobian of the coordinate transformation $y^i = y^i(x^j)$.

The compact form of governing differential equations can now be re-written as follows:

$$\frac{1}{J}\frac{\partial}{\partial x^j}(\rho u_m \beta^{mj}) = s_m \tag{3.3}$$

$$\frac{1}{J}\frac{\partial}{\partial x^j}[(\rho u_m u_i - T_{mi})\beta^{mj}] = s_i^u \tag{3.4}$$

$$\frac{1}{J}\frac{\partial}{\partial x^j}[(\rho u_m \phi - q_m)\beta^{mj}] = s_\phi \tag{3.5}$$

Each of these equations will now be expressed in its expanded form, which will serve as the starting point in discretisation procedure to follow.

3.2.1 Continuity equation

When expanded, the continuity equation (3.3) reads:

$$\frac{\partial}{\partial x^1}U_1 + \frac{\partial}{\partial x^2}U_2 + \frac{\partial}{\partial x^3}U_3 = s_m J \tag{3.6}$$

where the U_i are defined as follows:

$$\begin{aligned} U_1 &= \rho(u_1\beta_1^1 + u_2\beta_2^1 + u_3\beta_3^1) \\ U_2 &= \rho(u_1\beta_1^2 + u_2\beta_2^2 + u_3\beta_3^2) \\ U_3 &= \rho(u_1\beta_1^3 + u_2\beta_2^3 + u_3\beta_3^3) \end{aligned} \tag{3.7}$$

The bracketed expressions on the right-hand side of equations (3.7) for U_i are related to the velocity component normal to the x^i = const. surface.

3.2.2 Scalar equation

The flux vector $\vec{q}$ (Eq.(2.3) or (2.17)) can be now written as:

$$q_m = \Gamma_\phi \frac{\partial\phi}{\partial x^n}\frac{1}{J}\beta_m^n \tag{3.8}$$

or, in expanded form:

$$\begin{aligned} q_1 &= \frac{1}{J}\Gamma_\phi\left(\frac{\partial\phi}{\partial x^1}\beta_1^1 + \frac{\partial\phi}{\partial x^2}\beta_1^2 + \frac{\partial\phi}{\partial x^3}\beta_1^3\right) \\ q_2 &= \frac{1}{J}\Gamma_\phi\left(\frac{\partial\phi}{\partial x^1}\beta_2^1 + \frac{\partial\phi}{\partial x^2}\beta_2^2 + \frac{\partial\phi}{\partial x^3}\beta_2^3\right) \\ q_3 &= \frac{1}{J}\Gamma_\phi\left(\frac{\partial\phi}{\partial x^1}\beta_3^1 + \frac{\partial\phi}{\partial x^2}\beta_3^2 + \frac{\partial\phi}{\partial x^3}\beta_3^3\right) \end{aligned} \tag{3.9}$$

When these expressions are introduced into Eq.(3.5), taking into account (3.7), one gets:

$$\frac{\partial}{\partial x^j}\left(U_j\phi - \frac{\Gamma_\phi}{J}\frac{\partial\phi}{\partial x^m}B_m^j\right) = Js_\phi \tag{3.10}$$

or, in expanded form:

$$\begin{aligned}
&\frac{\partial}{\partial x^1}\left[U_1\phi - \frac{\Gamma_\phi}{J}\left(\frac{\partial\phi}{\partial x^1}B_1^1 + \frac{\partial\phi}{\partial x^2}B_2^1 + \frac{\partial\phi}{\partial x^3}B_3^1\right)\right] + \\
&\frac{\partial}{\partial x^2}\left[U_2\phi - \frac{\Gamma_\phi}{J}\left(\frac{\partial\phi}{\partial x^1}B_1^2 + \frac{\partial\phi}{\partial x^2}B_2^2 + \frac{\partial\phi}{\partial x^3}B_3^2\right)\right] + \\
&\frac{\partial}{\partial x^3}\left[U_3\phi - \frac{\Gamma_\phi}{J}\left(\frac{\partial\phi}{\partial x^1}B_1^3 + \frac{\partial\phi}{\partial x^2}B_2^3 + \frac{\partial\phi}{\partial x^3}B_3^3\right)\right] = J\, s_\phi
\end{aligned} \tag{3.11}$$

In the above equations the B's represent the following products of the β's:

$$B_j^i = \beta_k^j \beta_k^i \tag{3.12}$$

or, in expanded form:

$$\begin{aligned}
B_1^1 &= & &\beta_1^1\beta_1^1 + \beta_2^1\beta_2^1 + \beta_3^1\beta_3^1 \\
B_2^1 &= B_1^2 &=\ &\beta_1^2\beta_1^1 + \beta_2^2\beta_2^1 + \beta_3^2\beta_3^1 \\
B_3^1 &= B_1^3 &=\ &\beta_1^3\beta_1^1 + \beta_2^3\beta_2^1 + \beta_3^3\beta_3^1 \\
B_2^2 &= & &\beta_1^2\beta_1^2 + \beta_2^2\beta_2^2 + \beta_3^2\beta_3^2 \\
B_3^2 &= B_2^3 &=\ &\beta_1^3\beta_1^2 + \beta_2^3\beta_2^2 + \beta_3^3\beta_3^2 \\
B_3^3 &= & &\beta_1^3\beta_1^3 + \beta_2^3\beta_2^3 + \beta_3^3\beta_3^3
\end{aligned} \tag{3.13}$$

The additional complexity of Eq.(3.11), compared to its Cartesian counterpart (obtainable from (3.11) by replacing U_i, x^j and B_j^i with u_i, y^j and δ_j^i respectively), is obvious: cross-derivatives have emerged, and coefficients B contain many terms. However, all these terms are relatively simple to treat numerically, as they involve only the first derivatives of the grid coordinates.

3.2.3 Momentum equation

The stress tensor (Eq.(2.16)) can now be written in the following form:

$$T_{mi} = -p\delta_{mi} + \frac{1}{J}\mu\left[\frac{\partial u_i}{\partial x^n}\beta^n_m + \frac{\partial u_m}{\partial x^l}\beta^l_i\right] \tag{3.14}$$

or, when expanded:

$$\begin{aligned}
T_{11} &= -p + \frac{2\mu}{J}\left(\frac{\partial u_1}{\partial x^1}\beta^1_1 + \frac{\partial u_1}{\partial x^2}\beta^2_1 + \frac{\partial u_1}{\partial x^3}\beta^3_1\right)\\
T_{12} = T_{21} &= \frac{\mu}{J}\left[\left(\frac{\partial u_1}{\partial x^1}\beta^1_2 + \frac{\partial u_1}{\partial x^2}\beta^2_2 + \frac{\partial u_1}{\partial x^3}\beta^3_2\right) + \right.\\
&\qquad \left.\left(\frac{\partial u_2}{\partial x^1}\beta^1_1 + \frac{\partial u_2}{\partial x^2}\beta^2_1 + \frac{\partial u_2}{\partial x^3}\beta^3_1\right)\right]\\
T_{13} = T_{31} &= \frac{\mu}{J}\left[\left(\frac{\partial u_1}{\partial x^1}\beta^1_3 + \frac{\partial u_1}{\partial x^2}\beta^2_3 + \frac{\partial u_1}{\partial x^3}\beta^3_3\right) + \right.\\
&\qquad \left.\left(\frac{\partial u_3}{\partial x^1}\beta^1_1 + \frac{\partial u_3}{\partial x^2}\beta^2_1 + \frac{\partial u_3}{\partial x^3}\beta^3_1\right)\right]\\
T_{22} &= -p + \frac{2\mu}{J}\left(\frac{\partial u_2}{\partial x^1}\beta^1_2 + \frac{\partial u_2}{\partial x^2}\beta^2_2 + \frac{\partial u_2}{\partial x^3}\beta^3_2\right)\\
T_{23} = T_{32} &= \frac{\mu}{J}\left[\left(\frac{\partial u_2}{\partial x^1}\beta^1_3 + \frac{\partial u_2}{\partial x^2}\beta^2_3 + \frac{\partial u_2}{\partial x^3}\beta^3_3\right) + \right.\\
&\qquad \left.\left(\frac{\partial u_3}{\partial x^1}\beta^1_2 + \frac{\partial u_3}{\partial x^2}\beta^2_2 + \frac{\partial u_3}{\partial x^3}\beta^3_2\right)\right]\\
T_{33} &= -p + \frac{2\mu}{J}\left(\frac{\partial u_3}{\partial x^1}\beta^1_3 + \frac{\partial u_3}{\partial x^2}\beta^2_3 + \frac{\partial u_3}{\partial x^3}\beta^3_3\right)
\end{aligned} \tag{3.15}$$

With the above taken into account, the compact form of the momentum equations (2.12) can be written as:

$$\frac{\partial}{\partial x^j}\left[U_j u_i - \frac{\mu}{J}\left(\frac{\partial u_i}{\partial x^m}B^j_m + \frac{\partial u_k}{\partial x^m}\beta^m_i\beta^j_k\right) + p\,\beta^j_i\right] = J\,s^u_i \tag{3.16}$$

In expanded form three equations, for the Cartesian velocity components u_1, u_2 and u_3, are the following:

u_1-component

$$
\begin{aligned}
&\frac{\partial}{\partial x^1}\left[U_1 u_1 - \frac{\mu}{J}\left(\frac{\partial u_1}{\partial x^1}B_1^1 + \frac{\partial u_1}{\partial x^2}B_2^1 + \frac{\partial u_1}{\partial x^3}B_3^1 + \beta_1^1\omega_1^1 + \beta_2^1\omega_1^2 + \beta_3^1\omega_1^3\right) + p\beta_1^1\right] + \\
&\frac{\partial}{\partial x^2}\left[U_2 u_1 - \frac{\mu}{J}\left(\frac{\partial u_1}{\partial x^1}B_1^2 + \frac{\partial u_1}{\partial x^2}B_2^2 + \frac{\partial u_1}{\partial x^3}B_3^2 + \beta_1^2\omega_1^1 + \beta_2^2\omega_1^2 + \beta_3^2\omega_1^3\right) + p\beta_1^2\right] + \\
&\frac{\partial}{\partial x^3}\left[U_3 u_1 - \frac{\mu}{J}\left(\frac{\partial u_1}{\partial x^1}B_1^3 + \frac{\partial u_1}{\partial x^2}B_2^3 + \frac{\partial u_1}{\partial x^3}B_3^3 + \beta_1^3\omega_1^1 + \beta_2^3\omega_1^2 + \beta_3^3\omega_1^3\right) + p\beta_1^3\right] = Js_u^1
\end{aligned}
\tag{3.17}
$$

u_2-component

$$
\begin{aligned}
&\frac{\partial}{\partial x^1}\left[U_1 u_2 - \frac{\mu}{J}\left(\frac{\partial u_2}{\partial x^1}B_1^1 + \frac{\partial u_2}{\partial x^2}B_2^1 + \frac{\partial u_2}{\partial x^3}B_3^1 + \beta_1^1\omega_2^1 + \beta_2^1\omega_2^2 + \beta_3^1\omega_2^3\right) + p\beta_2^1\right] + \\
&\frac{\partial}{\partial x^2}\left[U_2 u_2 - \frac{\mu}{J}\left(\frac{\partial u_2}{\partial x^1}B_1^2 + \frac{\partial u_2}{\partial x^2}B_2^2 + \frac{\partial u_2}{\partial x^3}B_3^2 + \beta_1^2\omega_2^1 + \beta_2^2\omega_2^2 + \beta_3^2\omega_2^3\right) + p\beta_2^2\right] + \\
&\frac{\partial}{\partial x^3}\left[U_3 u_2 - \frac{\mu}{J}\left(\frac{\partial u_2}{\partial x^1}B_1^3 + \frac{\partial u_2}{\partial x^2}B_2^3 + \frac{\partial u_2}{\partial x^3}B_3^3 + \beta_1^3\omega_2^1 + \beta_2^3\omega_2^2 + \beta_3^3\omega_2^3\right) + p\beta_2^3\right] = Js_u^2
\end{aligned}
\tag{3.18}
$$

u_3-component

$$
\begin{aligned}
&\frac{\partial}{\partial x^1}\left[U_1 u_3 - \frac{\mu}{J}\left(\frac{\partial u_3}{\partial x^1}B_1^1 + \frac{\partial u_3}{\partial x^2}B_2^1 + \frac{\partial u_3}{\partial x^3}B_3^1 + \beta_1^1\omega_3^1 + \beta_2^1\omega_3^2 + \beta_3^1\omega_3^3\right) + p\beta_3^1\right] + \\
&\frac{\partial}{\partial x^2}\left[U_2 u_3 - \frac{\mu}{J}\left(\frac{\partial u_3}{\partial x^1}B_1^2 + \frac{\partial u_3}{\partial x^2}B_2^2 + \frac{\partial u_3}{\partial x^3}B_3^2 + \beta_1^2\omega_3^1 + \beta_2^2\omega_3^2 + \beta_3^2\omega_3^3\right) + p\beta_3^2\right] + \\
&\frac{\partial}{\partial x^3}\left[U_3 u_3 - \frac{\mu}{J}\left(\frac{\partial u_3}{\partial x^1}B_1^3 + \frac{\partial u_3}{\partial x^2}B_2^3 + \frac{\partial u_3}{\partial x^3}B_3^3 + \beta_1^3\omega_3^1 + \beta_2^3\omega_3^2 + \beta_3^3\omega_3^3\right) + p\beta_3^3\right] = Js_u^3
\end{aligned}
\tag{3.19}
$$

where $\omega_j^k = \frac{\partial u_k}{\partial x^m}\beta_j^m$, or, in expanded form, for the u_1-component:

$$
\begin{aligned}
\omega_1^1 &= \frac{\partial u_1}{\partial x^1}\beta_1^1 + \frac{\partial u_1}{\partial x^2}\beta_1^2 + \frac{\partial u_1}{\partial x^3}\beta_1^3 \\
\omega_2^1 &= \frac{\partial u_1}{\partial x^1}\beta_2^1 + \frac{\partial u_1}{\partial x^2}\beta_2^2 + \frac{\partial u_1}{\partial x^3}\beta_2^3 \\
\omega_3^1 &= \frac{\partial u_1}{\partial x^1}\beta_3^1 + \frac{\partial u_1}{\partial x^2}\beta_3^2 + \frac{\partial u_1}{\partial x^3}\beta_3^3
\end{aligned}
\tag{3.20}
$$

Similar expressions follow for the u_2 and u_3 components.

As can be seen, the momentum equations are even more complex than the scalar one: in particular, for any component and any direction all three derivatives of all three components are present. There are also three pressure gradients in the equation for each component. However, it is important to note that, apart from having a large number of terms, these equations are of relatively simple form and are easy to discretise. Since equations have a strong conservation form and do not contain any higher-order derivatives (curvature terms), it is expected that the increased number of terms will not significantly affect the accuracy of the numerical solution. These matters will be discussed in more detail in Chapter 8.

3.3 Discretisation Procedure

The differential equations which describe the transport of mass, momentum and an arbitrary scalar quantity were given in their expanded form, suitable for discretisation, in the previous section. In the finite volume approach, which has been adopted here, these equations are integrated over a finite number of control volumes (hereafter referred to as "computational cells", or just "cells"). Using Gauss' theorem* the volume integral terms on the left-hand sides of the equations (2.1) (without time derivative) can be converted into surface integrals over the six "faces" of computational cell (see Fig. 3.1), thus:

$$I = \int_V \mathrm{div}\vec{f}\, dV = \int_{A_e} f_e\, dA - \int_{A_w} f_w\, dA + \int_{A_n} f_n\, dA - \int_{A_s} f_s\, dA + \int_{A_t} f_t\, dA - \int_{A_b} f_b\, dA \qquad (3.21)$$

where, eg. f_e is the component of $\vec{f}$ normal to the "e" face of the cell, and A_e is the area of that cell-face.

Following Eq. (3.21) the transport equations (3.6), (3.11), (3.17), (3.18) and (3.19), integrated over control volume δV_P enclosing node P, can be now written in general form as:

---

* Also known as Ostrogradsky's theorem:

$$\int_V \mathrm{div}\,\vec{f}\, dV = \int_A \vec{f}\vec{n}\, dA$$

where A is the surface enclosing volume V, and $\vec{n}$ is the outward normal to that surface.

$$I_e - I_w + I_n - I_s + I_t - I_b = \int_{\delta V} s \, dV \tag{3.22}$$

where the surface integrals over the six cell faces have been denoted as I_e, I_w, I_n, I_s, I_t and I_b . Each of these terms is made of two distinct

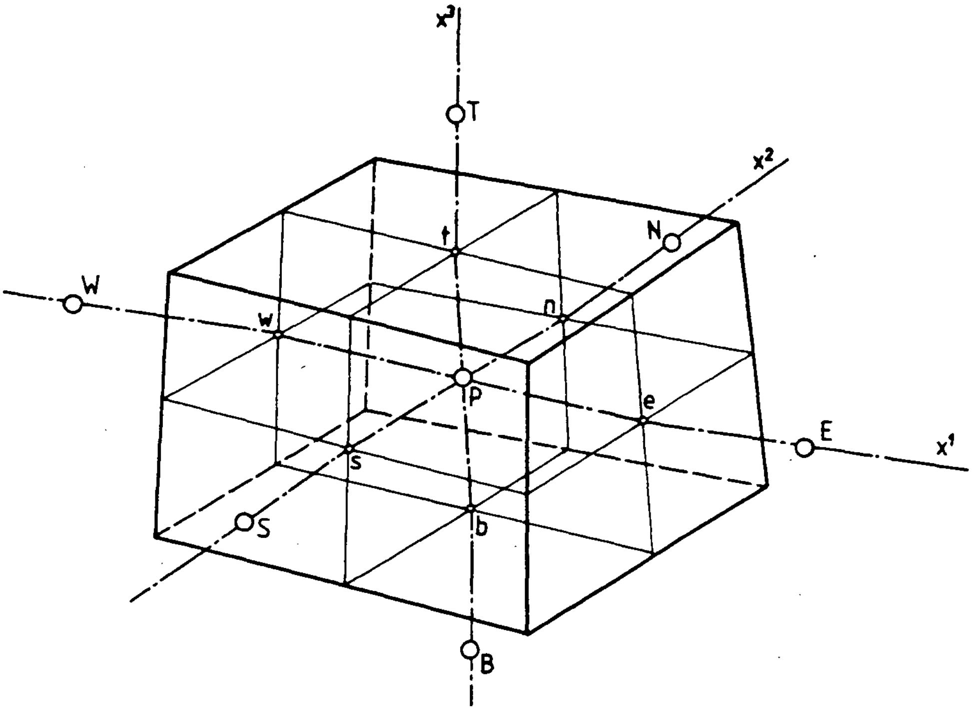


Fig. 3.1: Control volume surrounding computational node P

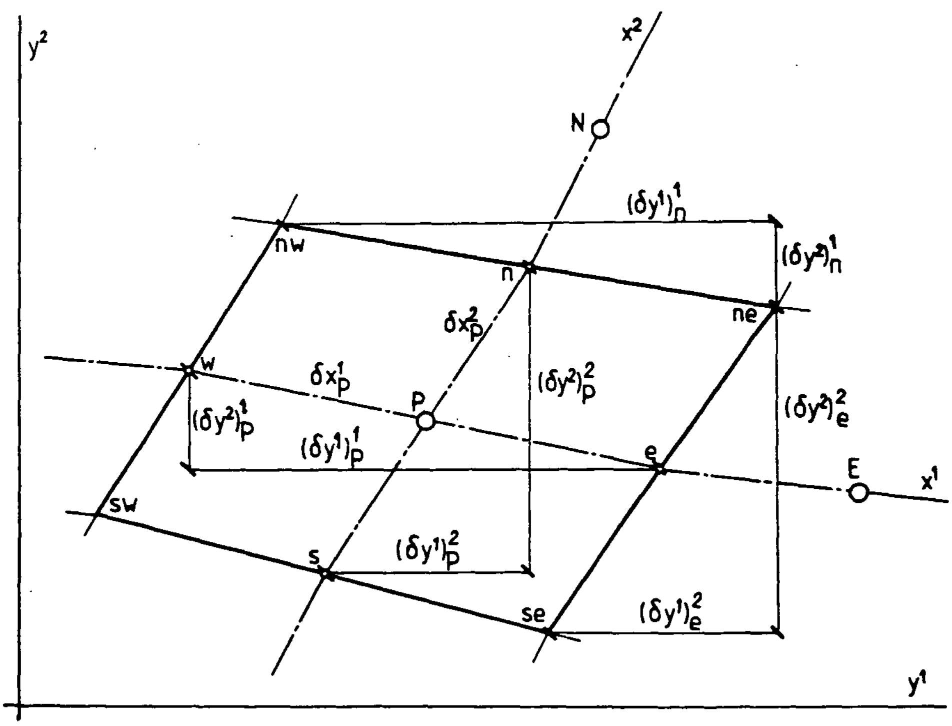


Fig. 3.2: Two-dimensional presentation of some basic geometrical quantities

parts: a convection contribution I^C and diffusion contribution I^D . Before each of them is given more attention, the basic geometrical quantities, entering the coefficients β , will be examined. For ease of presentation in figure 3.2 a two-dimensional projection of x^3=const surface is shown along with graphical representations of the basic geometrical quantities that will be used.

Figure 3.3 shows three cell faces (e, n and t) of a general three-dimensional cell, along with the notation used in this study. The differences in Cartesian coordinates at specific locations on cell faces, which are with the cell volumes the only necessary geometrical information about the grid, can be expressed as follows (see Fig. 3.3):

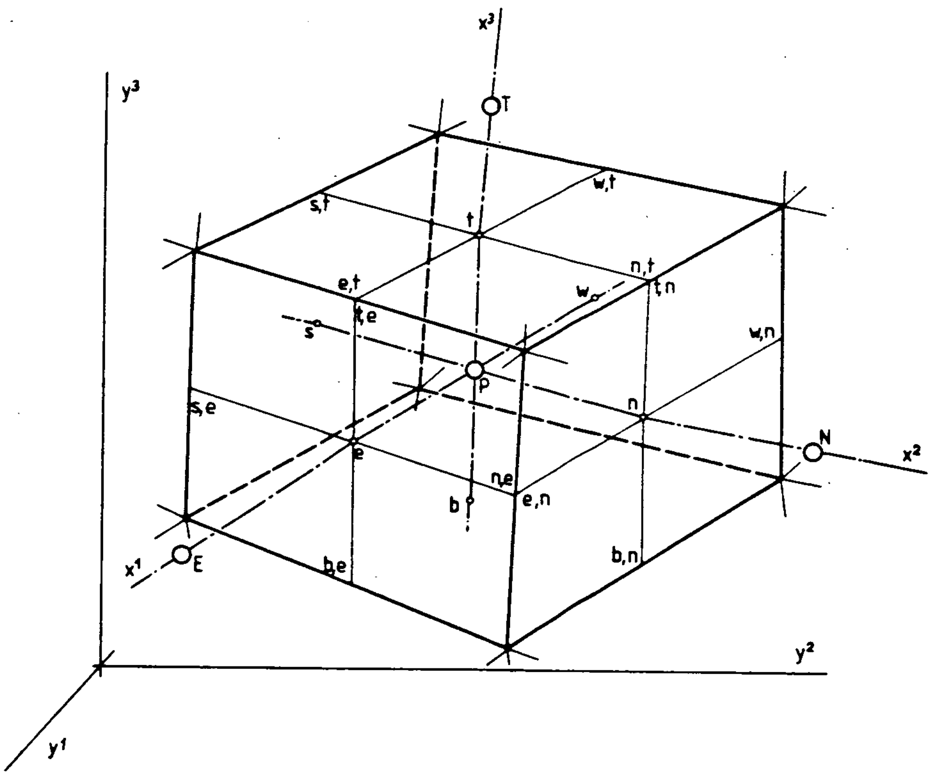


Fig. 3.3: Details of labeling scheme at e, n and t cell-faces

$$(\delta y^1)^1_e = y^1_E - y^1_P \;;\quad (\delta y^1)^2_e = (y^1_n - y^1_s)_e \;;\quad (\delta y^1)^3_e = (y^1_t - y^1_b)_e$$

$$(\delta y^1)^1_n = (y^1_e - y^1_w)_n \;;\quad (\delta y^1)^2_n = y^1_N - y^1_P \;;\quad (\delta y^1)^3_n = (y^1_t - y^1_b)_n \qquad (3.23)$$

$$(\delta y^1)^1_t = (y^1_e - y^1_w)_t \;;\quad (\delta y^1)^2_t = (y^1_n - y^1_s)_t \;;\quad (\delta y^1)^3_t = y^1_T - y^1_P$$

Here, $(\delta y^i)^j_e$ represent the increment in Cartesian coordinate y^i along arbitrary coordinate x^j at the "e" cell face. Similar expressions follow for the w, s and b cell-faces, and for other coordinates too. With computational cells being defined by segments of straight lines joining cell vertices, it follows that all differences in Eqs. (3.23) can easily be calculated if Cartesian coordinates of cell vertices are known.

Now the coefficients β^i_j can be discretised as follows:

$$\beta^1_1 \simeq \frac{1}{\delta x^2 \delta x^3}\left[(\delta y^2)^2(\delta y^3)^3 - (\delta y^2)^3(\delta y^3)^2\right] = \frac{1}{\delta x^2 \delta x^3}\, b^1_1$$

$$\beta^1_2 \simeq \frac{1}{\delta x^2 \delta x^3}\left[(\delta y^1)^3(\delta y^3)^2 - (\delta y^1)^2(\delta y^3)^3\right] = \frac{1}{\delta x^2 \delta x^3}\, b^1_2$$

$$\beta^1_3 \simeq \frac{1}{\delta x^2 \delta x^3}\left[(\delta y^1)^2(\delta y^2)^3 - (\delta y^1)^3(\delta y^2)^2\right] = \frac{1}{\delta x^2 \delta x^3}\, b^1_3$$

$$\beta^2_1 \simeq \frac{1}{\delta x^1 \delta x^3}\left[(\delta y^3)^1(\delta y^2)^3 - (\delta y^3)^3(\delta y^2)^1\right] = \frac{1}{\delta x^1 \delta x^3}\, b^2_1$$

$$\beta^2_2 \simeq \frac{1}{\delta x^1 \delta x^3}\left[(\delta y^1)^1(\delta y^3)^3 - (\delta y^1)^3(\delta y^3)^1\right] = \frac{1}{\delta x^1 \delta x^3}\, b^2_2 \qquad (3.24)$$

$$\beta^2_3 \simeq \frac{1}{\delta x^1 \delta x^3}\left[(\delta y^1)^3(\delta y^2)^1 - (\delta y^1)^1(\delta y^2)^3\right] = \frac{1}{\delta x^1 \delta x^3}\, b^2_3$$

$$\beta^3_1 \simeq \frac{1}{\delta x^1 \delta x^2}\left[(\delta y^2)^1(\delta y^3)^2 - (\delta y^2)^2(\delta y^3)^1\right] = \frac{1}{\delta x^1 \delta x^2}\, b^3_1$$

$$\beta^3_2 \simeq \frac{1}{\delta x^1 \delta x^2}\left[(\delta y^1)^2(\delta y^3)^1 - (\delta y^1)^1(\delta y^3)^2\right] = \frac{1}{\delta x^1 \delta x^2}\, b^3_2$$

$$\beta^3_3 \simeq \frac{1}{\delta x^1 \delta x^2}\left[(\delta y^1)^1(\delta y^2)^2 - (\delta y^1)^2(\delta y^2)^1\right] = \frac{1}{\delta x^1 \delta x^2}\, b^3_3$$

Here, δx^1, δx^2 and δx^3 represent the grid spacings along coordinates x^1, x^2 and x^3 respectively. As will be shown later, the actual values are immaterial here, and one can take it that in the transformed space (x^1, x^2, x^3) all these intervals are equal to unity. However, it is important to note that $J\, dx^1 dx^2 dx^3 = dV$, a differencial volume in physical space. Therefore

$$J \delta x^1 \delta x^2 \delta x^3 \simeq \delta V \qquad (3.25)$$

represents the actual volume of the computational cell. This consideration will lead to disappearance of the coordinate transformation in the final form of discretised equations. The cell volumes used in the calculations will not be determined from (3.25); instead, a different approach will be used, which will be described in Chapter 7.

One should also note here that all nine β's are needed at all six cell faces. The coordinate differences which enter expressions for the b's are easily calculated, as shown in Fig. 3.3.

3.4 Evaluation of Convection, Diffusion and Source Terms

The continuity equation (3.6) contains only convective terms and will be examined first. Here one has to determine values of the U's at all cell faces. By referring back to Eq. (3.7) and taking into account Eqs. (3.23), one arrives at:

$$I_e^C = \int_{A_e} (U_1 dA)_e \simeq F_{1e} = (U_1 \delta x^2 \delta x^3)_e = \rho_e (u_1 b_1^1 + u_2 b_2^1 + u_3 b_3^1)_e \quad (3.26)$$

This represents the mass flux through the "e" cell face (surface). The terms multiplying Cartesian velocity components u_1, u_2 and u_3 (ie. b_1^1, b_2^1 and b_3^1, defined in Eqs. (3.24)), represent projections of the "e"-surface onto Cartesian coordinate planes. One should note here that the flux F_{1e} does not depend on the shape of "e"-surface (which is bounded by four straight lines, joining four vertices); for as is well known the mass flux through any surface bounded by the same closed line would be the same. This justifies the approximation of cell surfaces as two planar triangles, as will be described later in Chapter 7.

On the "w"-surface, F_{1w} is calculated in the same way. Similarily, expressions for the mass fluxes in the other two directions are:

$$F_2 = U_2 \delta x^1 \delta x^3 = \rho(u_1 b_1^2 + u_2 b_2^2 + u_3 b_3^2) \quad (3.27)$$

$$F_3 = U_3 \delta x^1 \delta x^2 = \rho(u_1 b_1^3 + u_2 b_2^3 + u_3 b_3^3) \quad (3.28)$$

This gives the following discretised form of the continuity equation:

$$F_{1e} - F_{1w} + F_{2n} - F_{2s} + F_{3t} - F_{3b} = S_m \quad (3.29)$$

which is a well-known integral formulation of the mass-conservation law. Here S_m represents a volume integral of the volumetric mass source (or sink) over the control volume, and is calculated as:

$$\int_{\delta V} s_m J\, dx^1 dx^2 dx^3 \simeq s_{mP}\delta V_P = S_m \tag{3.30}$$

where s_{mP} is the value of volumetric source at node P , which is taken to represent the whole control volume, and δV_P is the actual volume of the computational cell containing node P .

The convection terms in Eqs. (3.11), (3.17), (3.18) and (3.19) are discretised in an analogous way, eg.:

$$I_e^C = \int_{A_e} U_1 \phi\, dA \simeq F_{1e}\phi_e \tag{3.31}$$

where ϕ_e is, according to the Mean Value Theorem, defined as:

$$\phi_e = \frac{\int_{A_e} U_1 \phi\, dA}{\int_{A_e} U_1 dA} \simeq \frac{\int_{A_e} U_1 \phi\, dA}{F_{1e}} \tag{3.32}$$

Similar expressions follow for the other cell faces and other transported quantities.

There remains to be defined how the mean values ϕ_e, ϕ_n , ϕ_t etc. of the dependent variables at the cell faces will be expressed in terms of values at the discrete nodes, for which a numerical solution is being sought. This will be delt with in the next chapter.

The diffusion-like terms in Eqs. (3.11), (3.17). (3.18) and (3.19) contain the β's and first-order derivatives. The discretisation of the β's , and thereby the B's , has been performed in the previous section. The first derivatives of ϕ in directions "normal" to the cell faces are simply central differenced, eg.:

$$\left(\frac{\partial \phi}{\partial x^1}\right)_e \simeq \frac{\phi_E - \phi_P}{\delta x_e^1} \tag{3.33}$$

This is equivalent to assuming a linear profile in ϕ between nodes P and E (or even quadratic, if "e" is midway between P and E). However, matters are not so clear for cross-derivatives, which are also required at each cell face (see Eq. (3.11)). With reference to Fig. 3.3 , the two cross-derivatives at the "e" cell face can, for example, be approximated as:

$$\left(\frac{\partial\phi}{\partial x^2}\right)_e \simeq \frac{(\phi_n - \phi_s)_e}{\delta x_e^2} \quad ; \quad \left(\frac{\partial\phi}{\partial x^3}\right)_e \simeq \frac{(\phi_t - \phi_b)_e}{\delta x_e^3} \tag{3.34}$$

Values like $\phi_{n,e}$, appearing in the above expressions, are to be expressed in terms of neighbouring nodal values by means of interpolation, as will be discussed in the next chapter.

The integrated and discretised diffusion terms in Eq. (3.11) over "e" cell face now look like:

$$I_e^D = -\int_{A_e} \left[\frac{\Gamma_\phi}{J}\left(\frac{\partial\phi}{\partial x^1}B_1^1 + \frac{\partial\phi}{\partial x^2}B_2^1 + \frac{\partial\phi}{\partial x^3}B_3^1\right)\right] dx^2 dx^3 \simeq I_e^{DN} + I_e^{DC} =$$

$$-\left(\frac{\Gamma_\phi}{\delta V}\right)_e D_{1e}^1(\phi_E - \phi_P) - \left(\frac{\Gamma_\phi}{\delta V}\right)_e \left[D_2^1(\phi_n - \phi_s) + D_3^1(\phi_t - \phi_b)\right]_e \tag{3.35}$$

where I_e^{DN} and I_e^{DC} represent normal- and cross-derivative contribution respectively, and the D's are determined from the b's in the analogous way as the B's were expressed through the β's , ie.:

$$\begin{aligned}
D_1^1 &= b_1^1 b_1^1 + b_2^1 b_2^1 + b_3^1 b_3^1 \\
D_2^1 = D_1^2 &= b_1^2 b_1^1 + b_2^2 b_2^1 + b_3^2 b_3^1 \\
D_3^1 = D_1^3 &= b_1^3 b_1^1 + b_2^3 b_2^1 + b_3^3 b_3^1 \\
D_2^2 &= b_1^2 b_1^2 + b_2^2 b_2^2 + b_3^2 b_3^2 \\
D_3^2 = D_2^3 &= b_1^3 b_1^2 + b_2^3 b_2^2 + b_3^3 b_3^2 \\
D_3^3 &= b_1^3 b_1^3 + b_2^3 b_2^3 + b_3^3 b_3^3
\end{aligned} \tag{3.36}$$

In arriving at Eq. (3.35), expression (3.25) has been utilized, and the b's are defined by Eq. (3.24).

The other diffusion-like terms of the differential equations (3.11), (3.17), (3.18) and (3.19) are discretised in a similar way.

The manner of integration of source terms has been demonstrated for the continuity equation, (3.30). If the volumetric source $S_{\phi P}$ depends on dependent variable ϕ itself, it is then linearized so that two distinct parts are obtained (see, eg., Patankar, 1980):

$$S_\phi = S_\phi' - S_\phi'' \phi_P \tag{3.37}$$

The reasons for doing so will be discussed further in Chapters 4 and 6. For example, the source terms for k and ε (Eqs. (2.30) and (2.31)) are linearized as:

$$
\begin{aligned}
S_k'' &= G_P \,\delta V_P \quad ; \quad S_k'' = \rho \frac{\varepsilon_P}{k_P} \delta V_P \\
S_\varepsilon' &= C_1 \frac{\varepsilon_P}{k_P} G_P \delta V_P \quad ; \quad S_\varepsilon'' = C_2 \rho \frac{\varepsilon_P}{k_P} \delta V_P
\end{aligned}
\tag{3.38}
$$

where the rate of production of turbulent kinetic energy G_P (Eq. (2.32)) is evaluated as:

$$
\begin{aligned}
G_P = \left(\frac{\mu_t}{\delta V}\right)_P \Big\{ & 2\left[(\delta u_1)^1 b_1^1 + (\delta u_1)^2 b_1^2 + (\delta u_1)^3 b_1^3\right]^2 + \\
& 2\left[(\delta u_2)^1 b_2^1 + (\delta u_2)^2 b_2^2 + (\delta u_2)^3 b_2^3\right]^2 + \\
& 2\left[(\delta u_3)^1 b_3^1 + (\delta u_3)^2 b_3^2 + (\delta u_3)^3 b_3^3\right]^2 + \\
& \left[(\delta u_1)^1 b_2^1 + (\delta u_1)^2 b_2^2 + (\delta u_1)^3 b_2^3 + (\delta u_2)^1 b_1^1 + (\delta u_2)^2 b_1^2 + (\delta u_2)^3 b_1^3\right]^2 + \\
& \left[(\delta u_1)^1 b_3^1 + (\delta u_1)^2 b_3^2 + (\delta u_1)^3 b_3^3 + (\delta u_3)^1 b_1^1 + (\delta u_3)^2 b_1^2 + (\delta u_3)^3 b_1^3\right]^2 + \\
& \left[(\delta u_2)^1 b_3^1 + (\delta u_2)^2 b_3^2 + (\delta u_2)^3 b_3^3 + (\delta u_3)^1 b_2^1 + (\delta u_3)^2 b_2^2 + (\delta u_3)^3 b_2^3\right]^2 \Big\}_P
\end{aligned}
\tag{3.39}
$$

Here $(\delta u_i)_P^j$ represents the difference in u_i velocity component, across the cell enclosing node P, in x^j coordinate direction. These differences are evaluated as:

$$
\begin{aligned}
(\delta u_i)_P^1 &= u_{ie} - u_{iw} \\
(\delta u_i)_P^2 &= u_{in} - u_{is} \\
(\delta u_i)_P^3 &= u_{it} - u_{ib}
\end{aligned}
\tag{3.40}
$$

Differences in Cartesian coordinates y^j, needed for the calculation of b's for the central location P (Eqs. (3.24)) are calculated in the same way. Values on cell faces, eg. u_{ie}, u_{iw} etc., are to be obtained from surrounding nodal values by means of interpolation. This will be demonstrated in section 4 of the next chapter.

The second part of the source term for k (Eq. (2.20)) has been "artificially" made dependent on k_P to produce S_k''. The reasons for doing so will become obvious in the next chapter.

The pressure terms in momentum equations are integrated as source terms, for reasons which will be discussed later in Chapter 6, when the manner of calculating the pressure is examined. For example, for the relevant terms in the u_1-component equation (3.17), one gets:

$$-\int_{\delta V}\left(\frac{\partial}{\partial x^1}p\,\beta_1^1 + \frac{\partial}{\partial x^2}p\,\beta_1^2 + \frac{\partial}{\partial x^3}p\,\beta_1^3\right)dx^1\,dx^2\,dx^3 \simeq$$

$$-(p_e-p_w)b_{1P}^1 - (p_n-p_s)b_{1P}^2 - (p_t-p_b)b_{1P}^3 \tag{3.41}$$

where the coefficients b_1^1, b_1^2 and b_1^3 are calculated at the central node P. Similar expressions follow for the other two equations.

3.5 The Form of Discretised Equations to Be Solved

After introducing expressions like (3.31) and (3.35) (in which (3.33) and (3.34) are utilized) into (3.22) and taking into account (3.30), (3.37) and (3.41), the typical discretised transport equation (eg. Eq. (3.11)) can be cast into the following quasi-linear form:

$$a_P\Phi_P = \sum_m a_m\Phi_m + C_P \tag{3.42}$$

where the a_m are coefficients multiplying the values of Φ at the "m" neighbour nodes surrounding the central node P, a_P is the coefficient of Φ_P, and C_P contains all other terms which are not expressed through the nodal values of the dependent variable (eg. the S_Φ' source term, the pressure gradients in the momentum equations, terms involving known values of Φ etc.). The coefficients a_m contain convective contribution (a_m^C) and contributions from "normal"- and "cross"-diffusion terms (a_m^{DN} and a_m^{DC}, resulting from expressions like (3.33) and (3.34) respectively), ie.:

$$a_m = a_m^C + a_m^{DN} + a_m^{DC} = a_m^C + a_m^D \tag{3.43}$$

The coefficient a_P also contains the linearized source term coefficient S''_ϕ , and is usually* given by:

$$a_P = \sum_m a_m + S''_\phi \tag{3.44}$$

The number of neighbour nodes "m" which appear in Eq. (3.42) depends on the interpolation practices used to evaluate the ϕ values at the cell face locations (eg. for "e" cell face: ϕ_e in (3.31), $\phi_{n,e}$, $\phi_{s,e}$, $\phi_{t,e}$ and $\phi_{b,e}$ in (3.34), and similar values at other cell faces). The minimum number of nodes involved, in case of an orthogonal grid and a linear interpolation scheme, would be 5 in two-dimensional cases and 7 in three-dimensional cases. The cross-derivative diffusion terms resulting from non-orthogonal coordinates bring in additional nodes (see Fig. 3.4), so that the minimum numbers now become 9 and 19 for the two- and three-dimensional cases respectively. If more complex interpolation practices are used, the number of nodes involved may rise to 13 or more for two-dimensional problems, and 27 or more for three-dimensional ones. The more commonly-employed interpolation practices and associated nodal structures will be discussed in more detail in the next chapter.

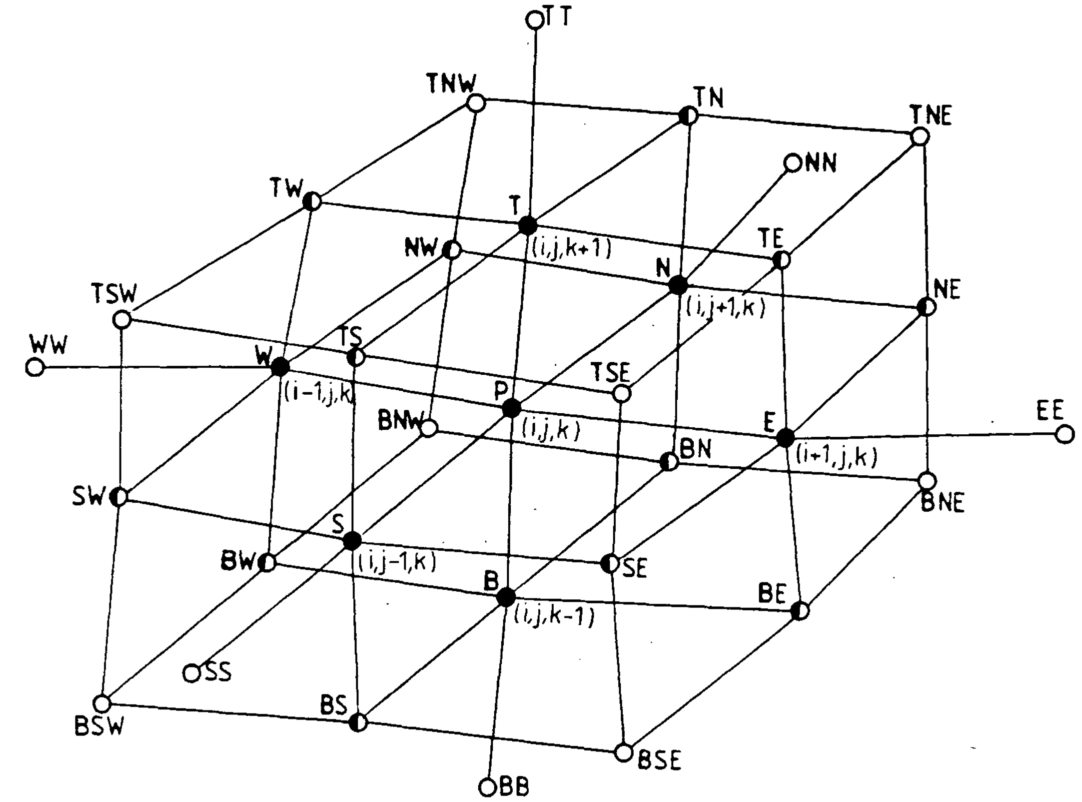
Since an equation like (3.42) exists for every interior computational node, the numerical solution procedure entails solving a system of M equations in M unknowns, where M is the total number of nodes. (Additional equations of a somewhat different form also exist for nodes on the solution domain boundaries where ϕ is unknown.) Such a system of equations can be written in matrix form as:

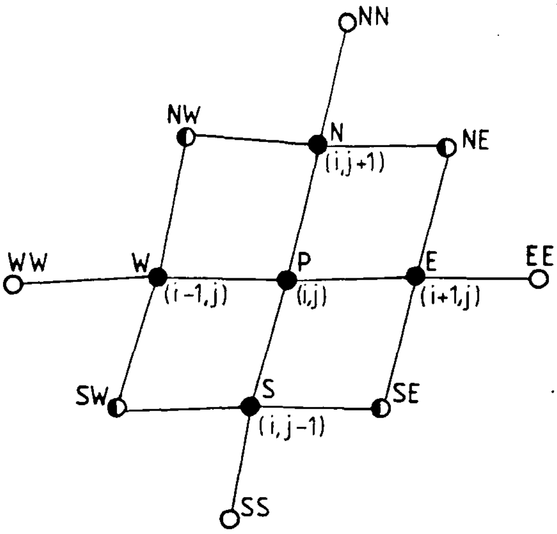
$$[A]\{\phi\} = \{C\} \tag{3.45}$$

Here, $[A]$ is the coefficient matrix, consisting of MxM coefficients A_{ij} , $\{\phi\}$ is a column matrix having M ϕ values, and $\{C\}$ is the matrix of the same form containing M C-terms. The diagonal coefficient A_{ii} of matrix $[A]$ corresponds to $a_{P,i}$ in Eq. (3.42), while the non-zero coefficients A_{ij} , $i \neq j$, correspond to the $-a_m$ coefficients in Eq. (3.42) (ie., the i-th row of Eq. (3.45) corresponds to Eq. (3.42) rewritten as: $a_P\phi_P - \sum_m a_m\phi_m = C_P$).

---

* Whether equation (3.44) holds depends on the type of interpolation practices employed; in this study only practices which satisfy Eq. (3.44) will be used, for reasons which will be discussed in the next chapter.





THE LEGEND:

● simplest molecule for orthogonal grid

●+◐ simplest molecule for non-orthogonal grid

○ additional nodes involving higher-order interpolations

Fig. 3.4: Computational molecules for two-dimensional and three-dimensional grids

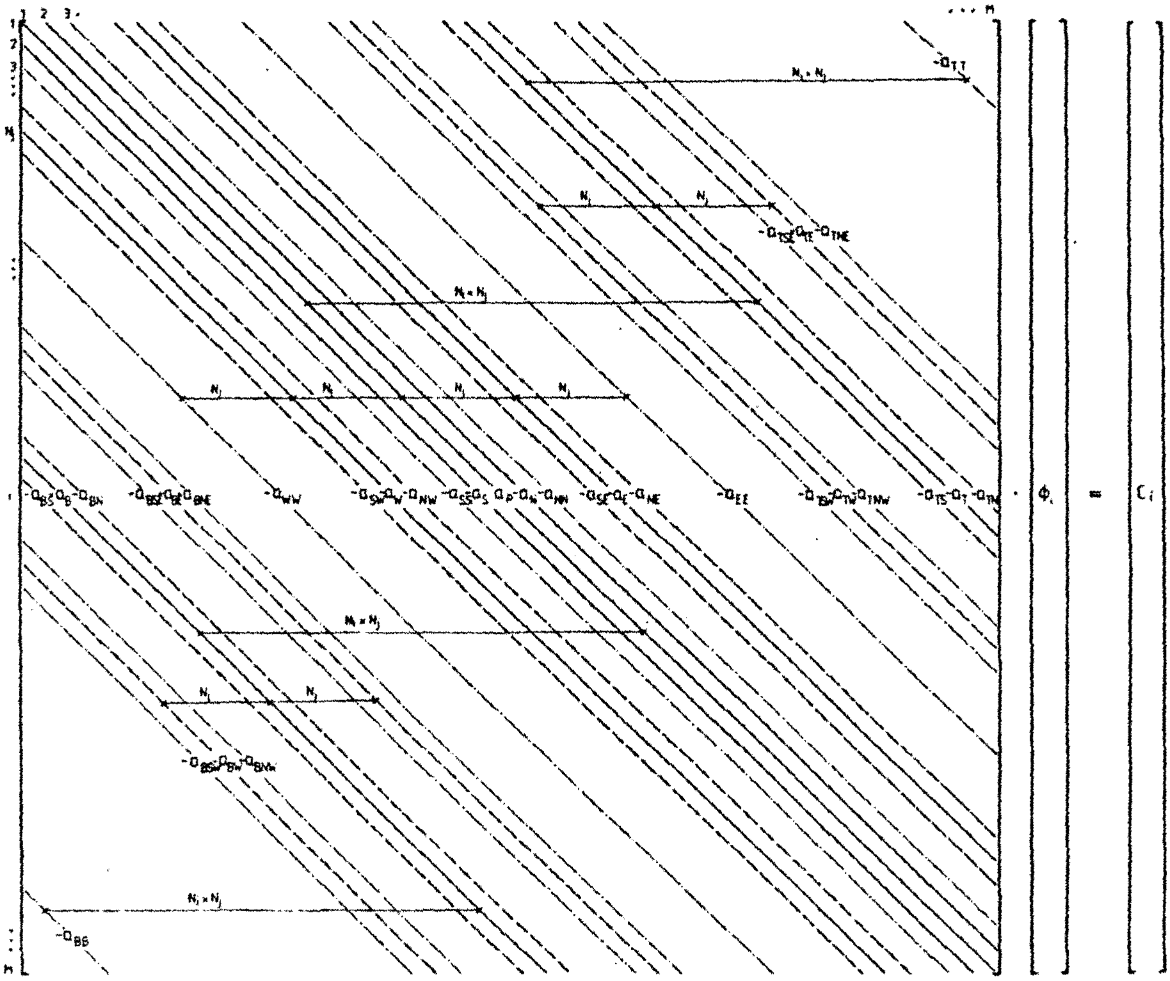


Fig. 3.5: Graphical presentation of matrix equation (3.45)

——— non-zero diagonals for orthogonal grid and linear interpolation

– – – – additional non-zero diagonals for non-orthogonal grid and linear interpolation

—·—·— additional non-zero diagonals for higher-order interpolation

In Fig. 3.5 the structure of the coefficient matrix [A] is shown for the case where $M = N_i \times N_j \times N_k$, where N_j , N_i and N_k are the numbers of nodes in each coordinate direction. The ordering of the nodes is such that index j (see Fig. 3.4) is increasing first, then i and en k . The diagonal lines represent the non-zero coefficients entries.

Equations of the form of (3.45) (or (3.42)) can be solved in several ways. Direct solution methods (ie. matrix inversion) offer exact solution to the algebraic equations for given coefficients (to within round-off error); however, in the present case they are not attractive, for several reasons. Firstly, such methods require the storage of a large number of auxiliary coefficients, which can easily exceed the memory capacity of largest available computers, especially for three-dimensional cases (eg., if the number of computational nodes is of the order of 10^4 - which would correspond to a $22 \times 22 \times 22$ nodes grid - the size of matrix [A] is of the order of 10^8, requiring about one Giga-byte of memory storage). Computing times for matrix inversion are also likely to be excessively large for any reasonable three-dimensional grid. Secondly, the equations of motion (and very often the scalar ones as well) are non-linear, ie. the coefficients a_m in Eq. (3.42) involve the dependent variable ϕ, and therefore an iterative approach is required in which the coefficients a_m are updated each time a new ϕ field is obtained. This renders direct methods even more expensive.

For those reasons wholly iterative solution methods, in which the equations are solved only to a certain level of convergence for a given set of coefficients, are to be preferred. There are numerous such methods (Varga, 1962) which are suitable for the solution of Eq. (3.42). In general, they require little additional storage capacity, and the computing times per iteration are normally only a fraction of those of direct solution methods. Because of the need for the "outer" iterations to update the coefficients it is usually enough to perform a few "inner" iterations in the equation solving algorithm*.

The "Strongly Implicit Procedure" (SIP) of Stone (1968) has been adopted as the iterative solver in this study. However, the present method is not restricted to use of this particular solver; other algorithms, like the widely used line iteration procedures (see Ames, 1977), the "alternating direction implicit" method (Peaceman and Rachford, 1955), the method of conjugate gradients (Hestens and Stiefel, 1952), the Gauss-Seidel method, and variants thereof, may be used as well. The original SIP method was, strictly speaking, developed for five- and seven-diagonal matrices. However, if the equations (3.42) being solved have more coeffi-

* How many "inner" iterations should be performed depends on the solution method employed, the equations being solved, and the problem under consideration. More details will be given in Chapter 6.

cients, one can still use this method by employing the so called "deferred correction" approach (Khosla and Rubin, 1974), ie. the extra terms can be absorbed into the C-term and evaluated at the previous iteration level.

All iterative methods start the solution process from a guessed distribution of the dependent variables and perform successive updates of ϕ and the coefficients until converged solution is obtained. However, depending on the initial guess and the structure of the coefficient matrix, the iterative process can sometimes fail to produce a solution, ie. it may diverge. Scarborough (1958) has shown that sufficient conditions for an iterative method to converge are that*:

$$\frac{\sum_m |a_m|}{|a_P|} \begin{cases} \leqslant & 1 \text{ at all nodes;} \\ < & 1 \text{ at one node at least;} \end{cases} \tag{3.46}$$

Although this is not a necessary requirement, it is still desirable that Eqs. (3.42) satisfy condition (3.46). It also has some other implications which will be discussed in section 2 of the next chapter.

Even satisfaction of the above criteria may not be sufficient to ensure convergence when a solution of the coupled non-linear equations, like the Navier-Stokes equations, is sought. In such cases under-relaxation is usually necessary in order to slow the changes in the ϕ, and hence in the coefficients, from iteration to iteration. This can be done in the following way.

Equation (3.42) can be rewritten as:

$$\phi_P = \phi_P^* + \left[\frac{\sum_m a_m \phi_m + C_P}{a_P} - \phi_P^* \right] \tag{3.47}$$

where ϕ_P^* is the value of ϕ at node P obtained in the previous iteration. The bracketed term represents the change in ϕ_P from one iteration to the next, ie. $\phi_P - \phi_P^*$. To reduce this change, an under-relaxation parameter α_ϕ, in the range between zero and unity, is used to multiply the bracketed term in (3.47), leading to the following relation:

$$\frac{a_P}{\alpha_\phi} \phi_P = \sum_m a_m \phi_m + C_P + \frac{1-\alpha_\phi}{\alpha_\phi} a_P \phi_P^* \tag{3.48}$$

---

* Matrices which possess these properties are often termed "diagonally dominant".

It should be noted that the bracketed term in (3.47) tends to zero when the converged solution is approached, so that the introduction of under-relaxation parameter α_ϕ does not affect the final solution.

Equation (3.48) can be recast into the form of Eq. (3.42), where now the central coefficient a_P becomes:

$$a_P = \frac{\sum_m a_m + S_\phi''}{\alpha_\phi} \tag{3.49}$$

and the last term in Eq. (3.48) is added to the other terms in C_P. This method of under-relaxation plays an important role in solving the Navier-Stokes equations, as will be shown in Chapters 5 and 6. Also, since α_ϕ is smaller than unity it makes the iterative procedure more stable, by enhancing the diagonal dominance of the coefficient matrix.

3.6 Closure

In this chapter the discretisation of the transport equations has been described. The complexity of these equations when written in general coordinates and strong conservation form as compared to their Cartesian counterparts is manifested in a substantially larger number of terms. However, the structure of these terms is of a similar kind, and they represent no problem from the discretisation point of view. The coordinate transformation is absorbed into the discretised equations, aided by the fact that the discretisation has been carried out in physical space. As a consequence all the necessary information about the computational grid is defined by, and easily calculated from, the Cartesian coordinates of the cell vertices.

The final form of discretised transport equations is a set of algebraic equations, one for each computational cell (node), in which a number of coefficients multiplying values of the dependent variable at neighbour nodes appear. The actual number of neighbour nodes involved depends on the interpolation practices, used to express the values of the dependent variable at various cell-face locations in terms of surrounding nodal values. The higher the order of these approximations, the greater number of nodes will be involved; this may result in increased numerical accuracy, but it will also increase the cost of obtaining solution, and may cause stability problems for iterative solution methods or numerical oscillations in the solution. These matters will be discussed in more detail in the next chapter.

CHAPTER 4

INTERPOLATION PRACTICES (DIFFERENCING SCHEMES)

4.1 Introduction

In the previous chapter the discretisation of the transport equations, in the finite volume manner, was performed. In the discrete form of the equations there appear values of the dependent variables at control volume boundary locations (cell faces): these have to be expressed in terms of the neighbour nodal values in order to obtain algebraic equations of the form (3.42). On the type of interpolation practices (also called "differencing schemes") used to accomplish this task depend both the stability of the numerical algorithm and the accuracy of the solutions obtained. These matters will receive attention in this chapter.

First, in section 2, the desired properties of the differencing schemes will be identified. This is followed, in section 3, by descriptions of some common differencing schemes for convective terms.

In section 4 alternative interpolation practices for the diffusion terms, and their properties, will be discussed.

In section 5 some existing methods for overcoming boundedness problems associated with higher-order differencing schemes, will be briefly described. Following this, in section 6, a new flux blending method, developed here for the same purpose, will be presented.

Finally, closing remarks will be given in section 7.

4.2 Desired Properties of Differencing Schemes

If the number of computational cells tends to infinity, one would expect to get the "exact" solution of the differential transport equation (eg. Eq.(3.5)), irrespective of the interpolation practices used to evaluate the convective and diffusive fluxes (eg. Eqs. (3.31) and (3.35)). However, in practice one can only employ a finite number of cells, and it is then important to ensure that the solution of discretised equations (3.42) satisfy certain important properties of the exact solution even on a coarse grid. These properties are outlined below.

4.2.1 Conservativeness

It has been demonstrated in the previous chapter that the discretised equations (3.42) are obtained by integrating the differential equations over a finite number of control volumes (cells). A set of integral equations like (3.22), involving fluxes through cell faces, results. These equations represent conservation balances over corresponding control volumes; if summed, all inner fluxes should cancel in pairs, thus giving the integral conservation equation for the solution domain as a whole (ie., the net transport across the solution domain boundaries should equal the total production - or consumption - by internal sources). It is, from the physical point of view, very important that the physically conserved quantities (eg. mass, heat, chemical species etc.) are conserved by a numerical solution method too, since otherwise unrealistic results may occur (eg., negative density or turbulent kinetic energy, species mass fractions less than zero or greater than one, etc.).

Conservation of transported quantities is ensured if the flux across any cell face is uniquely represented in the discretised equations for the two adjacent control volumes. In other words, with reference to Fig. 3.3, the expressions used to evaluate ϕ_e , $(\phi_n - \phi_s)_e$ and $(\phi_t - \phi_b)_e$ in Eqs. (3.31) and (3.35), which are written for the cell enclosing node P , should be the same as those used to evaluate ϕ_w , $(\phi_n - \phi_s)_w$ and $(\phi_t - \phi_b)_w$ respectively, in corresponding equations for the cell enclosing node E . A scheme which possesses this property is said to be conservative. Only such schemes will be considered here, for the reasons stated above.

4.2.2 Boundedness

The ensuring of conservation does not guarantee that all the other important properties of transport processes will be maintained within the solution domain. For example, in the absence of sources, the values of transported quantity ϕ in the interior of solution domain should be "bounded" by the values ϕ_B on the domain boundaries, ie. there should result no value larger than the maximum, nor smaller than the minimum, boundary value (Gosman and Lai, 1982):

$$\min(\phi_B) < \phi < \max(\phi_B) \tag{4.1}$$

Bounds will also exist for more general cases, but they are not so easily perceived.

It can be shown (eg. Lillington, 1981) that differencing schemes leading to the coefficient matrix which satisfies condition (3.46), ie. which is "diagonally dominant", will also satisfy the boundedness criterion (4.1) under stated conditions. From (3.46) it follows that all coefficients in Eq. (3.42) should have the same sign* (eg. positive). Thus an increase in the dependent variable value ϕ_m at one node, with other conditions remaining unchanged, would result in an increase (and not a decrease) of ϕ at neighbour nodes. This is a desirable feature of the discretised equation, although it is not easy to achieve in general, as will be demonstrated later in this chapter.

Boundedness is also influenced by the ratio between a_P, the coefficient of the central node P, and the neighbour coefficients a_m in Eq. (3.42). If (3.44) holds and S_ϕ'' is zero, then ϕ_P is a weighted mean of neighbour values ϕ_m (ie., if all the neighbour values ϕ_m are equal, ϕ_P is also equal to them). Although satisfaction of (3.44) does not by itself guarantee that the results will satisfy boundedness criterion (4.1), it is obviously a desired feature of the discretisation scheme.

If the discretised equations do not satisfy the boundedness requirement (4.1) under some conditions, the scheme used will be prone to produce "overshoots", "undershoots" or "wiggles", as will be demonstrated later.

4.2.3 Transportiveness

Figure 4.1 illustrates what is often termed the "transportive property" of a fluid flow (Roache, 1976). If there is a constant source of ϕ at a point P in a flow field with uniform velocity and diffusivity, then the shapes of contours of constant ϕ will be influenced by the ratio of convection to diffusion, ie. by the Peclet number ($Pe = \frac{\rho u L}{\Gamma}$, where L is a characteristic length) as shown in Fig. 4.1. When the fluid

---

* This is again a sufficient, not always necessary condition for the boundedness criterion to be satisfied. Whether the presence of a negative neighbour coefficient, say a_W, will drive ϕ_P out of bounds depends on the smoothness of the ϕ field around P. However, this is the only unconditional guarantee that bounded solution will be obtained.

is stagnant ($P_e = 0$), the contours will be circles with P as the centre; as P_e increases the contours become elliptical, shifted in the direction of flow until, for $P_e = \infty$ (no diffusion), they collapse into the streamline from P downstreams. The implication is that for high P_e flows events at node P will have a weak or no influence on upstream regions, while downstream regions will be strongly affected.

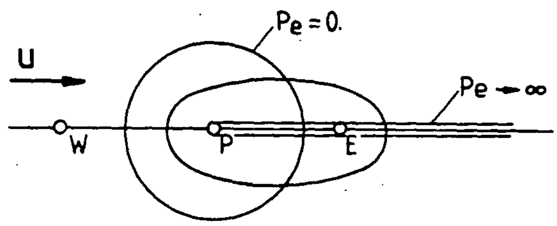


Fig. 4.1: Illustration of transportive property of a fluid flow

Failure of the discretised equations to account for these features of a fluid flow can give rise to unrealistic results as will be demonstrated in the next section.

4.3 Differencing Schemes for Convective Terms and their Properties

Here attention will be focussed on the convective fluxes (Eq. (3.31)), and the methods of expressing convected (cell face) value of the dependent variable (eg. ϕ_e in Eq. (3.32) for the "e" cell face) in terms of surrounding nodal values. Some common differencing schemes used for this purpose are briefly described below.

4.3.1 Central Differencing Scheme (CDS)

If the cell face value ϕ_e in Eq. (3.32) is evaluated by means of linear interpolation between nodes P and E (see Fig. 4.2), one gets:

$$\phi_e = \phi_E \cdot f_{1P} + \phi_P \cdot (1-f_{1P}) \tag{4.2}$$

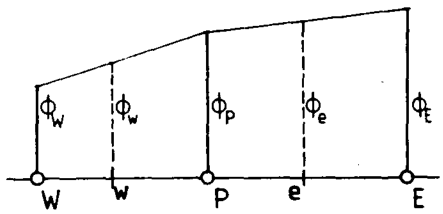


Fig. 4.2: Schematic representation of the CDS for one-dimensional case

Here, f_{1P} is a linear interpolation factor, associated with a particular node P and x^1 coordinate direction, and defined as:

$$f_{1P} = \frac{\overline{Pe}}{\overline{Pe}+\overline{eE}} = \frac{\delta x_P^{(1)}}{\delta x_P^{(1)} + \delta x_E^{(1)}} \tag{4.3}$$

where $\overline{Pe}$ and $\overline{eE}$ are the distances of cell face location "e" (cell face centre) from nodes P and E respectively; these are equal to the halves of the cell arc lengths along x^1 coordinate for P and E cells, $\delta x_P^{(1)}$ and $\delta x_E^{(1)}$ respectively (see Fig. 3.1 and 3.2). By analogy, linear interpolation coefficients for the other two coordinate directions, f_{2P} and f_{3P}, can also be defined.

With expressions similar to (4.2) following for the other cell faces, the convective parts of the coefficients a_m (Eqs. (3.42) and (3.43)) are:

$$
\begin{aligned}
a_E^C &= -F_{1,e} \cdot f_{1P} \\
a_W^C &= F_{1,w} \cdot (1-f_{1W}) \\
a_N^C &= -F_{2,n} \cdot f_{2P} \\
a_S^C &= F_{2,s} \cdot (1-f_{2S}) \\
a_T^C &= -F_{3,t} \cdot f_{3P} \\
a_B^C &= F_{3,b} \cdot (1-f_{3B}) \\
a_P^C &= \sum_m a_m^C \quad , \; m = E,W,N,S,T,B
\end{aligned}
\tag{4.4}
$$

From (4.4) it is obvious that convective contribution a_m^C to the coefficients of some neighbour nodes m is always negative, and if convective processes dominate, ie. if $|a_m^C| > |a_m^{DN} + a_m^{DC}|$ (see Eq. (3.43)), this can cause the whole coefficient a_m to be negative. In the limit of a uniform Cartesian grid this would happen if the grid Peclet number, defined as (see Eqs. (3.26) and (3.35)):

$$
P_e = \frac{F_i}{\Gamma_\phi \left(\frac{D_i^i}{\delta V} \right)} \tag{4.5}
$$

becomes greater than 2. In such a case the condition (3.46) would be violated, with the ultimate result of possibly unbounded solutions and difficulties in obtaining them (Spalding, 1972; Raithby and Torrance, 1974). The CDS exhibits this behaviour because it violates the transportive property at high P_e numbers, by employing downstream nodes in expressions like (4.2). Thus upstream propagation of any disturbance is possible even for $P_e = \infty$, which is physically unrealistic. However, the CDS is conservative and at small Peclet numbers ($P_e < 2$) it is both stable and accurate.

In terms of Taylor series truncation error analysis (TSTE), which is still widely used as a measure of accuracy in spite of its limitations (Cheng and Shubin, 1978; Stubley et al, 1980), the CDS is second order accurate; details about the TSTE analysis and resulting truncation error expressions for many differencing schemes, including those considered here, can be found in Lai (1982).

4.3.2 Upwind Differencing Scheme (UDS)

The upwind differencing scheme (also known as the "donor cell" scheme) is schematically represented in Fig. 4.3. The convected cell face value is taken to be that at the upstream node along the same coordinate line, eg.:

$$\phi_e = \begin{cases} \phi_P, \text{ if } F_{1,e} > 0 \\ \phi_E, \text{ if } F_{1,e} < 0 \end{cases} \tag{4.6}$$

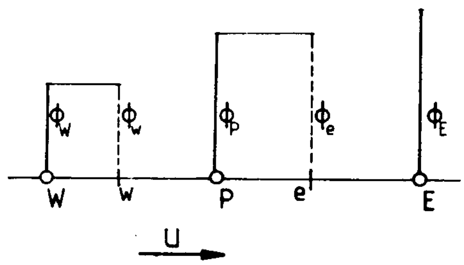


Fig 4.3: Schematic representation of the UDS in a one-dimensional case

Using similar expressions for other cell faces, the convective parts of the coefficients a_m in (3.42) are:

$$a_E^C = \max(0, -F_{1,e})$$

$$a_W^C = \max(0, F_{1,w})$$

$$a_N^C = \max(0, -F_{2,n})$$

$$a_S^C = \max(0, F_{2,s}) \tag{4.7}$$

$$a_T^C = \max(0, -F_{3,t})$$

$$a_B = \max(0, F_{3,b})$$

$$a_P = \sum_m a_m \quad , \quad m = E,W,N,S,T,B$$

From (4.7) it is obvious that convective contribution to the coefficients a_m is always non-negative (zero for downstream nodes and positive for the upstream ones). Thus, unless violated by diffusion contribution, the requirement (3.46) is unconditionally satisfied, which guarantees the boundedness of the solution and the ease of obtaining it by iterative techniques. This can be attributed to the fact that the transportive property is fully accounted for (no reference is made to downstream nodes in expressions like (4.6)). In one-dimensional cases the scheme is asimptotically exact as P_e tends to infinity (Spalding, 1972; Raithby and Torrance, 1974).

In terms of the TSTE the UDS is first-order accurate; the leading higher-order neglected term is a function of $\frac{\partial^2\phi}{\partial x^2}$, ie. diffusion like. This has led to the definition of the so called "false diffusion" concept, discussed among others by Patankar (1980) and Raithby (1976) . Namely, if the flow direction is not aligned with one set of grid lines and if there is a strong gradient of the dependent variable in the direction normal to the flow, the use of UDS results in severe numerical smearing, as if much stronger diffusion processes were present in the flow than actually is the case. The additional, numerical or "false" diffusion coefficient Γ_{false} has been worked out, among others, by Lai (1982). It is a function of the angle ψ between the flow and the grid lines, and is equal to zero if $\psi=0°$ or $90°$ (ie. flow aligned with one set of grid lines) and a maximum for $\psi=45°$.

However, relying on TSTE and "false diffusion" concept can often be misleading. Sometimes inaccuracies due to other sources (inadequate treatment of boundary conditions, for example) are - unfairly - attributed to "false diffusion" effects. For example, the presumption that the UDS is "over-diffusive" has thus resulted in "severe numerical diffusion" being claimed even in one-dimensional steady problems at large P_e (Leonard, 1979, for example), where it is not present (Patankar, 1980).

4.3.3 Linear Upwind Differencing Scheme (LUDS)

This scheme is, for a one-dimensional case, schematically represented in Fig. 4.4. The cell face values, used in the evaluation of convective fluxes, are obtained by linear extrapolation from the two closest upstream neighbour nodes on the same coordinate line, eg. for ϕ_e :

$$\phi_e = \begin{cases} \phi_P + (\phi_P - \phi_W)(1 - f_{1W}) \ , \text{ for } F_{1,e} > 0 \\ \phi_E + (\phi_E - \phi_{EE})\ f_{1E} \ , \text{ for } F_{1,e} < 0 \end{cases} \tag{4.8}$$

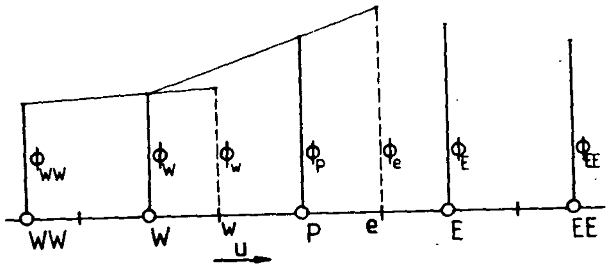


Fig. 4.4: Schematic presentation of the LUDS

Employing similar expressions for the other cell faces gives the following expressions for a_m^C :

$$\begin{aligned}
a_E^C &= \max(0,-F_{1e})(1 + f_{1E}) + \max(0,-F_{1w})\, f_{1P} \\
a_W^C &= \max(0,\ F_{1w})(2 - f_{1WW}) + \max(0,\ F_{1e})(1 - f_{1W}) \\
a_N^C &= \max(0,-F_{2n})(1 + f_{2N}) + \max(0,-F_{2s})\, f_{2P} \\
a_S^C &= \max(0,\ F_{2s})(2 - f_{2SS}) + \max(0,\ F_{2n})(1 - f_{2S}) \\
a_T^C &= \max(0,-F_{3t})(1 + f_{3T}) + \max(0,-F_{3b})\, f_{3P} \\
a_B^C &= \max(0,\ F_{3b})(2 - f_{3BB}) + \max(0,\ F_{3t})(1 - f_{3B}) \\
a_{EE}^C &= -\max(0,-F_{1e})\, f_{1E} \\
a_{WW}^C &= -\max(0,\ F_{1w})(1 - f_{1WW}) \\
a_{NN}^C &= -\max(0,-F_{2n})\, f_{2N} \\
a_{SS}^C &= -\max(0,\ F_{2s})(1 - f_{2SS}) \\
a_{TT}^C &= -\max(0,-F_{3t})\, f_{3T} \\
a_{BB}^C &= -\max(0,\ F_{3b})(1 - f_{3BB}) \\
a_P^C &= \sum_m a_m^C \qquad m = E,W,N,S,T,B,EE,WW,NN,SS,TT,BB
\end{aligned} \tag{4.9}$$

The scheme fully obeys the transportive property requirement, by interpolating in the region of influence only (no downstream nodes are involved in (4.8)). It is also conservative, but it does not satisfy the boundedness reqirement, as evidenced by the fact that the contributions to the coefficients of the most distant nodes EE, WW, NN, SS, TT and BB are non-positive (zero for downstream nodes, and negative for the upstream ones). These nodes are usually not involved in the diffusion term aproximations, and thus their total coefficients are always non-positive thus violating condition (3.46). In one-dimensional problems this feature will not result in unbounded solutions; a monotonic result will always be obtained, due to strict obedience of the transportive property (Wilkes and Thompson, 1983). However, in multi-dimensional problems unbounded solutions may be generated under certain conditions, which will be discussed in section 5.

In terms of the TSTE the LUDS is second order accurate, and unless boundedness problems arise, a given level of accuracy is achieved on far coarser grids than required by the UDS. The diagonal dominance of the resulting coefficient matrix is stronger than in case of CDS (this will be demonstrated in the summary for this section), and most iterative solvers handle it well. The LUDS is also attractive for its relative simplicity when employed with non-orthogonal and non-uniform grids, as compared to the more complex schemes to be described next.

4.3.4 Quadratic Upwind Differencing Scheme (QUDS)

This scheme is due to Leonard (1979), who named it "QUICK". For, say, the "e" cell face, the ϕ_e value is found from a quadratic function g_e passed through the two bracketing nodes, P and E , and the most distant node on the upstream side along the same coordinate line, ie. W for $F_{1e} > 0$ or EE for $F_{1e} < 0$ (see Fig 4.5), thus:

$$\phi_e = \begin{cases} g^1_{e1}\,\phi_E - g^2_{e1}\phi_W + (1 - g^1_{e1} + g^2_{e1})\phi_P \; , \text{ for } F_{1,e} > 0 \\ g^1_{e2}\,\phi_P - g^2_{e2}\phi_{EE} + (1 - g^1_{e2} + g^2_{e2})\phi_E \; , \text{ for } F_{1,e} < 0 \end{cases} \tag{4.10}$$

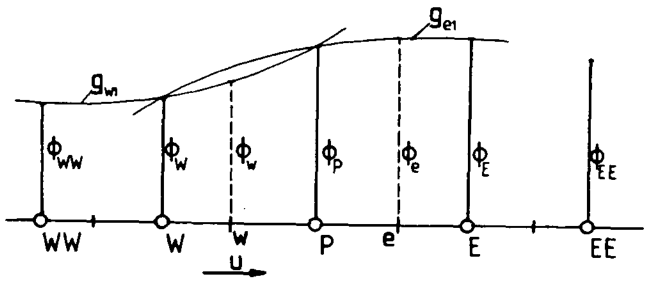


Fig. 4.5: Schematic presentation of the QUDS

Here, g_{e1}^1 and g_{e1}^2 are the coefficients arising from the quadratic function g_{e1} , and g_{e2}^1 and g_{e2}^2 are the similar coefficients arising from the parabola g_{e2} . These coefficients are functions of the distances between cell face location "e" and the participating nodes P , E , W and EE . For the case of a uniform grid $(f_1=f_2=f_3=\frac{1}{2})$ they are (for both g_{e1} and g_{e2}):

$$g_e^1 = \frac{3}{8} \quad ; \quad g_e^2 = \frac{1}{8} \tag{4.11}$$

and remain the same for all cell faces (other than those near boundaries). However, for non-uniform and non-orthogonal grids they vary from one cell face to another, and can be calculated, using the linear interpolation coefficients defined earlier, from the following expressions:

$$\begin{aligned}
g_{e1}^1 &= \frac{(2-f_{1W})(f_{1P})^2}{1 + f_{1P} - f_{1W}} \\
g_{e1}^2 &= \frac{(1-f_{1P})(1-f_{1W})^2}{1 + f_{1P} - f_{1W}} \\
g_{e2}^1 &= \frac{(1+f_{1W})(1-f_{1P})^2}{1 + f_{1E} - f_{1P}} \\
g_{e2}^2 &= \frac{(f_{1E})^2\, f_{1P}}{1 + f_{1E} - f_{1P}}
\end{aligned} \tag{4.12}$$

By applying the same treatment to the "w" cell face, there results:

$$\phi_w = \begin{cases} g^1_{w1}\phi_P - g^2_{w1}\phi_{WW} + (1 - g^1_{w1} + g^2_{w1})\phi_W\,, & \text{if} \quad F_{1w} > 0 \\ g^1_{w2}\phi_W - g^2_{w2}\phi_E + (1 - g^1_{w2} + g^2_{w2})\phi_P\,, & \text{if} \quad F_{1w} < 0 \end{cases} \tag{4.13}$$

where the coefficients g_w can be calculated as:

$$\begin{aligned} g^1_{w1} &= \frac{(2-f_{1WW})(f_{1W})^2}{1 + f_{1W} - f_{1WW}} \\ g^2_{w1} &= \frac{(1-f_{1W})(1-f_{1WW})^2}{1 + f_{1W} - f_{1WW}} \\ g^1_{w2} &= \frac{(1-f_{1W})^2(1+f_{1P})}{1 + f_{1P} - f_{1W}} \\ g^2_{w2} &= \frac{(f_{1P})^2\, f_{1W}}{1 + f_{1P} - f_{1W}} \end{aligned} \tag{4.14}$$

For "t" and "n" cell faces, expressions analogous to those derived for "e" cell face, (4.10) and (4.12), exist with indeces $_{1,E,P,W,EE}$ replaced by $_{2,N,P,S,NN}$ or $_{3,T,P,B,TT}$ respectively. Similarily, expressions for "s" and "b" cell faces follow from those for "w" cell face, (4.13) and (4.14), by replacing indeces $_{1,W,P,E,WW}$ with $_{2,S,P,N,SS}$ or $_{3,B,P,T,BB}$ respectively.

The convective contribution to the coefficients a_m in Eqs. (3.42) are now the following:

$$a^C_E = -\max(0,F_{1e})\, g^1_{e1} - \min(0,F_{1w})\, g^2_{w2} - \min(0,F_{1e})(1-g^1_{e2}+g^2_{e2})$$

$$a^C_W = \max(0,F_{1e})\, g^2_{e1} + \min(0,F_{1w})\, g^1_{w2} + \max(0,F_{1w})(1-g^1_{w1}+g^2_{w1})$$

$$a^C_N = -\max(0,F_{2n})\, g^1_{n1} - \min(0,F_{2s})\, g^2_{s2} - \min(0,F_{2n})(1-g^1_{n2}+g^2_{n2})$$

$$a^C_S = \max(0,F_{2n})\, g^2_{n1} + \min(0,F_{2s})\, g^1_{s2} + \max(0,F_{2s})(1-g^1_{s1}+g^2_{s1})$$

$$a^C_T = -\max(0,F_{3t})\, g^1_{t1} - \min(0,F_{3b})\, g^2_{b2} - \min(0,F_{3t})(1-g^1_{t2}+g^2_{t2})$$

$$a^C_B = \max(0,F_{3t})\, g^2_{t1} + \min(0,F_{3b})\, g^1_{b2} + \max(0,F_{3b})(1-g^1_{b1}+g^2_{b1})$$

$$
\begin{aligned}
a^C_{EE} &= \min(0,F_{1e})\, g^2_{e2} \\
a^C_{WW} &= -\max(0,F_{1w})\, g^2_{w1} \\
a^C_{NN} &= \min(0,F_{2n})\, g^2_{n2} \\
a^C_{SS} &= -\max(0,F_{2s})\, g^2_{s1} \\
a^C_{TT} &= \min(0,F_{3t})\, g^2_{t2} \\
a^C_{BB} &= -\max(0,F_{3b})\, g^2_{b1} \\
a^C_P &= \sum_m a^C_m \quad , \quad m = E,W,N,S,T,B,EE,WW,NN,SS,TT,BB
\end{aligned}
\tag{4.15}
$$

This scheme is also conservative, since the convected values of Φ are uniquely defined on each cell face. Unlike the LUDS, the QUDS obeys the transportive property only partially, for downstream nodes feature in interpolation formulae like (4.10) and (4.13) with a considerable weighting (see Eq. (4.11)). The implication is that even in one-dimensional cases "overstoots", "undershoots" or "wiggles" can result at high P_e , as shown by Leonard (1979) and Syed et al (1985). As with the LUDS, the contributions to the coefficients of the most distant nodes EE, WW, NN, SS, TT and BB are always non-positive (again zero for downstream nodes, negative for the upstream ones). However, the QUDS contains negative contributions for the "principal" downstream nodes too. For example, in case of a uniform flow with F_1, F_2 and F_3 all positive and uniform grid, the following convective contributions to the downstream nodes E, N and T coefficients would result:

$$a^C_E = -\frac{3}{8}F_1 \ ; \ a^C_N = -\frac{3}{8}F_2 \ ; \ a^C_T = -\frac{3}{8}F_3 \tag{4.16}$$

Thus, if the Peclet number is greater than $\frac{8}{3}$ in the case just described the total coefficients a_E, a_N and a_T are negative. This can give rise to both stability problems when an iterative solution procedure is used (Han et al, 1981) and unbounded solutions (Syed et al, 1985; Leschziner, 1980).

In terms of TSTE, the QUDS is third order accurate on uniform grids (Leonard, 1979). Indeed, it has proved to yield marginally better accuracy than other, lower-order schemes, in a number of cases; however, boundedness problems can offset the gain on interpolation accuracy, as will be demonstrated in Chapter 8 (also in Leschziner , 1980; Gosman and Lai, 1982; Han et al, 1981).

4.3.5 Skew Upwind Differencing Scheme (SUDS)

This scheme, due to Raithby (1976 b), is designed to take fully into account the transportive property of a fluid flow. It assigns to the cell face locations e, w, n, s, t and b the values found in surfaces containing four (2 in the 2D case) nodes immediately upstream from the particular cell face in question. These values are found by interpolating for the location where tangent to the streamline passing through the cell face centre intercepts the forementioned surface. Figure 4.6 shows a two-dimensional case, where eg. ϕ_n is found by linear inpolation between ϕ_W and ϕ_P for the location "un", where streamline passing through "n" crosses line $\overline{WP}$.

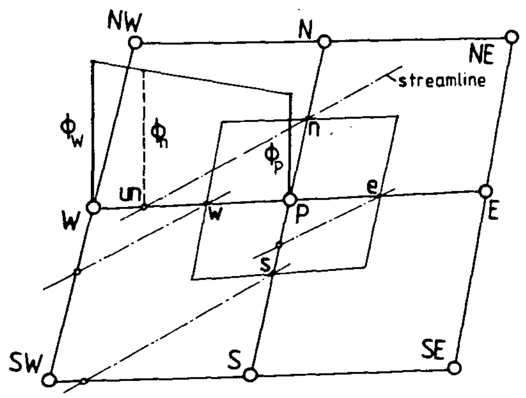


Fig. 4.6: Schematic presentation of the SUDS in a two-dimensional case

In a three-dimensional case, 26 nodes surrounding central node P are involved in such interpolations (see Fig. 3.4). The expressions for the convective contribution to the coefficients a_m in Eqs. (4.42) are very complex for general case and will not be presented here.
The SUDS is conservative and fully transportive; however, it does not satisfy unconditionally the boundedness requirement, since some coefficients can become negative, depending on the flow inclination towards grid lines and grid Peclet number (in a 2D case, a_E, a_W, a_N and a_S can become negative; see Lai (1982) for more details). It reduces to the UDS in case of flow along one set of grid lines. In terms of TSTE, the SUDS is, like UDS, first order "accurate". However, Lai (1982) has shown that the numerical diffusion coefficient is significantly smaller for SUDS than for UDS.

The scheme has been widely used in two-dimensional applications and discussions can be found in Leschziner (1980), Lillington (1981) and Lai (1982). Improved accuracy as compared to UDS has been reported, but unbounded results have also been encountered. For general three-dimensional applications the SUDS is less attractive than either LUDS or QUDS, for the expressions for coefficients a_m^c become extremly complex and produce a clumsy 27-diagonal matrix in (3.45).

4.3.6 Discussion

In addition to the five convective differencing schemes described here, there exist a number of other, less common schemes which could be used; these are reviewed by, eg. Gosman and Lai (1982) and Syed et al (1985).

All of five schemes described here are conservative, and thus possess at least one of the desired properties. Transportiveness is fully satisfied by the UDS, LUDS and SUDS, and only partially by QUDS. The boundedness property is unconditionally possessed only by the UDS; all other schemes either have some coefficients which are always negative (LUDS, QUDS), or some which can become negative under certain circumstances (CDS, SUDS, QUDS). The latter ones are also less diagonally dominant, although iterative solution methods have been successfully used (with more or less difficulty) on all but the CDS*. The degree of depar-

* The problems with the CDS can be, to some extent, surmounted by using the deferred-correction approach (Khosla and Rubin, 1974), where the CDS convective flux is expressed as: $I^{CDS} = I^{UDS} + (I^{CDS} - I^{UDS})$ and the bracketed right-hand side term is treated explicitly, as a pseudo source (Ghia et al, 1982).

ture from diagonal dominace, as measured by the ratio $|a_P^C|/\sum_m|a_m^C|$ and the maximum magnitude of the negative coefficients $\sum_m \max(0,-a_m^C)$ realtive a_P^C is shown in Table 4.1, for a uniform grid.

Scheme	CDS	UDS	LUDS	QUDS	SUDS
$\lvert a_P^C\rvert/\sum_m\lvert a_m^C\rvert$	0	1	$\frac{3}{5}$	$\frac{3}{11}$	depends on the angle between flow direction and grid lines
$\sum_m \max(0,-a_m^C)/a_P^C$	∞	0	$\frac{1}{3}$	$\frac{4}{3}$	

Table 4.1: Some scheme properties in the limit of uniform flow and grid

Experience has shown that the UDS, which is the only scheme which possesses all the desired properties, suffers from excessive numerical (false) diffusion (under conditions cited in subsection 2), which can outweigh the actual diffusion present in the flow (Leonard, 1979; Raithby, 1976; Leschziner, 1980; Lai, 1982; also to be demonstrated in Chapter 8). The cures for this are: (i) grid adaptation, to make it aligned with the flow direction, or (ii) grid refinement until Peclet number becomes low enough so that the numerical diffusion is small in comparision with the physical diffusion.

Evidence also exists (in references cited in this section, among others) that other so called "higher-order schemes" offer better accuracy and faster response to grid refinement. This, however, is true only if the solution is smooth and no unbounded regions are generated. If, on the other hand, "overshoots", "undershoots" or "wiggles" are present in the solution, it may be not only inaccurate, but also physically unrealistic. The cure in such cases is again grid refinement, until negative coefficients become non-negative (Pe<2 for CDS), or the solution is sufficiently "smooth" over the range of nodes covered by the particular interpolation formula (LUDS, QUDS). Another alternative exists, however, in the form of "flux correction" or "flux blending": this will be described in sections 5 and 6 of this chapter.

4.4 Interpolation Practices for Diffusion Terms and their Properties

The method of evaluation of gradients "normal" to the cell faces, which feature in the expressions for diffusion fluxes (Eq. (3.35)), was described in the previous chapter (Eq. (3.33)). The central-difference approximations employed there yield the following contributions a_m^{DN} to the coefficients of the discretised transport equation (3.42):

$$
\begin{aligned}
a_E^{DN} &= \Gamma_{\phi e}\left(\frac{D_1^1}{\delta V}\right)_e \\
a_W^{DN} &= \Gamma_{\phi w}\left(\frac{D_1^1}{\delta V}\right)_w \\
a_N^{DN} &= \Gamma_{\phi n}\left(\frac{D_2^2}{\delta V}\right)_n \\
a_S^{DN} &= \Gamma_{\phi s}\left(\frac{D_2^2}{\delta V}\right)_s \\
a_T^{DN} &= \Gamma_{\phi t}\left(\frac{D_3^3}{\delta V}\right)_t \\
a_B^{DN} &= \Gamma_{\phi b}\left(\frac{D_3^3}{\delta V}\right)_b \\
a_P^{DN} &= \sum_m a_m^{DN} \quad , \quad m = E,W,N,S,T,B
\end{aligned}
\tag{4.17}
$$

All these contributions are positive, and provided no larger negative contribution for any node arises from the treatment of the convection and cross-derivative diffusion terms, bounded solutions and a stable iterative procedure are guaranteed. In terms of the TSTE, this treatment is second order accurate, and it is almost universally used in numerical solution methods of all kinds (see eg. Gosman et al, 1969; Patankar, 1980; Hah, 1984; Demirdžić, 1982 and others).

If the computational grid is othogonal, then the cross-derivative diffusion terms will be zero, and the only diffusion contribution would be from a_m^{DN} just defined. However, when a non-orthogonal grid is employed, the cross-derivative terms like those in Eq. (3.34) are non-zero, and their contribution a_m^{DC} can sometimes outweigh a_m^{DN} , and be either positive or negative, as will be demonstrated below.

The central-difference type approximations for the cross-derivatives, given by equations like (3.34), require additional interpolation practices for the evaluation of the dependent variable values at locations on the cell bounderies (the cell corners in case of a two-dimensional grid), like those appearing in Eq. (3.35) for the "e" cell face. Two such practices will be described next.

SCHEME 1

The most obvious (and perhaps most economical, but not necessarily most accurate) approach is to employ linear interpolation and make use of the interpolation factors which were defined earlier (Eq. (4.3)) and used for the evaluation of convective terms. With reference to Fig. 4.7, the differences appearing in Eq. (3.35) can be evaluated as:

$$\begin{aligned}(\phi_n - \phi_s)_e &= (\phi_n - \phi_s)_P\,(1-f_{1P}) + (\phi_n - \phi_s)_E\, f_{1P} \\ (\phi_t - \phi_b)_e &= (\phi_t - \phi_b)_P\,(1-f_{1P}) + (\phi_t - \phi_b)_E\, f_{1P}\end{aligned} \qquad (4.18)$$

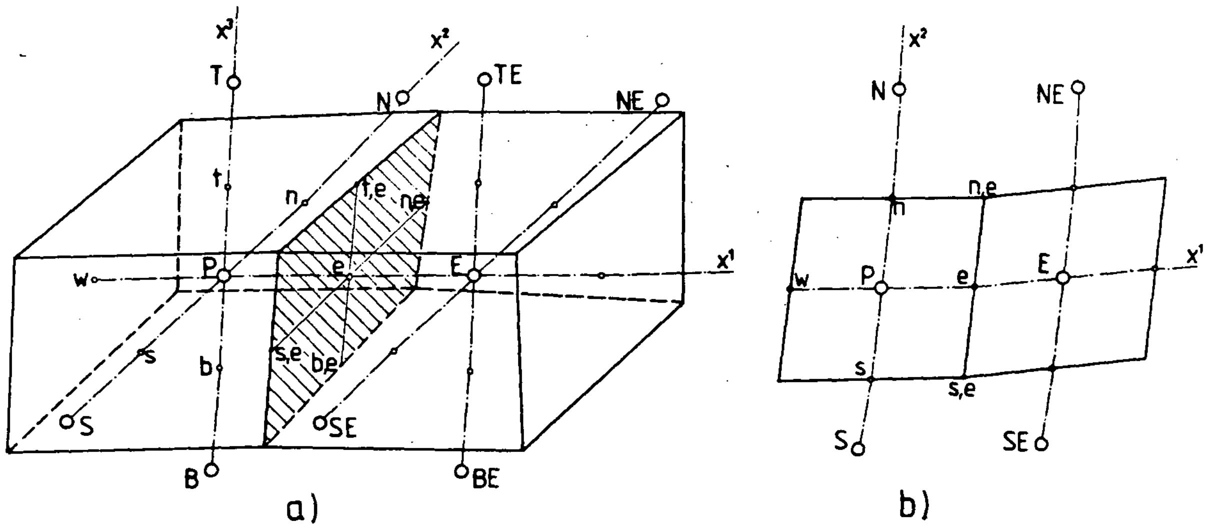


Fig. 4.7: Evaluation of cross-differences on "e" cell face.
a) 3D grid ; b) 2D grid

Expressions similar to (4.18) follow for the other cell faces. Using interpolation coefficients f_1, f_2 and f_3, the values at centres of cell faces surrounding node P can be calculated as:

$$\begin{aligned}
\phi_n &= \phi_N f_{2P} + \phi_P(1 - f_{2P}) \\
\phi_s &= \phi_P f_{2S} + \phi_S(1 - f_{2S}) \\
\phi_t &= \phi_T f_{3P} + \phi_P(1 - f_{3P}) \\
\phi_b &= \phi_P f_{3B} + \phi_B(1 - f_{3B}) \\
\phi_e &= \phi_E f_{1P} + \phi_P(1 - f_{1P}) \\
\phi_w &= \phi_P f_{1W} + \phi_W(1 - f_{1W})
\end{aligned} \tag{4.19}$$

Expressions of this form are also suitable for the evaluation of the pressure terms in momentum equations (Eq. (3.41)), and the rate of production of turbulent kinetic energy in equations for k and ε (Eq. (3.39)).

After introducing (4.19) (for central and neighbour nodes) into equations like (4.18) for all six cell faces, and then inserting these into flux expressions like (3.35), the following cross-derivative diffusion contributions a_m^{DC} to the coeficients of Eqs. (3.42) result:

$$\begin{aligned}
a_E^{DC} &= -E_P^1 f_{1P} + \left[E_e^2(1-f_{2E}-f_{2SE}) + E_e^3\,(1-f_{3E}-f_{3BE})\right] f_{1P} \\
a_W^{DC} &= E_P^1(1-f_{1W}) - \left[E_w^2\,(1-f_{2W}-f_{2SW}) + E_w^3\,(1-f_{3W}-f_{3BW})\right](1-f_{1W}) \\
a_N^{DC} &= -E_P^2 f_{2P} + \left[E_n^1(1-f_{1N}-f_{1NW}) + E_n^3(1-f_{3N}-f_{3BN})\right] f_{2P} \\
a_S^{DC} &= E_P^2(1-f_{2S}) - \left[E_s^1(1-f_{1S}-f_{1SW}) + E_s^3(1-f_{3S}-f_{3BS})\right](1-f_{2S}) \\
a_T^{DC} &= -E_P^3 f_{3P} + \left[E_t^1(1-f_{1T}-f_{1TW}) + E_t^2(1-f_{2T}-f_{2TS})\right] f_{3P} \\
a_B^{DC} &= E_P^3(1-f_{3B}) - \left[E_b^1(1-f_{1B}-f_{1BW}) + E_b^2(1-f_{2B}-f_{2BS})\right](1-f_{3B}) \\
a_{TE}^{DC} &= E_e^3 f_{1P} f_{3E} + E_t^1 f_{3P} f_{1T} \\
a_{BE}^{DC} &= -E_e^3 f_{1P}(1-f_{3BE}) - E_b^1(1-f_{3B}) f_{1B} \\
a_{TW}^{DC} &= -E_w^3(1-f_{1W}) f_{3W} - E_t^1 f_{3P}(1-f_{1TW}) \\
a_{BW}^{DC} &= E_w^3(1-f_{1W})(1-f_{3BW}) + E_b^1(1-f_{3B})(1-f_{1BW})
\end{aligned}$$

$$a_{TN}^{DC} = E_n^3 f_{2P} f_{3N} + E_t^2 f_{3P} f_{2T}$$

$$a_{BN}^{DC} = -E_n^3 F_{2P}(1-f_{3BN}) - E_b^2(1-f_{3B}) f_{2B}$$

$$a_{TS}^{DC} = -E_s^3(1-f_{2S}) f_{3S} - E_t^2 f_{3P}(1-f_{2TS})$$

$$a_{BS}^{DC} = E_s^3(1-f_{2S})(1-f_{3BS}) + E_b^2(1-f_{3B})(1-f_{2BS})$$

$$a_{NW}^{DC} = -E_w^2(1-f_{1W}) f_{2W} - E_n^1 f_{2P}(1-f_{1NW}) \tag{4.20}$$

$$a_{SW}^{DC} = E_w^2(1-f_{1W})(1-f_{2SW}) + E_s^1(1-f_{2S})(1-f_{1SW})$$

$$a_{NE}^{DC} = E_e^2 f_{1P} f_{2E} + E_n^1 f_{2P} f_{1N}$$

$$a_{SE}^{DC} = -E_e^2 f_{1P}(1-f_{2SE}) - E_s^1(1-f_{2S}) f_{1S}$$

$$a_P^{DC} = \sum_m a_m^{DC} \quad , \; m = E,W,N,S,T,B,TE,BE,TW,BW,TN,BN,TS,BS,NW,SW,NE,SE$$

The coefficients E^i , appearing in above expressions, are defined as:

$$E_e^2 = \left(\Gamma_\phi \frac{D_2^1}{\delta V}\right)_e$$

$$E_e^3 = \left(\Gamma_\phi \frac{D_3^1}{\delta V}\right)_e$$

$$E_w^2 = \left(\Gamma_\phi \frac{D_2^1}{\delta V}\right)_w$$

$$E_w^3 = \left(\Gamma_\phi \frac{D_3^1}{\delta V}\right)_w$$

$$E_n^1 = \left(\Gamma_\phi \frac{D_1^2}{\delta V}\right)_n$$

$$E_n^3 = \left(\Gamma_\phi \frac{D_3^2}{\delta V}\right)_n$$

$$E_s^1 = \left(\Gamma_\phi \frac{D_1^2}{\delta V}\right)_s$$

$$E_s^3 = \left(\Gamma_\phi \frac{D_3^2}{\delta V}\right)_s$$

$$E_t^1 = \left(\Gamma_\phi \frac{D_1^3}{\delta V}\right)_t$$

$$E_t^2 = \left(\Gamma_\phi \frac{D_2^3}{\delta V}\right)_t$$

$$E_b^1 = \left(\Gamma_\phi \frac{D_1^3}{\delta V}\right)_b$$

$$E_b^2 = \left(\Gamma_\phi \frac{D_2^3}{\delta V}\right)_b \tag{4.21}$$

$$E_P^1 = -E_n^1(1-f_{2P}) + E_s^1 f_{2S} - E_t^1(1-f_{3P}) + E_b^1 f_{3B}$$

$$E_P^2 = -E_e^2(1-f_{1P}) + E_w^2 f_{1W} - E_t^2(1-f_{3P}) + E_b^2 f_{3B}$$

$$E_P^3 = -E_e^3(1-f_{1P}) + E_w^3 f_{1W} - E_n^3(1-f_{2P}) + E_s^3 f_{2S}$$

This scheme results in cross-derivative diffusion contibutions to the coefficients of 18 neighbour nodes and the central node P . Since the cell face differences are uniquely defined at each cell face, the scheme is conservative, and the sum of neighbour coefficients a_m^{DC} remains equal to the central coefficient a_P^{DC} . However, boundedness of the solution is not guaranteed, since the contribution to some coefficients is negative.

It appears that a_m^{DC} is unconditionally negative if node "m" lies in the "sharp" corner (a corner with the interior angle <90°) of computational molecule. Figure 4.8 shows a two-dimensional grid (or, a two-dimensional projection of the surface x^3=0, see Fig. 3.4), with the angle

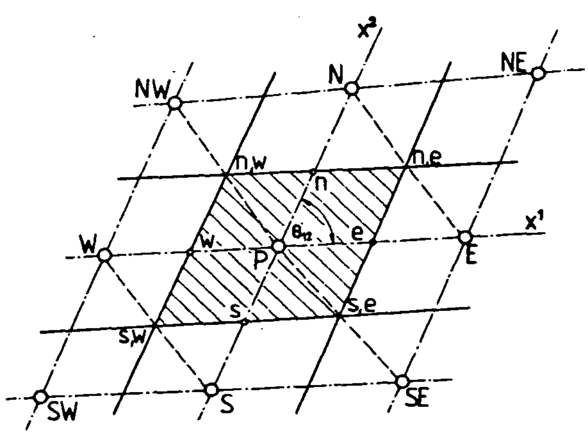


Fig. 4.8: On the evaluation of cross-derivative diffusion terms

between coordinate lines x^1 and x^2 , $\theta_{12} < 90°$. Here, a_{NE}^{DC} and a_{SW}^{DC} would turn out to be negative, while a_{NW}^{DC} and a_{SE}^{DC} would be positive, and the contribution to a_W, a_S, a_N and a_E could be either positive or negative (but zero for uniform grid). The situation would be just the opposite for $\theta_{12} > 90°$.

The magnitude of cross-derivative coefficients a_m^{DC}, relative to a_m^{DN}, depends on the ratio of cell dimensions (also called "cell aspect ratio") and the angle between the grid lines. These dependences are simultaneously determined by the ratio $|D_j^i/D_i^i|$, $i \neq j$ (no summation), which for a two-dimensional grid is:

$$\left|\frac{D_j^i}{D_i^i}\right|_{i\neq j} = \left|\frac{\delta x^{(i)}}{\delta x^{(j)}} \cos\theta_{ij}\right| \tag{4.22}$$

where $\delta x^{(i)}$, $\delta x^{(j)}$ are the cell "bisector" arc lengths measured along x^1 and x^2 coordinate lines (see Fig. 3.1 and 3.2). The larger are the aspect ratio, $\frac{\delta x^{(i)}}{\delta x^{(j)}}$, and the departure from orthogonality ($\theta_{ij} \neq 90°$), the more will be significant a_m^{DC} . Table 4.2 shows the a_m^{DC} and a_m^{DN} contributions for a two-dimensional case shown in Fig. 4.8, for a uniform diffusivity coefficient Γ_ϕ and a uniformly spaced grid ($f_1=f_2=\frac{1}{2}$) of parallel straight lines for each coordinate.

	m =	E	W	N	S	NE	NW	SE	SW
$\frac{\delta x^{(1)}}{\delta x^{(2)}} = 1$	a_m^{DN}	$\sqrt{2}\Gamma_\phi$	$\sqrt{2}\Gamma_\phi$	$\sqrt{2}\Gamma_\phi$	$\sqrt{2}\Gamma_\phi$	0	0	0	0
$\theta_{12} = 45°$	a_m^{DC}	0	0	0	0	$-\frac{\Gamma_\phi}{2}$	$\frac{\Gamma_\phi}{2}$	$\frac{\Gamma_\phi}{2}$	$-\frac{\Gamma_\phi}{2}$
$\frac{\delta x^{(1)}}{\delta x^{(2)}} = 10$	a_m^{DN}	$\frac{\sqrt{2}}{10}\Gamma_\phi$	$\frac{\sqrt{2}}{10}\Gamma_\phi$	$10\sqrt{2}\,\Gamma_\phi$	$10\sqrt{2}\,\Gamma_\phi$	0	0	0	0
$\theta_{12} = 45°$	a_m^{DC}	0	0	0	0	$-\frac{\Gamma_\phi}{2}$	$\frac{\Gamma_\phi}{2}$	$\frac{\Gamma_\phi}{2}$	$-\frac{\Gamma_\phi}{2}$

Table 4.2: Normal- and cross-derivative contributions to coefficients in Eq.(3.42) for a two-dimensional uniform grid and uniform Γ_ϕ , SCHEME 1

From table 4.2 one can see that, for θ_{12} =45°, the cross-derivative contribution to the "corner" coefficients outweighs the contribution to "principal" coefficients a_E and a_W for cell aspect ratio greater than $4\sqrt{2}$. Also a_{NE}^{D} and a_{SW}^{D} , which receive contributions from the cross-derivatives only, stay negative for any aspect ratio and $\theta_{12} < 90°$.

This feature of the scheme described (ie., having unconditionally negative "corner" coefficients) is undesirable, since negative diffusion coefficients can result in anomalous behaviour, as experienced by Demirdžić (1982). He proposed an alternative scheme, described below, which remedied the situation in his case.

SCHEME 2

This scheme, proposed by Demirdžić (1982), corresponds to using the following expressions instead of (4.18) and (4.19):

$$(\phi_n-\phi_s)_e = \begin{cases} (\phi_E-\phi_{SE})f_{1P} + (\phi_N-\phi_P)(1-f_{1P}) & , \text{ for } 0<\theta_{12}<90° \\ (\phi_{NE}-\phi_E)f_{1P} + (\phi_P-\phi_S)(1-f_{1P}) & , \text{ for } 90°<\theta_{12}<180° \end{cases}$$

$$(\phi_t-\phi_b)_e = \begin{cases} (\phi_E-\phi_{BE})f_{1P} + (\phi_T-\phi_P)(1-f_{1P}) & , \text{ for } 0<\theta_{12}<90° \\ (\phi_{TE}-\phi_E)f_{1P} + (\phi_P-\phi_B)(1-f_{1P}) & , \text{ for } 90°<\theta_{12}<180° \end{cases} \tag{4.23}$$

Here, θ_{ij} is the angle between positive parts of coordinate lines x^i and x^j. This treatment is equivalent to evaluating ϕ_{ne}, for the situation shown in Fig. 4.8, by interpolating between N and E nodes only, thus excluding "sharp corner" ones. The following cross-derivative contribution to the coefficients of discretised equation (3.42) result if this scheme is employed:

$$a_E^{DC} = E_e^2 f_{1P}(2c_e^{12}-1) + E_e^3 f_{1P}(2c_e^{13}-1) + E_n^1(1-f_{2P})c_n^{12} + E_t^1(1-f_{3P})c_t^{13} - E_s^1 f_{2S}(1-c_s^{12}) - E_b^1 f_{3B}(1-c_b^{13})$$

$$a_W^{DC} = E_w^2(1-f_{1W})(2c_w^{12}-1) + E_w^3(1-f_{1W})(2c_w^{13}-1) - E_n^1(1-f_{2P})(1-c_n^{12}) - E_t^1(1-f_{3P})(1-c_t^{13}) + E_s^1 f_{2S}c_s^{12} + E_b^1 f_{3B}c_b^{13}$$

$$a_N^{DC} = E_n^1 f_{2P}(2c_n^{12}-1) + E_n^3 f_{2P}(2c_n^{23}-1) + E_e^2(1-f_{1P})c_e^{12} + E_t^2(1-f_{3P})c_t^{23} - E_w^2 f_{1W}(1-c_w^{12}) - E_b^2 f_{3B}(1-c_b^{23})$$

$$a_S^{DC} = E_s^1(1-f_{2S})(2c_s^{12}-1) + E_s^3(1-f_{2S})(2c_s^{23}-1) - E_e^2(1-f_{1P})(1-c_e^{12}) - E_t^2(1-f_{3P})(1-c_t^{23}) + E_w^2 f_{1W}c_w^{12} + E_b^2 f_{3B}c_b^{23}$$

$$a_T^{DC} = E_t^1 f_{3P}(2c_t^{13}-1) + E_t^2 f_{3P}(2c_t^{23}-1) + E_e^3(1-f_{1P})c_e^{13} + E_n^3(1-f_{2P})c_n^{23} - E_w^3 f_{1W}(1-c_w^{13}) - E_s^3 f_{2S}(1-c_s^{23})$$

$$a_B^{DC} = E_b^1(1-f_{3B})(2c_b^{13}-1) + E_b^2(1-f_{3B})(2c_b^{23}-1) - E_e^3(1-f_{1P})(1-c_e^{13}) - E_n^3(1-f_{2P})(1-c_n^{23}) + E_w^3 f_{1W}c_w^{13} + E_s^3 f_{2S}c_s^{23}$$

$$a_{SE}^{DC} = -E_e^2 f_{1P}c_e^{12} - E_s^1(1-f_{2S})c_s^{12}$$

$$a_{NE}^{DC} = E_e^2 f_{1P}(1-c_e^{12}) + E_n^1 f_{2P}(1-c_n^{12})$$

$$a_{SW}^{DC} = E_w^2(1-f_{1W})(1-c_w^{12}) + E_s^1(1-f_{2S})(1-c_s^{12})$$

$$a_{NW}^{DC} = -E_w^2(1-f_{1W})c_w^{12} - E_n^1 f_{2P}c_n^{12}$$

$$a_{TE}^{DC} = E_e^3 f_{1P}(1-c_e^{13}) + E_t^1 f_{3P}(1-c_t^{13}) \tag{4.24}$$

$$a_{BE}^{DC} = -E_e^3 f_{1P}c_e^{13} - E_b^1(1-f_{3B})c_b^{13}$$

$$a_{TW}^{DC} = -E_w^3(1-f_{1W})c_w^{13} - E_t^1 f_{3P}c_t^{13}$$

$$a_{BW}^{DC} = E_w^3(1-f_{1W})(1-c_w^{13}) + E_b^1(1-f_{3B})(1-c_b^{13})$$

$$a_{TN}^{DC} = E_n^3 f_{2P}(1-c_n^{23}) + E_t^2 f_{3P}(1-c_t^{23})$$

$$a_{BN}^{DC} = -E_n^3 f_{2P}c_n^{23} - E_b^2(1-f_{3B})c_b^{23}$$

$$a_{TS}^{DC} = -E_s^3(1-f_{2S})c_s^{23} - E_t^2 f_{3P}c_t^{23}$$

$$a_{BS}^{DC} = E_s^3(1-f_{2S})(1-c_s^{23}) + E_b^2(1-f_{3B})(1-c_b^{23})$$

$$a_P^{DC} = \sum_m a_m^{DC} \quad , \ m = E,W,N,S,T,B,SE,NE,SW,NW,TE,BE,TW,BW,TN,BN,TS,BS$$

In these expressions the coefficients c^{ij} have the following meaning:

$$c^{ij} = \begin{cases} 1, \text{ if } 0<\theta_{ij}<90° \\ 0, \text{ if } 90°<\theta_{ij}<180° \end{cases} \tag{4.25}$$

This scheme is also conservative, but it still does not guarantee boundedness of the solution, since the contributions to some coefficients are negative. However, unlike the scheme previously described, this scheme does not produce negative "corner" coefficients, which never have contribution from the normal derivatives nor necessarily from the convective terms (eg., if UDS, LUDS or QUDS are employed). But, while the contributions to the corner coefficients are always non-negative, those to the "principal" coefficients (a_m ,m=E,W,N,S,T,B) are always negative and stronger than in the previous case. Table 4.3 shows the values that would result for the case shown in Fig. 4.8, under the same conditions as those for which the values in table 4.2 were calculated.

	m =	E	W	N	S	NE	NW	SE	SW
$\frac{\delta x^{(1)}}{\delta x^{(2)}} = 1$	a_m^{DN}	$\sqrt{2}\,\Gamma_\phi$	$\sqrt{2}\,\Gamma_\phi$	$\sqrt{2}\,\Gamma_\phi$	$\sqrt{2}\,\Gamma_\phi$	0	0	0	0
$\theta_{12} = 45°$	a_m^{DC}	$-\Gamma_\phi$	$-\Gamma_\phi$	$-\Gamma_\phi$	$-\Gamma_\phi$	0	Γ_ϕ	Γ_ϕ	0

Table 4.3: Diffusion contribution to the coefficients of Eq.(3.42) under the same conditions as for Table 4.2, but with SCHEME 2

From table 4.3 one can see that in this case, for any aspect ratio greater than $\sqrt{2}$, the total diffusion contribution to the coefficients of E and W nodes will be negative. It can be shown that, in the limit of grid being non-orthogonal in one plane only (which includes the two-dimensional case) and a uniform diffusivity coefficient Γ_ϕ , the diffusion contribution will stay positive for all nodes if the following condition is satisfied (Demirdžić, 1982):

$$|\cos\theta_{ij}|\max\left|\frac{\delta x^{(i)}}{\delta x^{(j)}}\right|_{i\neq j} \leq 1 \qquad (4.26)$$

Infringement of this condition may, even when a bounded scheme for the convective terms (eg. UDS) is used, cause the a_m coefficients to become negative due to the cross-derivative contributions, and unbounded (unrealistic) solutions may thereby be generated, as well as numerical instability.

The two schemes for the evaluation of cross-derivative diffusion terms described above are relatively simple, and in most cases with moderate non-orthogonality (45°<θ_{ij}<135°) and cell aspect ratio (< 5) they will yield bounded and stable results. As for the relative accuracy, it would be expected to be consistent with that of the linear interpolations used for the evaluation of the other terms.

It is possible to use higher-order interpolation practices, which may be desirable in the case of non-smooth grids, when linear interpolation along grid lines may be inaccurate, as will be demonstrated later in Chapter 8. In such cases interpolation formulae which give a continuous distribution of ϕ , fitted to the nodes in the region of interest (as is usually done in finite element methods) may be more appropriate. Such interpolation practices, on the other hand, result in considerably more complex formulae for the a_m^{DC} , especially in general three-dimensional cases, and for that reason they are not evaluated here. However, this subject deserves to be given more attention in the future, in order to assess the effects of the higher-order interpolation on the accuracy and stability of the present method.

4.5 Remedies for Unboundedness of Higher-Order Differencing Schemes

As was shown earlier, unconditionally bounded solutions will result only if all the coefficients of the discretised equations (3.42) are positive. This condition is satisfied only by the UDS for convective contribution, and under certain conditions (of Eq. (4.26)) by SCHEME 2 for the diffusion contribution.

As already noted, use of UDS can sometimes introduce problems of numerical diffusion. If sufficient grid refinement is not possible, the use of higher-order scheme becomes imperative if accuracy is to be improved; however, if this results in unbounded solutions* a cure must be sought which will remedy both problems, ie. unboundedness and numerical diffusion. In the past several methods have been devised which aim to achieve this. Some will be briefly described before a new method, developed in the present study, is presented.

4.5.1 Flux-Corrected Transport (FCT)

This method, due to Boris and Book (1973), was originally devised for the solution of the one-dimensional continuity equation in explicit time-marching methods, with the aim of maintaining the density field positive, and avoiding overshoots at peaks and oscillations elsewhere. It was later extended to multi-

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* Unbounded solution may result whenever flow direction is oblique to grid lines and/or there exist regions in which the rate of change of gradient of the dependent variable is high ("sharp" changes in profiles).

dimensional problems (Book et al, 1975) and further generalized (Zalesak, 1979), so that it could be used in conjunction with any two differencing schemes, one of which must be unconditionally bounded and non-oscillatory (invariably the UDS) and the other of higher accuracy but unbounded. The method goes through the following stages at each time step:

(1) A provisional updated ("transported and diffused") solution ϕ_i^{td} is obtained using the bounded scheme fluxes I_B ;

(2) A local antidiffusion flux correction I_{corr}, defined to be

$$I_{corr} = \gamma(I_U - I_B) \tag{4.27}$$

is sought (where I_U is flux obtained using unbounded, but more accurate scheme, and γ is a weighting factor between zero and unity) such that corrected solution does not exhibit extrema not found in the provisional solution ϕ_i^{td} and that of the previous time level, ϕ_i^n ;

(3) The provisional solution is corrected by applying I_{corr} .

The most critical step in the above is that of evaluation weighting factors γ . In multidimensional problems it is required that:

$$\phi_{min} < \phi_i^{n+1} < \phi_{max} \tag{4.28}$$

where the bounds are:

$$\begin{aligned} \phi_{max} &= \max(\phi_m^{td}, \phi_m^n) \\ \phi_{min} &= \min(\phi_m^{td}, \phi_m^n) \end{aligned} \tag{4.29}$$

and m covers range of neighbours (E,W,N,S,T,B, see Fig. 3.4). This practice does not guarantee that the physical bounds will always be obeyed, as will be demonstrated later. To overcome this Zalesak (1979) proposed that the correction procedure should be first applied in a one-dimensional manner, and then finally using multidimensional bounds (4.29). This, however, may result in "clipping" of physical peaks and "terracing" (ie. the extrema suppressed, but the profiles are not smooth) problems, as reported by Book et al (1975). Various remedies for these two problems have been suggested, but none can be said to have general validity (Book et al, 1975; Zalesak, 1979).

The FCT method has been employed on several discretisation schemes and applied to various problems, always giving improved results as compared to use of the "uncorrected" fluxes. More details, in addition to the references already cited can be found in Book (1981). However, the FCT method is fairly complex and strictly speaking only applicable to explicit time marching procedures (which possess the well known Courant restriction on the maximum time step) which are expensive of computer time and unsuitable if steady state solutions are sought.

4.5.2 Filtering Remedy and Methodology (FRAM)

A non-linear "filtering" methodology, called FRAM, has been developed by Chapman (1981). Unlike the FCT method, where the provisional solution is first calculated using a lower-order, "overdiffusive" scheme and then the "antidiffusion" correction is applied, FRAM operates the other way around, ie. a provisional solution at the new time level is first calculated using the higher-order scheme, and a "diffusion" correction is applied locally, only where necessary. The algorithm can be summarized as follows:

(i) Calculate a provisional time-advanced solution $\tilde{\Phi}_i^{n+1}$ using a higher-order (unbounded) scheme to evaluate the fluxes I_U;

(ii) Determine the local bounds on the advanced time solution;

(iii) Introduce a strong local corrective dissipation flux into the equations wherever the provisional solution is not within bounds, by switching to the bounded scheme at such locations.

The bounds are here determined by the local Lagrangian solutions Φ_i^* at the central and neighbouring points.

Seen in the light of FCT description, here a diffusion correction flux

$$I_{corr} = \gamma(I_B - I_U) \tag{4.30}$$

is applied, where γ is taken to be zero if $\tilde{\Phi}_i^{n+1}$ is within the bounds, and unity otherwise. In this respect FRAM is simpler than FCT and thus claimed to be more economical. However, the remarks made earlier about the "clipping" and "terracing" problems, encountered with FCT, apply also to FRAM, perhaps even more so because it uses the extreme cases, $\gamma=0$ or $\gamma=1$ only. Further, like FCT, FRAM is also designed for time-marching explicit solution algorithms, and cannot be directly implemented in implicit methods, whether time marching or steady state.

4.5.3 Flux blending methods

The idea of flux blending is similar to that exploited in the FCT and FRAM methods, and can be expressed as follows: at each cell face a resultant flux I is obtained by blending a flux I_B, obtained from a lower-order, unconditionally bounded scheme, with the flux I_U, determined from a higher-order, more accurate but not necessarily bounded scheme, thus

$$I = \gamma I_U + (1-\gamma)I_B \tag{4.31}$$

The main question to be answered is: how is the local value of γ to be determined for each cell face? Different approaches are possible, and there is also the possibility of applying the blending to convective, normal- and cross-derivative diffusive fluxes separately, or simultaneously, as will now be demonstrated.

Hybrid scheme (HS)

One of the first and perhaps most widely used methods that can be said to employ a form of flux blending technique is the so-called "hybrid scheme" of Spalding (1972). It operates on the convective fluxes, with the CDS as the U-scheme, and the UDS as the B-scheme. Strictly speaking, the fluxes are not blended; rather as in FRAM, the CDS (γ=1) is used wherever its contribution a_m^c does not cause coefficients in Eq. (3.42) to become negative (ie., wherever Pe<2), and the UDS (γ=0) is used otherwise. The scheme is simple and involves a single step in determining local values of γ, in which only the Peclet number has to be evaluated. However, in convection-dominated flows (Pe$\gg$1) the result of this scheme is that UDS is used everywhere, irrespective of whether boundedness problems arise or not, and thus the accuracy of the solution may be impaired by excessive numerical diffusion.

Bounded Skew Upwind Differencing Scheme (BSUDS)

This scheme, developed by Lai (1982), also aims to keep the coefficients of the discretised equations non-negative, in this instance by blending the SUDS and UDS schemes, although in principle the method could be applied to any two "compatible"* schemes. As was described earlier, SUDS has "corner" coefficients which are always positive, but the "principal" ones (a_E, a_W, a_N and a_S for a two-dimensional case) can become negative. On the other hand, UDS generates only principal coefficients and they are always positive, so it is possible to find a blending factor γ such that blended coefficients are non-negative. Lai (1982) achieves this by a modification of the SIMPLEX optimization algorithm (Dantzig, 1951), which results in a unique solution for the γ's. As this method of determining the blending factors does not depend on the solution itself, it is linear and therefore non-iterative.

The BSUDS requires little extra work for coefficient assembly as compared to SUDS, especially if scalar equations are being solved for a given flow field, when the γ's need be calculated only once. Moreover, since diagonal dominance is ensured by this blending procedure, the BSUDS is unconditionally stable as well as unconditionally bounded. The attractiveness of this scheme is however reduced by the facts that (i) its accuracy is lowered by the requirement that the coefficients be always positive since this is a sufficient, rather than necessary condition, and (ii) SUDS itself is significantly more complex and time consuming in coefficient assembly than eg. LUDS or QUDS, especially in the three-dimensional problems, where the computational molecule of the SUDS is a clumsy 27-point one.

4.6 A New Flux Blending Technique

4.6.1 Introduction

All bounding techniques presented in the previous section concentrated on treatment of convetive terms, chiefly because for discretisa-

* Two schemes are compatible in this respect if one produces positive contribution to those coefficients, to which the other scheme brings negative contribution, so that blending of the two can always result in non-negative coefficients. In general both schemes individually may be unbounded (higher-order), if the above criterion is met.

tion on orthogonal grids, to which they were directed, negative coefficients do not arise from the diffusion terms, at least when conventional second-order central differencing is used for them (see section 4 of this chapter and Eqs. (4.17)). In what follows a new method of suppressing spurious oscillations in solution of transport equations arising from the treatment of either the convective or diffusive terms will be described. It falls within the flux-blending category, and aims to use as much of the higher-order scheme as possible while retaining a bounded solution. As the method depends on the solution field itself to determine the blending factors, it is non-linear and therefore of an iterative nature: as a consequence it is more expensive than either of the component schemes on its own.

Implicit solution algorithms, like the one adopted for use in the present study, make it difficult to determine blending factors γ such that resultant fluxes produce a bounded solution everywhere. In particular, a change of γ at one cell face will provoke changes in the solution at more distant nodes, and it seems to be impossible to determine all the γ's in one step so that the new solution is everywhere bounded. For non-linear problems (like the Navier-Stokes equations) this is even more difficult to accomplish, as changes in the solution for one dependent variable result in changes in the solution for others too. Therefore, some sort of iterative procedure has to be devised. But first, one has to decide which quantities have to be bounded and what are the bounds to be. This is the subject of the next subsection.

4.6.2 Boundedness criterion

Taking into account Eqs. (3.43) and (3.44), the discretised transport equations (3.42) can be rewritten as:

$$\Big(\sum_m a_m^C + \sum_m a_m^D + S_\phi''\Big)\,\phi_P = \sum_m a_m^C \phi_m + \sum_m a_m^D \phi_m + C_P \tag{4.32}$$

When the set of these equations is solved for ϕ , an explicit expression of the following form will hold between ϕ_P and the components contributing to it:

$$\phi_P = \frac{1}{\sum_m a_m^C + \sum_m a_m^D + S_\phi''}\left(\sum_m a_m^C \phi_m + \sum_m a_m^D \phi_m + C_P\right) \tag{4.33}$$

This expression shows that the value of ϕ at node P is made up of contributions of three kinds: convection, diffusion and sources. One can define the following quantities which describe these contributions:

$$\phi_P^C = \frac{\sum_m a_m^C \phi_m}{\sum_m a_m^C}$$

$$\phi_P^D = \frac{\sum_m a_m^D \phi_m}{\sum_m a_m^D} \tag{4.34}$$

$$\phi_P^S = \frac{C_P}{\sum_m a_m^C + \sum_m a_m^D + S_\phi''}$$

When these are introduced into equation (4.33), the following relation between them results:

$$\phi_P = (f^C \phi_P^C + f^D \phi_P^D) f^{CD} + \phi_P^S = \phi_P^{CD} f^{CD} + \phi_P^S \tag{4.35}$$

where the weighting factors f^C, f^D and f^{CD} are given by:

$$f^C = \frac{\sum_m a_m^C}{\sum_m a_m^C + \sum_m a_m^D}$$

$$f^D = \frac{\sum_m a_m^D}{\sum_m a_m^C + \sum_m a_m^D} \tag{4.36}$$

$$f^{CD} = \frac{\sum_m a_m^C + \sum_m a_m^D}{\sum_m a_m^C + \sum_m a_m^D + S_\phi''}$$

It is obvious that $f^C + f^D = 1$; therfore, the combined convection-diffusion contribution ϕ_P^{CD} is formed by linear interpolation between ϕ_P^C and ϕ_P^D , as is schematically shown in Fig. 4.9. If the linearized source part S'' is zero, then f^{CD} is equal to unity, and the source contribution ϕ_P^S , if nonzero, adds to ϕ_P^{CD} to make the total value of ϕ_P . One can now examine these three contributions to ϕ_P separately, from the boundedness point of view.

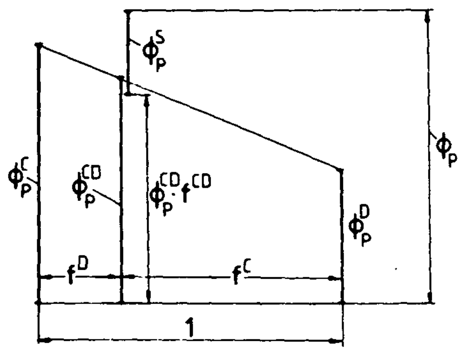


Fig. 4.9: Formation of ϕ_P from convection, diffusion and source contributions

Source contribution, ϕ_P^S. The evaluation of the source terms in finite-difference forms of transport equations is an important issue in itself, and can significantly influence both the accuracy and convergence rate of a numerical solution procedure. However, we shall assume here that these terms are adequately evaluated and that their contribution to ϕ_P is physically realistic. The source terms can make ϕ_P lie outside the range of its neighbours, and therefore the solution, when such sources are present, does not necessarily have to be monotonic. In formulating a bounding procedure it must therefore be recognized that local extrema due to the source effects are physically realistic, and should not be suppressed or clipped.

Diffusion contribution, ϕ_P^D. Since diffusion is a two-way process the value of ϕ at node P can be influenced by changes at any neighbour node. From the definition of the boundedness criterion in section 2 of this chapter (and Eq. (4.1)), it can be concluded that there are no physical grounds for ϕ_P^D to be larger than the maximum, nor smaller than the minimum value at immediate neighbour nodes (E,W,N,S,T and B for a three-dimensional case), ie. it should be bounded by:

$$\begin{aligned} \phi_{P,max} &= \max(\phi_E, \phi_W, \phi_N, \phi_S, \phi_T, \phi_B) \\ \phi_{P,min} &= \min(\phi_E, \phi_W, \phi_N, \phi_S, \phi_T, \phi_B) \end{aligned} \qquad (4.37)$$

If all a_m^D (ie. the sum $a_m^{DN}+a_m^{DC}$) are positive, then this will be unconditionally satisfied (this will occur when the grid is orthogonal, or when

SCHEME 2 is used for cross-derivative diffusion terms and conditions such as (4.26) are satisfied). Otherwise, ϕ_P^D may fall outside its bounds, and if f^D is not small, it can make ϕ_P unbounded. In such a case blending of the scheme used with one (if available) which produces unconditionally positive coefficients, and may be less accurate, could be performed to bring ϕ_P^D within the bounds. One of possible methods for evaluating the blending factors will be described in the next subsection. It can also be noted here that the contribution of ϕ_P^D to ϕ_P becomes smaller as Pe increases, since f^C tends to unity and f^D tends to zero as Pe tends to infinity: so in such circumstances the ϕ_P^D contribution is of lesser importance than ϕ_P^C.

Convective contribution, ϕ_P^C . It is usually this part of ϕ_P which drives it out of bounds when "higher-order" schemes like CDS, LUDS, QUDS are used for the convective terms in high Pe flows, under the circumstances described earlier in this chapter. While diffusion-schemes behave well on orthogonal grids, with the forementioned convective ones this makes no difference: negative coefficients are either always there, or their appearance depends on the flow conditions, but grid non-orthogonality playes no role. Therefore, the convective contribution ϕ_P^C is the most "dangerous" from the boundedness point of view, and deserves special attention.

As discussed earlier, from the transportive property of a fluid flow we know that, in the limit of high Pe , the value of ϕ at any node will be only influenced by values at upstream nodes - for when $Pe \to \infty$, the lines of constant ϕ will coincide with streamlines. It can, therefore, be concluded that by the action of convection only the value of ϕ at node P should not rise above the maximum, nor should it fall below the minimum value at upstream neighbour nodes. This is schematically shown in Fig. 4.10 for a two-dimensional case, with F_1 and $F_2 > 0$, where the bounds for ϕ_P^C would be:

$$\begin{aligned} \phi^C_{P,max} &= \max(\phi_W, \phi_{SW}, \phi_S) \\ \phi^C_{P,min} &= \min(\phi_W, \phi_{SW}, \phi_S) \end{aligned} \tag{4.38}$$

From these considerations the boundedness criterion for ϕ_P^C can be defined as follows: the convective contribution ϕ_P^C at any node should lie within the bounds given by the values ϕ_n at the n upwind neighbour nodes in the "compact" computational molecule (9-point in two-dimensional case, 27-point in a three-dimensional case, see Fig. 3.4). This is

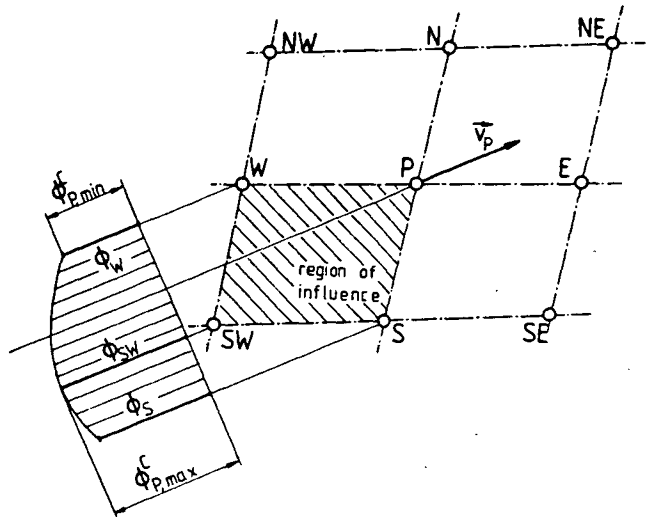


Fig. 4.10: The region of influence for node P

where the present method differs mostly from the FCT and FRAM methods, which both use bounds of the type (4.37) for convective fluxes, or require monotonic solution in each coordinate direction. According to the experience of the present study, bounds of the type (4.37) do not guarantee bounded solutions (this will be shown in Chapter8), and the latter approach is not appropriate when the solution is not monotonic due to either sources or initial conditions (an example illustrating this will be presented in Chapter 8).

Some remarks about the boundedness criterion for ϕ_P^C adopted here can now be given. Firstly, in the absence of sources and at high Pe, it will guarantee that interior nodal values are bounded by values at upstream boundaries, as expected. This will be later proved on some test cases. Secondly, in the one-dimensional case it will demand that $\phi_P^C = \phi_W$ for $u > 0$, ie. the use of UDS. This may seem like a drawback, but it should be borne in mind that ϕ_P^C arises from convection only, and in the one-dimensional case pure translation of upstream information would be expected. These comments are of course restricted to steady flows, which are the subject of the present study; however the analysis can readily be extended to the unsteady case.

It remains to evaluate the blending factors γ. One possible approach is described next.

4.6.3 Evaluation of Blending Factors γ

As already noted, in order to maintain the conservativeness of the blended scheme, the blending factors γ should be associated with cell faces rather than cell nodes. This makes it impossible to determine all the γ's uniquely in a single set of operations, such that the final converged solution is bounded everywhere. Therefore, the procedure must be iterative and the one devised here will now be described. It can be used if either convective, diffusive or total fluxes are to be blended, so coefficients a_m appear without superscript (but stand for either the convection part a_m^C, the cross-derivative diffusion part a_m^{DC}, the total diffusion part a_m^D or the complete coefficient, see Eq. (3.43)).

Consider the blended flux expression:

$$I = \gamma I_U + (1-\gamma)I_B = I_B + \gamma(I_U - I_B) \tag{4.39}$$

When particular U (unbounded, higher-order) and B (bounded, UDS for convective terms) differencing schemes are introduced in equation (4.39) for each cell face, the resulting coefficients of the discretised equation (3.42) will consist of two parts: one a_m' which does not depend on γ (the "B"-contribution), and one $\gamma a_m''$ which is γ-dependent (the "U-B" contribution):

$$a_m = a_m' + \gamma a_m'' \tag{4.40}$$

This leads to an expression for the contribution to ϕ_P which is to be "bounded" (and which could be ϕ_P^C, ϕ_P^D or ϕ_P^{CD}) of the following form (assuming that the same γ is applied to all cell faces enclosing node P):

$$\phi_P = \frac{\sum_m a_m' \phi_m + \gamma \sum_m a_m'' \phi_m}{\sum_m a_m' + \gamma \sum_m a_m''} \tag{4.41}$$

By requiring that $\phi_{min} < \phi_P < \phi_{max}$, one gets the following expression for <u>the maximum value of blending factor γ at all cell faces enclosing node P, which will keep ϕ_P bounded</u>:

$$\gamma_P \leqslant \frac{\phi^M \sum_m a_m' - \sum_m a_m' \phi_m}{\sum_m a_m'' \phi_m - \phi^M \sum_m a_m''} \tag{4.42}$$

Here ϕ^M is the violated bounding value, which is either ϕ_{max} or ϕ_{min}. With such a "provisional" γ_P value determined for each node, the final value of γ at each cell face is taken as the minimum of the two γ_P values calculated for the cells on either side of it, eg. for the "west" face:

$$\gamma_w = \min(\gamma_P, \gamma_W) \tag{4.43}$$

When all nodes are so checked for boundedness and the γ's are evaluated where necessary, the coefficients a_m are assembled again from Eq. (4.40), using the appropriate γ for each cell face, and a new solution is obtained. If the new solution is not bounded everywhere, this implies that more of the B-scheme has to be blended in at some locations, and new blending factors have to be evaluated. The procedure for this is the same as above, but now the B-scheme flux I_B is not blended with U-scheme flux I_U, but with the blended flux I^{n-1} from the previous blending step, ie.

$$I^n = \gamma^n I^{n-1} + (1-\gamma^n)I_B \tag{4.44}$$

Here, "n" denotes the new blending step, and "n-1" the previous one. This process is to be repeated as many times as necessary to produce a solution which is bounded everywhere. At this stage the final blended fluxes will be of the form:

$$I^n = I_B + \gamma^n\gamma^{n-1}\ldots\gamma^1(I_U-I_B) \tag{4.45}$$

This means that the current blending factor γ is always obtained by multiplying the one prevailing from the previous blending step with the newly determined one from Eq. (4.43). Therefore, the a''_m parts of the coefficients used in Eqs. (4.41) and (4.42) should contain the "old" γ's (starting with unity for the first blending step), and the "new" cell face γ should be taken to be the product of the value obtained from (4.43) and the "old" value at that cell face.

The overall procedure can now be summarized as follows:

(i) Assemble the coefficients of the discretised equations using $\gamma=1$ everywhere (full U-scheme);

(ii) Obtain the solution to the discretised equations;

(iii) Examine the solution for boundedness, by evaluating ϕ_P^C or ϕ_P^D (whichever may be unbounded) from (4.34) and comparing it with the appropriate bounds (Eqs. (4.37) or (4.38));

(iv) Where the checked value is not within bounds, calculate provisional γ's using (4.42) and assemble cell face blending factors by multiplying prevailing values with the newly obtained ones from (4.43);

(v) Assemble the coefficients of the discretised equations using the new blending factors;

(vi) Go to step (ii) and repeat the sequence until solution becomes bounded everywhere.

How many times the process will have to be repeated will depend on the problem involved. Obviously, if the U-scheme solution happens to be bounded, no blending will be needed. If, on the other hand, the solution happens to be unbounded in the majority of the solution domain (as in some model problems to be presented later), it may require more steps to get a bounded solution. However, in each new step less γ's will need to be changed, and therefore less iterations will be needed to obtain a new converged solution.

The above method will be assessed, by applying it to some standard test cases, in Chapter 8, where its advantages and disadvantages over the alternative methods will also be discussed.

4.7 Closure

It has been shown that interpolation practices (differencing schemes) used in the evaluation of fluxes in discretised transport equations can have a significant impact on both the accuracy and the stability of the numerical solution procedure. The three most important properties which differencing schemes should satisfy have been identified as: conservativeness, boundedness and transpotiveness. Five common differencing schemes have been investigated; four of them (UDS, CDS, LUDS, QUDS) have been found suitable for general three-dimensional applications, while the fifth one, SUDS, has been found too complex for three-, but suitable for two-dimensional cases. Only the UDS possesses all of desired properties, but also has a drawback in that its accuracy can be low when the grid and flow are not aligned. The other four schemes are more accurate in general, but on the other hand, they violate the transportiveness and/or boundedness requirements and can produce physically unrealistic solutions.

Two schemes, both involving linear interpolation, have been suggested for the evaluation of diffusion terms. One of them, SCHEME 1, produces some unconditionally negative coefficients and may be unbounded; the other one, SCHEME 2, can be made to be bounded under certain conditions.

Several methods which were designed to overcome the boundedness problems of higher-order schemes have been analysed; none of them has general applicability and validity. A new flux blending method has been devised here for that purpose, which is suitable for steady state implicit calculations. It guarantees a bounded solution of linear problems, with as much of higher-order scheme used as possible; it may not be, however, as efficient in case of non-linear problems (eg. the Navier-Stokes equations). An assessment of the method and more discussion about its features will be presented in Chapter 8.

CHAPTER 5

GRID AND VARIABLE ARRANGEMENTS AND CALCULATION OF PRESSURE

5.1 Introduction

In the two previous chapters the discretisation of the transport equations and the interpolation practices have been described. The discretisation has been carried out for cells enclosing the nodes at which a particular variable is "stored". However, the nodes and associated control volumes for different variables need not coincide, if this facilitates solution. In particular, existing methods of calculation of pressure, which does not have its "own" governing equation, depend strongly on the relative positions of the velocity and pressure nodes within the solution domain. These matters are addressed in this chapter.

First, in sections 2 and 3, the grid and variable arrangements are discussed. Several arrangements proposed and used by various research groups are analysed; their suitability for use in conjunction with general non-orthogonal grids and Cartesian velocity components is examined, and the choice for use in the present study is made.

Next, in section 4, the calculation of pressure is examined. An interpolation practice which facilitates the use of a family of pressure-calculation algorithms, originally devised and used in conjunction with orthogonal grids, is first introduced. This is followed by descriptions of four pressure-calculation algorithms: SIMPLE, SIMPLER, PISO and SIMPLEC, suitably modified here for the selected variable arrangement.

Finally, in section 5 some closing remarks are given.

5.2 Grid Arrangements

The two possible grid arrangements considered here are depicted (for the two-dimensional case) in figures 5.1 and 5.2 . In the first case a grid is drawn first and the computational nodes are placed at the crossings of the grid lines. Control volumes are then constructed around each node, by placing their boundaries midway between the nodes.

In the second case (Fig. 5.2) the grid first drawn defines the control volume boundaries and the computational nodes are then placed in the centre of each volume (cell).

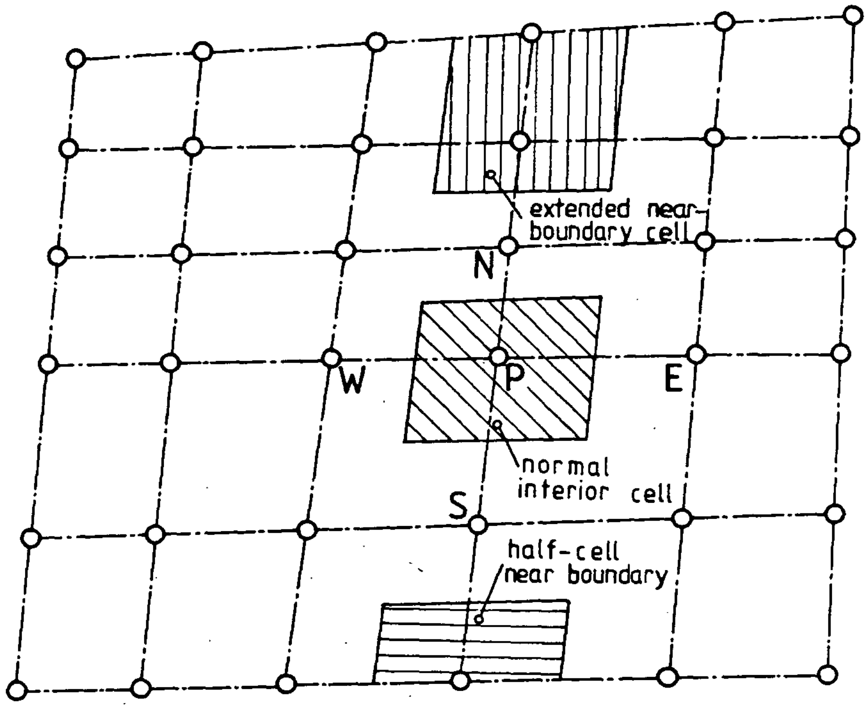


Fig. 5.1: Grid arrangement where nodes are defined at grid intersections

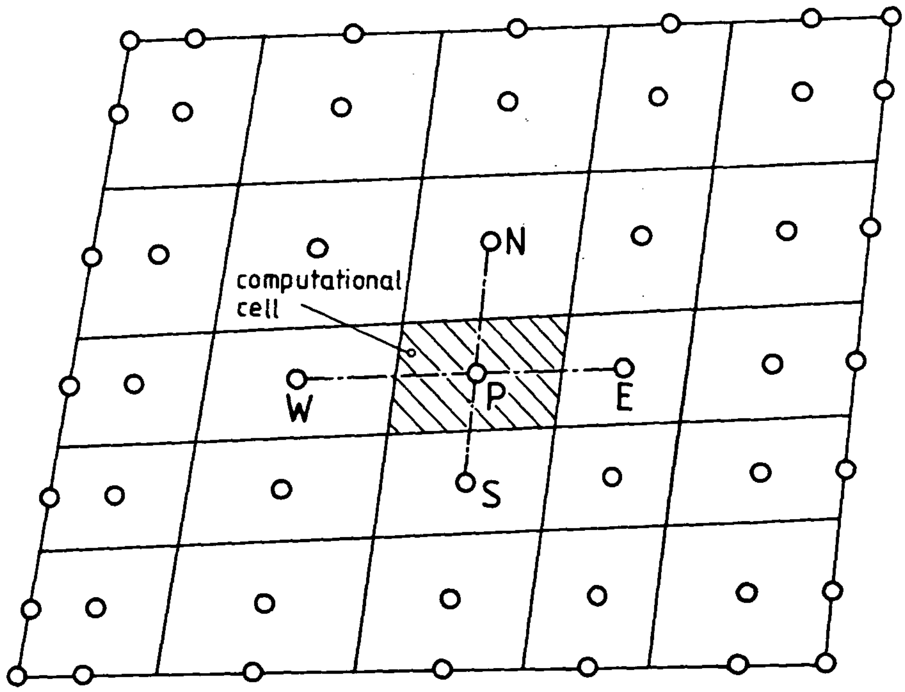


Fig. 5.2: Grid arrangement with nodes in the centre of control volumes

While the first practice may be more convenient for the finite-difference solution techniques, for the finite-volume approach used here the second practice is preferred, as will now be explained.

The main advantage of the arrangement shown in Fig. 5.2 is that the computational nodes reside in the centre of control volumes they represent. This is a very desirable feature, especially for the integration of the source terms. Secondly, the cell face locations e , w etc. are in the middle of the cell faces, which facilitates the derivation of accurate average flux approximations. Thirdly, with this arrangement irregular cells are not generated adjacent to the solution domain boundaries, whereas in the first practice it is necessary either to insert half cells near the boundary, or to extend the adjacent cells to the boundary, as shown in Fig. 5.1 . These irregular cells require special treatment, which is inconvenient and sometimes cumbersome. Finally, with the second arrangement it is easier to treat different types of boundary conditions, different shapes of boundaries and solid bodies submerged within the solution domain, as illustrated in Fig. 5.3 . The treatment of these features in general is simpler than with the other arrangement because the boundary nodes can be placed at the centre of boundary cell faces in what can be imagined as zero-thickness cells, which facilitates implementation of the boundary conditions, as will be demonstrated in the next chapter.

The only advantage that the first practice has over the second is that the linear interpolation used in approximating normal derivatives at cell faces (ie. the "diffusion" terms) is third order accurate (in terms of the TSTE) even if the grid is non-uniform. This is because cell

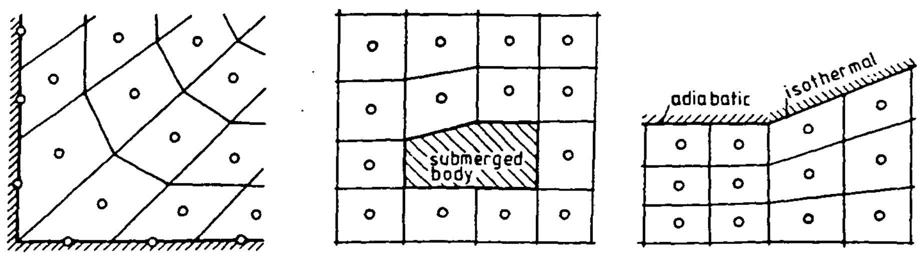


Fig. 5.3: Some examples of grid construction, with irregular boundaries and submerged bodies, using the practice of Fig. 5.2

face locations are midway between the nodes, and therefore the slope of a linear profile at that location is the same as the slope of a quadratic profile. However, since the cell face location where gradient is calculated does not (for a non-uniform grid) lie in the centre of that cell face, the flux thus calculated is not as representative as it would be if the location was centered, as in the second arrangement.

For the reasons listed above, the grid arrangement shown in Fig. 5.2 is selected for use in the present study.

5.3 Variable Arrangements

In the forementioned grid arrangements the "nodes" could be taken to be the locations where all dependent variables ϕ are stored. However, if the fluid flow equations are being solved, problems may arise with this arrangement. The reason for this is connected with the fact that pressure, which does not have its "own" governing equation, has to be extracted somehow from the continuity and momentum equations.

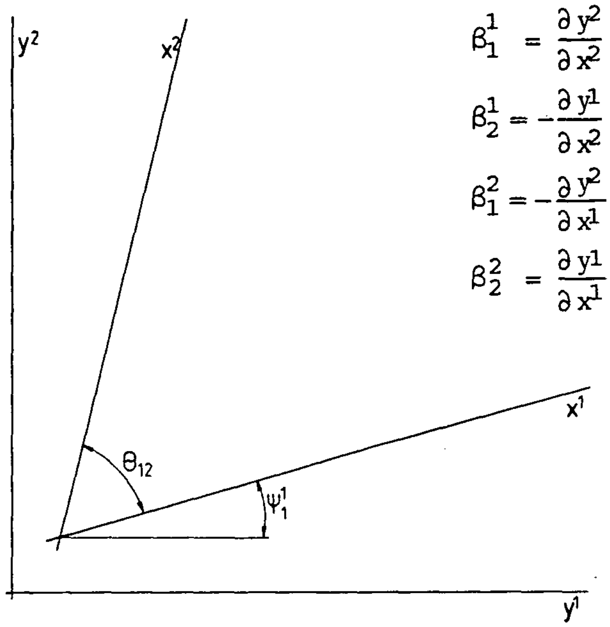
In case of compressible, high Mach number flows, the continuity equation (3.3) can be taken as an equation for density, and pressure can then be obtained via an equation of state. Since, however, the present study is concerned with incompressible - or low Mach number - flows only, such methods will not be considered here, for they can not be applied in these circumstances (a survey of methods which use this approach is given by Hollanders and Viviand, 1980).

Several other variable arrangements have been proposed in order to facilitate the calculation of pressure. Some will now be discussed by considering, for the sake of simplicity but without loss of generality, the two-dimensional momentum equations of inviscid fluid and the continuity equation. These equations are obtained by omitting the source and viscous terms from Eqs. (3.6), (3.17) and (3.18), and run:

$$\frac{\partial U_1}{\partial x^1} + \frac{\partial U_2}{\partial x^2} = 0 \tag{5.1}$$

$$\begin{aligned}\frac{\partial}{\partial x^1}(U_1 u_1) + \frac{\partial}{\partial x^2}(U_2 u_1) &= -\frac{\partial p}{\partial x^1}\beta_1^1 - \frac{\partial p}{\partial x^2}\beta_1^2 \\ \frac{\partial}{\partial x^1}(U_1 u_2) + \frac{\partial}{\partial x^2}(U_2 u_2) &= -\frac{\partial p}{\partial x^1}\beta_2^1 - \frac{\partial p}{\partial x^2}\beta_2^2\end{aligned} \tag{5.2}$$

where U_i are defined by Eq. (3.7).



$$\beta_1^1 = \frac{\partial y^2}{\partial x^2} = \sin(\theta_{12}+\psi_1^1)$$
$$\beta_2^1 = -\frac{\partial y^1}{\partial x^2} = -\cos(\theta_{12}+\psi_1^1)$$
$$\beta_1^2 = -\frac{\partial y^2}{\partial x^1} = -\sin\psi_1^1$$
$$\beta_2^2 = \frac{\partial y^1}{\partial x^1} = \cos\psi_1^1$$

Fig. 5.4: Geometrical representation of coefficients β for a simple two-dimensional case

In Fig. 5.4 an arbitrary coordinate system (x^1,x^2) and a reference Cartesian coordinate system (y^1,y^2) are shown, together with the relations between the β coefficients and the angles θ_{12} and ψ_1^1. To simplify the equations further (again without loss of generality in the conclusions derived), the case $\theta_{12}=90°$ will be considered, ie. the coordinate system (x^1,x^2) is taken to be orthogonal, but translated and rotated by ψ_1^1 relative to (y^1,y^2). In this case Eqs. (5.2) reduce to:

$$\frac{\partial}{\partial x^1}(U_1u_1) + \frac{\partial}{\partial x^2}(U_2u_1) = -\frac{\partial p}{\partial x^1}\cos\psi_1^1 + \frac{\partial p}{\partial x^2}\sin\psi_1^1$$
$$\frac{\partial}{\partial x^1}(U_1u_2) + \frac{\partial}{\partial x^2}(U_2u_2) = -\frac{\partial p}{\partial x^2}\cos\psi_1^1 - \frac{\partial p}{\partial x^1}\sin\psi_1^1 \qquad (5.3)$$

It should be noted in the above that because Cartesian velocity components are employed, two pressure gradients are always present if the coordinate lines x^1 and x^2 do not coincide with Cartesian coordinate directions y^1 and y^2.

When integrated and discretised in the manner of Chapter 3 equations (5.1) and (5.3) become:

$$F_{1e} - F_{1w} + F_{2n} - F_{2s} = 0 \tag{5.4}$$

$$a_P u_{1P} = \sum_m a_m u_{1m} - (p_e - p_w)\,\delta x^2 \cos\psi_1^1 + (p_n - p_s)\,\delta x^1 \sin\psi_1^1$$

$$a_P u_{2P} = \sum_m a_m u_{2m} - (p_e - p_w)\,\delta x^2 \sin\psi_1^1 - (p_n - p_s)\,\delta x^1 \cos\psi_1^1 \tag{5.5}$$

where the F's and a's are defined by relations analogous to (3.26) and eg. (4.7).

In addition to a "non-staggered" variable arrangement (ie., one where the velocities and pressure are all stored at the centres of the computational cells), the alternative "staggered" arrangements employed in the (i) MAC method of Harlow and Welch (1965), (ii) ICED-ALE method of Hirt et al (1974) and (iii) TURF method of Wachspress (1979) will be examined. The arrangement of variables employed in these four alternatives is shown in Fig. 5.5 , for Cartesian grid (ie. $\psi_1^1 = 0$).

5.3.1 Non-staggered arrangement

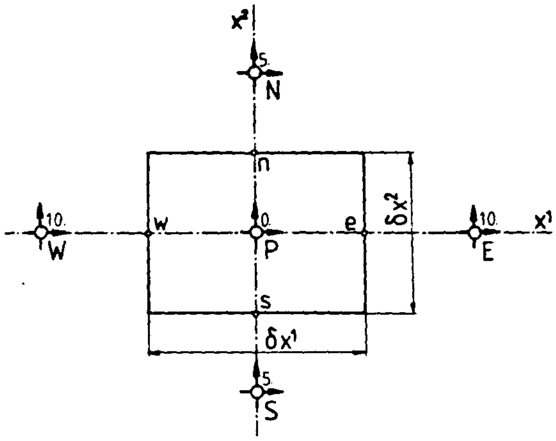
In case of a non-staggered arrangement (Fig. 5.5 a) all the variables share the same control volume, which facilitates efficient computer programming, since in this case the convection coefficients a_m^C (see Eq. (3.43)) are the same for all variables, and for those variables which have the same diffusivity coefficient, the diffusion coefficients a_m^{DN} are also the same. Further, this arrangement minimizes the amount of geometrical information required about the computational grid and the amount of interpolation which needs be performed. Unfortunately, however, this arrangement has one major drawback which can cause difficulties, as will now be described.

Equation (5.5), for the case $\psi_1^1 = 0$, reduces to:

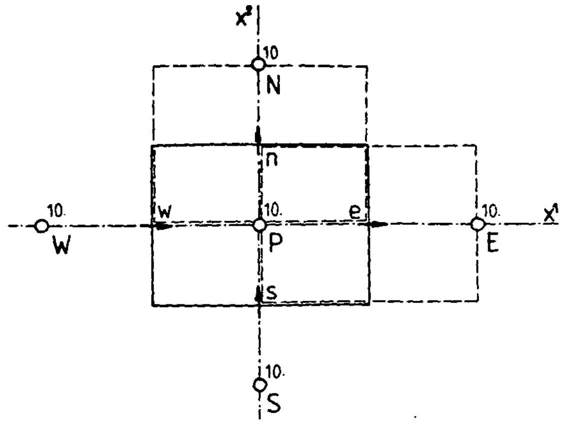
$$a_P u_{1P} = \sum_m a_m u_{1m} - (p_e - p_w)\,\delta x^2 \tag{5.6}$$

Now if the pressures at the "e" and "w" locations are obtained by linear interpolation (the obvious choice), the above equation becomes (for a uniform grid):

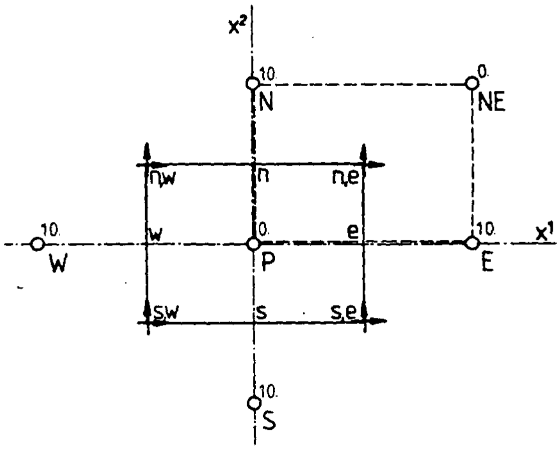
$$a_P u_{1P} = \sum_m a_m u_{1m} - \frac{1}{2}\,\delta x^2 (p_E - p_W) \tag{5.7}$$



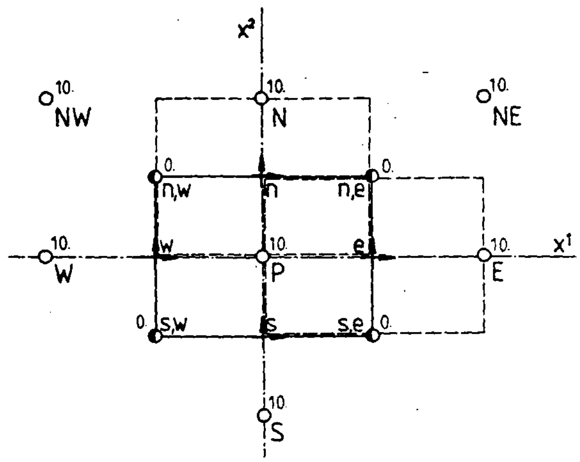
a) non-staggered arrangement



b) the MAC staggered arrangement



c) the ICED-ALE arrangement



d) the TURF arrangement

———	boundaries of main (scalar) control volumes
– – – –	boundaries of velocity control volumes
o	locations of scalar nodes
→	locations of u_1 velocity component nodes
↑	locations of u_2 velocity component nodes
◐	additional pressure nodes

Fig. 5.5: Variable arrangements employed in various solutions methods

As can be seen from the result, the pressure p_P at the central node P does not appear in the equation for u_{1P}. Similarily, the pressure term in the equation for u_{2P} contains only p_N and p_S. The consequence of this anomaly can be appreciated by considering an inviscid flow in a straight channel, in which velocity and pressure fields are uniform. If, however, in the numerical simulation the correct velocity field, but a "checkerboard" pressure field are specified as initial conditions (in the manner indicated in Fig. 5.5 a) the pressure term in Eq. (5.7) would be everywhere zero. Thus, unless a special treatment is employed, such anomalous pressure fields could persist in the final numerical solutions (Patankar, 1980; Rhie and Chow, 1982).

5.3.2 The MAC staggered arrangement

A remedy to the above deficiency was suggested by Harlow and Welch (1965), in the form of displacement of the velocities (and associated control volumes) to lie between the pressures which drive them. In their arrangement, first implemented in their "MAC" algorithm, the u_1 velocity component is stored at the centres of the "e" and "w" cell faces of the "scalar" control volumes, and u_2 is stored at the centres of the "n" and "s" faces, as shown in Fig. 5.5 b. In this case the pressure terms in Eqs. (5.5) turn out to be (for $\psi_1^1=0$):

$$\begin{aligned} a_e u_{1e} &= \sum_m a_m u_{1m} - \delta x^2 (p_E - p_P) \\ a_n u_{2n} &= \sum_m a_m u_{2m} - \delta x^1 (p_N - p_P) \end{aligned} \qquad (5.8)$$

Here a_e stands for the a_P coefficient of the u_1 momentum equation, obtained by integrating over cells constructed around the displaced velocities. In this case only the (correct) uniform velocity and pressure fields will satisfy Eqs. (5.8) and (5.4) for the inviscid channel flow problem, ie. no spatial oscillation in the pressure field can pass undetected. Another advantage of this arrangement is that the velocity components which determine the cell face mass fluxes for scalar cells are directly available; furthermore the continuity equation (5.4) can be simply written as:

$$\rho\, \delta x^2 u_{1e} - \rho\, \delta x^2 u_{1w} + \rho\, \delta x^1 u_{2n} - \rho\, \delta x^1 u_{2s} = 0 \qquad (5.9)$$

These features of the staggered MAC arrangement have been exploited to advantage in several methods as will be discussed in the next section.

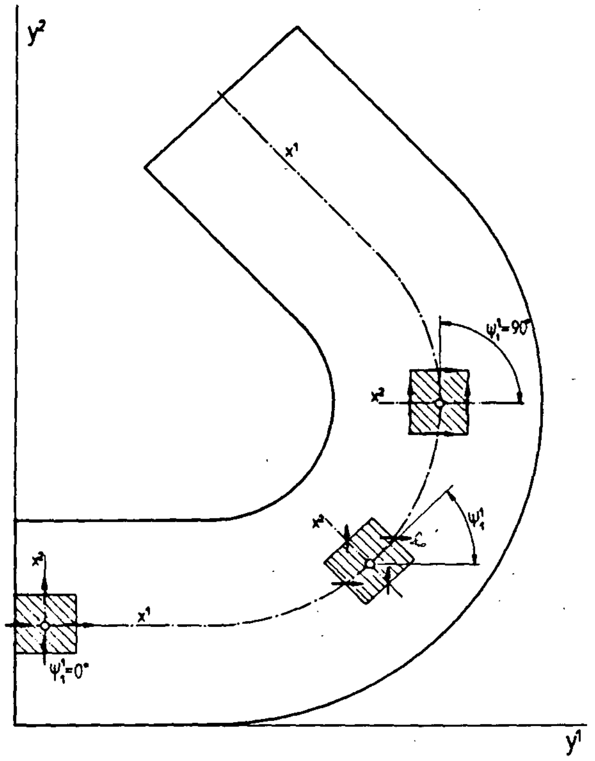


Fig. 5.6: MAC variable arrangement on non-Cartesian grid

This arrangement does, however,possess a serious deficiency when grid ceases to be Cartesian. If, for example, a grid is fitted to a duct as in Fig. 5.6 , and the MAC variable arrangement is used, then at some cells (at the 90° bend location) the situation will be as shown in Fig. 5.7 , where the velocities now lie parallel, rather than perpendicular, to the cell faces. Moreover, equations (5.5) reduce in this case (for a uniform grid and linear interpolation) to:

$$
\begin{aligned}
a_e u_{1e} &= \sum_m a_m u_{1m} + 0.25\,\delta x^1 (p_{NE}+p_N-p_{SE}-p_S) \\
a_n u_{2n} &= \sum_m a_m u_{2m} - 0.25\,\delta x^2 (p_{NE}+p_E-p_{NW}-p_W)
\end{aligned}
\qquad (5.10)
$$

showing that the pressures are only coupled at alternate nodes. Consequently a "zig-zag" pressure distribution like that indicated in Fig. 5.7, would satisfy the momentum equations and therefore be accepted as the final solution.

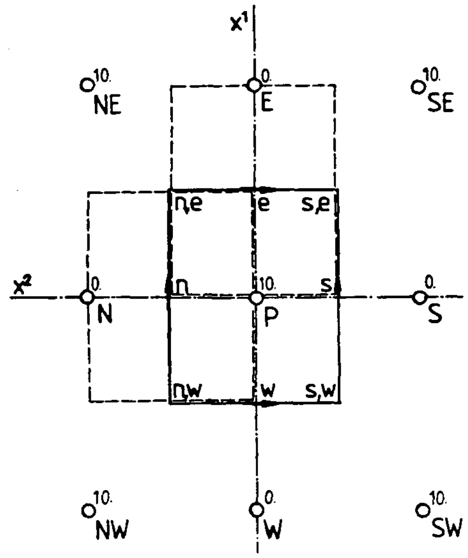


Fig. 5.7: The MAC variable arrangement for $\psi_1^1=90°$ and $\theta_{12}=90°$

A further problem arising from the orientation of velocities is that the continuity constraint is no longer easily enforced; because both velocity components are now needed at all cell faces to evaluate the mass fluxes, and two always have to be obtained by interpolation from four neighbour nodes. This not only may introduce additional inaccuracies but also causes additional problems in the pressure calculation, as will become clear in the next section.

Thus, although successful for rectilinear grids (as evidenced by its use in the solution methods of Gosman and Ideriah (1976), Patankar and Spalding (1972) and others), the MAC arrangement may not work in circumstances like those just described, at least when the Cartesian velocity components are employed. However, if grid-oriented curvilinear velocity components are used, all the advantages that the MAC method has on Cartesian grids are retained, as shown by Pope (1978), Demirdžić et al (1980), Demirdžić (1982), and others. On the other hand, the strong conservation form of governing equations is then lost, bringing in the other disadvantages which have been discussed in Chapter 2. (A minor disadvantage of the MAC arrangement, even on Cartesian grids, is that the velocities are, in case of a non-uniform grid, not stored in the centres of their control volumes.)

Despite the problems with the MAC arrangement and Cartesian velocities just described, several applications to non-orthogonal grids have been reported (Hah, 1983 and 1984; Nakayama, 1985). However, in the published applications the angles ψ_1^1 were not too large, so that situations like that shown in Fig. 5.7 were probably not encountered.

5.3.3 The ICED-ALE arrangement

In the ICED-ALE method, developed at the Los Alomos Scientific Laboratory by Hirt et al (1974), the velocity locations are also staggered in relation to the scalar ones, but here both velocity components, u_1 and u_2, are stored at the same locations, coinciding with the scalar cell vertices, as shown in Fig. 5.5 c. This was intended to overcome the problems with the MAC scheme when ψ_1^1 approaches 90° and/or the grid is non-orthogonal. However, this arrangement is also not free from deficiencies. Specifically, for the inviscid channel flow problem, the pressure term in the discretised form of Eq. (5.5) for u_1 is now (see Fig. 5.5 c):

$$a_{ne}u_{1ne} = \sum_m a_m u_{1m} - 0.5\,\delta x^2(p_{NE}+p_E-p_N-p_P) \qquad (5.11)$$

Now if the correct velocity and equal pressures along alternate diagonals were specified as initial fields (as numbers assigned to nodes in Fig. 5.5 c show) equation (5.11) would be satisfied,which is again clearly unsatisfactory. This feature of the ICED-ALE arrangement may result in oscillatory solutions being generated, as there appears to be no mechanism of detecting and correcting them by the method itself. Such problems indeed have been reported by Hirt et al (1974) and Rhie and Chow (1982). To avoid the decoupling of adjacent nodes Hirt et al (1974) introduced an artificial restoring force at each velocity node to keep them in line with those at neighbour nodes - a treatment which does not have a physical basis and is not guaranteed to work in all circumstances.

Nonetheless the ICED-ALE method, which has been developed for general, two-dimensional (Hirt et al, 1974) or three-dimensional (Pracht, 1975), Eulerian calculations, has seen numerous successful applications, frequently to the high Mach number flows. A similar variable arrangement has also been employed by Vanka et al (1980) in their method for the solution of three-dimensional partially-parabolic flows in ducts of arbitrary cross-section.

5.3.4 The TURF arrangement

Another velocity staggering technique has been proposed by Wachspress (1979). Realizing that both velocity components are needed at each scalar cell face to calculate the mass fluxes (unless the grid is Cartesian), he decided to store both there. To match this arrangement of velocity locations, he introduced additional pressure nodes, situated in the corners of the scalar cells, as shown in Fig. 5.5 d . The consequence of these additional velocity and pressure nodes is that there are now two sets of momentum equations and associated (overlapping) control volumes for the velocity components. The coupling between the equations for the two sets (ie. the coupling between variables stored at odd and even columns) is, according to Wachspress (1979), very weak, arises only from the non-orthogonality of grid lines, and therefore vanishes in the case of an orthogonal grid. This can be seen from discretised momentum equations (5.5) at "e" and "n" locations, for the situation shown in Fig. 5.5 d:

$$
\begin{aligned}
a_e u_{1e} &= \sum_m a_m u_{1m} - \delta x^2 (p_E - p_P) \\
a_e u_{2e} &= \sum_m a_m u_{2m} - \delta x^1 (p_{ne} - p_{se}) \\
a_n u_{1n} &= \sum_m a_m u_{1m} - \delta x^2 (p_{ne} - p_{nw}) \\
a_n u_{2n} &= \sum_m a_m u_{2m} - \delta x^1 (p_N - p_P)
\end{aligned}
\tag{5.12}
$$

where it is evident that u_{1e} and u_{2n} are (like in the MAC method) linked with pressure at P , E and N nodes, whereas u_{2e} and u_{1n} are linked with the pressures at the 'ne', 'nw' and 'se' nodes: thus two overlapping, staggered solution may result. For example, in the inviscid channel flow problem uniform pressure values along alternate lines/columns, as suggested by the numbers assigned to the nodes in Fig. 5.5 d, would satisfy Eqs. (5.12). This is a serious drawback of the TURF method, in addition to the fact that there are twice as many velocity and pressure nodes as there are scalar ones. There appears to be no published results of calculations performed with the TURF method.

5.3.5 Discussion

All three of the staggered variable arrangements considered above are significantly more complex, from the computational point of view, than the non-staggered one , especially when non-orthogonal and non-uniform grids are being used. The fact that, for a three-dimensional grid, up to four different sets of control volumes may be involved in the staggered arrangements, results in significant coefficient-assembly time and storage overloads. Also the treatment of boundaries becomes more complex because of the irregular cells which result there due to staggering, as was shown in Fig. 5.1 and discussed in section 2 of this chapter.

From the above discussion it can also be concluded that all of the arrangements studied suffer from some decoupling problems. The non-staggered arrangement would be the most convenient one, for the reasons stated earlier in this section. However, due to the problems which may arise, and the success of the MAC scheme on Cartesian and general orthogonal grids, it has seen little use in the past. Early attempts by Rhie (1981) to employ non-staggered arrangement failed, due to the (expected) spatial oscillations produced in the pressure field. However, Rhie (1981) and Rhie and Chow (1982) seem to have been the first to find a successful cure for this and other problems associated with the non-staggered variable arrangement. It was this work which provided the inspiration for the present approach, which will be described in the next section.

Therefore, the non-staggered arrangement of variables has been adopted for use in the present study. The way of calculating pressure and avoiding oscillations will be discussed in the next section.

5.4 Calculation of Pressure

5.4.1 Introduction

It has been noted earlier that in case of incompressible or low Mach number flows, the effects of pressure on density are negligible, but the effects on velocity are of great importance. Thus in these circumstances the continuity equation ceases to be a conservation equation for density; rather it is an additional constraint on the velocity field, which can be satisfied, together with momentum conservation, through the pressure field. However, the way of deriving pressure from continuity and momentum equations is not so obvious.

One of the first methods for pressure calculation was the forementioned MAC-method of Harlow and Welch (1965), developed originally for the Cartesian grids. The staggered velocity locations enable them to obtain an equation for pressure by combining the continuity and momentum equations. Several improvements on this approach, all based on the MAC staggered variable arrangement, have been subsequently developed: a survey of the most widely used ones has been made by Raithby and Schneider (1979). The SIMPLE method of Patankar and Spalding (1972) is perhaps the most extensively used: it also forms a basis of the algorithm employed in the present study. In what follows SIMPLE, and three other methods believed to offer faster convergence than SIMPLE, will be described and the steps required to adapt them to the non-staggered arrangement of variables will be outlined. First, however, a technique will be described which prevents decoupling problems when the non-staggered variable arrangement is employed. The key lies in the interpolation practice used to evaluate the mass fluxes, which will be described next.

5.4.2 Evaluation of mass fluxes (continuity equation)

The discretised continuity equation (for the two-dimensional case shown in Fig. 5.5 a , represented by Eq. (5.4)) requires the values of the velocity components at cell faces for the evaluation of the mass fluxes. In earlier unsuccessful attempts to use the non-staggered variable arrangement these values were obtained by linear interpolation. This practice led to pressure oscillations, as reported by Rhie (1981) and discussed in the previous section. Rhie (1981) suggested a different interpolation practice, which avoids this drawback. It can be interpreted as follows*: to evaluate a velocity component at a cell face, use the discretised momentum equations (5.6) for the two neighbouring nodes as the interpolation formulae, but replace the terms representing pressure gradient across the cell face, by one which is centered about it (as done in the MAC staggered scheme), ie.:

---

* This slightly differs from the formulation given by Rhie (1981) and Rhie and Chow (1982), in that they interpolate quantities U_i , defined by Eqs. (3.7), which they evaluate at central point P . The two practices differ when grid lines are not parallel; the above is believed to be more appropriate, since U_i's involve also areas and are not directionally invariant as u_i's .

$$u_{1e} = \overline{\left(\frac{\sum_m a_m u_{1m}}{a_P}\right)}_e - \overline{\left(\frac{1}{a_P}\right)}_e \delta x^2 (p_E - p_P) \tag{5.13}$$

Here the overbar denotes linear interpolation between the two neighbour nodes. Now a checkerboard pressure distribution in the inviscid flow example used earlier will result in non-zero values of the pressure term in (5.13); hence mass fluxes calculated using these velocity components will not satisfy the continuity constraint, even when the u_i imposed at central nodes are correct. The "mass source" thus produced will, in the course of pressure "correction" (using one of the methods to be described in the following subsections), lead to the correct fields of both velocity and pressure. This approach has been extensively tested in the present study on one- and two-dimensional problems with various initial field values, including zig-zag and checkerboard pressure distributions, and it always results in correct solutions being obtained. More details about these tests and the influence of various factors on the accuracy of numerical solutions will be presented in Chapter 8.

In case of non-Cartesian and non-orthogonal grids more pressure gradients are present in momentum equations, as can be seen in Eqs. (5.5). This represents no problem: again, for a particular cell face, the velocity components are assembled by interpolating from their discretised momentum equations at the two neighbour nodes all terms but those representing pressure gradient across the cell face (ie., including the "cross-derivative" pressure terms), which are evaluated from neighbour pressure nodes, as in Eq. (5.13).

The interpolation practice just described couples efficiently values at neighbour nodes, without the need to resort to artificial damping of oscillations as eg. in the ICED-ALE method.

5.4.3 The SIMPLE algorithm for pressure calculation

As a starting point for describing the derivation of the SIMPLE method of obtaining pressure the discretised momentum equations (3.17), (3.18) and (3.19) will be first written in the following form, using (3.41) and (3.42):

$$u_{iP} = H_P(u_{im}) + (Q_i^1 \delta p^1 + Q_i^2 \delta p^2 + Q_i^3 \delta p^3)_P + C_P'^{\,i} \tag{5.14}$$

Here, the following new notation has been employed:

$$H_P(u_{im}) = \frac{\sum_m a_m u_{im}}{a_P^{u_i}}$$

$$Q_i^j = -\frac{b_i^j}{a_P^{u_i}} \tag{5.15}$$

where $a_P^{u_i}$ is the coefficient of u_{iP} in Eq. (3.42) and given by (3.49), and C_P' is the "source term" C_P from Eq. (3.42) after extraction of the pressure terms, which feature explicity in Eqs. (5.14). The pressure differences δp^j in the coordinate directions x^j are calculated using linear interpolation formulae (4.19).

The solution method consists of two stages: a predictor and a corrector, which are described in turn below.

Predictor stage

In this stage an estimate of the pressure field is used, denoted by p^*, and by solving the system of equations (5.14) the velocity field u_i^* is obtained, ie.:

$$u_{iP}^* = H_P(u_{im}^*) + (Q_i^1 \delta p^{*1} + Q_i^2 \delta p^{*2} + Q_i^3 \delta p^{*3})_P + C_P'^i \tag{5.16}$$

This velocity field u_i^* satisfies the momentum equations for the given pressure field and coefficients a_m (which, it should be noted, also involve velocities), but in general does not satisfy the continuity constraint. To check this, the velocity components at the cell faces are required. These are evaluated, as described in the previous sub-section, by the following interpolation formulae:

$$\begin{aligned}
u_{ie}^* &= \overline{H_e(u_{im}^*)} + (Q_i^1 \delta p^{*1} + \overline{Q_i^2 \delta p^{*2}} + \overline{Q_i^3 \delta p^{*3}} + \overline{C_P'^i})_e \\
u_{in}^* &= \overline{H_n(u_{im}^*)} + (\overline{Q_i^1 \delta p^{*1}} + Q_i^2 \delta p^{*2} + \overline{Q_i^3 \delta p^{*3}} + \overline{C_P'^i})_n \\
u_{it}^* &= \overline{H_t(u_{im}^*)} + (\overline{Q_i^1 \delta p^{*1}} + \overline{Q_i^2 \delta p^{*2}} + Q_i^3 \delta p^{*3} + \overline{C_P'^i})_t
\end{aligned} \tag{5.17}$$

where the overbar denotes linear interpolation between two nodes on either side of the cell face. For example, for the "e" cell face:

$$\overline{H_e(u^*_{im})} = (1-f_{1P})\ H_P(u^*_{im}) + f_{1P}\ H_E(u^*_{im})$$

$$\overline{(Q_i^2\,\delta p^{*2})_e} = (1-f_{1P})\ (Q_i^2\delta p^{*2})_P + f_{1P}\ (Q_i^2\delta p^{*2})_E \tag{5.18}$$

etc.

Equations similar to (5.17) follow for w, s and b cell faces. The pressure gradients across the cell faces, and the corresponding coefficients Q_i^j, are to be evaluated as:

$$\begin{aligned}
(\delta p^*)^1_e &= p^*_E - p^*_P \quad ; \quad (Q_i^1)_e = -(b_i^1)_e \cdot \overline{\left[\frac{1}{a_P^{u_i}}\right]_e} \\
(\delta p^*)^2_n &= p^*_N - p^*_P \quad ; \quad (Q_i^2)_n = -(b_i^2)_n \cdot \overline{\left[\frac{1}{a_P^{u_i}}\right]_n} \\
(\delta p^*)^3_t &= p^*_T - p^*_P \quad ; \quad (Q_i^3)_t = -(b_i^3)_t \cdot \overline{\left[\frac{1}{a_P^{u_i}}\right]_t}
\end{aligned} \tag{5.19}$$

Similar expressions follow the w, s and b cell faces.

With the cell face velocity components so defined, the predictor stage mass fluxes can be calculated using Eqs. (3.26), (3.27) and (3.28), ie.:

$$F^*_{ic} = \rho(u^*_1 b^i_1 + u^*_2 b^i_2 + u^*_3 b^i_3)_c \tag{5.20}$$

where subscript "c" stands for cell face locations e, w, n, s, t and b .

Corrector stage

Since the mass fluxes calculated from (5.20) do not necessarily satisfy continuity, it is necessary to introduce corrections to the field values obtained in the predictor stage. Thus new field values, denoted here by u_i^{**} and p^{**}, are sought such that:

$$\begin{aligned}
u_i^{**} &= u_i^* + u_i' \\
p^{**} &= p^* + p'
\end{aligned} \tag{5.21}$$

The corrections u_i' and p' should be such that the u_i^{**} aid p^{**} fields satisfy both the continuity and momentum equations. One way of achieving the first goal is as follows.

Approximate momentum equations can be written for the advanced velocities u_i^{**}, in which only pressure field is updated, as follows:

$$u_{iP}^{**} = H_P(u_{im}^{*}) + (Q_i^1 \delta p^{**1} + Q_i^2 \delta p^{**2} + Q_i^3 \delta p^{**3})_P + C_P^{\prime i} \qquad (5.22)$$

By subtracting Eqs. (5.16) from Eqs. (5.22), the following expressions for the velocity corrections result:

$$u_{iP}' = (Q_i^1 \delta p'^1 + Q_i^2 \delta p'^2 + Q_i^3 \delta p'^3)_P \qquad (5.23)$$

In a similar way equations for the cell face velocity corrections can be obtained, eg.:

$$\begin{aligned} u_{ie}' &= (Q_i^1 \delta p'^1 + \overline{Q_i^2 \delta p'^2} + \overline{Q_i^3 \delta p'^3})_e \\ u_{in}' &= (\overline{Q_i^1 \delta p'^1} + Q_i^2 \delta p'^2 + \overline{Q_i^3 \delta p'^3})_n \\ u_{it}' &= (\overline{Q_i^1 \delta p'^1} + \overline{Q_i^2 \delta p'^2} + Q_i^3 \delta p'^3)_t \end{aligned} \qquad (5.24)$$

and accordingly for w, s and b cell face locations. Using the above expressions (5.24) cell face mass flux corrections F_{ic}' are defined as:

$$F_{ic}' = \rho(u_1' b_1^i + u_2' b_2^i + u_3' b_3^i)_c \qquad (5.25)$$

where c again stands for cell face locations e, w, n, s, t and b . The continuity equation (3.29) can now be written as:

$$(F_1^*+F_1')_e - (F_1^*+F_1')_w + (F_2^*+F_2')_n - (F_2^*+F_2')_s + (F_3^*+F_3')_t - (F_3^*+F_3')_b = S_m \qquad (5.26)$$

If the F_i^* values are calculated from Eqs. (5.20), expressions (5.24) are introduced into (5.25), and these into (5.26), then there results an equation of the form of (3.42) in which the p' are the only unknowns. The coefficients a_m and C_P are defined as follows:

$$a_E = \rho\left[E_e^1 + E_P^1 f_{1P} + E_E^2(1-f_{2E}-f_{2SE}) + E_E^3(1-f_{3E}-f_{3BE})\right]$$

$$a_N = \rho\left[E_n^2 + E_P^2 f_{2P} + E_N^1(1-f_{1N}-f_{1NW}) + E_N^3(1-f_{3N}-f_{3BN})\right]$$

$$a_T = \rho\left[E_t^3 + E_P^3 f_{3P} + E_T^1(1-f_{1T}-f_{1TW}) + E_T^2(1-f_{2T}-f_{2TS})\right]$$

$$a_W = \rho\left[E_w^1 - E_P^1(1-f_{1W}) - E_W^2(1-f_{2W}-f_{2SW}) - E_W^3(1-f_{3W}-f_{3BW})\right]$$

$$a_S = \rho\left[E_s^2 - E_P^2(1-f_{2S}) - E_S^1(1-f_{1S}-f_{1SW}) - E_S^3(1-f_{3S}-f_{3BS})\right]$$

$$a_B = \rho\left[E_b^3 - E_P^3(1-f_{3B}) - E_B^1(1-f_{1B}-f_{1BW}) - E_B^2(1-f_{2B}-f_{2BS})\right]$$

$$a_{NE} = \rho\left[E_E^2 f_{2E} + E_N^1 f_{1N}\right]$$

$$a_{SE} = -\rho\left[E_E^2(1-f_{2SE}) + E_S^1 f_{1S}\right]$$

$$a_{NW} = -\rho\left[E_W^2 f_{2W} + E_N^1(1-f_{NW})\right]$$

$$a_{SW} = \rho\left[E_W^2(1-f_{2SW}) + E_S^1(1-f_{1SW})\right]$$

$$a_{TE} = \rho\left[E_E^3 f_{3E} + E_T^1 f_{1T}\right] \tag{5.27}$$

$$a_{BE} = -\rho\left[E_E^3(1-f_{3BE}) + E_B^1 f_{1B}\right]$$

$$a_{TW} = -\rho\left[E_W^3 f_{3W} + E_T^1(1-f_{1TW})\right]$$

$$a_{BW} = \rho\left[E_W^3(1-f_{3BW}) + E_B^1(1-f_{1BW})\right]$$

$$a_{TN} = \rho\left[E_N^3 f_{3N} + E_T^2 f_{2T}\right]$$

$$a_{BN} = -\rho\left[E_N^3(1-f_{3BN}) + E_B^2 f_{2B}\right]$$

$$a_{TS} = -\rho\left[E_T^2(1-f_{2TS}) + E_S^3 f_{3S}\right]$$

$$a_{BS} = \rho\left[E_S^3(1-f_{3BS}) + E_B^2(1-f_{2BS})\right]$$

$$a_P = \sum_m a_m \quad , \quad m = E,W,N,S,T,B,NE,NW,SE,SW,TE,TW,BE,BW,TN,BN,TS,BS$$

$$C_P = F_{1e}^* - F_{1w}^* + F_{2n}^* - F_{2s}^* + F_{3t}^* - F_{3b}^* - S_m$$

In these expressions the coefficients E^i represent the following quantities:

$$E^1_P = (b^2_{1n}Q^1_{1P}+b^2_{2n}Q^1_{2P}+b^2_{3n}Q^1_{3P})(1-f_{2P}) - (b^2_{1s}Q^1_{1P}+b^2_{2s}Q^1_{2P}+b^2_{3s}Q^1_{3P})f_{2S} + (b^3_{1t}Q^1_{1P}+b^3_{2t}Q^1_{2P}+b^3_{3t}Q^1_{3P})(1-f_{3P}) - (b^3_{1b}Q^1_{1P}+b^3_{2b}Q^1_{2P}+b^3_{3b}Q^1_{3P})f_{3B}$$

$$E^2_P = (b^1_{1e}Q^2_{1P}+b^1_{2e}Q^2_{2P}+b^1_{3e}Q^2_{3P})(1-f_{1P}) - (b^1_{1w}Q^2_{1P}+b^1_{2w}Q^2_{2P}+b^1_{3w}Q^2_{3P})f_{1W} + (b^3_{1t}Q^2_{1P}+b^3_{2t}Q^2_{2P}+b^3_{3t}Q^2_{3P})(1-f_{3P}) - (b^3_{1b}Q^2_{1P}+b^3_{2b}Q^2_{2P}+b^3_{3b}Q^2_{3P})f_{3B}$$

$$E^3_P = (b^1_{1e}Q^3_{1P}+b^1_{2e}Q^3_{2P}+b^1_{3e}Q^3_{3P})(1-f_{1P}) - (b^1_{1w}Q^3_{1P}+b^1_{2w}Q^3_{2P}+b^1_{3w}Q^3_{3P})f_{1W} + (b^2_{1n}Q^3_{1P}+b^2_{2n}Q^3_{2P}+b^2_{3n}Q^3_{3P})(1-f_{2P}) - (b^2_{1s}Q^3_{1P}+b^2_{2s}Q^3_{2P}+b^2_{3s}Q^3_{3P})f_{2S}$$

$$E^1_e = (b^1_1Q^1_1 + b^1_2Q^1_2 + b^1_3Q^1_3)_e$$

$$E^1_w = (b^1_1Q^1_1 + b^1_2Q^1_2 + b^1_3Q^1_3)_w$$

$$E^2_n = (b^2_1Q^2_1 + b^2_2Q^2_2 + b^2_3Q^2_3)_n$$

$$E^2_s = (b^2_1Q^2_1 + b^2_2Q^2_2 + b^2_3Q^2_3)_s$$

$$E^3_t = (b^3_1Q^3_1 + b^3_2Q^3_2 + b^3_3Q^3_3)_t$$

$$E^3_b = (b^3_1Q^3_1 + b^3_2Q^3_2 + b^3_3Q^3_3)_b$$

$$E^2_E = (b^1_{1e}Q^2_{1E} + b^1_{2e}Q^2_{2E} + b^1_{3e}Q^2_{3E})\ f_{1P}$$

$$E^3_E = (b^1_{1e}Q^3_{1E} + b^1_{2e}Q^3_{2E} + b^1_{3e}Q^3_{3E})\ f_{1P}$$

$$E^2_W = (b^1_{1w}Q^2_{1W} + b^1_{2w}Q^2_{2W} + b^1_{3w}Q^2_{3W})(1-f_{1W})$$

$$E^3_W = (b^1_{1w}Q^3_{1W} + b^1_{2w}Q^3_{2W} + b^1_{3w}Q^3_{3W})(1-f_{1W})$$

$$E^1_N = (b^2_{1n}Q^1_{1N} + b^2_{2n}Q^1_{2N} + b^2_{3n}Q^1_{3N})\ f_{2P}$$

$$E^3_N = (b^2_{1n}Q^3_{1N} + b^2_{2n}Q^3_{2N} + b^2_{3n}Q^3_{3N})\ f_{2P}$$

$$E^1_S = (b^2_{1s}Q^1_{1S} + b^2_{2s}Q^1_{2S} + b^2_{3s}Q^1_{3S})(1-f_{2S})$$

$$E^3_S = (b^2_{1s}Q^3_{1S} + b^2_{2s}Q^3_{2S} + b^2_{3s}Q^3_{3S})(1-f_{2S})$$

$$E_T^1 = (b_{1t}^3 Q_{1T}^1 + b_{2t}^3 Q_{2T}^1 + b_{3t}^3 Q_{3T}^1)\ f_{3P}$$

$$E_T^2 = (b_{1t}^3 Q_{1T}^2 + b_{2t}^3 Q_{2T}^2 + b_{3t}^3 Q_{3T}^2)\ f_{3P} \qquad (5.28)$$

$$E_B^1 = (b_{1b}^3 Q_{1B}^1 + b_{2b}^3 Q_{2B}^1 + b_{3b}^3 Q_{3B}^1)(1-f_{3B})$$

$$E_B^2 = (b_{1b}^3 Q_{1B}^2 + b_{2b}^3 Q_{2B}^2 + b_{3b}^3 Q_{3B}^2)(1-f_{3B})$$

When the pressure-correction equation is solved, the p' so obtained are used to correct the cell face mass fluxes, making use of Eqs.(5.24) and (5.25). The F's now satisfy the continuity requirement, and are used for the convective parts of coefficients in the momentum and scalar equations at the next iteration level. The velocity components at central nodes are corrected using Eqs. (5.23) and (5.21). The pressure changes usually have to be under-relaxed (Patankar, 1981), as:

$$p^{**} = p^* + \alpha_p p' \qquad (5.29)$$

where α_p is an under-relaxation factor in the range between zero and unity.

5.4.4 The SIMPLER method

Patankar (1980, 1981) has recognized that although the pressure corrections p' correctly drive the velocities towards satifying continuity, they provide poor approximations to the correct pressure field. The reason for the poor performance of SIMPLE if the "optimum" α_p is not used is that the velocities u_i^{**} no longer satisfy their momentum equations. This is because p' is derived primarily from the continuity constraint, and hence there is no guarantee that the momentum balance will be satisfied by u_i^{**} and p^{**} fields. Realizing this Patankar (1980) suggested that p' be used only to correct velocity components, to make them satisfy continuity, and that the pressure be calculated from another equation to match the velocities, so that the momentum equations are also satisfied. A suitable pressure equation can be obtained, for the non-orthogonal grid and non-staggered arrangement considered here, after rewriting the momentum equations (5.23) as:

$$u_{iP} = \hat{u}_{iP} + (Q_i^1\delta p^1 + Q_i^2\delta p^2 + Q_i^3\delta p^3)_P \qquad (5.30)$$

where the pseudovelocities $\hat{u}_{iP}$ are given by:

$$\hat{u}_{iP} = H_P(u_{im}) + C_P'^{i} \qquad (5.31)$$

and then proceeding as follows.

By using Eqs. (5.30) and (5.17), the expressions given below for the cell face velocity components result, eg. for "e" cell face:

$$u_{ie} = \hat{u}_{ie} + (Q_i^1\delta p^1 + \overline{Q_i^2\delta p^2} + \overline{Q_i^3\delta p^3})_e \qquad (5.32)$$

where

$$\hat{u}_{ie} = \hat{u}_{iP}(1-f_{1P}) + \hat{u}_{iE}f_{1P} \qquad (5.33)$$

Similar expressions can be derived for other cell faces. When these are introduced into the flux expressions (5.20), and the fluxes are required to satisfy the continuity equations (3.29), then there results the pressure equation

$$a_P p_P = \sum_m a_m p_m + C_P \qquad (5.34)$$

in which coefficients a_m are the same as in p' equation, ie. those from Eqs. (5.27), and C_P is now given by:

$$C_P = \hat{F}_{1e} - \hat{F}_{1w} + \hat{F}_{2n} - \hat{F}_{2s} + \hat{F}_{3t} - \hat{F}_{3b} - S_m \qquad (5.35)$$

where $\hat{F}_i$ are pseudo-fluxes, obtained from Eqs. (5.20) by substituting u_i^* by $\hat{u}_i$.

The solution procedure is now slightly different from that of SIMPLE. Here the pressure equation (5.34) is solved first (using an estimate for the velocity field) and then solutions are obtained to the momentum and pressure-correction equations. Because no approximations like that in SIMPLE corrector stage, in Eqs. (5.22), are made in deriving the pressure equation, if the initial velocity field happens to be the correct one, the p equation (5.34) will then yield the correct pressure* . It is

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* Strictly speaking, this is only true for MAC staggered variable arrangement and orthogonal grids. In case of non-staggered variable arrangement, due to linear interpolation of non-linear quantities $\hat{u}_i$, some iteration will in general be always necessary.

from this characteristic of the pressure equation that SIMPLER derives its effectiveness, which results in significantly faster convergence than when SIMPLE is used (Patankar, 1981).

5.4.5 The PISO method

Issa (1982) developed another improved pressure-velocity calculation method, aimed initially at non-iterative computations of unsteady compressible flows. The simplified iterative version for steady-state calculations can be interpreted as an extension of the SIMPLE method, in which a second corrector stage is added. In it both left- and right-hand side of Eq. (5.22) are further updated, ie.:

$$u^{***}_{iP} = H_P(u^{**}_{im}) + (Q^1_i \delta p^{***1} + Q^2_i \delta p^{***2} + Q^3_i \delta p^{***3}) + C'^{i}_{P} \tag{5.36}$$

Thus new field values, denoted here by u^{***}_i and p^{***}, are sought. For this purpose additional corrections u''_i and p'' are defined, thus:

$$\begin{aligned} u^{***}_i &= u^{**}_i + u''_i \\ p^{***} &= p^{**} + p'' \end{aligned} \tag{5.37}$$

By subtracting Eq. (5.22) from (5.36) there result equations for u''_i, in terms of u'_i and p'', of the following kind:

$$u''_{iP} = H_P(u'_{im}) + (Q^1_i \delta p''^1 + Q^2_i \delta p''^2 + Q^3_i \delta p''^3)_P \tag{5.38}$$

It should be noted that the difference between Eqs. (5.23) and (5.38) is in the appearance of $H_P(u'_{im})$ term in the latter. The expressions for the cell face velocity corrections and mass flux corrections are obtained in the same way as in the first corrector stage. Since the F^{**}_i mass fluxes already satisfy continuity as a result of the first corrector stage, the continuity equation (3.29) now reduces to:

$$F''_{1e} - F''_{1w} + F''_{2n} - F''_{2s} + F''_{3t} - F''_{3b} = 0 \tag{5.39}$$

From (5.39), when the F''_i are expressed through u''_i and these through u'_i and p'' as in Eq. (5.38), there results an equation for p'', in which the coefficients a_m are the same as those in the p'-equation, ie. they are given by (5.27), and only C_P differs, as follows:

$$C_P = \tilde{F}_{1e} - \tilde{F}_{1w} + \tilde{F}_{2n} - \tilde{F}_{2s} + \tilde{F}_{3t} - \tilde{F}_{3b} \qquad (5.40)$$

Here the $\tilde{F}_i$ are pseudofluxes obtained from Eqs. (5.20) by substituting u^*_{ic} for a particular cell face "c" by $\overline{H_c(u'_{im})}$ evaluated, for the same cell face, according to (5.18) and (5.15) (in which the u_i are substituted by u'_i defined by (5.23)).

The second pressure correction p" brings the velocity and pressure fields closer to satisfying both the momentum and continuity equations, for given coefficients, without the need to under-relax the pressure changes (Issa, 1982; Issa et al, 1984).

When implemented with the MAC staggered variable arrangement on Cartesian-like grids PISO provides considerable savings on computational time as compared with standard iterative techniques like SIMPLE (Issa et al, 1984). The applicability of PISO to non-iterative calculations of unsteady flows on non-staggered grids may be affected by the interpolations like (5.17), used to evaluate the mass fluxes. However, no adverse effects are expected in calculations of steady flows.

5.4.6 The SIMPLEC method

Another enhancement of the SIMPLE method has been proposed by Van Doormaal and Raithby (1984). The rationale behind their method, called SIMPLEC (SIMPLE Consistent) is the following.

In the SIMPLE corrector stage, in Eqs. (5.22) only the pressure on the right-hand side has been updated, leaving the term $H(u^*_{im})$ unchanged. Had this approximation not been done, the expressions for the velocity corrections would have been:

$$u'_{iP} = H_P(u'_{im}) + (Q^1_i\,\delta p'^1 + Q^2_i\,\delta p'^2 + Q^3_i\,\delta p'^3)_P \qquad (5.41)$$

The reason for omitting the term $H_P(u'_{im})$ in the SIMPLE method is that the pressure-correction equation, derived from (5.41) and the continuity constraint, would have been too complex and impractical to solve. However, Van Doormaal and Raithby (1984) have found that a more "consistent" omission can be made to simplify Eq. (5.41). To show this, Eq. (5.41) will be rewritten as:

$$a_P u'_{iP} = \sum_m a_m u'_{im} - (b_i^1 \delta p'^1 + b_i^2 \delta p'^2 + b_i^3 \delta p'^3)_P \qquad (5.42)$$

Now, to introduce a consistent approximation, the term $\sum_m a_m u'_{iP}$ is substracted from both sides of Eq. (5.42), thus leading to:

$$(a_P - \sum_m a_m) u'_{iP} = \underline{\sum_m a_m (u'_{im} - u'_{iP})} - (b_i^1 \delta p'^1 + b_i^2 \delta p'^2 + b_i^3 \delta p'^3)_P \qquad (5.43)$$

Assuming that all u'_{im} are of about the same order as u'_{iP} , the underlined term on the right-hand side of Eq. (5.43) can be omitted, thus giving:

$$u'_{iP} = (\tilde{Q}_i^1 \delta p'^1 + \tilde{Q}_i^2 \delta p'^2 + \tilde{Q}_i^3 \delta p'^3)_P \qquad (5.44)$$

which has the same form as Eq. (5.23) in SIMPLE , only now the coefficients $\tilde{Q}_i^j$ are replacing Q_i^j , and are evaluated as:

$$\tilde{Q}_i^j = - \frac{b_i^j}{a_P^{u_i} - \sum_m a_m} \qquad (5.45)$$

The coefficients of the pressure-correction equation resulting from introduction of (5.44) into (5.25) and (5.26), are the same as those in (5.27), if the Q_i^j in (5.28) are substituted by the $\tilde{Q}_i^j$. The pressure corrections obtained by solving the pressure-correction equation are used to correct the velocities and pressure through (5.44) and (5.21). No under-relaxation of the pressure changes is needed, but u_i should be under-relaxed, either according to (3.48) or according to the practice of Van Doormaal and Raithby (1984). Significant savings on computational time have been reported by Van Doormaal and Raithby (1984) for several applications (on Cartesian grids with the MAC staggered arrangement), as compared with the SIMPLE and SIMPLER methods.

5.5 Closure

It has been shown that all four of the variable arrangements which have been suggested for use with Cartesian velocity components and general grids suffer from some sort of decoupling problem. The simplest arrangement, and perhaps the most economical in general situations from the numerical solution point of view, is the non-staggered one, in which all the dependent variables are stored at the centres of the control volumes. A suitable interpolation practice, first suggested by Rhie (1981), enables avoidance of decoupling problems when the non-staggered arrangement is employed. This practice also facilitates the use of a family of efficient pressure-velocity coupling algoritms, built around the SIMPLE algorithm of Patankar and Spalding (1972). Therefore, the non-staggered arrangement of variables has been adopted for use in the present study.

Apart from SIMPLE, three more advanced pressure-velocity coupling algorithms have been described and the way they could be adapted for use in conjunction with the non-staggered variable arrangement has also been suggested. Their relative merits will be discussed in the next chapter, where one will be selected for use in the present study.

CHAPTER 6

SOLUTION ALGORITHM AND BOUNDARY CONDITIONS

6.1 Introduction

In the previous two chapters various differencing schemes and pressure-velocity coupling algorithms have been described. At this stage a selection will be made from these, along with some other decisions concerning the numerical solution procedure. First, in section 2,the discretised equations will be rearranged into a form suitable for iterative solution techniques, by treating some parts of the convection and diffusion fluxes explicitly . The motive for doing so is to reduce computer storage requirements.

Next, in section 3, the differencing schemes to be employed are selected. Following this, in section 4, the chosen pressure-calculation algorithm will be specified, and the possibility of further simplification, to yield a form of pressure-correction equation which can be solved efficiently, will be investigated.

In section 5 the under-relaxation of pressure changes and the selection of under-relaxation factors will be discussed. Next, in section 6, implementation of boundary conditions will be described.

In section 7 the sequence of operations which make up a typical iteration cycle will be presented, followed by a specification of the convergence criteria employed in the present study.

Finally, closing remarks will be given in section 8.

6.2 Rearrangement of Discretised Equations

As already noted in section 5 of Chapter 3, due to the non-linearity of the equations to be solved, an iterative solution technique is employed in the present study. Because of the need to update the coefficients a_m (Eq. (3.42)) from one iteration to another, there is no incentive to solve the equations exactly at each iteration level. This fact enables some terms in the discretised equations to be treated in an "explicit" manner, ie. they are evaluated at the previous iteration level, and incorporated as known quantities into the "sources"* . Eventually, as the solution con-

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* This procedure is sometimes refered to as "matrix preconditioning" or "deferred correction", and is used not only to reduce the size of the coefficient matrix, but also to enhance the stability of iterative solution method (Khosla and Rubin, 1974).

verges, the differences between the two iteration levels disappear, so the full equations are solved.

If the grid is orthogonal (eg. Cartesian), all the cross-derivative terms vanish. Therefore, if the grid lines do not depart much from orthogonality, these terms are relatively small compared with the "principal" terms (ie., $a_m^{DC} \ll a_m^{DN}+a_m^C$). In these circumstances explicit treatment will not influence significantly the rate of convergence of the overall solution procedure.

Based on these arguments and supported by the evidence from numerous test calculations performed so far, it has been decided that all cross-derivative terms (ie., those involving B's with unequal indeces in Eqs. (3.11), (3.17), (3.18) and (3.19)) and also selected stress terms in momentum equations - specifically, those involving the ω operators in Eqs. (3.17), (3.18) and (3.19) - will be treated explicitly. Also the contributions in convective terms from nodes other than the principal ones (ie. P,E,W,N,S,T and B) will be treated in the same way. This has been found to improve stability and convergence rate when schemes like the QUDS are employed (Han et al, 1981).

In the light of the above decisions, the final form of algebraic equations to be solved is* :

$$a_P\phi_P = \sum_m a_m\phi_m + C_P \tag{6.1}$$

where:

$$a_m = a_m^C + a_m^{DN} \quad , \qquad m=E,W,N,S,T,B$$

$$a_P = \sum_m a_m + \sum_n a_n^C + S_\phi'' \quad , \qquad n=EE,WW,NN,SS,TT,BB \tag{6.2}$$

$$C_P = S_\phi' + (\sum_l a_l^{DC}\phi_l^* - a_P^{DC}\phi_P^*) + \sum_n a_n^C\phi_n^* \quad ,$$

$$l=m + NE,NW,SE,SW,TE,TW,BE,BW,TN,TS,BN,BS$$

---

* In the computer programming of these equations the a_m^{DC} are not calculated as such; rather the whole cross-derivative cell-face fluxes I_e^{DC}, I_w^{DC} etc. are assembled (see Eq. (3.35)) and the "source" term C_P is evaluated as:

$$C_P = I_e^{DC}-I_w^{DC}+I_n^{DC}-I_s^{DC}+I_t^{DC}-I_b^{DC}+S_\phi'+\sum_n a_n^C\phi_n^* \tag{6.3}$$

This also facilitates the implementation of boundary conditions, as will be seen later in this chapter.

Here, the coefficients a_m^C , a_m^{DN} and a_m^{DC} have already been defined for the differencing schemes considered in Chapter 4, and the ϕ^* are the values of the dependent variable ϕ obtained in the previous iteration. If under-relaxation is necessary, it is performed according to Eqs. (3.48) and (3.49).

In the case of the momentum equations, the additional terms, which are absorbed into C_P , can be identified by comparison of the u_1, u_2 and u_3 momentum equations (Eqs. (3.17), (3.18) and (3.19)) with the scalar equation (3.11): the "extra" terms are then seen as those stress terms involving the ω operator, and the pressure terms. These terms, when discretised, are as follows:

$$S_{ui}^* = I_e^s - I_w^s + I_n^s - I_s^s + I_t^s - I_b^s + S_p =$$

$$\left(\frac{\mu}{\delta V}\right)_e \cdot \left(b_1^1\Omega_i^1 + b_2^1\Omega_i^2 + b_3^1\Omega_i^3\right)_e - \left(\frac{\mu}{\delta V}\right)_w \cdot \left(b_1^1\Omega_i^1 + b_2^1\Omega_i^2 + b_3^1\Omega_i^3\right)_w +$$

$$\left(\frac{\mu}{\delta V}\right)_n \cdot \left(b_1^2\Omega_i^1 + b_2^2\Omega_i^2 + b_3^2\Omega_i^3\right)_n - \left(\frac{\mu}{\delta V}\right)_s \cdot \left(b_1^2\Omega_i^1 + b_2^2\Omega_i^2 + b_3^2\Omega_i^3\right)_s + \tag{6.4}$$

$$\left(\frac{\mu}{\delta V}\right)_t \cdot \left(b_1^3\Omega_i^1 + b_2^3\Omega_i^2 + b_3^3\Omega_i^3\right)_t - \left(\frac{\mu}{\delta V}\right)_b \cdot \left(b_1^3\Omega_i^1 + b_2^3\Omega_i^2 + b_3^3\Omega_i^3\right)_b -$$

$$\left[(p_e - p_w)b_i^1 + (p_n - p_s)b_i^2 + (p_t - p_b)b_i^3\right]_P$$

In the above the Ω_i^j terms represent the discretised form of ω_i^j operators (see Eq. (3.20)), and are calculated as follows:

$$\Omega_{ie}^j = \left[(u_{jE} - u_{jP})b_i^1 + (u_{jn} - u_{js})b_i^2 + (u_{jt} - u_{jb})b_i^3\right]_e$$

$$\Omega_{in}^j = \left[(u_{je} - u_{jw})b_i^1 + (u_{jN} - u_{jP})b_i^2 + (u_{jt} - u_{jb})b_i^3\right]_n \tag{6.5}$$

$$\Omega_{it}^j = \left[(u_{je} - u_{jw})b_i^1 + (u_{jn} - u_{js})b_i^2 + (u_{jT} - u_{jP})b_i^3\right]_t$$

Similar expressions follow for w, s and b cell faces.

It follows that the C_P term of the momentum equations becomes:

$$C_P = S'_{ui} + S_{ui}^* + (\sum_l a_l^{DC}u_{il}^* - a_P^{DC}u_{iP}^*) + \sum_n a_n^C u_{in}^* \tag{6.6}$$

The implications of these decisions are the following: firstly, the coefficients of nodes other than E, W, N, S, T, B and P (ie., those marked by ● in Fig. 3.4) need not be stored, thus resulting in a considerable saving on storage (only seven coefficients per node need be stored, instead of the 18 to 33, required if all terms were treated "implicitly"). This is of particular importance in three-dimensional applications, where storage capacity is often the limiting factor. Secondly, the coefficient matrix in Eq. (3.45) has been reduced to a seven-diagonal one (five-diagonal for a two-dimensional case), which is suitable for selected method of solving the algebraic equations (the SIP method of Stone, 1968).

The above treatment is, however, only appropriate if the grid departure from orthogonality is not too severe in the major part of the solution domain (ie., if on average $45° < \theta_{ij} < 135°$ approximately), and if the grid aspect ratios are not too large ($\sim < 10$). Otherwise, as discussed in section 4 of Chapter 4, the magnitude of coefficients a_m^{DC} may exceed the magnitude of a_m^{DN}, thus resulting in important parts of the equations being treated explicitly: this would in turn probably require substantial under-relaxation for stability, which in the limit is equivalent to working with the time-step restrictions of explicit time-marching solution methods* (Roache, 1976 and Patankar, 1980).

6.3 Choice of Differencing Schemes

6.3.1 Convective terms

It has been shown in section 3 of Chapter 4 that the UDS is the only scheme which possesses all of the desirable properties: conservativeness, transportiveness and boundedness. It also satisfies condition (3.46) unconditionally and thus guarantees that a solution can always be obtained by iterative solution methods. For those reasons it is taken to be the

* It can be shown (Raithby and Schneider, 1979 and Issa et al, 1984) that the under-relaxation procedure described in section 5 of Chapter 3 is equivalent to solving the time-dependent form of equations with a time step δt related to the under-relaxation factor α_ϕ as:

$$\delta t = \frac{\alpha_\phi}{1-\alpha_\phi} \frac{\rho\, \delta V}{a_P} \qquad (6.7)$$

Thus lover value of α_ϕ is equivalent to smaller time step δt. It should be noted, however, that for a fixed α_ϕ, δt must vary from one node to another.

basic scheme employed in the present numerical solution method. However, due to the numerical diffusion problems discussed in Chapter 4, arrangements are made for a higher-order scheme to be employed whenever it does not prevent convergence and produces realistic solutions. This is done by adding to the UDS convective flux the difference between the higher-order scheme flux and the UDS flux according to Eq. (4.39), in which I_B is the UDS flux, and I_U is the higher-order scheme flux. Multiplication of this difference by a factor γ enables switching from one scheme (UDS for $\gamma=0$) to the other ($\gamma=1$), or blending of the two ($0<\gamma<1$).

Of the four higher-order schemes described in Chapter 4, the SUDS and the CDS are regarded as unsuitable for a general solution method. The CDS is neither transportive nor bounded, and the structure of its coefficient matrix is not suitable for iterative solution techniques (see table 4.1). Only at low Peclet number, typically less than 2, does the CDS become stable and bounded. The SUDS, on the other hand, is considerably more complex than the alternatives, especially in three-dimensional cases, while offering no obvious advantages.

The LUDS and the QUDS are here believed to be the most suitable for implementation in general three-dimensional applications. For the present study the LUDS is adopted, for the following reasons. Firstly, its coefficients are significantly simpler than those of the QUDS, and hence it is more economical from the computational point of view. Secondly, the structure of its coefficient matrix is more suitable of the two for iterative solution algorithms (ie., it is more "diagonally dominant" than that of QUDS, see table 4.1). Thirdly, the rate of convergence of the LUDS equations is usually similar to that of the UDS, while QUDS usually converges slower (see Han et al, 1981, and also Chapter 8 of the present study). Finally, as the test calculations to be presented in Chapter 8 show, the difference in accuracy between LUDS and QUDS is far smaller than the difference between either of them and the UDS.

For the majority of the calculations to be presented in Chapters 8 and 9 the full LUDS-scheme was used, although usually UDS solutions were also obtained for comparison. Therefore, the coefficients a_m^c, appearing in (6.2), (6.3) and (6.5) were obtained either from Eqs. (4.9) or (4.7).

6.3.2 Diffusion terms

The coefficients a_m^{DN} in Eq. (6.2), which arise from diffusion terms involving "normal" derivatives, are calculated according to Eqs. (4.17). Their contribution is always positive, thus leading to bounded solutions and stable iterative solution procedure.

The cross-derivative contribution to the coefficients of algebraic equation (6.1), a_m^{DC} , are in the present study calculated using SCHEME 1 of section 4, Chapter 4. This results in negative coefficients being generated at the "sharp corner" nodes in computational molecule (some of those nodes labeled ◐ in Fig. 3.4), which is a potential danger; however, in cases where the grid is moderately non-orthogonal and where the Peclet number is high, these coefficients are small relative to the principal ones (see table 4.2 for an example), and it is anticipated that no problems will occur. Nonetheless, in other circumstances these negative coefficients may result in unrealistic solutions; in such cases SCHEME 2 will have to be used, and possibly the grid will have to be rearranged so that conditions of the form (4.25) are satisfied in affected regions, as discussed by Demirdžić (1982). In the present applications no problems attributable to negative a_m^{DC} where encountered.

6.4 Choice of Pressure-Calculation Algorithm

Of the four pressure-calculation algorithms described in the previous chapter, SIMPLEC had not been published at the time the present study was undertaken. Two of the three remaining algorithms, namely PISO and SIMPLER, were known to give significantly faster convergence and thus lower cost of computation than the "standard" SIMPLE, at least when applied in conjunction with Cartesian grids and the MAC staggered arrangement of variables. However, there are reasons which make these algorithms unattractive for steady flow predictions on non-orthogonal grids. Before, however, these reasons are discussed, a possible simplification of the SIMPLE algorithm for such circumstances will be described.

From Eqs. (5.28) it is evident that, even when simple linear interpolation is employed to evaluate the pressure terms in the momentum equations (see Eqs. (3.41) and (4.19)), the coefficient matrix of the pressure-correction equation will have 18 diagonals (9 in a two-dimensional case), entailing substantial storage requirements. Earlier, to avoid this problem with the momentum and scalar equations, the cross-derivatives were treated explicitly (see Eqs. (6.1), (6.2) and (6.4)). The same approach could be used here, but it would still be both inconvenient and costly, since - unlike the other equations- the pressure-correction equation has to be solved to a certain tolerance for reasons of stability, as will be described later. To achieve this, the explicit pseudo-sources would have to be updated for each iteration in the pressure-correction equation, thus increasing the computing time. Since the solution of this equation consumes the majority of the total computing time, it would clearly be advantageous to avoid such updates.

Avoidance is possible through recognition that: (i) the cross-derivative terms vanish if the grid becomes orthogonal, and are samll compared to the principal ones for moderate non-orthogonality ($45° < \theta_{ij} < 135°$) and grid aspect ratios (less than 10), and (ii) as the calculation converges, the corrections in Eqs. (5.22) all tend to zero. From these it can be concluded that an approximated pressure-correction equation in which the cross-derivative terms are dropped might be satisfactory*. This has indeed been confirmed in numerous calculations performed in the present study (some of which will be presented in Chapters 8 and 9), as well as in the earlier works of Rhie (1981), Rhie and Chow (1982), Demirdžić (1982) and Hah (1983, 1984), who used this approach.

With this simplification the coefficients for the pressure-correction equation become, instead of (5.28):

$$
\begin{aligned}
a_E &= \rho E_e^1 \\
a_W &= \rho E_w^1 \\
a_N &= \rho E_n^2 \\
a_S &= \rho E_s^2 \\
a_T &= \rho E_t^3 \\
a_B &= \rho E_b^3 \\
a_P &= \sum_m a_m \quad , \qquad m = E,W,N,S,T,B
\end{aligned}
\tag{6.8}
$$

---

* As the converged solution is approached the source terms C_p in the pressure-correction equation tends to zero at every node, since mass fluxes will then satisfy continuity. Thus both the simlpified and the full version of equation will be satisfied, ie. both will yield negligible pressure corrections p' .

while the source term C_P is calculated in the same way as in Eqs. (5.28). With this simplification the coefficient matrix is again reduced to a convenient seven-diagonal one, as in the momentum and scalar equations.

Under-relaxation of pressure changes remains essential for the stability and efficiency of the SIMPLE algorithm, as will be discussed in the next section.

The SIMPLER and PISO algorithms do not allow for the above simplifications. In the SIMPLER algorithm the "final" pressure is calculated from the correct velocity field by solving Eq. (5.35) for pressure rather than p' : hence the equation cannot be analogously simplified.

In the PISO algorithm if the simplified version of p' and p'' equations are employed the "exactness" of the method will be impaired, and divergence may occur unless the pressure changes are under-relaxed* .

The consequent necessity to solve the two pressure-correction equations in their full form to a tight tolerance results in a dramatic increase in the computational time per "outer" iteration when these two algorithms are used, as compared to the simplified SIMPLE algorithm described earlier. Thus the reduction in the total number of outer iterations may not be enough to compensate for extra work needed. Also, the implementation of boundary conditions for the full form of pressure-correction equation is significantly more complex than for the simplified version, as will be discussed later in this chapter.

The SIMPLEC algorithm seems to be a more attractive alternative to SIMPLE than PISO and SIMPLER, for it requires only one pressure-correction equation to be solved per outer iteration. However, if the latter equation is simplified, under-relaxation will again become essential if grid is significantly non-orthogonal.

For the reasons listed above the "simplified" SIMPLE algorithm has been adopted for use in the present study. A further incentive has been the development in the present study of a method of calculating optimum under-relaxation factors for SIMPLE, which makes it far more efficient. This will be described in the next section.

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* It is also expected that, in order to retain the full efficiency of PISO on non-orthogonal grids, the momentum equations should also be solved in their fully implicit form, ie. the coefficients of all neighbour nodes should be included in the H-operator of Eqs. (5.15).

6.5 Optimum Relation Between Under-Relaxation Factors for Pressure and Velocity

It was noted in section 5 of Chapter 3 that under-relaxation of the momentum equations is essential to ensure convergence, due to the inter-linkage through the pressure field (Patankar, 1980). Experience with the SIMPLE algorithm has also shown that the pressure changes also should be under-relaxed in order to improve convergence (Patankar, 1980, 1981; Raithby and Schneider, 1979; Demirdžić, 1982). This is due to the fact that the p' equation has been derived from approximate momentum equations, as discussed in section 4.4 of Chapter 5. Patankar (1980, 1981) suggested that the pressure under-relaxation factor α_p should be set at 0.8 and the velocity under-relaxation factors α_u at 0.5 . However, in the present study these values have been found not to be the optimum ones, for reasons which will now be explained.

The rationale is as follows. Pressure corrections p' make the velocity components u_i^{**} and pressure p^{**} , obtained via Eqs. (5.22), satisfy the continuity equation (5.27) and the approximate momentum equations (5.23). However, it would be preferable to have fields which satisfy the "exact" momentum equations, which can be written as:

$$u_{iP}^{**} = H_P(u_{im}^{**}) + (Q_i^1\delta p^{***1} + Q_i^2\delta p^{***2} + Q_i^3\delta p^{***3})_P + C_P' \qquad (6.9)$$

Here p^{***} is the pressure field which ensures that the above equations are satisfied. By subtracting Eq. (5.23) from Eq. (6.9), the following expression is obtained* :

$$0 = H_P(u_{im}') + (Q_i^1\delta p''^1 + Q_i^2\delta p''^2 + Q_i^3\delta p''^3)_P \qquad (6.10)$$

in which the new unknowns are the second pressure corrections p'' , defined as:

$$p'' = p^{***} - p^{**} \qquad (6.11)$$

and p^{**} is defined by Eqs. (5.22).

---

* The similarity between equations (6.9), (6.10) and (5.37), (5.39) of the PISO algorithm is noteworthy; here only no second correction of velocity components is required.

It can be argued that there is little incentive to find the "exact" solution for p'' out of equations like (6.10) (if this is possible at all) since the momentum equations have been linearized and further iteration will in any case be needed irrespective of the goodness* of the solution. A satisfactory approximation can be obtained if: (i) it is assumed that at every node the following relation between p'' and p' holds:

$$p'' + p' = \alpha_p p' \tag{6.12}$$

(ii) the u'_{iP} are expressed in terms of p' according to Eqs (5.24), and (iii) the operator H_P is expressed as in (5.16). These steps allow equation (6.10) to be transformed into an equation for α_p, thus

$$\alpha_p = 1 - \frac{\sum_m a_m u'_{im}}{a_P^{u_i} u'_{iP}} \tag{6.13}$$

When $a_P^{u_i}$ is expressed via neighbour coefficients a_m and the under-relaxation factor α_u according to Eq. (3.49), equation (6.13) becomes (for $S''_{ui} \ll \sum_m a_m$):

$$\alpha_p = 1 - \alpha_u \frac{\sum_m a_m u'_{im}}{u'_{iP} \Sigma a_m} \tag{6.14}$$

If it is now further assumed that (iv) the velocity correction at the central node P , u'_{iP} , is approximately equal to the a_m-weighted mean of the velocity corrections u'_{im} at the m neighbour nodes, then equation (6.14) finally reduces to**:

$$\alpha_p = 1 - \alpha_u \tag{6.15}$$

It follows therefore that the pressure should be corrected according to:

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* This will be demonstrated on some test calculations in Chapter 8.

** It has been drawn to the attention of the present author that the PS3 scheme of Raithby and Schneider (1979) embodies the same relation as (6.15) between the under-relaxation factors. However, they arrived at this result by a different route, in which they required consistency of the pseudo-time steps implied by α_u and α_p , in a way which is not obvious. It is believed that the present derivation is more transparent; the fact that the same result can be derived in another way only reinforces its validity.

$$p^{***} = p^* + p' + p'' = p^* + \alpha_p p' \qquad (6.16)$$

in order that the momentum and continuity equations are satisfied by the corrected velocity u_i^{**} and pressure p^{***} fields. The equation (6.16) is identical to the equation (5.30), and hence α_p defined by Eq. (6.15) is the optimum under-relaxation factor for pressure, for a given value of α_u.

The derivation presented above is valid for orthogonal and non-orthogonal grids, and non-staggered and staggered variable arrangements. However, if the simplified version of the pressure-correction equation presented in the previous section is used and the grid is substantially non-orthogonal (so that the neglected terms are significant), expression (6.15) may no longer represent the optimum relation. Experience suggests, as will be demonstrated in Chapter 8, that in such cases the value of α_p should be reduced below the value suggested by Eq. (6.15) to obtain the maximum rate of convergence.

6.6 Boundary Conditions and their Implementation

The most frequently encountered boundary conditions in fluid flow problems and the treatment necessary to incorporate them into the discretised equations are described below.

6.6.1 Inlet boundaries

The values of all dependent variables are normally known at inlet boundaries. These are usually obtained by reference to experimental data, or analysis, or - if neither practice is possible - estimated. The last practice is strictly only justifiable when inlet conditions are not expected to significantly influence the downstream region of interest. It is, however, usually necessary to calculate the turbulence dissipation rate ε from the measured kinetic energy of turbulence k and estimates of the distribution of the length scale l, via the relationship $\varepsilon \propto k^{3/2}/l$.

Figure 6.1 shows a two-dimensional cell next to an inlet boundary, for illustrative purposes. As noted in section 2 of Chapter 5, boundary nodes are placed at the centre of boundary cell faces, and the known va-

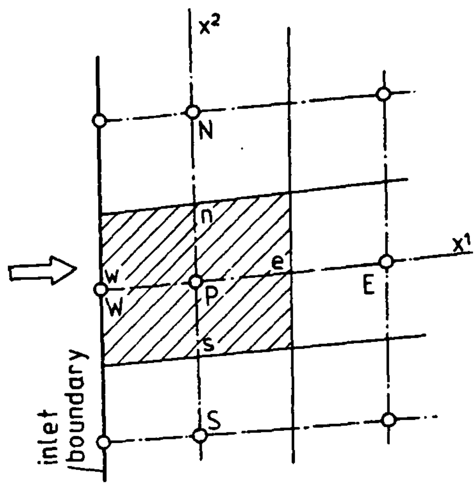

Fig. 6.1: Location of boundary nodes

lues of dependent variables are assigned to them. If a zero-thickness boundary cells are imagined to enclose these nodes, then according to Eq. (4.3) the linear interpolation factor f_{iB} will turn out to be zero on the w, s or b boundaries, if B denotes the boundary node (W in Fig. 6.1). Further, f_{iP} will turn out to be unity for nodes in the centre of cells next to e, n and t boundaries. It follows that no explicit modifications are necessary to the discretised equations for the cells neighbouring boundaries, since the weighting factors ensure the correct evaluation of diffusion terms, and the convection inlet flux is directly calculated, since the cell face values in the expression like (3.31) correspond to the boundary nodal values (see Fig. 6.1). Only when the LUDS is employed is special action necessary: eg. for the situation shown in Fig. 6.1 , the coefficient a^C_{WW} must be set equal to zero, and a^C_W is given by Eqs. (4.7).

Since the inlet boundary mass fluxes are known (or treated as such), they need not be corrected and so the flux corrections, given by Eqs. (5.16), for boundary cell faces are suppressed. This is equivalent to prescribing zero-gradient boundary conditions for the pressure-correction equation, as can be inferred from Eq. (5.25), which is readily implemented by setting to zero the coefficient of Eq. (6.8), corresponding to the boundary node. The same treatment also applies for outlet, symmetry and wall boundaries.

It should be noted that, if the full form of the pressure-correction equation is used (ie. Eq. (5.28)), the imposition of the zero boundary flux correction condition requires changes to the coefficients for the neighbour cells, which complicates computer programming. The practice employed for the simplified pressure-correction equation is actually equivalent to setting to zero the gradient in the coordinate direction x^j. This, for reasons explained earlier, has no effect on the converged solution but may inhibit convergence as the degree of non-orthogonality is increased.

6.6.2 Outlet boundaries

If the values of the dependent variables are known at exit boundaries, these are then fixed as such. However, it is rare that such information is known in advance. In such cases the outlet boundary should be placed far downstream from the region of interest, at a location where the flow is everywhere directed outwards (see Fig. 6.2) so that any inaccuracy in estimating the outlet conditions will not propagate far upstream, provided that the Reynolds number is large.

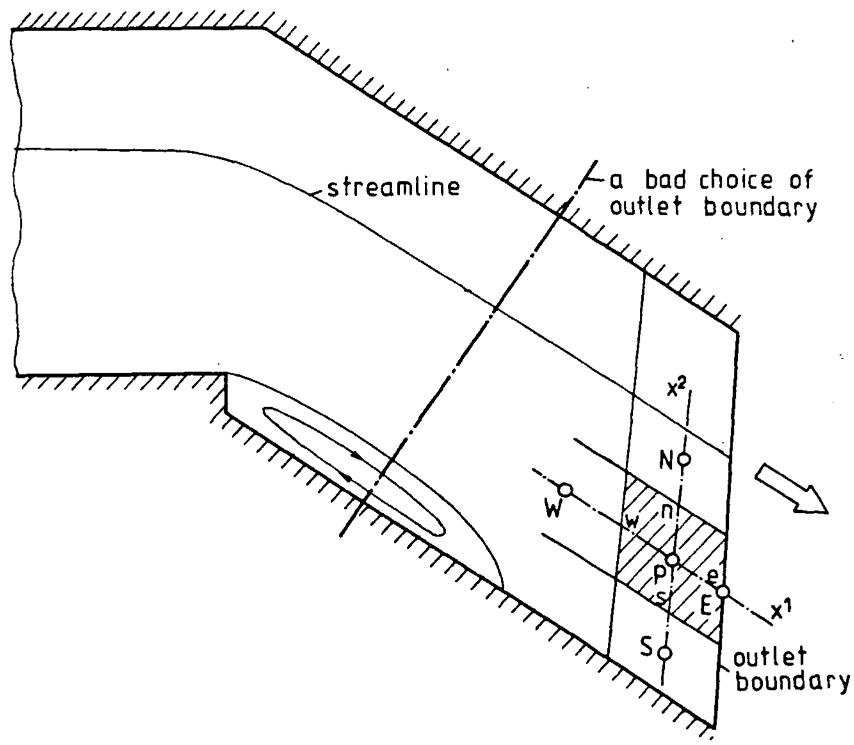

Fig. 6.2: Outlet boundary

Since both the UDS and LUDS do not refer to downstream nodes, the convetive part of coefficients, a_m^c, corresponding to the outlet boundary nodes (a_E^C for the situation shown in Fig. 6.2) will be zero for all neighbour cells. Thus only the diffusion part of the coefficients will remain, and this will be small for high Reynolds numbers. It is therefore appropriate to evaluate outlet boundary values by extrapolation from upstream. It is often adequate to set boundary values equal to the values at immediate upstream neighbour, eg. $\phi_E = \phi_P$ for the situation shown in Fig. 6.2 (which is equivalent to zero gradient in x^1 direction). Otherwise, linear (Eq. (4.8)) or quadratic extrapolation may be used* .

In the case of the velocity components, it is also necessary to ensure that they satisfy overall continuity, ie. that the sum of outlet mass fluxes F^O equals the sum of inlet mass fluxes F^I . This can be done by calculating outlet mass fluxes using newly estimated boundary values of velocity components, and correcting these velocity components by multiplying them with the ratio of F^I/F^O .

6.6.3 Symmetry boundaries

The conditions applying at an axis or plane of symmetry are:

- no cross-flow (zero convective flux);
- zero diffusion flux I^D of dependent variables in the normal direction.

Figure 6.3 shows a cell whose "b" face is a part of a symmetry plane. The above conditions can be imposed according to Eqs. (3.28), (3.35) and (6.3) by setting

$$F_{3b} = 0$$

$$I_b^{DC} = 0 \tag{6.17}$$

$$a_b^{DN} = 0$$

and, in addition, by setting in the momentum equations to zero that part

* It should be noted, however, that a good estimate of outlet normal velocities is important, since its effects can propagate upstream through the pressure.

of S^*_{ui} pertaining to the "b" face (see Eq. (6.4)):

$$I^s_b = 0 \tag{6.18}$$

The values of the dependent variables at symmetry boundary nodes are needed for evaluation of fluxes through cell faces which intersect this boundary (for the situation shown in Fig. 6.3 , these are e, w, n and s cell faces). These values can be calculated from the interior nodal values by imposing the condition of zero normal gradient (which is equivalent to specifying the zero diffusion flux), ie.:

$$\frac{\partial \phi}{\partial n_s} = 0 \tag{6.19}$$

Here n_s denotes the coordinate normal to the symmetry plane or axis. For the situation shown in Fig. 6.3 , the above equation is equivalent to:

$$\left(\frac{\partial \phi}{\partial x^3}\right)_b = \left(\frac{\partial \phi}{\partial x^1}\right)_b \cdot \cos\theta_{13} + \left(\frac{\partial \phi}{\partial x^2}\right)_b \cdot \cos\theta_{23} \tag{6.20}$$

where θ_{13} and θ_{23} are the angles between positive parts of coordinate lines x^1 and x^2 and the inward part of x^3 , as shown in Fig. 6.3 .

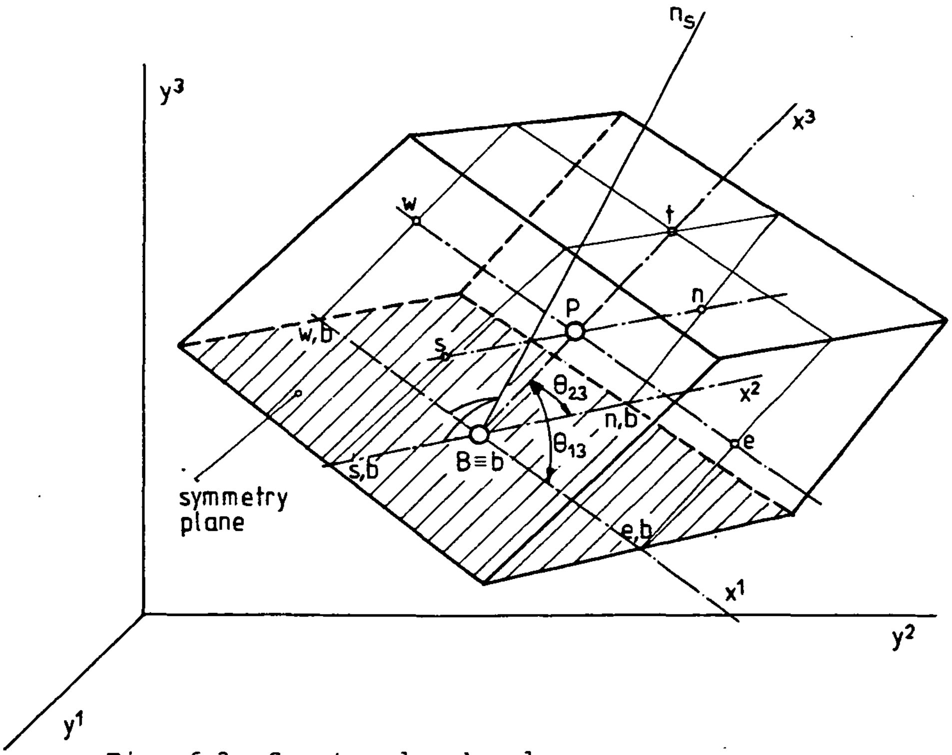

Fig. 6.3: Symmetry plane boundary

By employing the discretisation practices described in section 4 of Chapter 3 (notably Eq. (3.34)) and some analytic geometry, the above expression can be transformed to:

$$\phi_B = \phi_P - \lambda_1(\phi_e - \phi_w)_b - \lambda_2(\phi_n - \phi_s)_b \tag{6.21}$$

where the coefficients λ_1 and λ_2 are given by:

$$\lambda_1 = \frac{\sum_j [(y_P^j - y_B^j)(y_{eb}^j - y_B^j)]}{2\sum_j (y_{eb}^j - y_B^j)^2} \qquad j=1,2,3$$
$$\lambda_2 = \frac{\sum_j [(y_P^j - y_B^j)(y_{nb}^j - y_B^j)]}{2\sum_j (y_{nb}^j - y_B^j)^2} \tag{6.22}$$

Similar expressions follow for the other cell faces. The values of ϕ_e, ϕ_w etc. on the boundary cell face can be calculated from the boundary nodal values obtained in the previous iteration with the aid of the interpolation formulae (4.18). Alternatively, the set of equations (6.21) for all symmetry boundary nodes could be solved simultaneously; however this is not warranted, in view of the facts that (i) the overall procedure is iterative, and (ii) λ_1 and λ_2 tend to zero as θ_{13} and θ_{23} tend to 90°, ie. as the grid becomes orthogonal.

6.6.4 Impermeable walls

Rigid impermeable walls occur in most fluid flow problems. The values of the velocity components u_i, and hence the mass fluxes F_i, are known to be zero there. These conditions are easy to impose; however, in the case of turbulent flows the calculation of the stresses on the wall needs special treatment, since expressions (3.35) and (6.4) become inappropriate for the wall cell faces (Launder and Spalding, 1974; Gosman and Ideriah, 1976). This is due to existence of boundary layers, across which steep variations of flow properties occur and the standard k-ε model of turbulence becomes inadequate (Launder and Spalding, 1974). In order to adequately avoid these problems it would be necessary to employ a fine grid and use a low Reynolds number form of the k-ε model of turbulence (Jones and Launder, 1972), which would be expensive. An alternative and widely employed approach is to use formulae known as "wall functions"

(Launder and Spalding, 1974) to bridge this region: this has been followed in the present study. It is based on a one-dimensional Couette flow analysis, which is assumed to be valid in the near-wall region* , and will be summarized briefly in what follows.

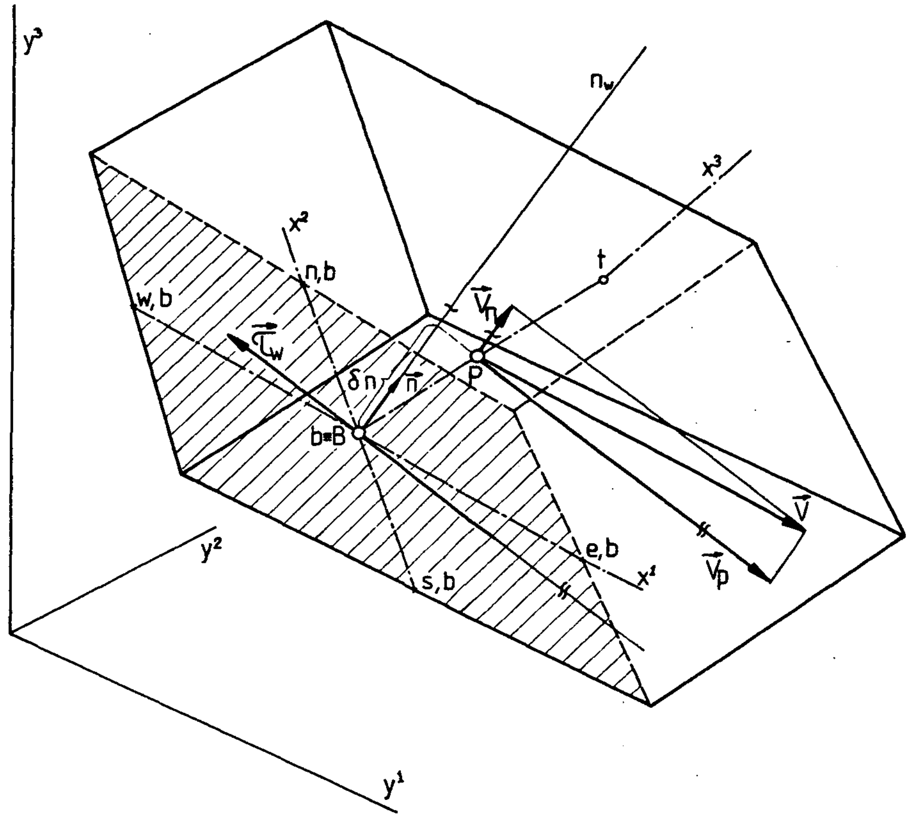

Fig. 6.4: Wall boundary

The wall shear stress $\vec{\tau}_w$ is expressed as a function of the adjacent nodal velocity component parallel to the wall, $\vec{v}_p$, thus (see Fig. 6.4):

$$\vec{\tau}_w = -\lambda_w(\vec{v}_p)_P \tag{6.23}$$

where the coefficient λ_w is determined from a two-part "universal" velocity profile expression (Gosman and Ideriah, 1976), ie.:

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* The Couette flow assumption implies constant shear stress ($\tau = \tau_w$, where τ_w is the wall shear stress) and hence no streamwise pressure gradient. Although this is a reasonable assumption for parallel near-wall flows, it may be greatly in error in case of recirculating flows, where large gradients along the wall are present and the flow near wall is not parallel (eg. in separation and reattachment zones).

a) laminar sublayer:

$$\lambda_w = \frac{\mu}{\delta n} \qquad \text{if} \quad y_P^+ < 11.6$$

b) fully turbulent layer: (6.24)

$$\lambda_w = \frac{\rho C_\mu^{1/4} k_P^{1/2} \kappa}{\ln(E y_P^+)} \qquad \text{if} \quad y_P^+ > 11.6$$

where y_P^+ is calculated as:

$$y_P^+ = \frac{\rho C_\mu^{1/4} \, k_P^{1/2} \, \delta n}{\mu} \tag{6.25}$$

In the above equations κ is the von Karman's constant (κ =0.4187) and E is an integration constant which depends on the wall roughness, and through which other effects such as pressure gradients and mass transfer can be accounted for (Launder and Spalding, 1974). For smooth impermeable walls E is assigned a constant value of 9.0 (Launder and Spalding, 1974).

More elaborate multilayer wall functions have been proposed, eg. by Chieng and Launder (1980) and more recently by Amano (1984), in which some of the restrictions implied in the above equations are relaxed. These result in much more complex formulae, and have not yet been sufficiently tested: hence it was decided not to implement them in the present solution procedure at this stage. However, they deserve to be considered as future improvements, as will be discussed later in Chapter 9.

Since the momentum equations are resolved in the directions of the Cartesian coordinates, the resultant shear wall force $\vec{T}_w$ needs also be expressed in terms of its Cartesian components. For simplicity* , it is assumed here that $\vec{T}_w$ acts in the direction opposite to $\vec{v}_p$, as shown in Fig. 6.4. This force is equal to the product of the wall shear stress $\vec{\tau}_w$ and the area A_w , ie.:

$$\vec{T}_w = \vec{\tau}_w A_w = -\lambda_w A_w (\vec{v}_p)_P \tag{6.26}$$

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* The validity of this assumption is rather limited in complex three-dimensional flows, where substantial velocity vector rotation through near-wall region may occur.

Its components in the y^i directions can readily be obtained if the $\vec{v}_p$ is expressed in terms of its components in these directions, ie.:

$$T_{wi} = -\lambda_w A_w (v_{pi})_P \tag{6.27}$$

$\vec{v}_p$ can be obtained by substracting from the total velocity $\vec{v}$ its component in the direction normal to the wall, $\vec{v}_n$, which is deducible from the scalar product of $\vec{v}$ and the unit normal vector $\vec{n}$. If the Cartesian components of $\vec{n}$ are n_1, n_2 and n_3, then the following expressions for shear force components result:

$$\begin{aligned}
T_{w1} &= -\lambda_w A_w \{[1-(n_1)^2]\, u_{1P} - n_1 n_2 u_{2P} - n_1 n_3 u_{3P}\} \\
T_{w2} &= -\lambda_w A_w \{-n_1 n_2 u_{1P} + [1-(n_2)^2] u_{2P} - n_2 n_3 u_{3P}\} \\
T_{w3} &= -\lambda_w A_w \{-n_1 n_3 u_{1P} - n_2 n_3 u_{2P} + [1-(n_3)^2]\, u_{3P}\}
\end{aligned} \tag{6.28}$$

The normal vector $\vec{n}$ is defined by the vector product of two cell face "central lines", eg. for the situation shown in Fig. 6.4, from $(\overrightarrow{ew})_b$ and $(\overrightarrow{ns})_b$.

Thus, for the u_i momentum equations the shear stress integrals over boundary cell faces, appearing in equations like (3.35) and in (6.4), are replaced by the T_{wi}, defined in (6.28). These are conveniently introduced through the source term C_P, and then linearized according to (3.37).

The equations for k and ε also need special treatment for the near wall cells. In the equation for k, the diffusion flux through the wall cell face, I_w^D, is set to zero (Launder and Spalding, 1974) in the manner indicated earlier, while the source terms are modified as follows.

Within the turbulent layer a balance between production and dissipation of turbulent kinetic energy is assumed (Launder and Spalding, 1972), giving the following relation for ε:

$$\varepsilon = \frac{C_\mu^{3/4}\, k^{3/2}}{\kappa\, y} \tag{6.29}$$

The dissipation of k, given by Eq. (2.30), is now obtained by integration over the near-wall cell (Gosman and Ideriah, 1976):

$$-\int_{\delta V}\rho\varepsilon dV \simeq -S_k'' k_P = -\frac{\rho C_\mu^{3/4} k_P^{*1/2} u_P^+ \delta V_P}{y_P} k_P \tag{6.30}$$

where u_P^+ is given by:

$$\begin{aligned} u_P^+ &= y_P^+ \quad , \quad \text{if} \quad y_P^+ < 11.6 \\ u_P^+ &= \frac{1}{\kappa}\ln(Ey_P^+) \quad , \quad \text{if} \quad y_P^+ > 11.6 \end{aligned} \tag{6.31}$$

In the above k_P^* is evaluated at the previous iteration, which enables linearization according to Eq. (3.37).

The rate of generation of turbulent kinetic energy G, under the Couette flow assumptions, is given by:

$$G \simeq \tau_w \frac{\partial v_P}{\partial n} \tag{6.32}$$

Integration over the cell volume gives the S_k' part of the linearized source term (Gosman and Ideriah, 1976) as:

$$\int_{\delta V} G \cdot dV \simeq S_k' = \tau_w \left(\frac{v_P}{\delta n}\delta V\right)_P \tag{6.33}$$

The transport equation for ε is abandoned for the next-to-wall cells due to its inapplicability there. Instead, ε is fixed to the value given by Eq. (6.29) by means of linearized source term, ie. by setting:

$$\begin{aligned} S_\varepsilon' &= L\frac{C_\mu^{3/4} k_P^{3/2}}{\kappa\, y_P} \\ S_\varepsilon'' &= L \end{aligned} \tag{6.34}$$

where L is a large number, typically 10^{30}.

Wall functions provide also the means of bridging the near-wall region for scalar variables ϕ like temperature. The following expressions (Launder and Spalding, 1974; Gosman and Ideriah, 1976) for the diffusion flux through the wall cell faces result:

$$I_w^D = \begin{cases} \dfrac{\rho C_\mu^{1/4}\ k_P^{1/2}\ c_P(\phi_w - \phi_P)A_w}{\phi_P^+} \quad , \quad \text{if} \quad y_P^+ > 11.6 \\ \\ \dfrac{\mu}{\sigma_\phi}(\phi_w - \phi_P)\dfrac{A_w}{\delta n} \quad , \quad \text{if} \quad y_P^+ < 11.6 \end{cases} \tag{6.35}$$

where

$$\phi_P^+ = \sigma_\phi^t \left[u_P^+ + P\left(\frac{\sigma_\phi}{\sigma_\phi^t}\right)\right] \tag{6.36}$$

In the above equations w referes to the wall boundary, σ_ϕ and σ_ϕ^t are the molecular and turbulent Prandtl numbers respectively, c_P is the constant-pressure specific heat, and P is an integration constant which depends on the ratio of Prandtl numbers σ_ϕ and σ_ϕ^t, and involves additional empirical constants to account for wall roughness (Jayatillaka, 1969).

Cells surrounding a submerged solid body within the solution domain are treated in the same way.

6.6.5 Other boundaries

Various other types of boundaries can be encountered in fluid flow problems, like free stream boundaries, porous walls, periodic boundaries etc. Each of these requires special treatment, as can be seen from examples appearing in the literature (see, eg. Antonopoulos, 1979; Demirdžić, 1982; Hah, 1983, 1984; and others). In all cases information will be available from which the boundary values or boundary fluxes, or relations between the two, can be extracted. If the boundary values are known, the treatment is the same as for inlet boundaries. If the flux is known, it can be usually expressed (after linearization, if necessary) as a function of the near-wall nodal value, eg.:

$$I_b = \lambda_2 - \lambda_1 \phi_P \tag{6.37}$$

In such case the flux expressions for corresponding boundary cell faces in the discretised equations should be suppressed, and the flux from Eq. (6.37) introduced instead, through the source term in the usual way (ie. λ_2 inserted in S' and λ_1 in S'').

6.7 Solution Algorithm

It is now possible to assemble the complete solution algorithm, ie. the sequence of operations which, when repeated as many times as necessary, leads to the final, converged solution, satisfying all the governing equations involved. It can be summerized as follows:

(i) Initialize all field values by an initial guess;

(ii) Solve the momentum equations (5.17) to obtain u_i^* , using the prevailing pressure as p^* ;

(iii) Solve the pressure-correction equation to obtain p' ;

(iv) Correct the velocities using expressions (5.24) and (5.22), the mass fluxes using Eq. (5.26) and the pressures using Eq. (5.30), to obtain the u_i^{**} , F_i^{**} and p^{**} fields;

(v) If the flow is turbulent, solve the equations for k and ε , and calculate the new μ_{eff} ;

(vi) Solve additional equations for further scalar quantities ϕ , if necessary;

(vii) Using the u_i^{**} , F_i^{**} and p^{**} fields as the prevailing fields for the new iteration cycle, return to step (ii) .

The sequence of steps (ii) to (vi) is to be repeated until the convergence criterion defined below is satisfied.

Convergence can be defined in various ways. One is to monitor the changes in the field values at all locations, and a second is to monitor the "residual sources" (defined below) in the discretised equations: in both cases the monitored quantities are required to fall below prescribed levels. The second approach is believed to be more appropriate, as slow changes in the field values can sometimes be misleading (eg., the changes may be small due to low values of under-relaxation factors being used). In the present study the sum of absolute residuals at all computational nodes is used as the primary test. The residual at node P is calculated as (see Eq. (6.1)):

$$R_P^n = \left|\sum_m a_m^n \phi_m^{n-1} + C_P^{n-1} - a_P^n \phi_P^{n-1}\right| \tag{6.38}$$

and the sum is given by:

$$SR_\phi = \sum_l R_l^n \quad , \quad l=1,\ldots,N \tag{6.39}$$

where N is the total number of nodes, and n is an iteration counter.

These sums are calculated at each iteration, for each dependent variable, after the new sets of coefficients a_m are assembled but before equations are solved. The sums SR_ϕ are then normalized by the appropriate inlet fluxes SN_ϕ , given in table 6.1 , to give non-dimensional values:

$$r_\phi = \frac{SR_\phi}{SN_\phi} \tag{6.40}$$

It is then required that:

$$\max(r_\phi) \leqslant \lambda_s \tag{6.41}$$

where λ_s is the prescribed maximum allowable value of normalized residual sums, and is typically $1 \cdot 10^{-3}$ for all equations.

In addition, the values of the dependent variables are also monitored at one or more locations, chosen in the regions which can be considered as most sensitive, to ensure that there are no appreciable changes in them when convergence is declared.

Equation	Normalization quantity SN_ϕ
Continuity (p'-eq.)	Total mass inflow, $F^I = \Sigma F_i^{in}$
Momentum equations	Total inlet momentum, $\Sigma(F_i^{in} \cdot v^{in})$
Scalar equations	Total inlet ϕ flux, $F^I \cdot \overline{\Phi}_{in}$
Turbulence kinetic energy k	Total inlet flow of mean kinetic energy, $F^I \cdot k_{norm}$, where $k_{norm} = (\overline{v}_{in})^2$
Turbulence dissipation ε	Nominal total inlet flow $F^I \varepsilon_{norm}$, $\varepsilon_{norm} = k_{norm}^{3/2}/l$, l is typical length scale, usually fraction of duct width eg. $0.1 \cdot L$

Table 6.1: List of quatities used to normalize residual sums

For each dependent variable (u_i , ϕ , k, ε) usually only one "inner" iteration (ie. application of the SIP solver) is performed before the coefficients a_m and C_P in Eq. (6.1) are updated. The exception is the pressure-correction equation, for which the sufficient inner iterations are performed to satisfy the following criterion (Van Doormaal and Raithby, 1984):

$$SR_P^n \leqslant \lambda_p \cdot SR_P^o \tag{6.42}$$

where SR_P^o is the absolute sum of the mass sources C_P in the pressure-correction equation (defined in Eqs. (5.28)), and λ_p is typically 0.3.

6.8 Closure

In the final arrangement of the discretised equations it has been decided to treat the cross-derivative diffusion terms and some of the stress terms explicitly, ie. to evaluate them from the values of the dependent variables obtained in the previous iteration. There are several important advantages which arise from this treatment; these are:

(i) the coefficient matrix is reduced to a convenient seven-diagonal (five-diagonal in two-dimensional cases) form, thus significantly reducing storage requirements;

(ii) the coefficients a_m in all momentum equations become the same, thus reducing computational effort;

(iii) relegation of (possibly negative) cross-derivative coefficients into source term improves stability and sometimes convergence of the numerical solution process.

It has also been found that the pressure-correction equation in the SIMPLE algorithm, chosen for use in the present study, can be simplified by neglecting cross-derivative terms, thus achieving the same convenient coefficient matrix as in other equations.

The optimum relation between under-relaxation factors for velocity and pressure has been found, resulting in much faster convergence than when standard values, found in the literature, are used.

The most frequently encountered boundary conditions and the treatment necessary to incorporate them into the discretised equations have been described. Standard wall functions have been employed in the treatment of turbulence near-wall flow.

Finally, the solution algorithm and the convergence criteria have been defined, thus completing the description of the present numerical solution method. Its assessment and applications will be presented in Chapters 8 and 9, after some more details about the numerical grids are given in the next chapter.

CHAPTER 7

COMPUTATIONAL GRID AND ASSOCIATED PARAMETERS

7.1 Introduction

In simple, regular geometries, the question of grid generation is trivial; however, in complex geometries it becomes an important issue. For reasons already discussed in Chapter 1, the computational grid should be boundary-fitted and have sufficient resolution in regions of interest. To be able to achieve this for a wide range of problems, a general numerical grid generation technique is needed. It should, ideally, be easy to use and cost effective, and allow for user control over particular grid properties, which will be identified later. In this chapter some simple methods for the generation of grids suitable for use with the present solution method will be described.

First, in section 2, the desired properties of the computational grid will be discussed, and the possibility of achieving them will be outlined.

In section 3 the generation of grids for two-dimensional problems will be addressed, and several approaches will be suggested, which were appropiate in all the cases considered in this study. References will also be given to other, more general methods which might be more appropriate in other circumstances. The same will be done for three-dimensional problems in section 4.

In section 5 the calculation of geometrical quantities related to the grid which feature in discretised form of governing equations will be described, for both two- and three-dimensional grids.

Finally, in section 6, closing remarks on this topic will be given.

7.2 Desired Properties of Computational Grid

As already noted, the accuracy of the numerical solutions, and the stability and the rate of convergence of the solution procedure all depend to a certain degree on particular properties of the computational grid. These properties, the way of achieving them and the degree of their influence will be described in this section.

7.2.1 Orthogonality

The very reason for developing a method of the present kind is to enable working with non-orthogonal grids, which is desirable, if not essential in most pratical situations, as discussed in Chapter 1. However, it is still useful to identify the advantages which arise from minimizing the departure from orthogonality, all other factors being equal.

First of all, if orthogonality prevails, then all cross-derivative terms in the governing equations vanish (see Eq. (3.35)). This means that there will be no explicit "sources" in the discretised equations arising from these terms, and the "neglected" cross-derivative terms in flux-corrections used to derive the pressure-correction equation are also formally absent. The result is a more stable and faster converging numerical solution procedure.

The accuracy of numerical solution is not affected by grid non-orthogonality alone, but also depends rather strongly on some other grid properties, as will be discussed in the next chapter. However, strong non-orthogonality does affect the stability and convergence rate of the present method, and therefore should be avoided if possible* .

7.2.2 Grid spacing

The accuracy of chosen interpolation practices (differencing schemes) is usually judged by means of the Taylor Series Truncation Analysis (TSTA). This shows that all schemes loose one order of accuracy if the grid spacing is non-uniform. From this consideration it could be concluded that uniform spacing is a desirable property of a computational grid. However, this goal is neither easy nor economical to achieve in every situation. In practical flow configurations there are regions of very steep gradients where very fine resolution is needed, and regions of low gradients, where more coarse spacing is adequate. For this reason a non-uniform grid is usually preferable, and it is desirable that a grid generation procedure should facilitate the achievement of such an arrangement.

In constructing non-uniform grids it is desirable to keep the grid

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* This, however, is not a shortcoming of the method in general, but a consequence of the adopted treatment of the cross-derivative terms.

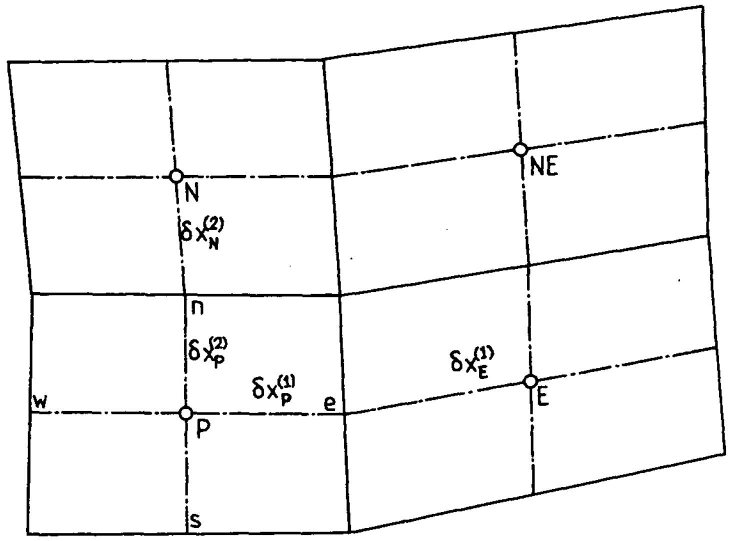

Fig. 7.1: Definition of grid expansion and aspect ratios

expansion ratios in each coordinate direction, defined as (see Fig. 7.1)

$$r_e^1 = \frac{\delta x_E^{(1)}}{\delta x_P^{(1)}} \quad ; \quad r_e^2 = \frac{\delta x_N^{(2)}}{\delta x_P^{(2)}} \quad ; \quad r_e^3 = \frac{\delta x_T^{(3)}}{\delta x_P^{(3)}} \tag{7.1}$$

under control, for their departure from unity adversely affects the accuracy of finite-difference approximations (at least in terms of the TSTA). Also the grid aspect ratios, defined as

$$r_a = \left(\frac{\delta x_P^{(i)}}{\delta x_P^{(j)}} \right)_{i \neq j} \tag{7.2}$$

should not exceed ~ 10 . These ratios influence the stability of the present solution method (and also the boundedness of solutions) due to the fact that they can generate negative cross-derivative coefficients a_m^{DC} (see Eqs. (4.22) and (4.26) and related discussion in section 4 of Chapter 4, and also Demirdžić, 1982). The rate of convergence of some iterative solvers, including the one employed in the present study, is especially adversely affected by the high aspect ratios. This is due to the fact that the coefficients a_m of Eq. (3.42) contain the cell face areas (see Eq. (3.26)) and in case of high aspect ratios they thus become larger in one coordinate direction than in the other. The pressure-correction equation is most sensitive, since its coefficients (Eq. (6.8)) are proportional to the squares of these areas: for example, if the ratio

$\frac{\delta x^{(1)}}{\delta x^{(2)}} \simeq 10$, than the coefficients a_N and a_S of Eq. (6.8) will be approximately 100 times larger than a_E and a_W . The result is very weak coupling in the x^1 coordinate direction, as will be further discussed in Chapter 9.

7.2.3 Smoothness

As noted earlier in Chapter 2, methods which use curvilinear velocity components, and therefore require the calculation of curvature terms, suffer significantly from sensitivity to grid smoothness* . In the present method, however, such terms do not appear. The geometrical information about the grid needed by the present method is limited to the Cartesian coordinates of the cell vertices only, and hence these may and are taken to be connected by segments of straight lines. Therefore, the grid lines are not - and need not to be - continuous.

Here, however, the term smoothness is taken to refer to the relative change in grid line direction from one cell to another, characterized by the angle θ shown in Fig. 7.2 . This does have an influence on the accuracy of the interpolation formulae used to obtain dependent variable values at locations other than the principal nodes. The way the accuracy of the linear interpolation used in this study can be affected by this

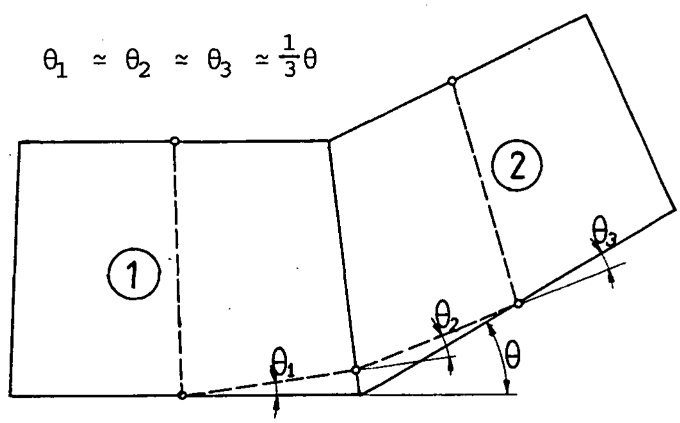

Fig. 7.2: Definition of grid smoothness

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* "Smooth" in this respect assumes continuity of the grid lines and their first derivatives.

property of the grid will be examined in an example in the next chapter. Higher-order interpolation practices may help in reducing the errors introduced by non-smoothness; however, they are computationally more expensive and can generate more negative coefficients* , thus affecting the stability of solution process and unboundedness of the solutions. Often it is possible to avoid large θ's (ie. to obtain a smoother grid) by local refinement. Figure 7.2 shows how large θ between cells 1 and 2 can be reduced by one-third by halving the two cells.

It follows that, at least when linear interpolation is used, it is desirable that the grid should be as smooth as geometry of the solution domain allows. The method of grid generation should clearly offer the possibility of controlling this grid property. The effects of grid non-smoothness on the solutions of some complex flows will be demonstrated later in Chapter 9.

7.2.4 Alignment of grid lines and streamlines

It has been shown in Chapter 4 that "numerical diffusion", which significantly affects the accuracy of a numerical solution, is a function of the skewness of grid and streamlines, and is reduced to minimum or completely eliminated if one set of grid lines is closely aligned with the streamlines (see also Spalding, 1972; Raithby and Torrance, 1974; Patankar, 1980 and Lai, 1982). This, therefore is a very desirable property of a computational grid, and is probably the one which most affects numerical accuracy, especially if lower-order differencing schemes are used.

One of great advantages of the kind of general calculation procedure presented here over those which employ regular (Cartesian, polar cylidrical) or orthogonal curvilinear grids, is that this property usually can be procured to a higher degree. However, in order to achieve alignment, it is necessary at the time of grid generation to know the streamline pattern, which is usually not the case. Therefore, the generation of an ideal grid should be an adaptive process, carried out during the course of the flow calculation (see eg. Dwyer, 1984; Eisenman, 1985). Adaptive grid generation is, however, very complex and difficult to achieve.

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* As illustrated by the higher-order convective differencing schemes examined in section 3 of Chapter 4 .

A crude but effective alternative is to perform the adaptation manually, by generating a solution on an initial grid and then using it to improve alignment for a further calculation. Of course, this is possible only if the grid generation procedure employed offers this flexibility, which is obviously desirable.

All of the forementioned desirable properties of a computational grid cannot, in general, be satisfied simultaneously. However, the two which are of the highest importance for the present solution method - alignment of grid lines and streamlines and high resolution in regions of high gradients - do not conflict, and usually can be satisfied to a much higher degree than when regular or orthogonal grids are used. In order, however, to exploit this flexibility to the fullest extent, a versatile grid generation procedure is needed. This will be discussed in the next two sections.

7.3 Generation of Grid for Two-Dimensional Problems

The type of computational grid employed in the present method has been described earlier, in Chapter 5. It has also been shown that the only information about the grid, needed in the present method, are the Cartesian coordinates of cell vertices. In recent years a number of methods for the generation of such information have emerged. A collection of most important ones has been published by Thompson (1982), and many other publications have appeared since (eg. Shieh, 1984, and others).

Most methods generate the coordinates of the grid points or cell vertices by solving differential equations of the form:

$$\begin{aligned} \frac{\partial^2(x^1)}{\partial(y^1)^2} + \frac{\partial^2(x^1)}{\partial(y^2)^2} &= P(x^1,x^2) \\ \frac{\partial^2(x^2)}{\partial(y^1)^2} + \frac{\partial^2(x^2)}{\partial(y^2)^2} &= Q(x^1,x^2) \end{aligned} \qquad (7.3)$$

with Dirichlet boundary conditions, where user-prescribed functions P and Q are employed for control over the distribution of grid lines in the interior of solution domain and degree of non-orthogonality. Any of these procedures could be used to generate grids for the present solution method. However, the solution of the differential equations (7.3) can be quite complicated and time consuming; and since the requirements for smoothness and orthogonality of grid lines are not so strict here, simpler methods of grid generation might be equally applicable.

A class of such methods, named "transfinite mappings" or "transfinite interpolations", exists and many have found application in finite-element methods (eg. Gordon and Hall, 1973; Haber et al, 1981; Gordon and Thiel, 1982). Here two simplified versions of discrete transfinite mapping (Haber et al, 1981) were used, and they will be described below. For more information about other general grid generation methods that could be employed, interested readers can refer to the forementioned references.

Practice 1

One of the simplest methods of grid generation, which is nonetheless quite versatile, has been used and described by Demirdžić (1982). In this, the Cartesian coordinates of grid points which lie on two opposite boundaries (x^2_{min} and x^2_{max}) are prescribed* along with a function which gives the distribution of points along the straight lines joining these, defined as (see Fig. 7.3):

$$f_j = \frac{l_j}{L} \tag{7.4}$$

Then the Cartesian coordinates of any interior grid point can be found from:

$$\begin{aligned} y^1_{i,j} &= y^1_{i,1} + f_j\,(y^1_{i,N_j} - y^1_{i,1}) \\ y^2_{i,j} &= y^2_{i,1} + f_j\,(y^2_{i,N_j} - y^2_{i,1}) \end{aligned} \tag{7.5}$$

Here, N_i and N_j denote the number of grid points in x^1 and x^2 directions respectively and f_j has limiting values $f_1=0$, $f_{N_j}=1$ (it should be noted that the number of computational cells in each direction is N_i-1 and N_j-1 ; the total number of nodes, including the boundary ones, is thus N_i+1 and N_j+1). If the boundary lines x^2_{min} and x^2_{max} are smooth, this method will always produce smooth interior grid lines.

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* This is equivalent to linear lofting mapping of Haber et al (1981).

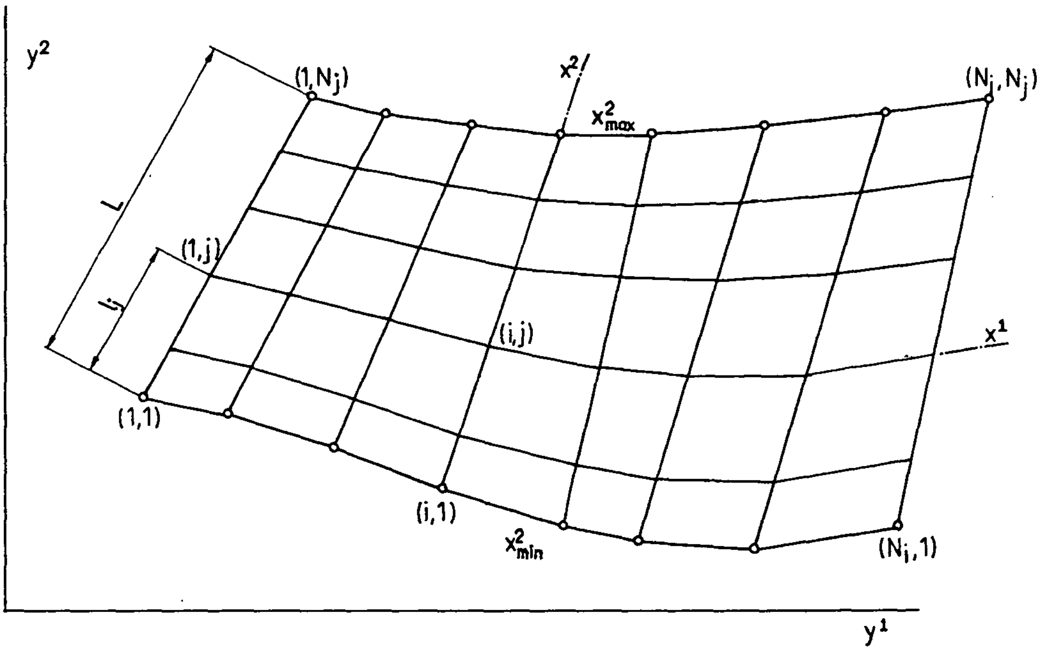


Fig. 7.3: Schematic presentation of Practice 1 for grid generation

This grid generation procedure is very simple, non-iterative and cost-effective, and in most cases involving duct-like flows, it is quite satisfactory. The grid lines can easily be concentrated in regions of high gradients by appropriately specifying the f_j function, and the possibility also exists of varying the latter from line to line to achieve reasonable alignment of the grid and streamlines. However, it is restrictive in that one set of grid lines (x^2 in the present case) is taken to be straight. This sometimes can result in severe non-orthogonality and ill-shaped control volumes, in which case it is necessary to use alternative approaches like that described below.

Practice 2

The shortcomings of Practice 1 can be avoided by a more general, but equally simple, method which requires that the Cartesian coordinates of the cell vertices on all four boundaries (x^1_{min} , x^1_{max} , x^2_{min} , x^2_{max}) of the solution domain be given* . Four distribution functions are then calculated (see Fig. 7.4):

---

* This practice is a simplification of the bilinear transfinite mapping of Haber et al (1981).

$$f^1_{i,1} = \frac{y^1_{i,1} - y^1_{1,1}}{y^1_{N_i,1} - y^1_{1,1}} \quad ; \quad f^1_{i,N_j} = \frac{y^1_{i,N_j} - y^1_{1,N_j}}{y^1_{N_i,N_j} - y^1_{1,N_j}}$$

$$f^2_{j,1} = \frac{y^2_{1,j} - y^2_{1,1}}{y^2_{1,N_j} - y^2_{1,1}} \quad ; \quad f^2_{j,N_i} = \frac{y^2_{N_i,j} - y^2_{N_i,1}}{y^2_{N_i,N_j} - y^2_{N_i,1}} \tag{7.6}$$

which describe the distribution of the boundary grid points. The coordinates of interior grid points are then calculated from the following interpolation formulae:

$$y^1_{i,j} = y^1_{1,j} + f^1_{i,j}\,(y^1_{N_i,j} - y^1_{1,j})$$

$$y^2_{i,j} = y^2_{i,1} + f^2_{i,j}\,(y^2_{i,N_j} - y^2_{i,1}) \tag{7.7}$$

where values of the functions $f^1_{i,j}$ and $f^2_{i,j}$ for interior points are calculated by interpolating between the two boundary functions defined in Eq. (7.6). The interpolation practice employed in the present study is as follows:

$$f^1_{i,j} = \frac{(N_j - j)f^1_{i,1} + (j-1)f^1_{i,N_j}}{N_j - 1}$$

$$f^2_{i,j} = \frac{(N_i - i)f^2_{1,j} + (i-1)f^2_{N_i,j}}{N_i - 1} \tag{7.8}$$

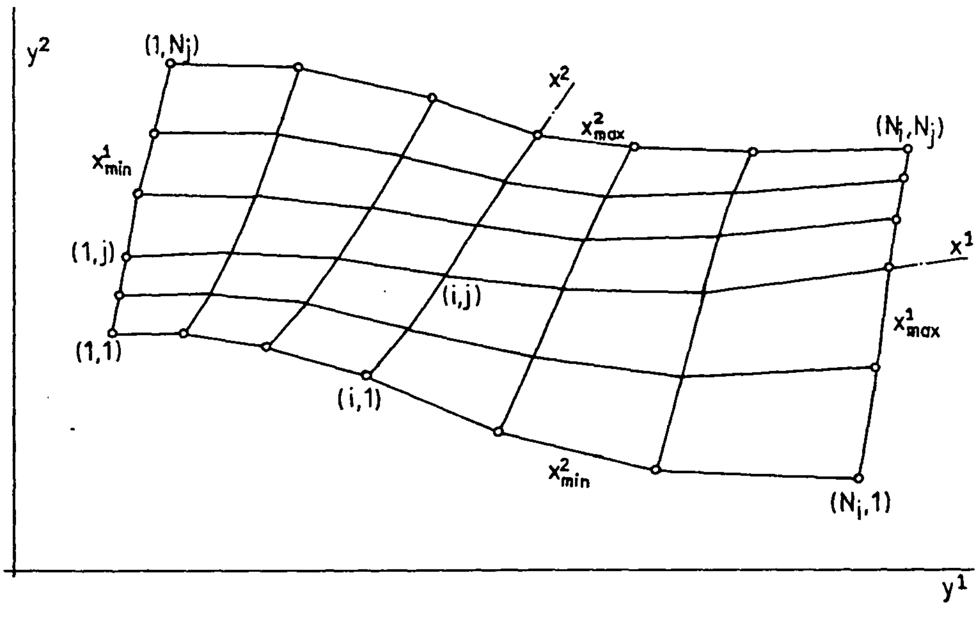


Fig. 7.4: Schematic presentation of Practice 2 for grid generation

This practice offers greater control over the distribution of the grid lines than the previous one. Depending on the boundary distributions, various shapes of interconnecting lines can result, as shown in Fig. 7.5 . In practice, the grid lines are usually concentrated near solid walls, which results in good interior distributions in quite complex cases, as will be seen in the applications presented in Chapter 9 .

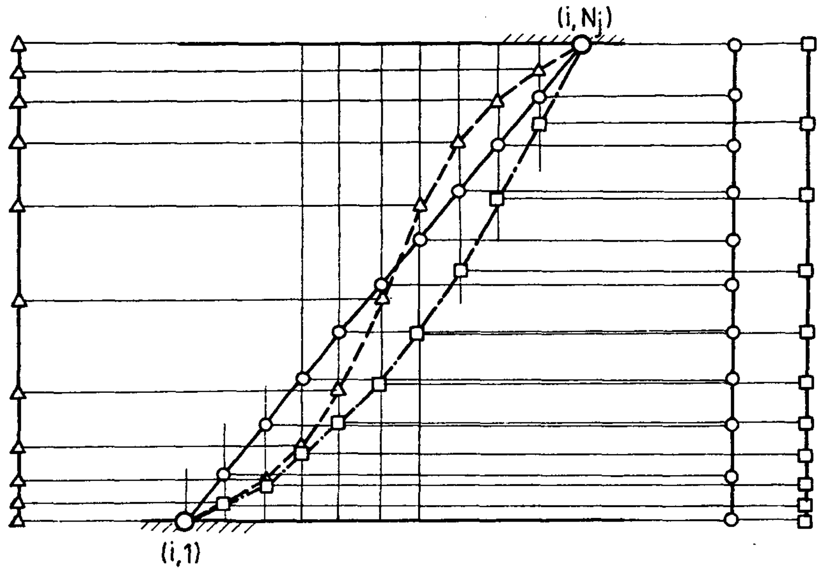


Fig. 7.5: Various possible shapes of grid line connecting two boundary points

When a submerged body is present within the solution domain, the problem of grid generation is more complicated. The grid lines have to be forced to coincide with, and be concentrated near, the boundaries of the body. In many cases this can be achieved by subdividing the solution domain into several regions in which one of the practices described above gives a good grid, and then merge these subgrids to get the grid for the solution domain as a whole. An example is shown in Fig. 7.6, where three subdomain grids, 1 , 2 and 3 were separately generated and then merged.

The same treatment can be used whenever particular grid line(s) are required to pass through particular point(s), or when the above practices produce an unsatisfactory grid if applied to the solution domain as a whole. For example, when the boundaries are highly irregular, the grid lines generated by either of the two practices may stray outside the solution domain, giving what is sometimes termed the "overspill" problem. Haber et al (1981) also use the multiple-region approach to handle this problem, while Gordon and Hall (1973) employ the "constraint curve" technique, in which higher-order mappings are used to force the grid lines to pass through the constraint curves.

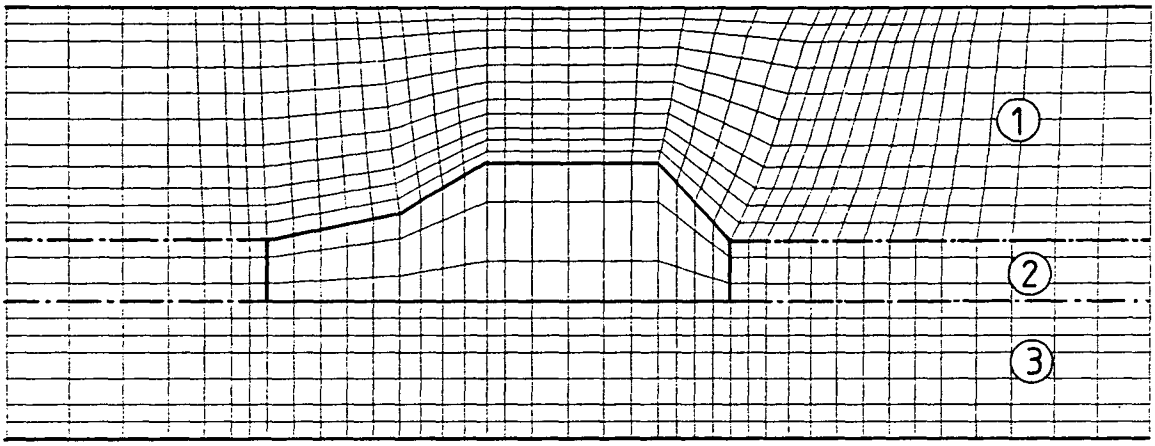


Fig. 7.6: Generation of grid around submerged bodies
region (1) - Practice 2; regions (2) and (3) - Practice 1

The multiple-region approach was also used in the present study in the generation of a grid for the problem of flow around a bluff body confined in a pipe, illustrated in Fig. 7.7 . There the solution domain is divided into two regions, the boundary between which is the outer surface of the bluff body, and - to some extent - the deviding streamline enveloping the recirculation region (in this case known from experimental results). The aim of this manipulation was to achieve good alignment of grid

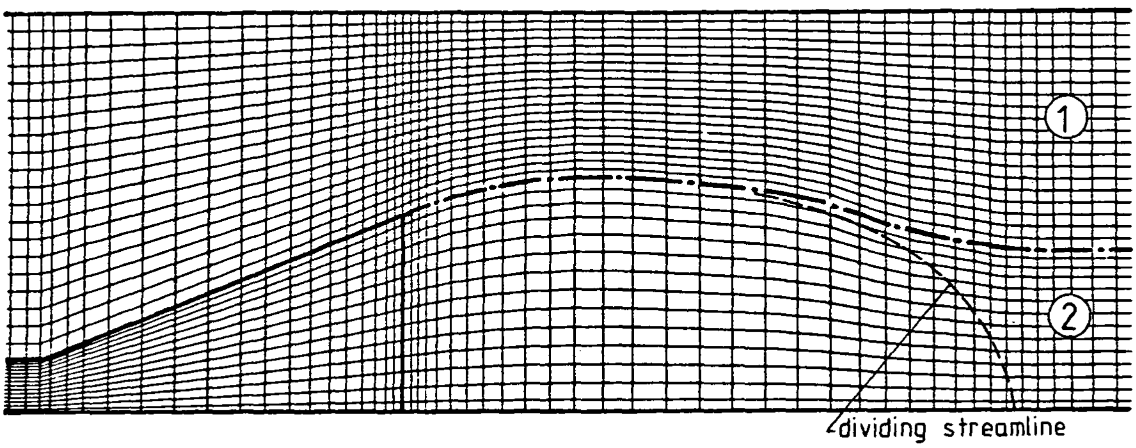
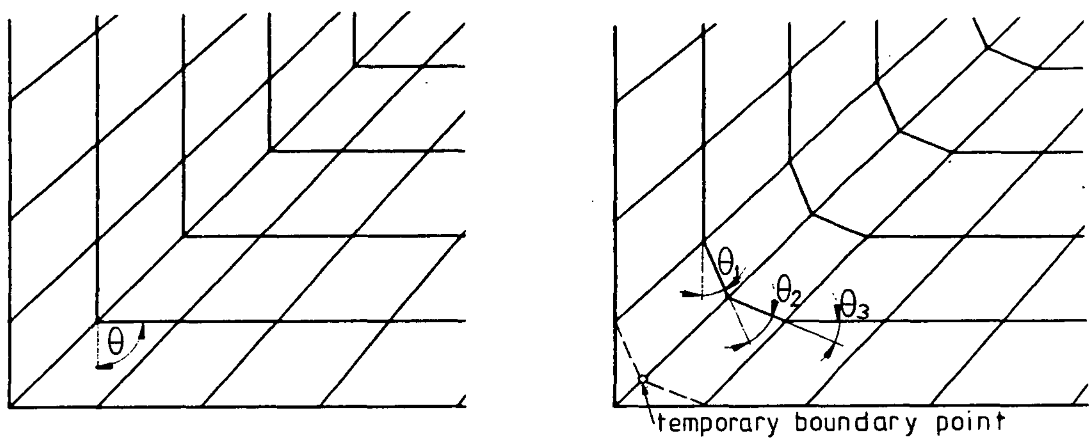


Fig. 7.7: Multiple-region approach in grid generation

lines and streamlines, which was indeed accomplished in the most of computational domain, thus resulting in reduction of numerical diffusion. Further details will be given in section 2 of Chapter 9.

Another area where manual intervention in grid generation is effectively used in this study is in smoothing severe discontinuities. These usually result from sharp discontinuities in the boundaries, which is often the case in practical situations as will be seen in many applications to be shown later. As was discussed in the previous section, sudden changes in the grid direction introduce some inaccuracies and are therefore undesirable. To minimize this effect a provisional "smoothing" is introduced, as illustrated in the right-angle bend example of Fig. 7.8 , in which temporary boundary points are used to round off the discontinuities.



a) non-smooth grid b) smoothed grid

Fig. 7.8: Smoothing of grid discontinuities

The interior grid points are then generated by the usual practices, after which the temporary points are moved back to the real boundaries. This helps create smoother interior grid lines, as can be seen in Fig. 7.8: instead of a sharp change of direction for $\theta=90°$ (grid a), it now happens gradually through three θ's of $\sim 30°$.

In all applications in the present study in which this practice was employed only one boundary point was manipulated in this fashion, as this was found to be sufficient. If the local grid refinement, as suggested in the previous section (see Fig. 7.2) is also used, it may be desirable to introduce more than one temporary point, as this will further reduce the θ's and thus make the grid even smoother.

7.4 Generation of Grid for Three-Dimensional Problems

The task of generating grids for general three-dimensional problems is significantly more complex than in two-dimensional cases. As in two-dimensional applications several methods based on the solution of differential equations of the form:

$$
\begin{aligned}
\frac{\partial^2(x^1)}{\partial(y^1)^2} + \frac{\partial^2(x^1)}{\partial(y^2)^2} + \frac{\partial^2(x^1)}{\partial(y^3)^2} &= P(x^1,x^2,x^3) \\
\frac{\partial^2(x^2)}{\partial(y^1)^2} + \frac{\partial^2(x^2)}{\partial(y^2)^2} + \frac{\partial^2(x^2)}{\partial(y^3)^2} &= Q(x^1,x^2,x^3) \qquad (7.9) \\
\frac{\partial^2(x^3)}{\partial(y^1)^2} + \frac{\partial^2(x^3)}{\partial(y^2)^2} + \frac{\partial^2(x^3)}{\partial(y^3)^2} &= R(x^1,x^2,x^3)
\end{aligned}
$$

have been devised (in Thompson, 1982; Shieh, 1984 and other). They require that the coordinates of boundary grid points be given, usually employ some adjustable parameters (or forcing functions) which can be tuned to optimize the grid distribution and solve Eq. (7.9) iteratively. The method of Shieh (1984) is attractive, since it generates the forcing functions P, Q and R automatically. This is achieved by enforcing orthogonality of grid lines on the boundaries and using exponential interpolation functions to obtain the interior values of P, Q and R. Two adjustable parameters are employed, which influence the distribution of the interior grid points for given coordinates of the boundary ones.

The solution of Eq. (7.9), however, requires considerable effort, especially if an optimized grid is sought. In the present study, in which complex ducts are of primary interest, a logical extension of Practice 2 is developed. This approach requires that in general the coordinates of the cell vertices lying on all boundary surfaces of solution domain be given. However, if the inlet and outlet surfaces are plane, which is often the case, then only the coordinates on the other four boundary surfaces are necessary: those on the inlet and outlet planes can be determined using Practice 2 .

When coordinates of the grid points on all boundary surfaces (labelled E, W, N, S, T and B in the usual notation) are known, then the coordinates of interior points can be obtained from the following equations (see Fig. 7.9):

$$
\begin{aligned}
y^1_{i,j,k} &= y^1_{1,j,k} + f^1_{i,j,k}\,(y^1_{N_i,j,k} - y^1_{1,j,k}) \\
y^2_{i,j,k} &= y^2_{i,1,k} + f^2_{i,j,k}\,(y^2_{i,N_j,k} - y^2_{i,1,k}) \qquad (7.10) \\
y^3_{i,j,k} &= y^3_{i,j,1} + f^3_{i,j,k}\,(y^3_{i,j,N_k} - y^3_{i,j,1})
\end{aligned}
$$

where N_k is the number of grid lines in the x^3 direction, and the functions f^1, f^2 and f^3 for interior grid points (i,j,k) are defined as:

$$
\begin{aligned}
f^1_{i,j,k} &= \frac{(N_j - j)\, f^1_{i,1,k} + (j-1)\, f^1_{i,N_j,k}}{N_j - 1} \\
f^2_{i,j,k} &= \frac{(N_k - k)\, f^2_{i,j,1} + (k-1)\, f^2_{i,j,N_k}}{N_k - 1} \qquad (7.11) \\
f^3_{i,j,k} &= \frac{(N_i - 1)\, f^3_{1,j,k} + (i-1)\, f^3_{N_i,j,k}}{N_i - 1}
\end{aligned}
$$

Values of f^1, f^2 and f^3 on the boundary surfaces are defined according to Eq. (7.6).

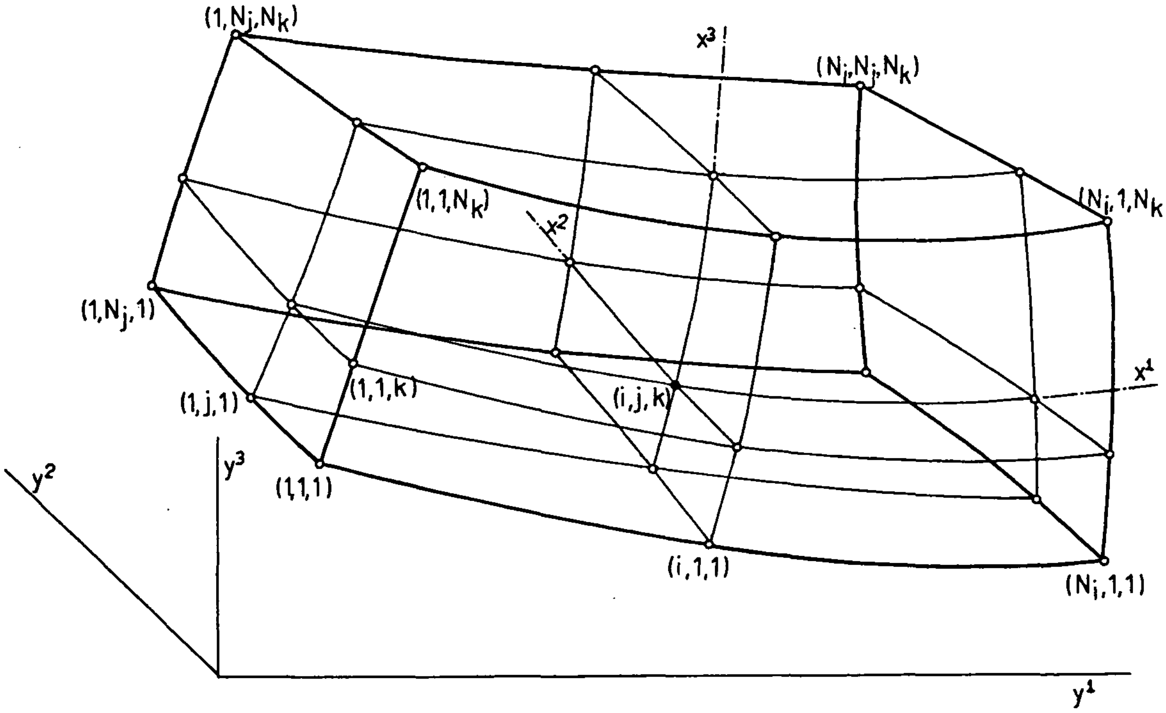


Fig. 7.9: Generation of a three-dimensional grid

In situations where two opposite duct walls are plane, one set of coordinate surfaces (eg. x^3=const.) can be taken to be plane too, and in those planes Practice 1 or 2 can be used without modification. This was the case in all three-dimensional applications to be presented later in Chapter 9. In more complex cases neither of practices described here may be successful; in those circumstances either more general transfinite mapping methods (like that of Gordon and Thiel, 1982) or methods which solve differential equations of the type (7.9) (like that of Shieh, 1984) would have to be used.

The coordinates of the cell vertices are the only basic information about the grid, needed by the present solution method. All other quantities related to the grid geometry can subsequently be calculated from this information, irrespective of how it was generated. These quantities can either be recalculated within the iteration process each time they are needed, or calculated once for all and stored, to save on computing time. The nature of the calculations required is described in the next section.

7.5 Calculation of Geometrical Quantities Featuring in Discretised Equations

7.5.1 Two-dimensional planar geometries

In the case of two-dimensional planar geometries, the third dimension is taken to be unity, and thus the cell volume is equal to its area in the plane considered. Since the cell faces normal to this plane are projected as straight lines joining two vertices, the cell volumes can be calculated as:

$$\delta V = \frac{1}{2} | \vec{a} \times \vec{b} | \tag{7.12}$$

where $\vec{a}$ and $\vec{b}$ are the two vector diagonals, shown in Fig. 7.10 . When expressed through the coordinates of the cell vertices, expression (7.12) becomes:

$$\delta V = \frac{1}{2} | (y^1_{ne} - y^1_{sw})(y^2_{nw} - y^2_{se}) - (y^1_{nw} - y^1_{se})(y^2_{ne} - y^2_{sw}) | \tag{7.13}$$

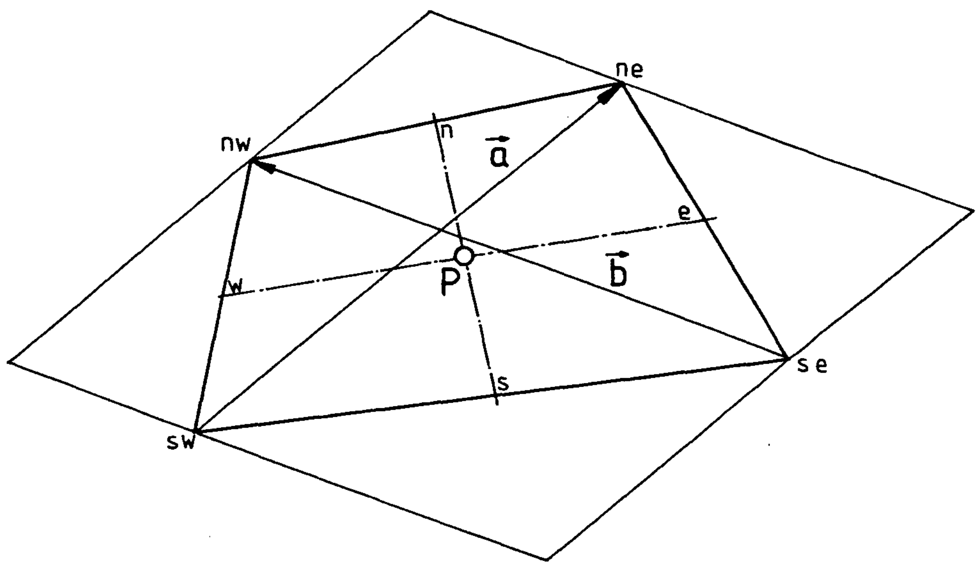


Fig.7.10: Calculation of cell volume for two-dimensional planar grid

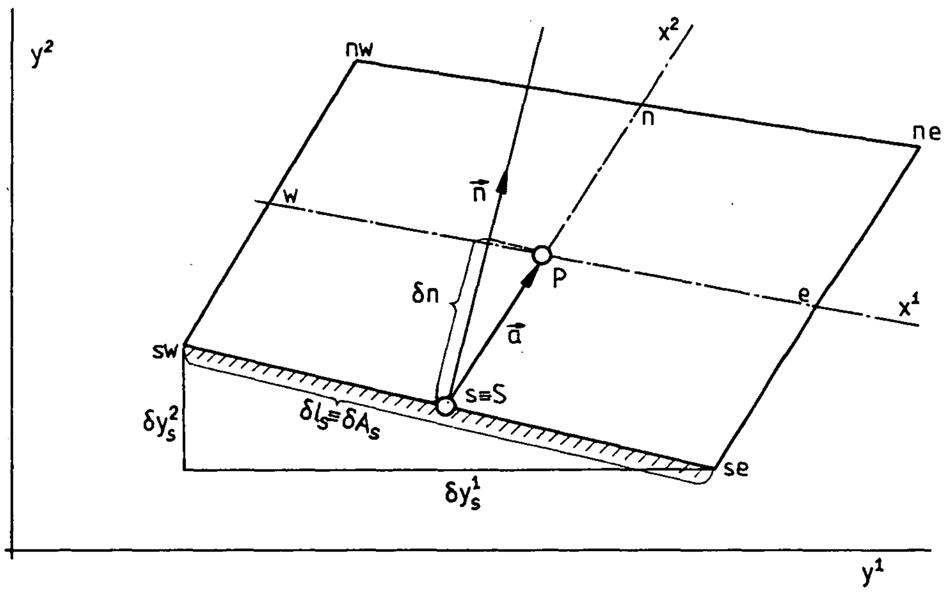


Fig.7.11: Calculation of cell face area and distance between near wall node and the wall

Another two quantities of interest are the perpendicular distance from a near-boundary node to the boundary surface and the area of the boundary cell face, which feature in the boundary condition treatment, as described in the previous chapter. The approach used is illustrated by reference to the general near-boundary cell shown in Fig. 7.11 . The coordinates of points ne, nw, se and sw are known, and those of nodes S and P are easily obtained by linear interpolation. Now δn , the normal distance between node P and the boundary can be found as the scalar product of a vector connecting the boundary node S with its near-boundary neighbour (eg. $\vec{a}=\overrightarrow{SP}$ in Fig. 7.11) and the unit vector $\vec{n}$, normal to the boundary surface:

$$\delta n = \vec{a}\cdot\vec{n} \tag{7.14}$$

The unit vector is defined as:

$$\vec{n} = -\frac{\delta y_s^2}{\delta l_s}\vec{i}_1 + \frac{\delta y_s^1}{\delta l_s}\vec{i}_2 \tag{7.15}$$

Here δl_s is the lenght of boundary cell face projection, which is also the area δA_s of that cell face, ie:

$$\delta l_s = \delta A_s = \sqrt{(\delta y_s^1)^2+(\delta y_s^2)^2} \tag{7.16}$$

and δy_s^1 and δy_s^2 are the differences in coordinates of two boundary cell vertices, eg. for the situation shown in Fig. 7.11 , $\delta y_s^1 = y_{se}^1 - y_{sw}^1$, and $\delta y_s^2 = y_{se}^2 - y_{sw}^2$. The same expressions are valid for any other external boundary, as well as for the surfaces of a submerged body present within the solution domain.

The linear interpolation factors f_1 and f_2 used to calculate values of the dependent variables at locations other than cell centres, were defined earlier in Chapter 4 . The lenghts of segments of coordinate lines x^i between two opposite cell face centres, needed for the calculation of f_i's , are calculated as (see Eq. (4.3)):

$$\begin{aligned} \delta x_P^{(1)} &= \sqrt{(y_e^1-y_w^1)^2+(y_e^2-y_w^2)^2} \\ \delta x_P^{(2)} &= \sqrt{(y_n^1-y_s^1)^2+(y_n^2-y_s^2)^2} \end{aligned} \tag{7.17}$$

The coordinates of cell centres are, for two-dimensional case, calculated by linear interpolation from the coordinates of vertices, eg.:

$$y_e^i = 0.5(y_{ne}^i + y_{se}^i)$$
$$y_n^i = 0.5(y_{ne}^i + y_{nw}^i) \qquad (7.18)$$

7.5.2 Two-dimensional axisymmetric geometries

In the case of an axisymmetric geometry , the third dimension is unit angle, 1 rad. The coordinate transformation is performed in one plane only, ie. between r, z and x^1, x^2 , while the third coordinate, the circumferential angle θ , remains the same. Therefore allowance is made in the calculation of the areas and volumes which appear in the discretised equations for the axisymmetric geometry.

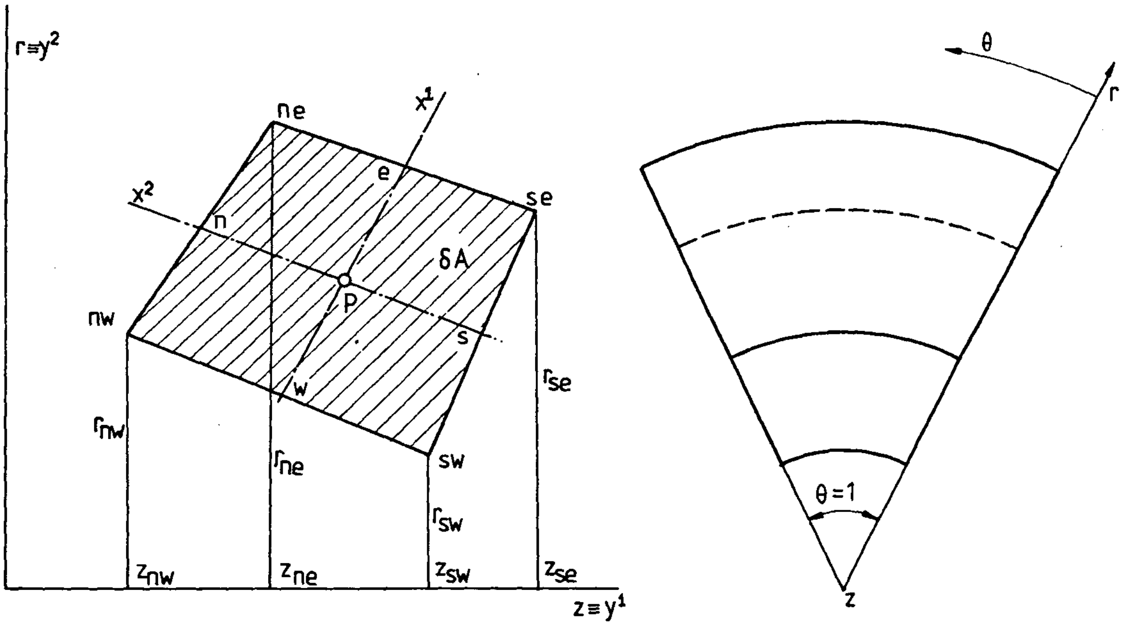


Fig. 7.12: Calculation of volume of axisymmetric cells

The volume δV of a cell surrounding a typical node P (Fig. 7.12) is defined by rotating area δA for unit angle around the symmetry axis z (where area δA is bounded by straight lines), and is given by:

$$\delta V = \frac{1}{6}[(z_{se} - z_{ne})(r_{se}^2 + r_{ne}^2 + r_{se} r_{ne}) + (z_{ne} - z_{nw})(r_{ne}^2 + r_{nw}^2 + r_{ne} r_{nw}) +$$
$$(z_{nw} - z_{sw})(r_{nw}^2 + r_{sw}^2 + r_{nw} r_{sw}) + (z_{sw} - z_{se})(r_{sw}^2 + r_{se}^2 + r_{sw} r_{se})] \qquad (7.19)$$

The area of a particular cell face is equal to its projection in the cross-sectional plane (which is a straight line, eg. $\overline{ne,se}$ for e-cell face) times the mean radius in that cell face. The distances of near-wall nodes from the wall are calculated in the same way as for planar geometry, and so are the interpolation factors. The coordinate frames can be arranged so that z coincides with the Cartesian coordinate y^1 ; in this case r is the same as y^2 in the planar case (see Fig. 7.12). This makes computer programming easier, and the same code can be used for both planar and axisymmetric two-dimensional problems.

7.5.3 Three-dimensional problems

To calculate the volume of an arbitrary three-dimensional cell, bounded by six surfaces which are in general non-planar, it is necessary to make certain approximations. Here the method suggested by Kordulla and Vinokur (1983) will be used.

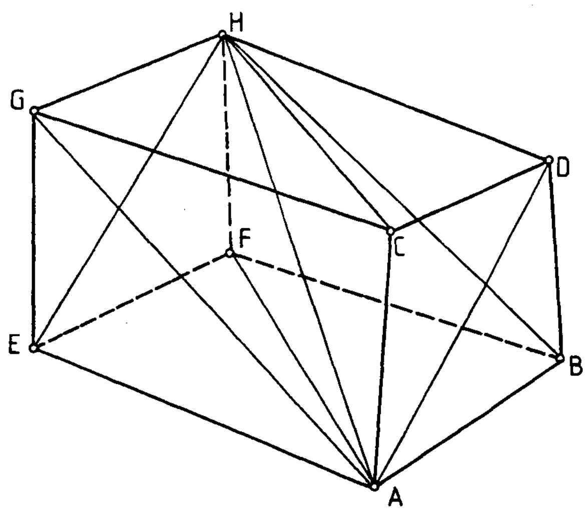


Fig. 7.13: A typical three-dimensional computational cell

A typical three-dimensional computational cell, as shown in figure 7.13, is enclosed by six surfaces, each being defined by four cell vertices, which in general do not lie in the same plane, but are connected by straight lines. For computational purposes, each of these surfaces can be approximated by two planar triangles, which are arranged such that the surface diagonals on opposite sides of the cell connect corresponding vertices (and are therefore parallel in transformed space). This ensures that each surface is uniquely defined for the neighbouring cells it divides, and in the end that the volumes of all cells, when summed, give exactly the total volume of solution domain (ie., there is no overlapping of neighbouring cells). One more diagonal joining two opposite corners of computational cell (eg. A-H in figure 7.13) enables division of the cell into six tetrahedra, bounded by planar surfaces. Their volumes can easily be calculated when the coordinates of the vertices are known. These six tetrahedra, obtained by decomposing the cell of Fig.7.13, are shown in figure 7.14:

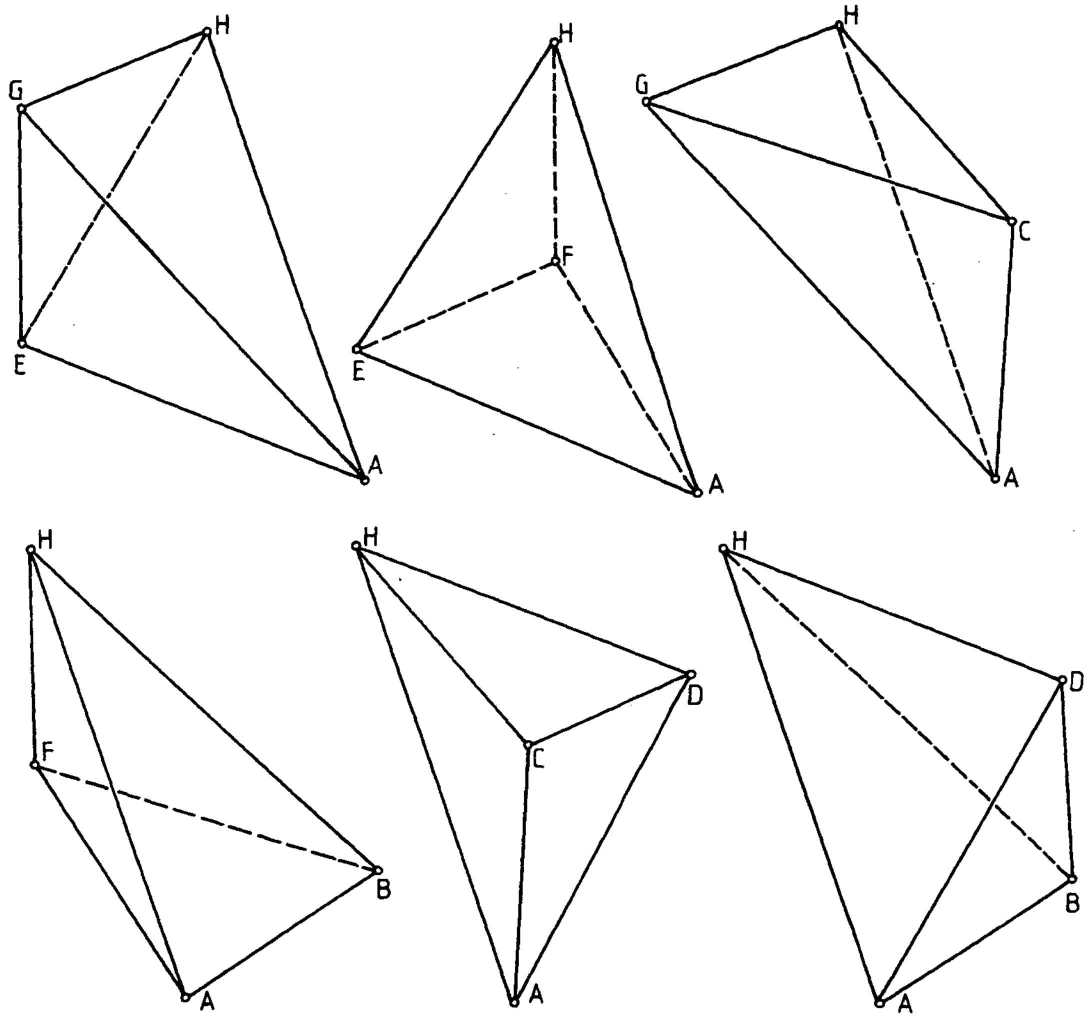


Fig. 7.14: Control volume from Fig. 7.13 decomposed into six tetrahedra

Now, the total volume of the cell can be expressed, in vector form, as:

$$\delta V = \frac{1}{6}\left[(\vec{AD}\times\vec{AB}) + (\vec{AC}\times\vec{AD}) + (\vec{AG}\times\vec{AC}) + (\vec{AB}\times\vec{AF}) + (\vec{AF}\times\vec{AE}) + (\vec{AE}\times\vec{AG})\right]\cdot\vec{AH} \tag{7.20}$$

In terms of the coordinates of cell vertices, the components of the above are given by:

$$
\begin{aligned}
(\vec{AD}\times\vec{AB})\vec{AH} = {} & [(y_D^2-y_A^2)(y_B^3-y_A^3) - (y_B^2-y_A^2)(y_D^3-y_A^3)](y_H^1-y_A^1) + \\
& [(y_B^1-y_A^1)(y_D^3-y_A^3) - (y_D^1-y_A^1)(y_B^3-y_A^3)](y_H^2-y_A^2) + \\
& [(y_D^1-y_A^1)(y_B^2-y_A^2) - (y_B^1-y_A^1)(y_D^2-y_A^2)](y_H^3-y_A^3)
\end{aligned}
\qquad (7.21)
$$

where, eg. vector $\vec{AH}$ is defined as $(y_H^1-y_A^1)\vec{i}_1 + (y_H^2-y_A^2)\vec{i}_2 + (y_H^3-y_A^3)\vec{i}_3$.

To calculate the distance of a near-wall node from a boundary, the same approach as employed in two-dimensional case can be used. However, since the boundary surface is in general non-planar, the direction of the normal vector varies from point to point. Since, however, for any grid of reasonable finess, the variation in direction over a cell face will not be large, a mean normal direction can be taken as that of the vector product of two cell-face diagonal vectors $\vec{a}$ and $\vec{b}$ which connect opposite vertices on boundary cell face, as shown in Fig. 7.15 , thus

$$\vec{N} = \vec{a}\times\vec{b} \qquad (7.22)$$

The unit normal vector is then $\vec{n} = \frac{\vec{N}}{|\vec{N}|}$, and the distance δn can be found as the scalar product of a vector connecting the boundary node B and its near-boundary neighbour P , $\vec{BP}$, and the unit normal vector $\vec{n}$, thus:

$$\delta n = \vec{BP}\cdot\vec{n} \qquad (7.23)$$

In terms of the nodal coordinates the above becomes:

$$\delta n = (x_P-x_B)n_1 + (y_P-y_B)n_2 + (z_P-z_B)n_3 \qquad (7.24)$$

where the coordinates of nodes B and P are obtained from the vertex coordinates by linear interpolation, eg.:

$$
\begin{aligned}
y_b^i &= y_B^i = 0.25(y_{bse}^i + y_{bne}^i + y_{bsw}^i + y_{bnw}^i) \\
y_t^i &= 0.25(y_{tse}^i + y_{tne}^i + y_{tsw}^i + y_{tnw}^i) \qquad (7.25) \\
y_P^i &= 0.5(y_t^i + y_b^i)
\end{aligned}
$$

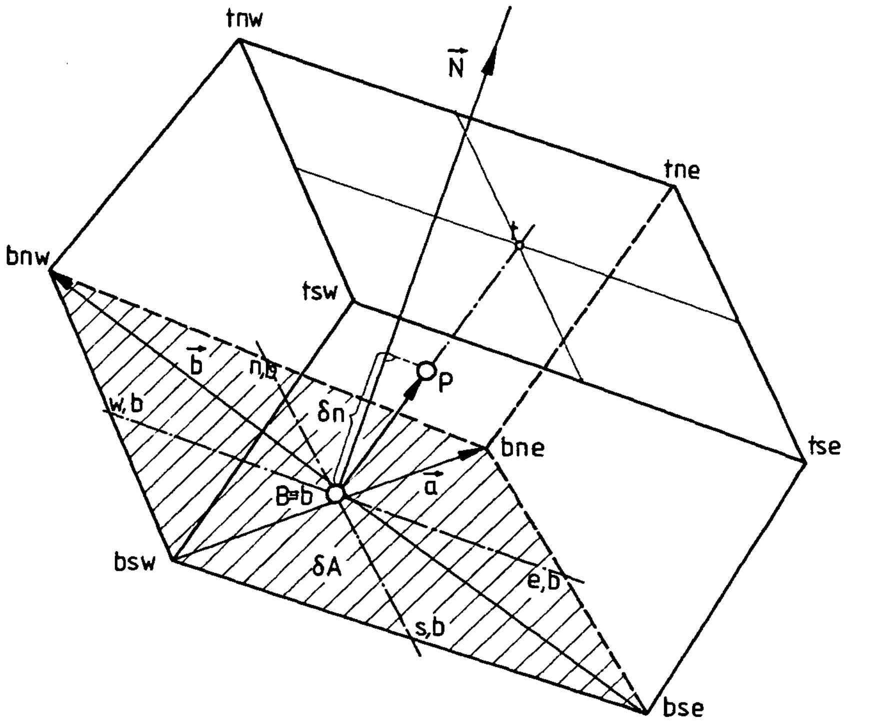


Fig. 7.15: Distance between node P and the boundary

and the n_1 , n_2 and n_3 are Cartesian components of the unit vectors $\vec{n}$. The area of a boundary cell face δA_B is equal to the half of $|\vec{N}|$, and in terms of vertex coordinates can be calculated as (see Fig. 7.15):

$$\delta A_B = \frac{1}{2}|\vec{N}| = \frac{1}{2}\sqrt{(\delta y_N^1)^2+(\delta y_N^2)^2+(\delta y_N^3)^2} =$$

$$\frac{1}{2}\{[(y^2_{bne}-y^2_{bsw})(y^3_{bnw}-y^3_{bse}) - (y^2_{bnw}-y^2_{bse})(y^3_{bne}-y^3_{bsw})]^2 +$$

$$[(y^1_{bnw}-y^1_{bse})(y^3_{bne}-y^3_{bsw}) - (y^1_{bne}-y^1_{bsw})(y^3_{bnw}-y^3_{bse})]^2 +$$

$$[(y^1_{bne}-y^1_{bsw})(y^2_{bnw}-y^2_{bse}) - (y^2_{bne}-y^2_{bsw})(y^1_{bnw}-y^1_{bse})]^2\}^{1/2} \qquad (7.26)$$

The Cartesian components of $\vec{n}$ can now be conveniently expressed as:

$$n_i = \frac{\delta y_N^i}{2\delta A_B} \qquad (7.27)$$

where δy_N^i are calculated from vertex coordinates as shown in Eq. (7.26).

The distances between the two opposite cell face centres, needed for the calculation of interpolation factors f_i (see Eq. (4.3)), are now calculated as, eg.:

$$\delta x_P^{(3)} = \sqrt{(y_t^1 - y_b^1)^2 + (y_t^2 - y_b^2)^2 + (y_t^3 - y_b^3)^2} \tag{7.28}$$

The coordinates of cell face centres are calculated as in Eq. (7.25). Similar expressions follow for $\delta x_P^{(1)}$ and $\delta x_P^{(2)}$.

Thus, all geometrical quantities which feature in the discretised transport equations have been defined.

7.6 Closure

In this section four desirable grid properties were identified as: orthogonality, uniformity of spacing, smoothness and alignment with streamlines. The three last named properties most affect the accuracy of numerical solutions; fortunately, they are not conflicting and usually can be satisfied to much higher degree by sacrificing the first one, orthogonality. However, as will be shown in the next chapter, non-orthogonality of grid lines mainly slows down convergence, and does not introduce significant inaccuracies, which justifies giving higher priority to the other three properties.

The numerical grid generation method employed in solving fluid flow problems should allow control over the above-named grid properties. However, general methods capable of creating grids for any situation would have to be very complex. In many applications, like those considered later in this study, the two simple grid generation practices described here, which belong to a family of transfinite mapping methods, give satisfactory results and are very simple to use and cost effective, and yet allow good control over grid properties.

The calculation of quantities related to grid geometry in the present solution method is independent of the way in which coordinates of the cell vertices are generated: indeed they may even be generated by hand. The volumes and areas of the cells are calculated in the way which ensures conservation of space, ie. the sum of all cell volumes gives exactly the total volume of computational domain, which is important for assuring true conservation of transported quantities. All these calculations are described in a general, vector approach, thus making programming simple and easy to follow.

In the next chapter the present method will first be assessed on some test problems for which solutions are known, before applying it to complex flows, which will be reported in Chapter 9.

CHAPTER 8

ASSESSMENT OF THE SOLUTION METHOD

8.1 Introduction

The numerical solution method described in previous chapters has been embodied in two computer programs: for two-dimensional (plane or axisymmetric) and general three-dimensional problems respectively. Before applying them to complex flows, a series of test calculations on model problems has been performed in order to assess the performance of the method and influence of various factors. The problems selected are ones for which either analytic solutions, or well established experimental or numerical results exist.

First, in section 2, the performance of the selected differencing schemes and the newly-developed flux blending method are examined.

In section 3 the effects of certain important grid properties, namely the non-orthogonality and non-smoothness of grid lines, on the accuracy and stability of the present method are investigated.

In section 4 effects of under-relaxation of velocity components and pressure on the stability and rate of convergence of the solution method are analysed. Attention is also paid to combined effects of grid non-orthogonality and under-relaxation factors on convergence.

Finally, in section 5 closing remarks are given.

8.2 Differencing Schemes and Flux Blending Method

The assessment of the three differencing schemes for convection terms selected earlier is here performed by considering several one- and two-dimensional model problems for which exact solutions are known. In all cases the transport of a scalar quantity ϕ in a given flow field is considered. The flux blending method presented earlier in Chapter 4 is applied whenever boundedness problems occur. These tests, and the conclusions derived from them, are described below.

One-dimensional test problem

This problem involves the one-dimensional transport of a scalar quantity ϕ (eg. temperature), with the inlet boundary value prescribed and a constant source one cell wide, located seven cells downstream from inlet. Figure 8.1 presents the profile of ϕ predicted using the UDS, LUDS and QUDS, for three grid Peclet numbers, namely 2, 5 and ∞. Because of the sudden rise in ϕ due to source, the QUDS produces substantial undershoot and oscillations immediatly upstream of the source location for all Peclet numbers greater than $\frac{8}{3}$ (see Eq. (4.16) in section 4.3.4).

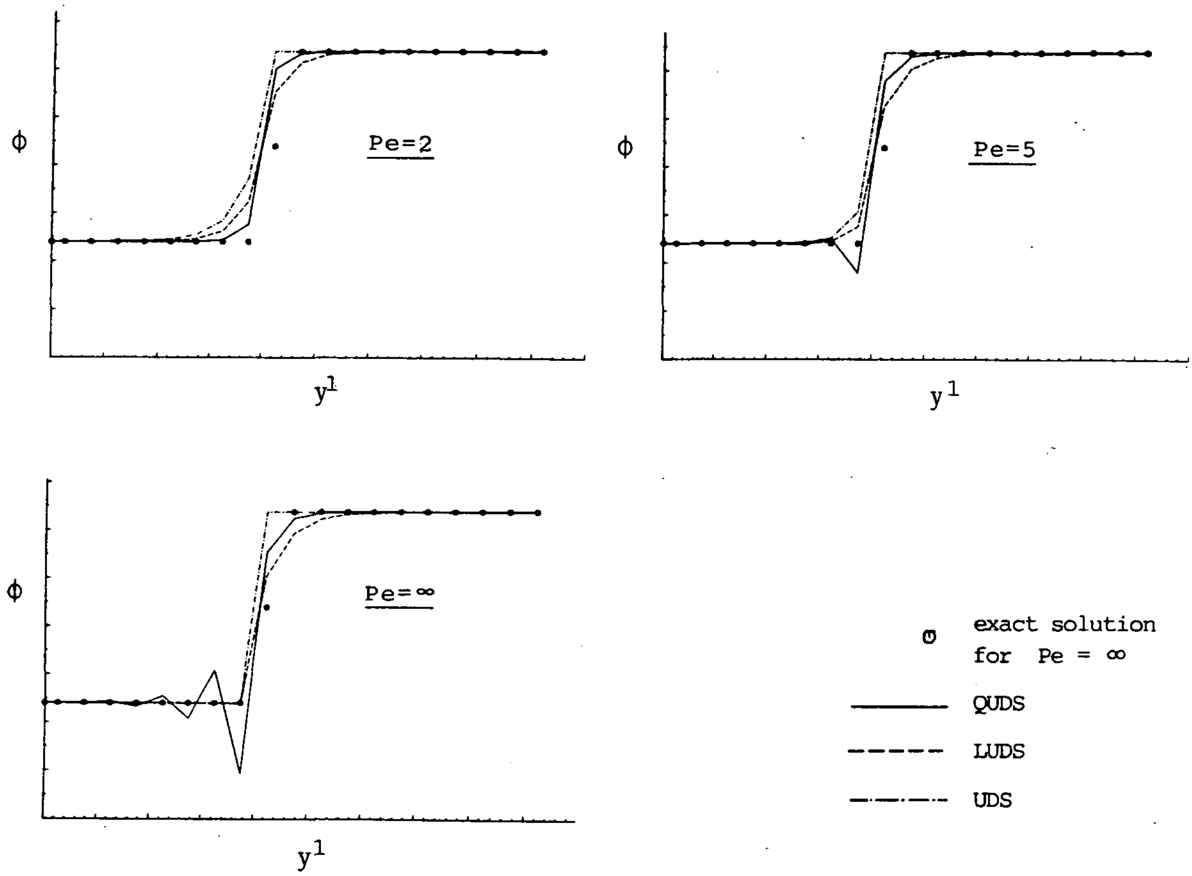


Fig. 8.1: One-dimensional transport of ϕ with constant source

This is due to the fact that the QUDS employs downstream nodes in evaluating the convected ϕ value, and thus violates the transportiveness requirement. On the other hand, the LUDS produces always a monotonic solution, although it too has negative coefficients: however, these occur only for upstream nodes, because the scheme is fully transportive. The UDS produces, for Pe=∞ , the exact solution shifted upstream half a cell.

When the flux blending method is applied to the QUDS, it results in the UDS being used in the affected region, as discussed earlier in Chapter 4.

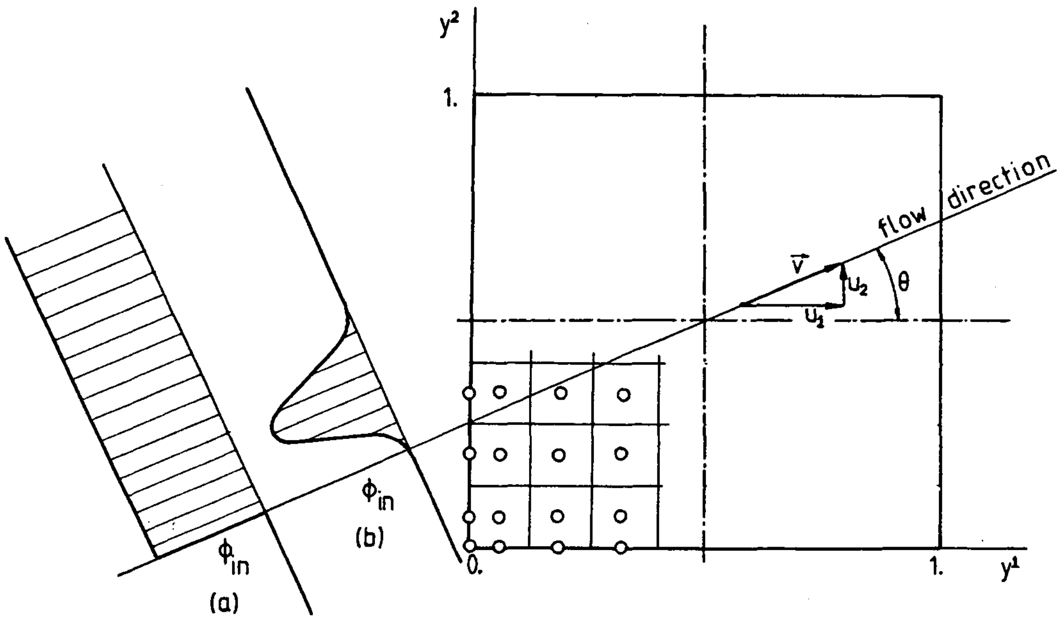


Fig. 8.2: Test problems with uniform velocity field

Two-dimensional test problems

One frequently used (eg. by Raithby, 1976; Lillington, 1981; Leschziner, 1980; Lay, 1982) test case involves the convective transport of a scalar profile in a uniform velocity field, with the flow direction inclined at an angle θ to the Cartesian grid lines, as shown in Fig. 8.2. Of particular interest are the effects it reveals about the "numerical diffusion" introduced by the differencing schemes (ie. smearing of steep gradients normal to the flow direction), as well as any unboundedness problem that may arise. A Cartesian grid consisting of 20×20 control volumes (CV) was used here, with uniform spacing in both directions. Calculations were performed first for the step profile shown in Fig. 8.2 (a), at $Pe=\infty$ and five inclination angles defined by the ratio* u_2/u_1 : 0.2, 0.4, 0.6, 0.8 and 1.0 (the corresponding angles θ are 11.3°, 21.8°, 31°, 38.6° and 45°). In Fig. 8.3 (a)-(e) the ϕ profiles produced by the LUDS and BLUDS (Bounded LUDS) schemes along the vertical centerline of the solution domain are shown and compared with the exact result (which is in all cases the initial step profile). As can be seen from these figures, the LUDS produces overshoots at all flow angles, and as θ increases, undershoots also appear. These extrema are about 8% of the step hight, several cells wide and spread, on both sides, all

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* For $u_2/u_1=0$, ie. when streamlines and grid lines are aligned, all three schemes yield the exact solution.

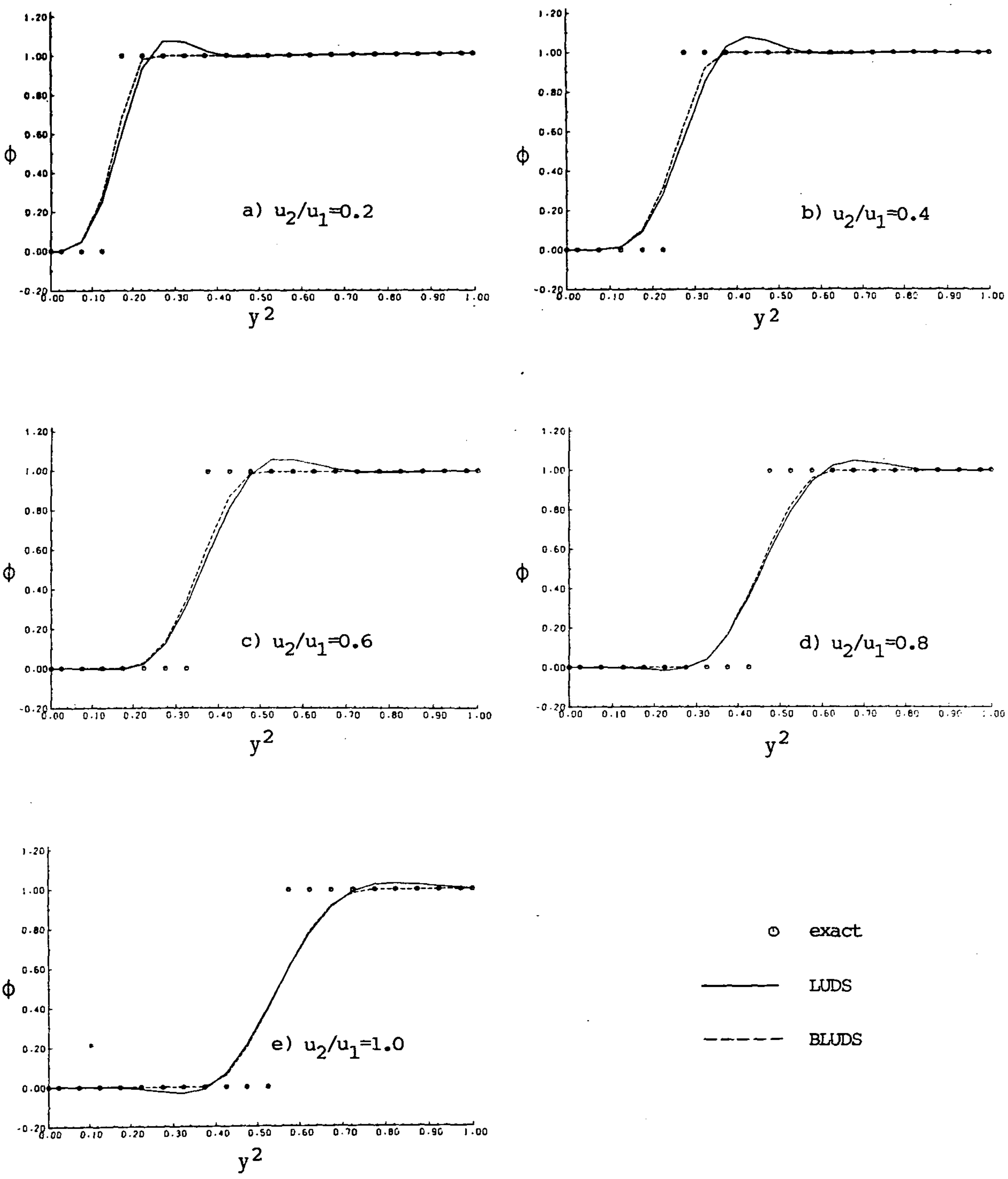

Fig. 8.3: Transport of a step profile in ϕ, in uniform two-dimensional flow field: vertical centerline profiles, LUDS and BLUDS predictions, Pe= ∞

along the step.

The BLUDS results obtained using the new blending technique are free from both overshoots and undershoots, while the step resolution of the LUDS is retained (the change in profile steepness, apparent at lower angles, is due to the conservativeness of the blended scheme, as the inte-

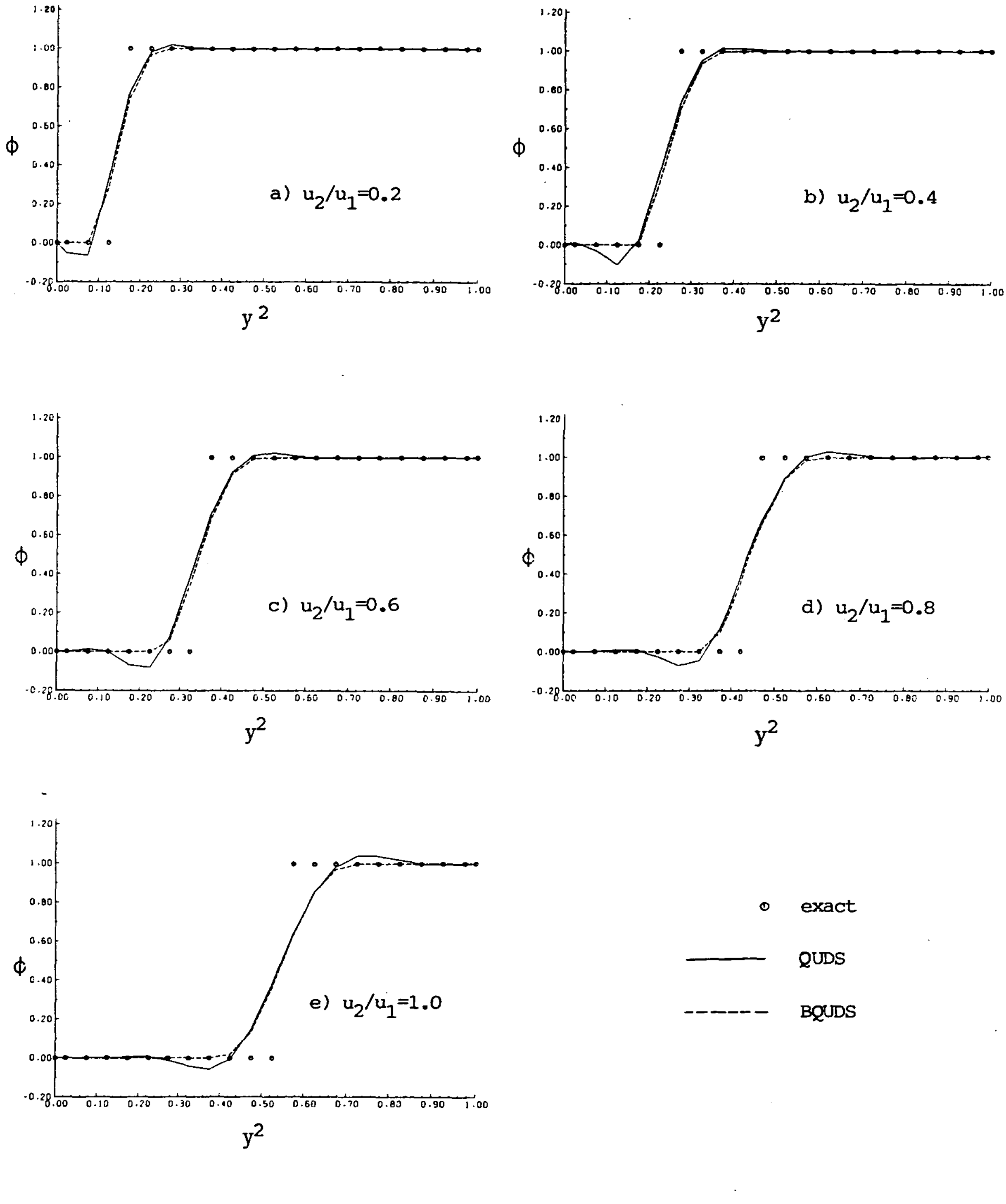

Fig. 8.4: Transport of a step profile of ϕ in uniform two-dimensional flow field: vertical centerline profiles, QUDS and BQUDS predictions, $Pe=\infty$

grals under both curves - LUDS and BLUDS - should be the same). This is a very important feature of the blending technique, as it shows that the amount of UDS blended in is just sufficient to remove non-physical oscillations, while as much of the LUDS (the "U-scheme" here) as possible is retained.

Solutions of the same problem obtained using QUDS and BQUDS (Bounded QUDS) are shown in figure 8.4. Contrary to LUDS, the QUDS always produces undershoots, while overshoots appear at larger angles of flow inclination. The undershoots are of a larger magnitude at lower angles, and amount up to 12 % of the step hight. The efficiency of blending technique in again removing them without destroying the profile resolution is obvious.

It should be noted here that the use of the boundedness criteria given by Eq. (4.37) would not remove undershoots spread along the step, for the solution is still monotonic in one coordinate direction at each node.

Fig. 8.5 shows results for the transport of a Gausian-like inlet profile on the same grid, with Pe= ∞ and $u_2/u_1=0.5$ ($\theta=26.5°$). Both the QUDS and LUDS produce undershoots along either side of the profile, but neither overshoots the exact peak value. The blending technique, in both cases, removes these undershoots, but it does not "clip" the peak of profile because it is able to recognize that it does not represent a numerical overshoot.

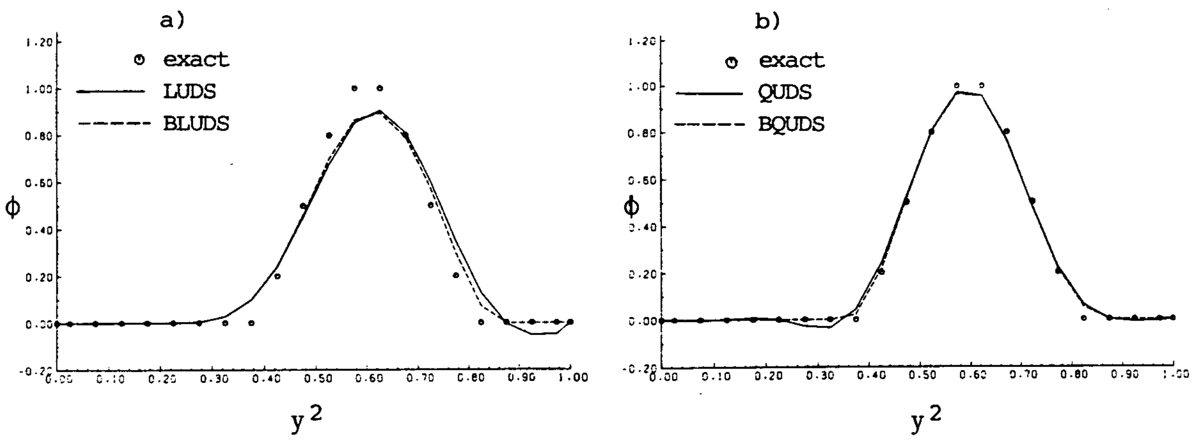

Fig. 8.5: Transport of a Gaussian-like profile of ϕ in uniform two-dimensional flow field: vertical centerline profiles, $u_2/u_1=0.5$

The third test case, which involves both a variable velocity field and inclination of streamlines to grid lines is the convective transport of a scalar profile in an inviscid stagnation flow field, given by $u_1=y^1$ and $u_2=-y^2$. Three different scalar profiles were prescribed on the inlet boundary, as shown in insets (a) - (c) in Fig. 8.6 , and the resulting profiles at the exit plane are compared with the exact values.

Figure 8.7 shows results obtained with the LUDS and BLUDS with in-

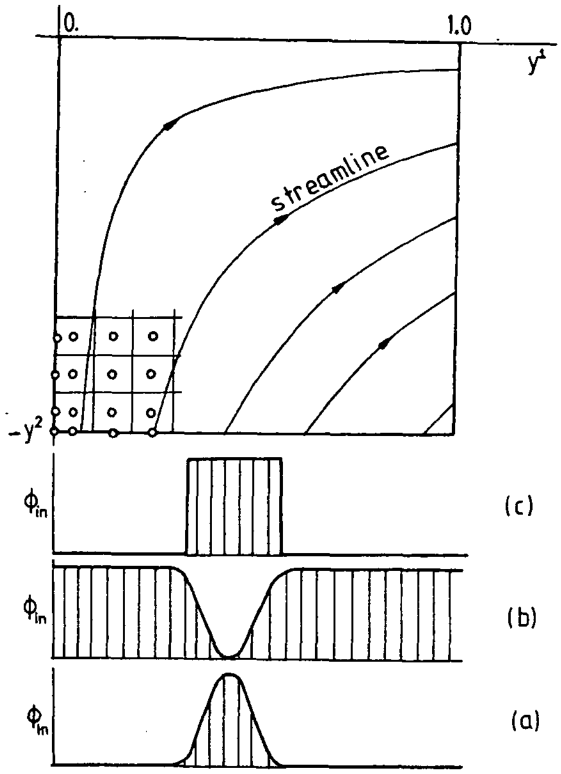

Fig. 8.6: Test problems with variable velocity field

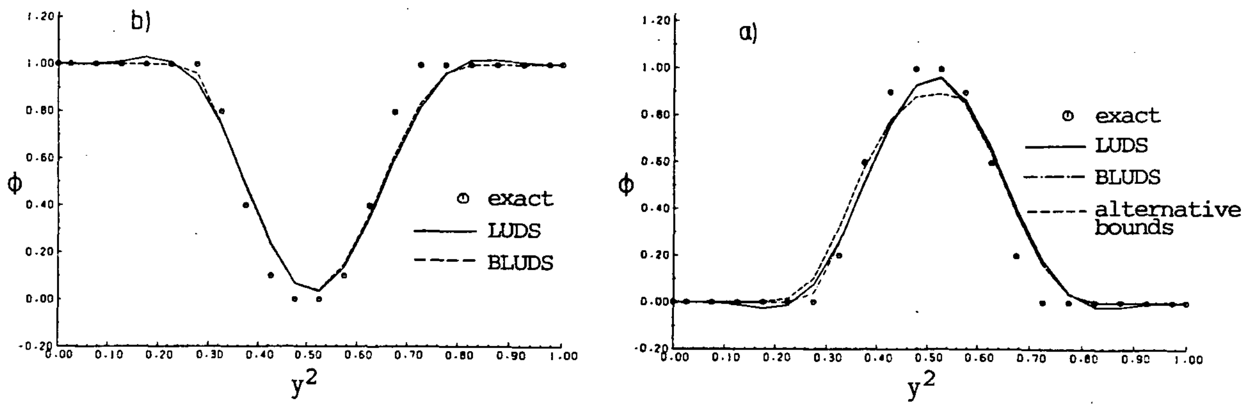

Fig. 8.7: Transport of a scalar profile in stagnation flow field: exit profiles, LUDS and BLUDS solutions

let profiles (a) and (b). Again, only the numericaly generated peaks (overshoots and undershoots) are removed by the BLUDS, whereas the "real" peaks are left as with the LUDS, since they were not out of bounds.

Also shown in Fig. 8.7(b) is the result obtained by requiring a monotonic solution in both coordinate directions, which is equivalent to Zalesak's (1979) proposal for a variant of FCT. The undershoots are evidently efficiently removed, but the peak (which does not with the present "U-scheme", LUDS, overshoot the exact value) is also clipped-off and rounded, giving about a 6.5% lower peak value than LUDS alone. While these results are still much better than those of the UDS (not shown), they are inferior to the results obtained using the new bounding method.

Figure 8.8 shows results for the same problems obtained with the QUDS and BQUDS. The profiles of QUDS are closer to the exact ones than those of LUDS, but they also exhibit overshoots or undershoots at the peaks. Here, they are recognized as such by the bounding method, and removed in the BQUDS solution, together with the oscillations on both sides of the peaks. Otherwise, the profile shape is the same as that of QUDS solution, ie. the accuracy of profile resolution has not deteriorated when blending is applied.

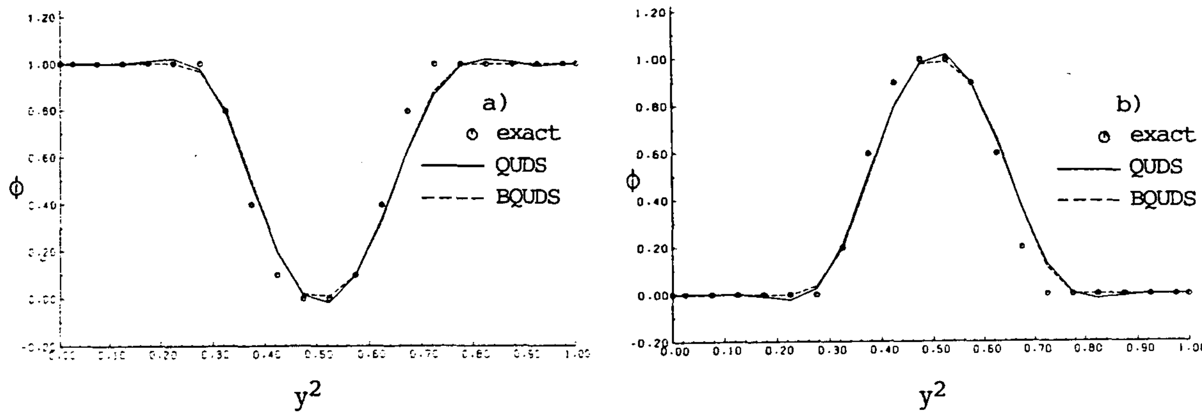

Fig. 8.8: Transport of scalar profile in stagnation flow field: exit profiles, QUDS and BQUDS solutions

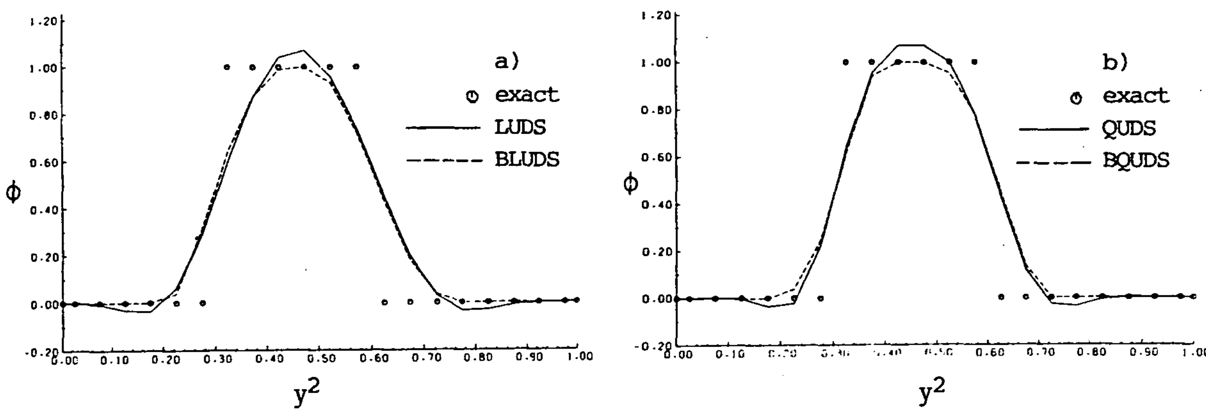

Fig. 8.9: Transport of a square wave, stagnation flow field: exit profiles, Pe= ∞

Fig. 8.9 shows results for the convection of a square wave (profile (c) of Fig. 8.6) in the inviscid stagnation flow field, on the same 20 x 20 CV grid, at Pe= ∞. Both the LUDS and QUDS overshoot the peak and produce undershoots on both sides of wave at the exit of the calculation domain. These are again efficiently removed by the bounding method; the peaks are reduced just to the correct value, and not clipped further, as would happen with the alternative way of determining bounds mentioned earlier.

In Fig. 8.10 comparisons are made between the UDS, BLUDS and BQUDS solutions for two cases: one is a uniform flow field and a step inlet profile, with $u_2/u_1=0.4$, and the other is square wave transport in an inviscid stagnation flow field. It is obvious from these diagrams that both bounded higher-order schemes give much better results than the UDS. The difference between the BLUDS and BQUDS results is much smaller than the difference between either of them and UDS. The BQUDS is marginally the most accurate, which is expected, since it employes interpolation of one order higher than the LUDS.

Some surface plots of the solution fields and their projection onto a surface normal to the computational plane are shown in Fig. 8.11 . They offer better insight into the problem of unboundedness than do profiles at one cross-section only. The unbounded solution by QUDS (plot (b)) looks rather unrealistic; the smooth, properly bounded solution of BQUDS (c) represents a much better approximation to the exact solution. These plots

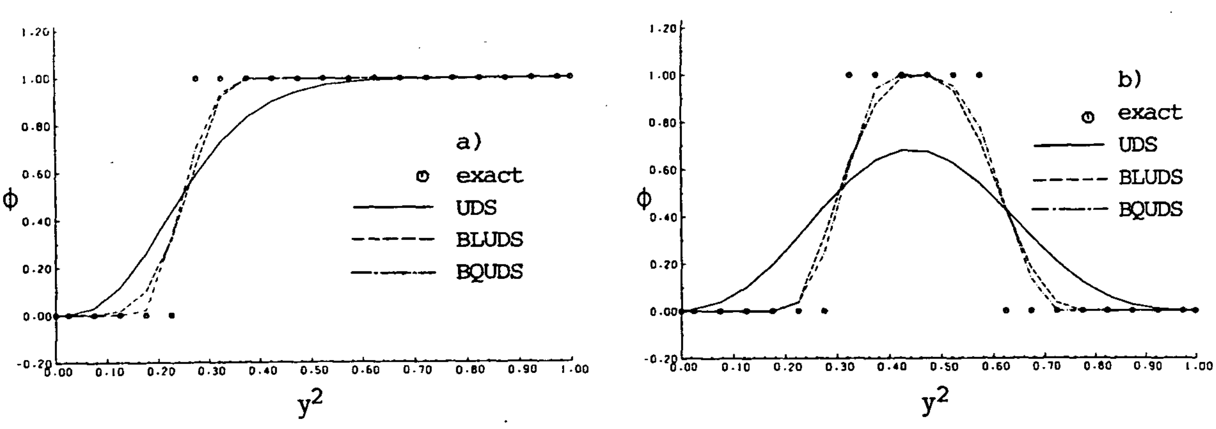

Fig. 8.10: Comparison of UDS, BLUDS and BQUDS results
a) step profile transport, result on vertical centerline
b) square wave transport in stagnation flow field, exit profiles

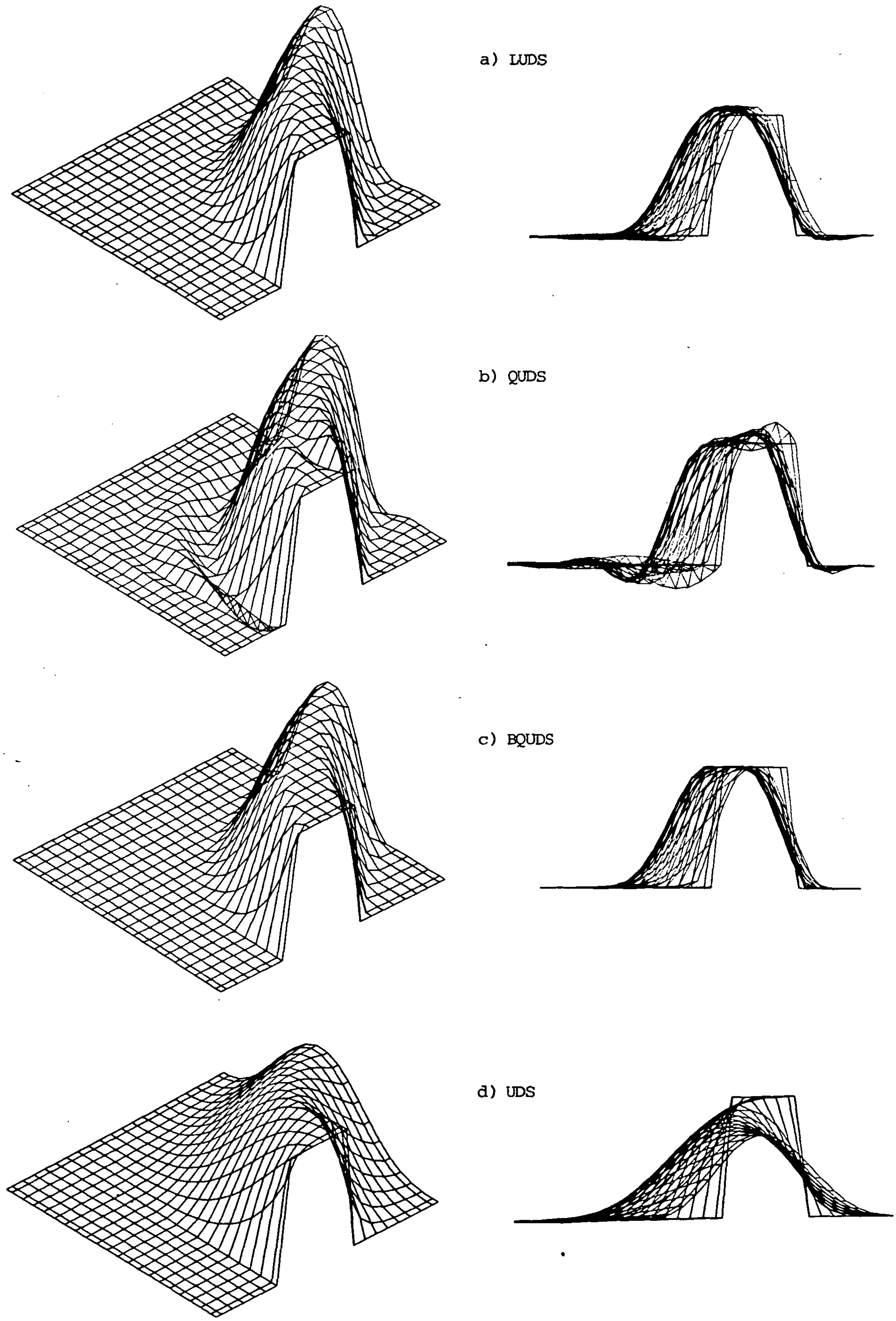

Fig. 8.11: Transport of a square wave in stagnation flow field

also show that the LUDS (a) is less oscillatory than the QUDS.

Discussion

From the above results the following conclusions can be drawn:

(i) The UDS offers good accuracy if the streamlines and one set of grid lines are aligned, but suffers from excessive numerical diffusion when the flow is oblique to the grid lines and steep gradients across the flow are present.

(ii) The LUDS is less affected by numerical diffusion than the UDS, but it is prone to produce overshoots or undershoots when high second derivatives in ϕ are present across the flow.

(iii) The QUDS is also far less affected by numerical diffusion than the UDS, and slightly less than the LUDS. However, it is prone to produce overshoots or undershoots whenever strong second derivatives exist in any direction, due to the fact that it is not fully transportive. It also requires more computational effort to obtain solution, because of the unfavourable structure of its coefficient matrix.

(iv) The new flux blending method is successful in removing numerical oscillations from the LUDS and QUDS solutions, while retaining their accuracy. This is, however, guaranteed only for linear problems and when sources are prescribed, as was the case in problems considered here. Also it requires that the higher-order scheme solution be obtained first; hence the computational effort needed to obtain bounded solution exceeds that for component schemes alone* .

8.3 Grid Effects

The desired grid properties have been described in the previous chapter. The effect of grid and flow alignment is obvious. Here the attention will be focussed on two other important grid properties, namely non-orthogonality and smoothness.

* How much more effort is needed to obtain the bounded solution depends on the proportion of the solution domain affected by numerical oscillations. In the cases considered above (which were extreme ones with steep gradients and $Pe=\infty$) the ratio of total work for the bounded calculation to that of higher-order scheme alone was in the range 2 to 6 (higher values for BQUDS).

8.3.1 Non-orthogonality of grid lines

The effects of grid non-orthogonality were first studied on inviscid flows. Since in such cases the viscosity is zero, only convection and pressure terms feature in momentum equations: thus the influence of these terms on non-orthogonal grids can be examined in isolation.

First flow in a straight channel was considered, in which both pressure and velocity fields are uniform. Various grids and initial field values were used, for both two- and three-dimensional cases. The angle θ_{ij} between x^i and x^j grid lines was varied in the range $45^\circ < \theta_{ij} < 135^\circ$, and linear, "zig-zag" and random initial pressure distributions were imposed. The solution process was always stable and converged to the exact solution (to within prescribed convergence tolerance) in all cases. The effect of grid non-othogonality was to increase the number of iterations needed to reach converged solution by up to 100% for $\theta_{ij}=45^\circ$, as compared to $\theta_{ij}=90^\circ$. This is due to the omission of the p' cross-derivatives in the process of deriving the approximate pressure-correction equation. However, these tests have shown that the presence of more than one pressure gradient in the momentum equations, and omission of p' cross-derivatives in flux corrections, does not affect the stability of the present solution method, at least for the range of grids considered.

Next laminar flows between parallel plates and in a circular pipe were considered. These belong to a small number of flows for which analytic solutions exist (see eg. Schlichting, 1979).

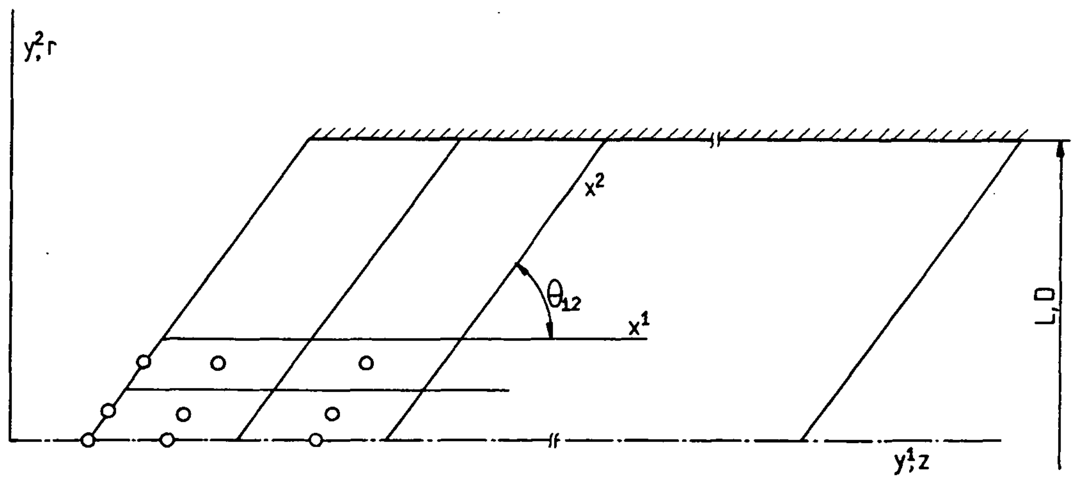

Fig. 8.12: Grid used for prediction of flows between parallel plates and in a pipe

Grids consisting of 20×10 control volumes (CV), covering region of 50 L in the flow direction and the half width (because of symmetry, see Fig. 8.12), were used to predict these flows at a Reynolds number ($Re=\frac{\rho \bar{u} L}{\mu}$, where $\bar{u}$ is the mean velocity) of 100 , for different angles between the grid lines ($45° < \theta_{12} < 135°$). Uniform velocity profiles at inlet and zero axial gradient at outlet were prescribed, with the initial interior values set to: $u_1=\bar{u}$, $u_2=0$, $p=0$. Both the predicted velocity profiles (see Fig. 8.13 and 8.14) and pressure drops in the fully developed region were in very good agreement with analytic solution. The number of iterations was again increased as θ_{12} departed from $90°$, being up to three times higher at $\theta_{12}=45°$ than at $\theta_{12}=90°$. However, the solution procedure was always stable, and no effect of grid non-orthogonality on accuracy was observed. No detrimental effects attributable to the extra terms in the axi-symmetric case (see Appendix) were observed on non-orthogonal grids.

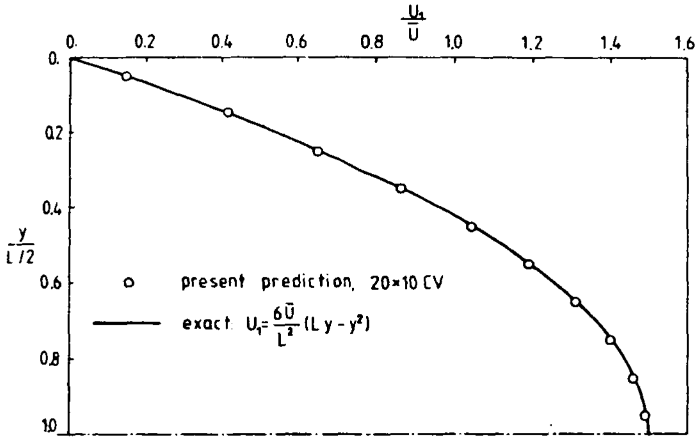

Fig. 8.13: Comparison of predicted and exact velocity profile in fully developed region of flow between infinite parallel plates at Re=100

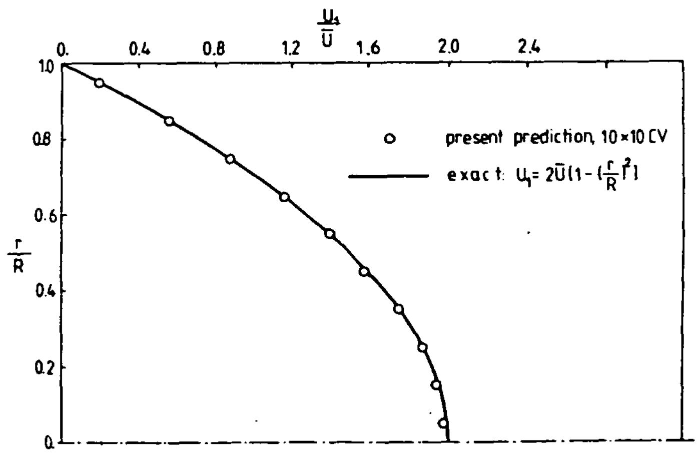

Fig. 8.14: Comparison of predicted and exact velocity profile in fully developed region of flow in a circular pipe at Re=100

The parallel plate and circular pipe cases were also predicted for the turbulent regime, in order to assess the effect of grid non-orthogonality* when the additional scalar equations for k and ε are solved. The computational grids used were the same as in laminar cases, with uniform inlet profiles prescribed:

$$u_{1in} = \bar{u} \quad , \quad k_{in} = I_T \bar{u}^2 \quad , \quad \varepsilon_{in} = \frac{k_{in}^{3/2}}{0.1L}$$

Here I_T is the turbulent intensity, taken to be 5% .

For these cases no increase in the number of iterations nor any instability was observed with non-orthogonal grids: on the contrary, the numbers of iterations required were found to be lower than for laminar ones. This could be explained as the result of more pronounced dominance of convection, as the cell Peclet numbers were higher than those of the laminar case. However, the iteration ratios remained about the same. Again, no difference was observed between the solutions obtained on the various non-orthogonal grids and the orthogonal one, to within the convergence tolerance.

The velocity profile obtained in the fully developed regime between parallel plates** is shown in Fig. 8.15(a) and compared there with the numerical solution of Demirdžić (1982). Figure 8.15(b) shows a similar comparison of the fully developed profile of k normalized with u_τ^2 , where $u_\tau = \sqrt{\frac{\tau_w}{\rho}}$, and τ_w is the wall shear stress. The agreement in both cases can be said to be satisfactory for the grids used (20 x 10 CV in the present case, and 10 x 10 CV in Demirdžić, 1982).

The fully developed velocity profile for turbulent pipe flow is shown in Fig. 8.16 , where it is compared with the experimental data of Flügge (1982). Good agreement can be seen everywhere but near the wall: however, in this region experimental uncertainity was rather high (about 15%, according to Flügge, 1982), and in this light the present results can be

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* These flows belong to a small group which can be predicted by using either orthogonal or non-orthogonal grids; other problems which might be more representative of the range for which the present method is designed usually do not allow this freedom.

** The same results were obtained by three-dimensional computer code, taking the two side boundaries to be symmetry planes.

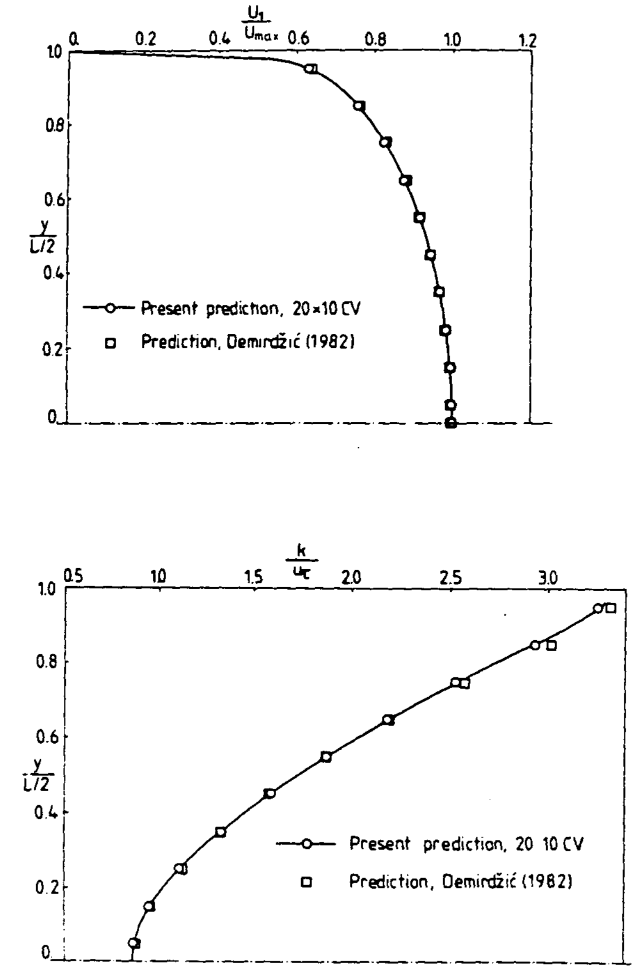

Fig. 8.15: Predicted velocity (a) and normalized kinetic energy of turbulence (b) for flow between parallel plates at $Re = 2 \cdot 10^4$

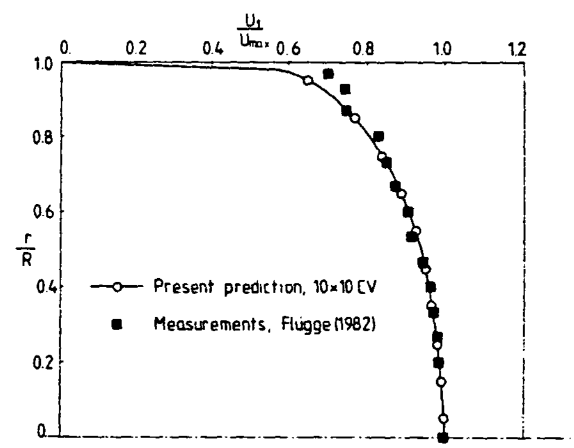

Fig. 8.16: Fully developed velocity profile in a circular pipe at $Re = 3.5 \cdot 10^4$, prediction compared with experiment

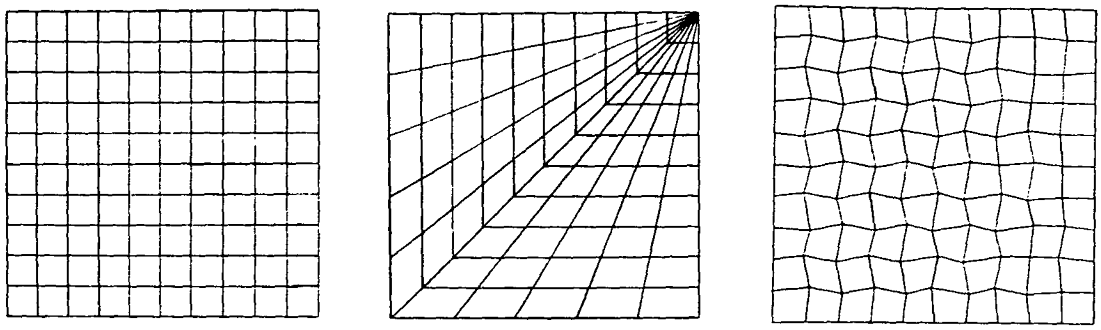

Fig. 8.17: Computational grids used in predictions of inviscid stagnation flow

taken as quite satisfactory. Also the pressure drop coefficient obtained from the numerical results was within 1% of the value obtained from Moody's diagram for the same Re and smooth pipes. Thus no adverse effects of grid non-orthogonality, other than slower convergence, were encountered in these calculations.

8.3.2 Smoothness of grid lines

The possibility of an influence of the "smoothness" of the grid lines on the solution process was outlined in the previous chapter. These effects will now be demonstrated. Inviscid stagnation flow has been chosen as the most suitable test case, since it possesses an analytic solution (which is $u_1=y^1$, $u_2=-y^2$, $p=p_0-\frac{1}{2}\rho[(u_1)^2+(u_2)^2]$, where p_o is the pressure at stagnation point), and allows various grids to be used. Figure 8.17 shows the three types of grid here considered: uniform Cartesian (diagram (a)), polar (b) and an arbitrary grid (c), generated by hand. The accuracy of the numerical solution is assessed by evaluating at each node errors

$$E^{\Phi} = \frac{\Phi_c-\Phi_e}{\Phi_{ref}}\cdot 100\ (\%) \tag{8.1}$$

where Φ_c is the numerically calculated value, Φ_e is the exact value, and Φ_{ref} is a reference value, taken to be $u_{ref}=L$, or $p_{ref}=p_o$. Table 8.1 shows maximum values of the errors in the u_1 , u_2 , p and p_t

predicted fields: here p_t is the total pressure, which should be constant and equal to p_o . The results are for 10×10 CV grids (as shown in Fig. 8.17), and for cases (a) and (b) also for 20×20 CV grids.

Grid	"a", 10x10 CV	"b", 10x10 CV	"c", 10x10 CV	"a", 20x20 CV	"b", 20x20 CV
$\lvert E^{u_1}\rvert_{max}$	3.15%	4.79%	2.77%	1.19%	3.64%
$\lvert E^{u_2}\rvert_{max}$	2.19%	3.76%	4.81%	0.93%	2.54%
$\lvert E^{p}\rvert_{max}$	5.99%	8.61%	5.64%	3.17%	4.76%
$\lvert E^{p_t}\rvert_{max}$	5.55%	8.96%	5.09%	3.09%	6.14%
No. of iter.	15	25	17	21	43

Table 8.1: Maximum errors in predictions of inviscid stagnation flow

The Cartesian grid calculations revealed that E^{u_1} and E^{u_2} are only significant at the cells adjacent to boundaries. The reason is that with the non-staggered variable arrangement knowledge of the pressure on the boundaries is required in order to evaluate pressure gradients for adjacent cells, and the boundary pressures were obtained here by linear extrapolation from interior nodes. In cases like the present one where the pressure variation is highly non-linear this procedure introduces errors. When grid was refined by halving the cell size, these errors were also halved. The maximum errors on both grids occured in the corner cell adjacent to the stagnation point, where two boundaries meet.

In case of the irregular grid (c) the average E^{u_1} and E^{u_2} were higher than on Cartesian grid (a), and they are more evenly spread, although still the $\lvert E^{u_i}\rvert_{max}$ are situated near the stagnation point. The larger E's are due to interpolation errors in the calculation of cross-derivative pressure terms; these result from the non-smoothness of the grid lines, rather than non-orthogonality (these effects were not present in the channel and pipe flow predictions because both sets of grid lines were always specified to be straight and parallel). However, the increase in E was not large (under 1%), and only $\lvert E^{u_2}\rvert_{max}$ was larger than the Cartesian grid level.

Grid (b) was designed to give better insight into the effect of non-smoothness. In many practical situations, as will be shown in the next chapter, the boundaries of solution domain exhibit discontinuities (corners), in which case the adjacent grid lines have to change direction suddenly. This has been simulated in grid (b), where x^1 grid lines sharply change their direction, from vertical to horizontal, at the diagonal of solution domain, thus producing a rather extreme case.

As evident in table 8.1 , the errors E were larger on grid (b) than in the two previous cases. This is partly due to the non-optimal distribution of grid lines (a coarser grid is produced along the $y^1=0$ and $y^2=0$ boundary lines) but more significantly to the sudden change in their direction at the diagonal. The latter is influential because the linear interpolation along coordinate lines used to calculate the pressure at the cell faces is inadequate for cells along the diagonal; for in this particularly extreme case, the contours of constant pressure are circles with centre at stagnation point, while the "centre" of streamline curvature and the grid pole are situated diagonally opposite. For example, for location "e" shown in Fig. 8.18 , linear interpolation along the x^1 line gives:

$$p_e = 0.5(p_E+p_P) \tag{8.2}$$

Since in the exact solution p_E is equal to p_P , this interpolated value for p_e actually corresponds to the value at location e' , as shown

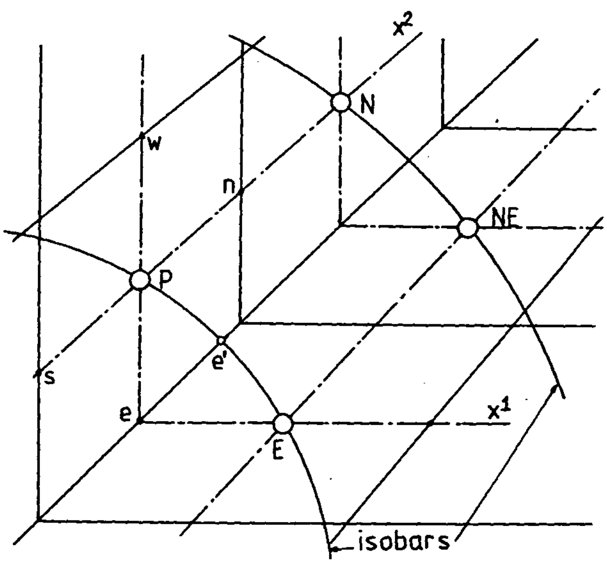

Fig. 8.18: Interpolation at the diagonal for grid (b)

in Fig. 8.18 . Thus, significant pressure errors are produced at cells along the diagonal, which leads to the forementioned increase in the E^{Φ} as compared to those for Cartesian grid (up to two times larger $|E^{\Phi}|_{max}$, according to table 8.1). Halving the cell spacing reduced the maximum errors by a third. In Fig. 8.19 (a,b) the profiles of the u_1 and u_2 velocity components along the two centerlines of the solution domain, predicted with the aforementioned grids, are shown and compared with the exact solution. Appreciable discrepancies can be seen only for grid (b), and they are largest near boundaries for the same reasons as before. However, no obvious effect of the discontinuity of grid (b) can be seen in the velocity profiles.

Figure 8.20 showns a 22 x 20 CV grid similar to 8.17 (b) , and corresponding plots of the predicted streamlines and isobars. The effect of the discontinuity is obvious only in the pressure contours, where it is manifested as kinks in otherwise smooth curves. Results obtained on a similar but smoothed grid (using technique described in section 3 of the previous chapter) are also presented in Fig. 8.20. The interpolation errors, although not eliminated, are significantly smaller on this grid, as can be seen from the pressure contours.

These test calculations show that the accuracy of numerical solutions is affected by non-smoothness of the grid lines (which, it should be noted, is not a shortcoming of the present method in general, but a

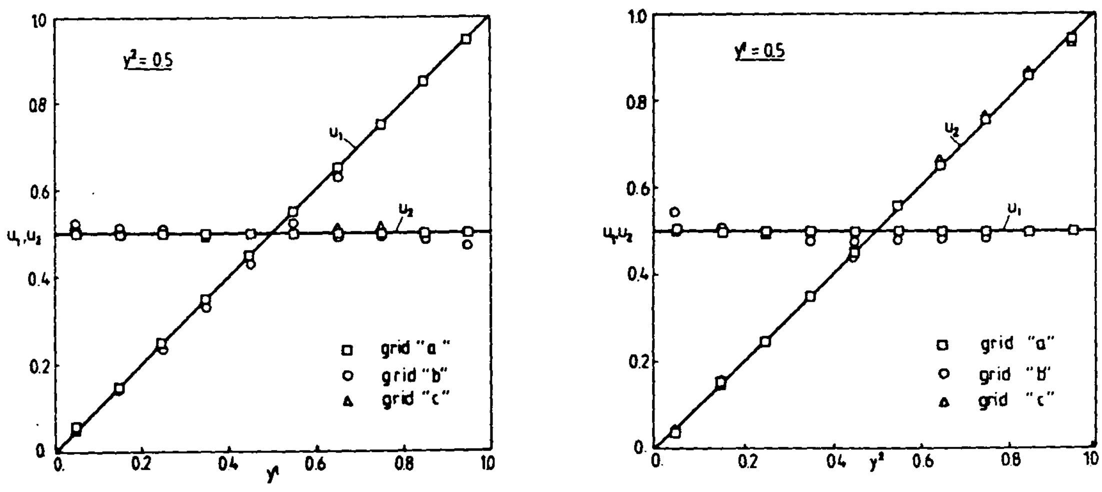

Fig. 8.19: Profiles of u_1 and u_2 velocity components along two centerlines of solution domain for inviscid stagnation flow problem

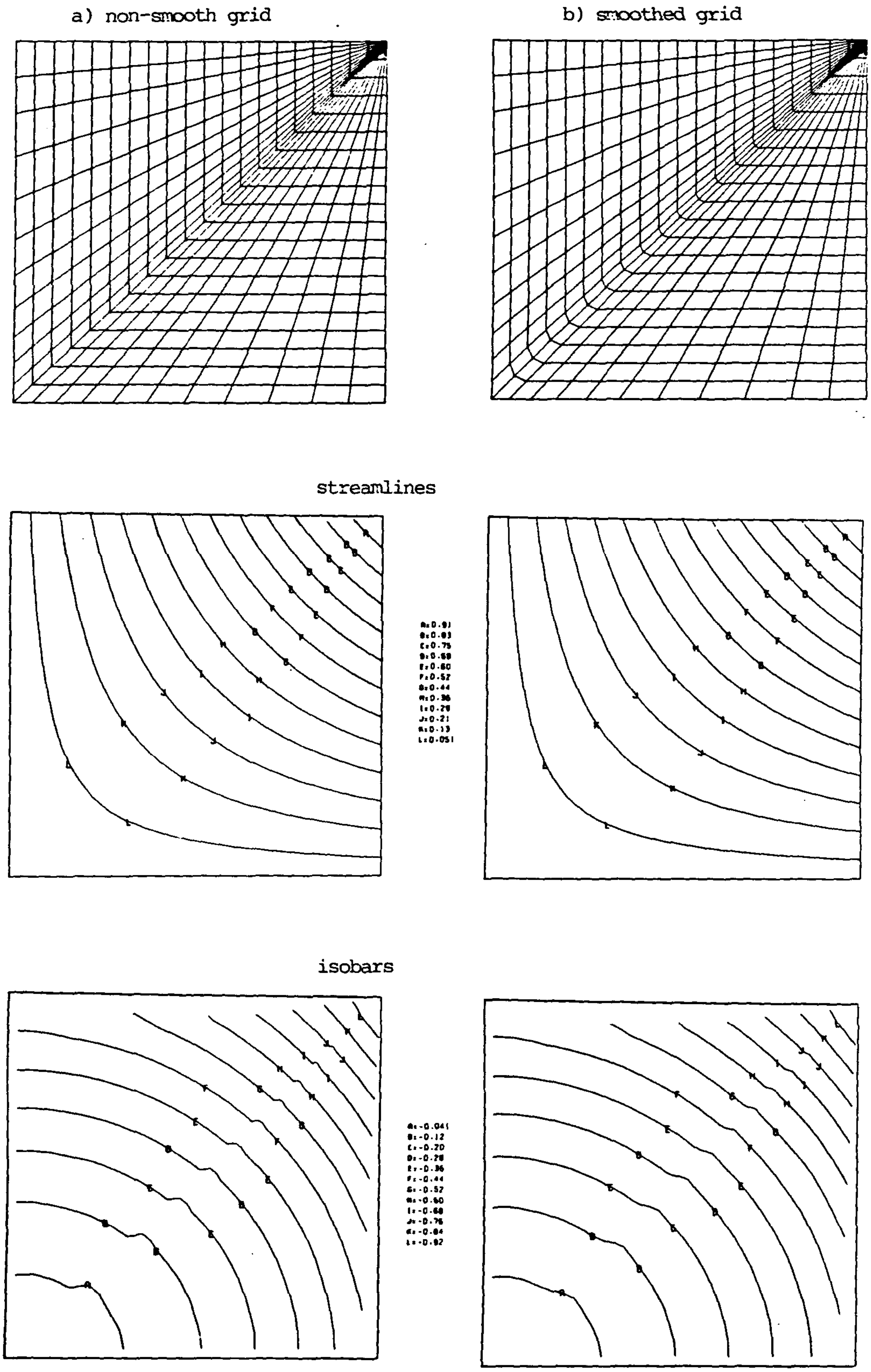

Fig. 8.20: Effects of grid non-smoothness on predicted inviscid stagnation flow field

consequence of interpolation practices currently employed). The effect is, however, of limited extent, tends to be apparent only in the pressure field and can be brought to within acceptable limits by simple actions like local grid refinement and smoothing. This conclusion will be supported by practical applications to some complex flows to be presented in the next chapter.

8.4 Effects of Under-Relaxation

In section 5 of Chapter 6 an optimal relation between the under-relaxation factors for velocity and pressure was derived of fairly general validity.

In this section the influence of α_u and α_p on the rate of convergence is studied on two laminar flow problems, which are schematically presented in Fig. 8.21 and 8.22 , together with the boundary conditions employed. The first is flow in a plane channel with a sudden expansion, and the second flow in a cavity with moving lid. The two side walls of cavity can be inclined, and here three angles θ_{12} were considered: θ_{12} =90° (the square cavity), 60° and 45° . First results obtained on orthogonal grids will be discussed.

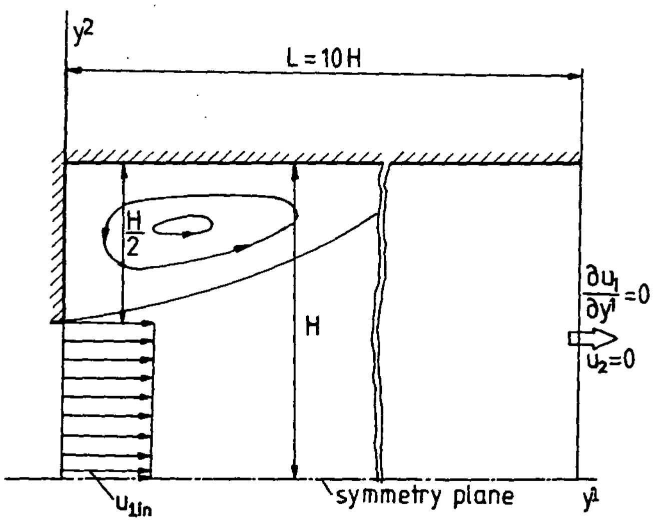

Fig. 8.21: Channel with sudden expansion

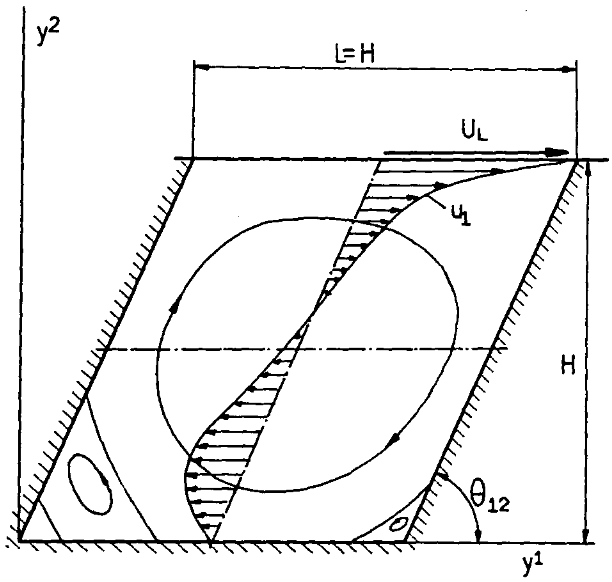

Fig. 8.22: Cavity with moving lid

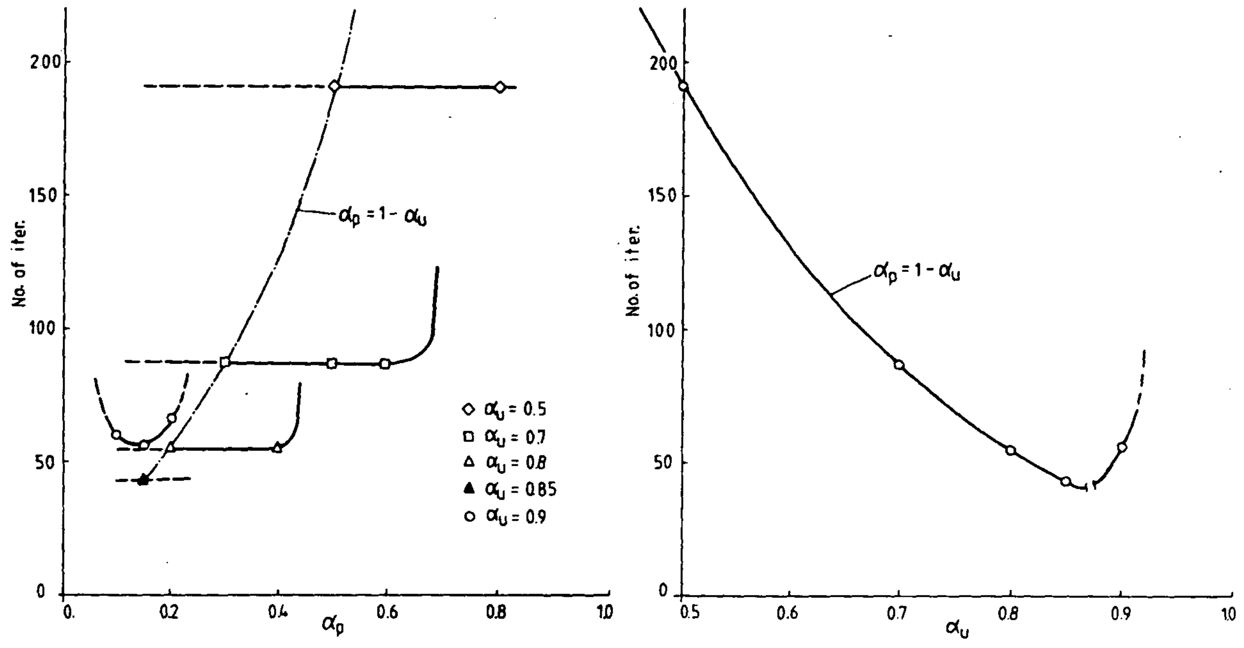

Fig. 8.23: Computational effort as a function of α_u and α_p for flow in channel with a sudden expansion at Pe =50 , 20 x 20 CV grid

Figures 8.23 and 8.24 show the dependence on α_u and α_p of the number of iterations required to obtain converged solutions for channel and square cavity flows. In all cases the criterion for convergence of the pressure-correction equation was λ_p=0.3* , while the residual sums in continuity and momentum equations were normalized with respect to $u_L \cdot H$ and $u_L^2 \cdot H$ respectively, and $\lambda_s = 1 \cdot 10^{-3}$ was employed (see section 7 of Chapter 6). These diagrams show that the rate of convergence is primarily determined by the value of α_u: the closer to unity it is, the fewer the number of required iterations. This trend holds up to a certain limiting value, which is problem-dependent, after which it reverses, as Fig. 8.23 (b) and 8.24 (b) show. For given α_u , a range of α_p values can be used without affecting the convergence rate. This "band" of suitable α_p values extends for α_u's not too close to the aforementioned limiting value on both sides of the value given by Eq. (6.15), ie. $\alpha_u = 1 - \alpha_p$, and becomes narrower as α_u increases.

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* It should be noted that this tolerance is necessary to ensure that the corrected mass fluxes reasonably satisfy continuity. Higher λ_p would necessitate use of lower values of α_u for stability reasons, while tighter tolerance usually does not result in improvement of the overall convergence rate (Van Doormaal and Raithby, 1984).

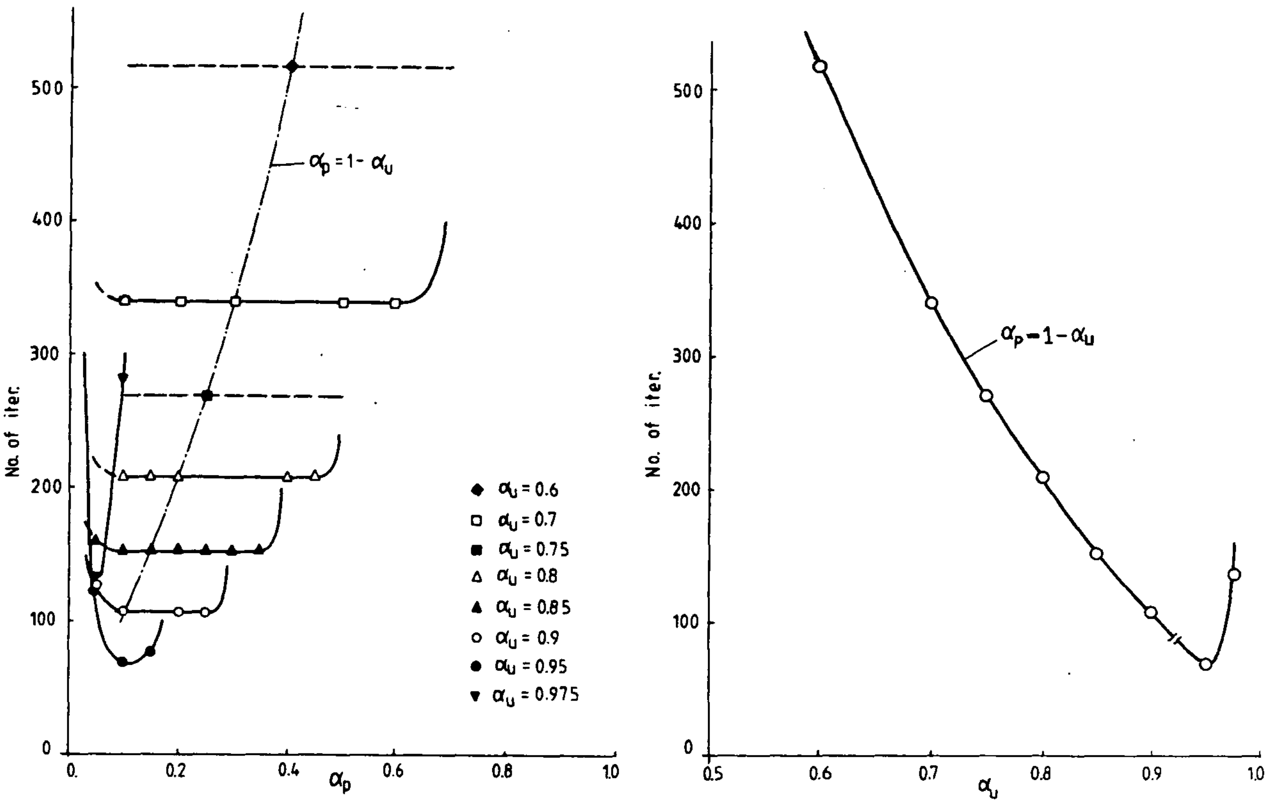


Fig. 8.24: Computational effort as a function of α_u and α_p for square cavity flow at Re=100 , 40 x 40 CV grid

From Fig. 8.23 and 8.24 it is also evident that the optimum set of under-relaxation factors α_u and α_p which provides fastest convergence, is problem dependent: for the channel case the set $\alpha_u=0.85$, $\alpha_p=0.15$ is best, while for square cavity it is $\alpha_u=0.95$, $\alpha_p=0.1$. However, it appears that the values of $\alpha_u \approx 0.8$ and $\alpha_p \approx 0.2$ offer good convergence and are relatively safe. Most of calculations to be presented in the next chapter were performed with this set.

Figure 8.25 shows the dependence of number of iterations to convergence on α_u and α_p for prediction of flows in cavities with inclined walls. It is evident that the number of iterations for $\alpha_u<0.8$ is lower for the inclined case (and thus more non-orthogonal grid) than for the square one. This seems contradictory to earlier claims that grid non-orthogonality slows down the rate of convergence. The explanation lies in the fact that flows in inclined cavities are physically different from that in a square cavity, as will be shown later, so that direct comparisons in this respect are inappropriate.

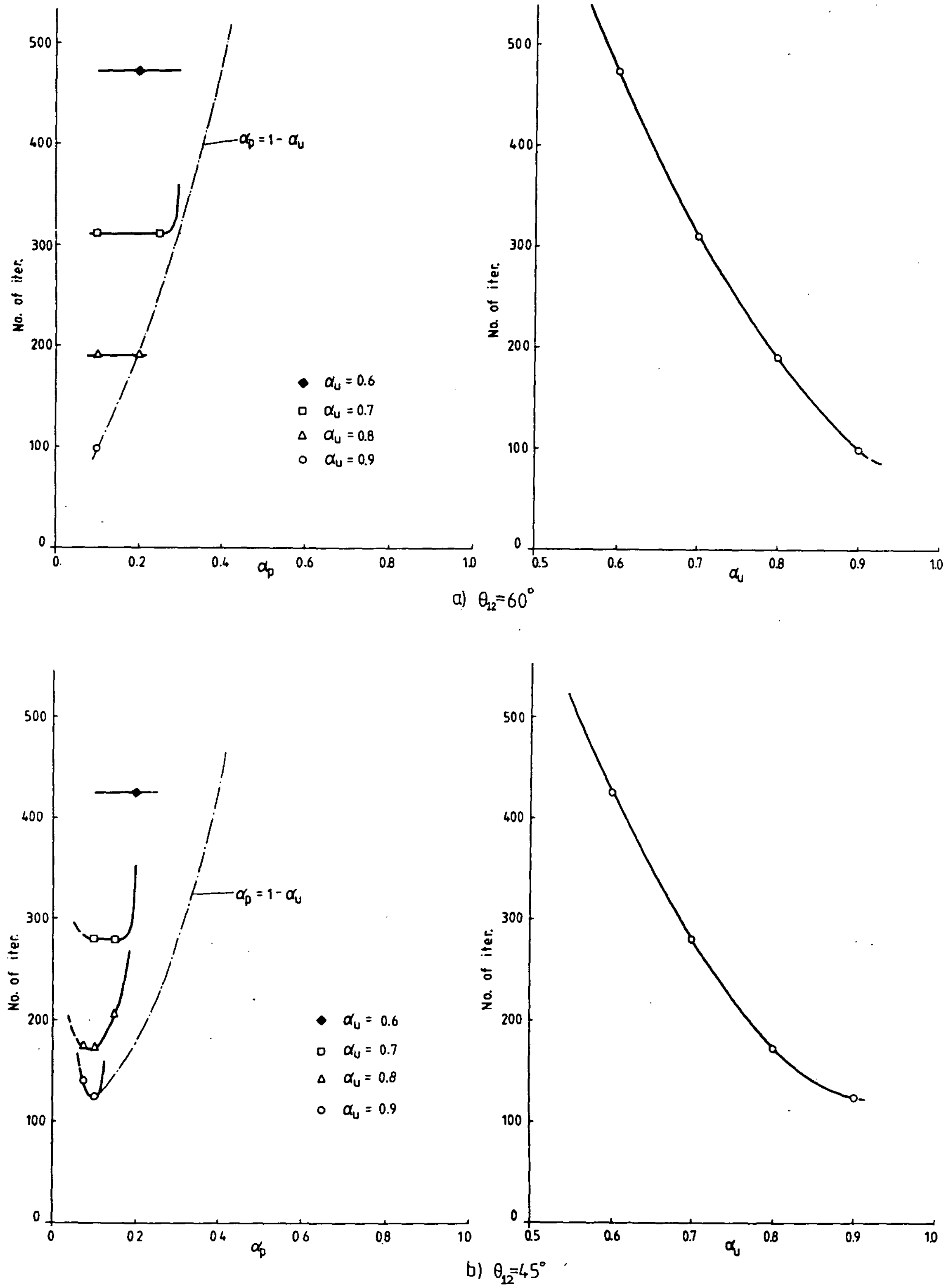


Fig. 8.25: Computational effort as a function of α_u and α_p for cavity with inclined walls, at Re=100 and 40 x 40 CV grid

Another important fact to note in Fig. 8.25 is that the width of the "band" of α_p values which can be used with given α_u becomes narrower as the non-orthogonality of grid increases. One consequence is that for $\theta_{12}=60°$ the α_p values given by Eq. (6.15) are on the edge of the band, while for $\theta_{12}=45°$ they are outside it. The optimum values are lower than for the square cavity, and for larger α_u the range of permissible α_p is not wide.

The reasons for the above effects are: (i) the cross-derivative terms are treated explicitly in the momentum equations, and thus they do not feature in corrector stage of SIMPLE algorithm, making the corrections far more approximate, and (ii) the cross-derivatives in the pressure-correction equation are omitted in the present method.

The trends observed suggest that for grids with θ_{ij} less than 60° lower values of α_u should be chosen, and α_p should be set smaller than the value suggested by relation (6.15).

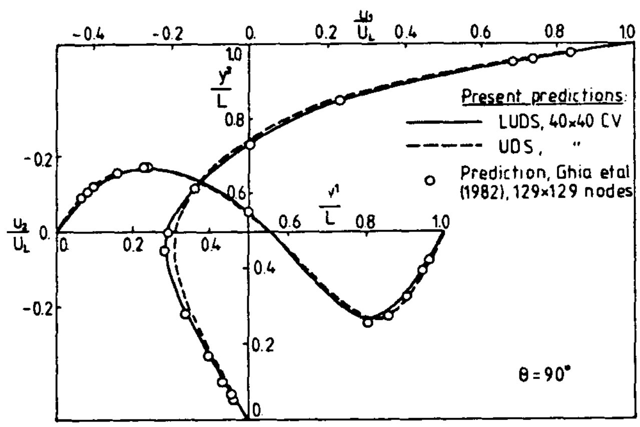


Fig. 8.26: Profiles of u_1 and u_2 along two centerlines of square cavity with moving lid

In Fig. 8.26 the predicted profiles of u_1 and u_2 along the two centerlines of square cavity using the UDS and LUDS are shown and compared with the results of Ghia et al (1982). The good agreement between all three sets of data suggests that grid-independent solutions were achieved.

For interest's sake in Fig. 8.27 the predicted flow field in cavities with $\theta_{12}=90°$, $\theta_{12}=63.4°$ and $\theta_{12}=116.5°$ are compared. While in the square cavity two bottom corner vortices are present, in cavities with inclined walls only the "sharp corner" vortex persists, while the other one is suppressed. No other predictions of flow in cavities with inclined walls seem to have been published.

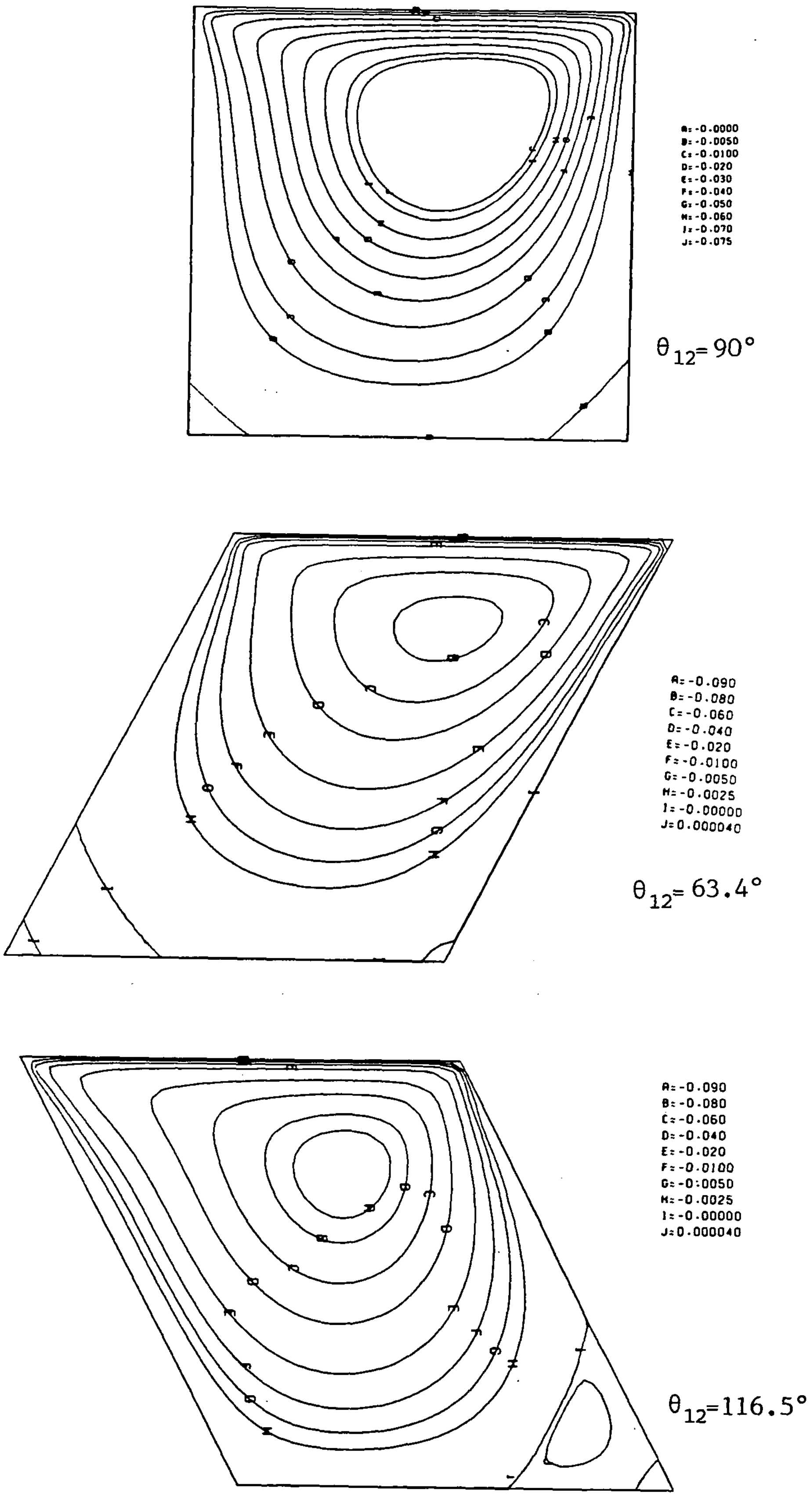


Fig. 8.27: Predicted streamlines for flows in cavities with moving lid

8.5 Closure

From the test calculations presented in this chapter the following conclusions can be drawn:

(i) Both the LUDS and QUDS are substantially less numerically diffusive than the UDS; however, both higher-order schemes suffer from numerical oscillations under certain conditions and hence can produce unrealistic solutions. The LUDS is less oscillatory and converges faster with the solver employed here than the QUDS.

(ii) The new flux blending technique is successful in overcoming unboundedness of higher-order schemes, while retaining their high accuracy in unaffected regions. This is, however, only guaranteed for linear problems and where the source field is prescribed, ie. independent of ϕ .

(iii) Non-orthogonality of grid lines and the consequent presence of cross-derivative terms in the equations does not, alone, introduce additional inaccuracies: however, other grid features may do so, as noted below.

(iv) Explicit treatment of cross-derivative terms in momentum and scalar equations, and omission of such terms in pressure-correction equation, does not cause stability problems provided that the non-orthogonality is not too severe ($45° < \theta_{ij} < 135°$) and proper under-relaxation parameters are employed. It may, however, slow down the rate of convergence.

(v) Sudden changes in grid direction do lower the accuracy of the present method, due it is believed to the inadequacy of linear interpolation along the grid lines in such cases. The effect is, however, of limited extent and can be reduced to acceptable levels by the simple interventions described in previous chapter.

(vi) The optimum relation $\alpha_u = 1 - \alpha_p$ between the under-relaxation factors for pressure and velocities derived earlier is, in the present method, valid for orthogonal and moderately non-orthogonal grids.

(vii) The optimum set of α_u and α_p is problem dependent, but in most cases where the above relation is valid, the set $\alpha_u = 0.8$, $\alpha_p = 0.2$ will be safe and give significantly (up to four times) faster convergence than when the "conventional" set ($\alpha_u = 0.5$, $\alpha_p = 0.8$) found in the literature (eg. Patankar, 1980,1981) is used.

(viii) In cases of substantial grid non-orthogonality, the optimum set involves a lower value of α_u than would apply for the orthogonal case, and a lower value of α_p than that suggested by the above relation. These reductions are believed to be necessitated by the explicit treatment of cross-derivatives.

The above conclusions will be supported by the results of applications of the method to complex flows which will be presented in the next chapter.

CHAPTER 9

APPLICATIONS OF THE METHOD

9.1 Introduction

In this chapter the present method is applied to some complex laminar and turbulent flows, and the results are discussed and compared with experimental data where available. These discussions focus on the assessment of the present numerical solution method, rather than the detailed study of the particular flows. Also, in view of the large number of applications considered here the discussions are deliberately kept short and concise.

First, in section 2, applications of the method to two-dimensional, plane or axisymmetric flows are presented. These include: laminar flow in a circular pipe with sudden contraction, laminar and turbulent crossflow in staggered tube banks and turbulent flow around a bluff body confined in a circular pipe.

Next, in section 3, applications to three-dimensional flows are presented. These include: laminar and turbulent flow in a S-shaped diffuser, and turbulent flow in a C-shaped diffuser.

In section 4 some results of applications to more complex flows, for which no experimental data or other numerical results have been found, are presented. They demonstrate the range of possible applications of the present solution method.

Finally, in section 5 some closing remarks are given.

9.2 Applications to Two-Dimensional Flows

9.2.1 Laminar flow in a circular pipe with sudden contraction

Here the present method is used to predict laminar flow in a circular pipe with sudden contraction of its cross-sectional area. This type of flow has been studied, both numerically and experimentally, by many researchers;. a survey can be found in Durst and Loy (1985).

Although standard numerical techniques which employ rectangular grids can be used to predict this type of flow, as done by Durst and Loy (1985), this approach has the disadvantage that the number of computational cells across the pipe after contraction is significantly lower than before it, and the proportion decreases as the contraction ratio increases. Also, since to achieve sufficient resolution a very fine grid near the wall of the smaller diameter pipe is needed, an excessive fine grid results in a larger pipe in a region where it is not required. The foregoing deficiencies can be avoided if a boundary fitted grid of the form shown in Fig. 9.2 (a) is used, as in the present study. This grid, which was generated using the Practice 2 described in section 3 of Chapter 7, is highly non-orthogonal and non-smooth around the contraction area. In particular, The "north" boundary of the solution domain (ie. the pipe wall) has two sharp corners, where x^1 grid lines change their direction through 90° . This problem, therefore, represents a very severe test of the numerical accuracy and stability of the present solution method.

Results

The case considered here is the same as that which Durst and Loy (1985) studied both experimentally and numerically, using the TEACH computer code (Gosman and Ideriach, 1976). The upstream and downstream pipes have diameters $D=19.1mm$ and $d=10.2mm$ respectively: the contraction ratio D/d is thus nearly 2 . The experimental data will be used here to assess the accuracy of numerical solutions.

The measurements were performed for four Reynolds numbers (defined with respect to the mean velocity in, and diameter of, the upstream tube), $Re_D=23$, 196, 372 and 968 . The results showed that the influence of contraction, for this range of Re_D , does not extend upstream more than one diameter D , and that the flow recovers to almost fully developed state about $1D$ downstream from the contraction. Based on this information, the computational grid was constracted such that it covers approximately $1.3D$ upstream and $1.25D$ downstream from the contraction (see Fig. 9.1(a)).

The calculations were performed for the three highest Reynolds numbers, using two grids (80 x 30 CV and 80 x 54 CV) and two differencing schemes (the LUDS and UDS). The following boundary conditions were imposed:

- fully developed flow at inlet;
- standard wall and symmetry boundary conditions;
- zero-gradient in streamwise direction at the exit boundary.

These conditions were imposed in the usual way, as described in section 6 of Chapter 6 .

As an indication of the grid dependence of the numerical solutions obtained, in Fig. 9.1 (b), (c) the dimensions of the downstream recirculation region, predicted using the two grids and the two differencing schemes, are presented and compared with the measured values. Based on this evidence, as well as comparison of velocity profiles at various cross-sections, it was concluded that the LUDS results on the finest grid were almost grid independent for the two lower Re_D (196 and 372), and nearly so at Re_D=968 .

To obtain converged solutions on the finest grid typically 200 iterations were needed, using under-relaxation factors α_u=0.7 and α_p =0.2 .

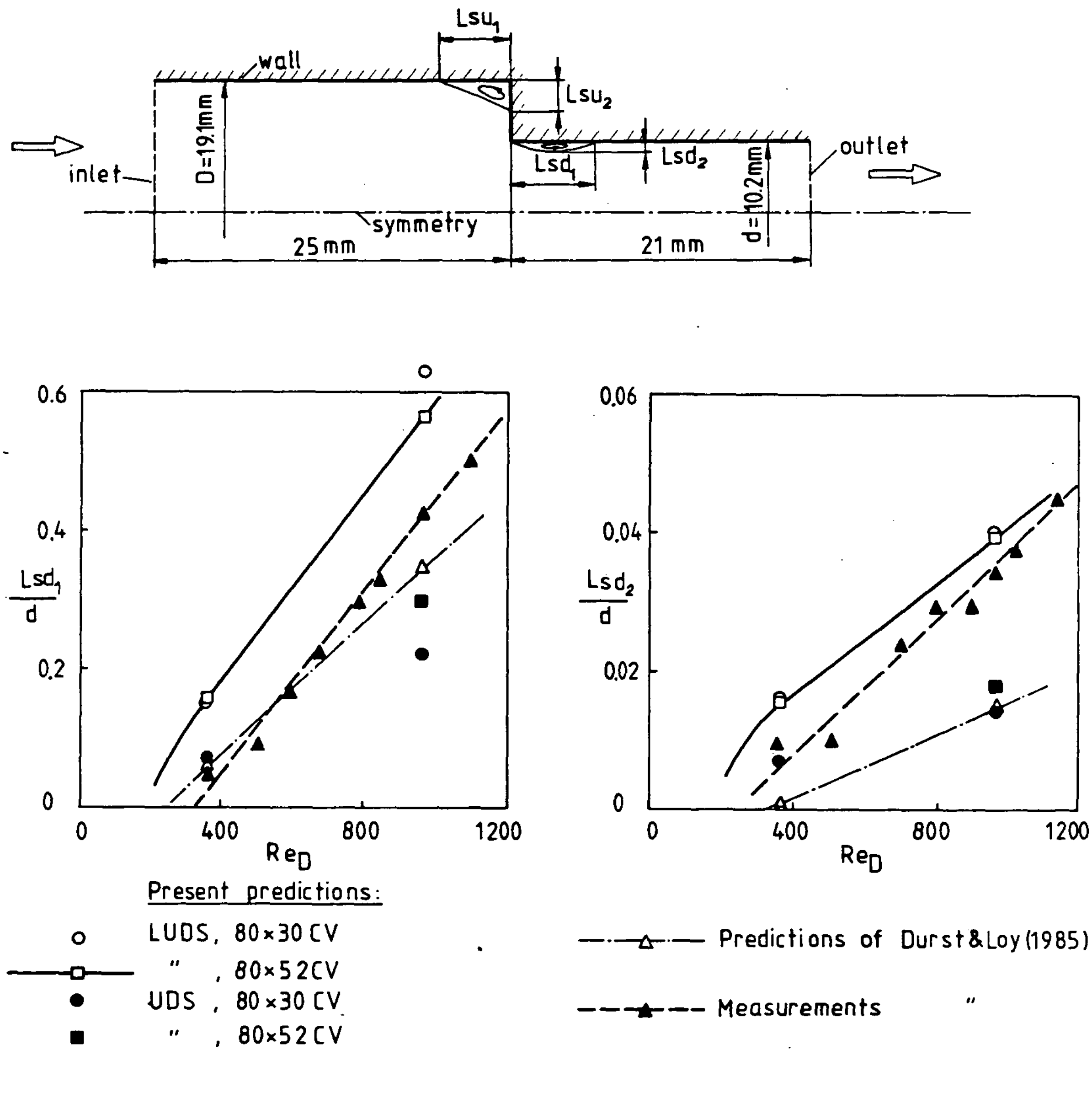


Fig. 9.1: Grid dependence tests
a) solution domain dimensions and labeling
b) dimensions of downstream recirculation region

Various features of the predictions for $Re_D=968$ are presented in Fig. 9.2 (b)-(f) . The plot (e) of the isobars shows that a very sharp pressure drop is caused as the flow accelerates around the contraction. The pressure then rises again immediatly downstream, until it recovers: thereafter the pressure starts falling again as in fully developed pipe flow. This adverse pressure gradient in the entrance region causes separation there, which happens, according to the experimental observations, for all Re_D greater than 300 . Separation also occurs in the corner of the larger pipe, as can be seen in the plot (f) of the streamlines.

Comparison with experimental data

The comparison of predicted and measured dimensions of the downstream recirculation zone were made in Fig. 9.1 (b) and (c) . Both length Lsd_1 and width Lsd_2 were overpredicted by the LUDS for both Re_D (372 and 968). For highest Re_D the length Lsd_1 reduced with grid refinement, suggesting that some numerical "overshoots" may be generated there.

Figure 9.3 shows comparison between the predicted and measured profiles of the streamwise velocity component for $Re_D=196$ at locations upstream (z/D=-1.047, -0.157 and -0.039) and downstream (z/d =0.074, 0.147, 0.294 and 1.961) of the contraction, with z measured along the symmetry axis from the contration plane. The agreement is excellent upstream of contraction, and reasonably good further downstream. However, at the station z/d=0.074 immediatly after contraction there are differences of up to 8% which, it is believed, may be due in part to measurement errors or flow asymmetries. The justification for this assertion is that the predicted profiles satisfy overall continuity at every cross-section, whereas the measurements at the forementioned station clearly do not, for the discrepancies are always in the same direction.

Elsewhere the agreement can be said to be quite satisfactory, especially in the region upstream of contraction, in which computational grid is highly non-orthogonal and also non-smooth (see Fig. 9.2 (a)).

One of the characteristics of the velocity profiles immediately after the contraction is the peak near the wall. The experiments suggest that this occurs at all $Re_D>125$, and it was clearly visible in predicted profiles at all three Re_D considered. The location and magnitude of this "overshoot" at $Re_D=196$ are correctly predicted (see Fig. 9.3).

Figure 9.4 shows a comparison of the predicted and measured radial velocity profiles at $Re_D=196$, at five cross-sections. This component becomes significant as the flow approaches the contraction. The profile in the upstream section has a peak at about mid-radius whose magnitude reaches a maximum just upstream of the plane of the contraction. Downstream, in the smaller pipe, the shape changes: the maximum is initially closer to the wall, and it moves towards the symmetry axis as the flow advances, reducing in magnitude until it eventually vanishes when the flow reaches its fully developed state again (see also Fig. 9.2 (d)). The agreement between predicted and measured values is reasonably good, except immediatly upstream of the contraction, where predictions were about 30% larger near the peak of profile. The same sort of discrepancies were reported by Durst and Loy (1985) in their calculations.

Figure 9.5 shows streamwise velocity comparisions for $Re_D=372$. The agreement is in general very good, with the maximum discrepancies of about 2% near the contraction. The magnitude of the overshoot immediately downstream of contraction is somewhat smaller than experimental data show, but its position is correctly predicted.

In figure 9.6 the development of the streamwise velocity component along the pipe centerline is shown and compared with the experimental data for the three Reynolds numbers considered. The agreement is good for the two lower values of Re_D. However, for $Re_D=968$ ($Re_d=1831$ in the smaller pipe) the predictions are consistently higher everywhere. The shape of profile is, though, well predicted, including the fact that a small maximum appears about 7mm downstream from contraction. After falling slightly, the velocity starts growing again gradually; this was seen on both predicted and measured data.

The accuracy of numerical solutions for $Re_D=196$ and 372, where almost grid independent solutions were obtained, can be said to be satisfactory, in the light of their agreement with the experimental data. Direct comparison with the numerical results of Durst and Loy (1985) was not possible, since they presented predicted profiles for Re_D values for which no tabulated experimental data was given. However, inspection of their results suggests that the levels of agreement obtained here are comparable, if not better than theirs.

These results support the previous evidence that the accuracy of the present numerical solution method is satisfactory even when highly non-orthogonal and non-smooth grids are employed.

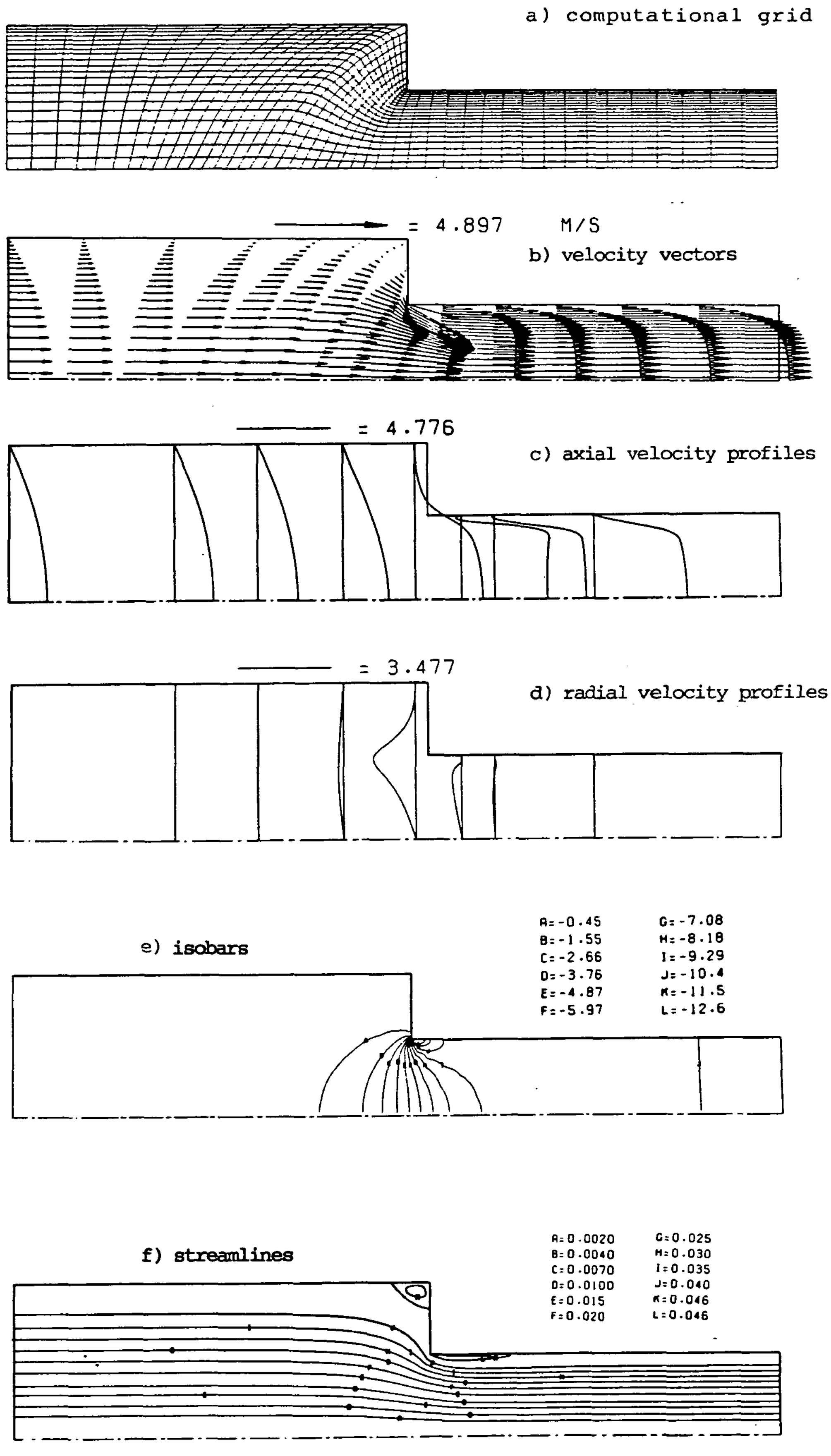


Fig. 9.2: Predicted features of laminar flow in a circular pipe with sudden contraction

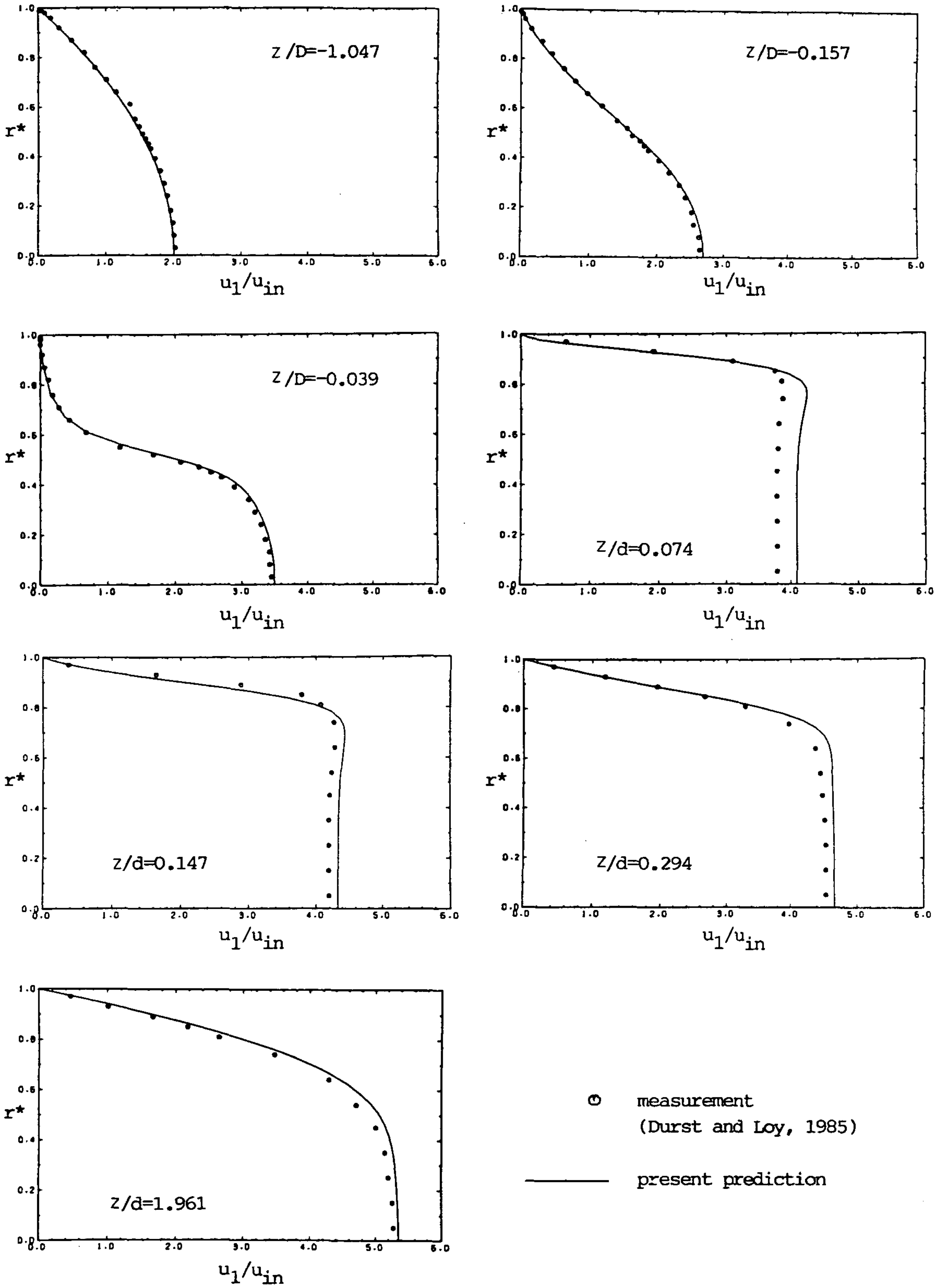


Fig. 9.3: Comparison of predicted and measured profiles of axial velocity component for flow at Re_D = 196.

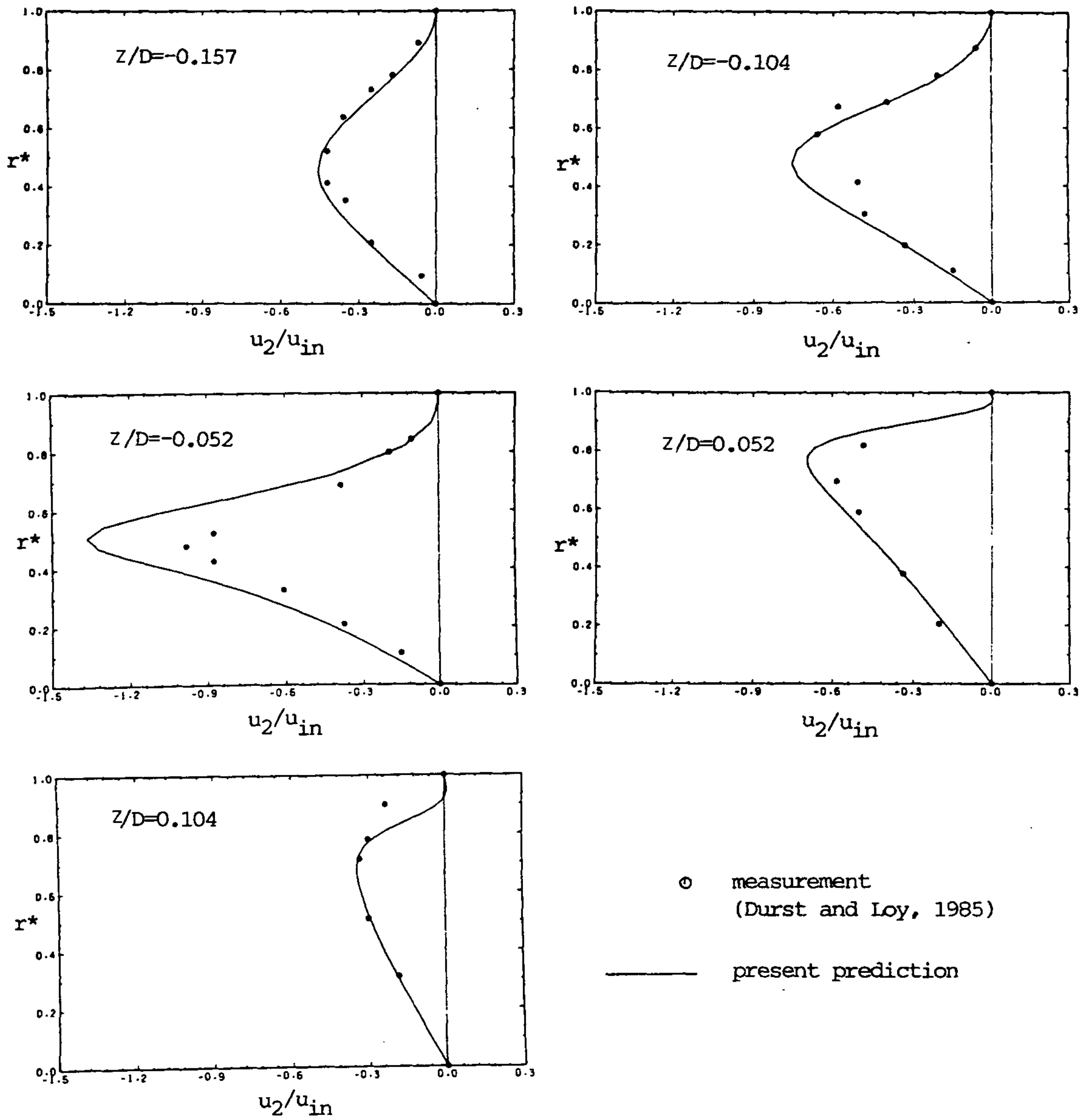


Fig. 9.4: Comparison of predicted and measured profiles of radial velocity component for flow at Re_D = 196.

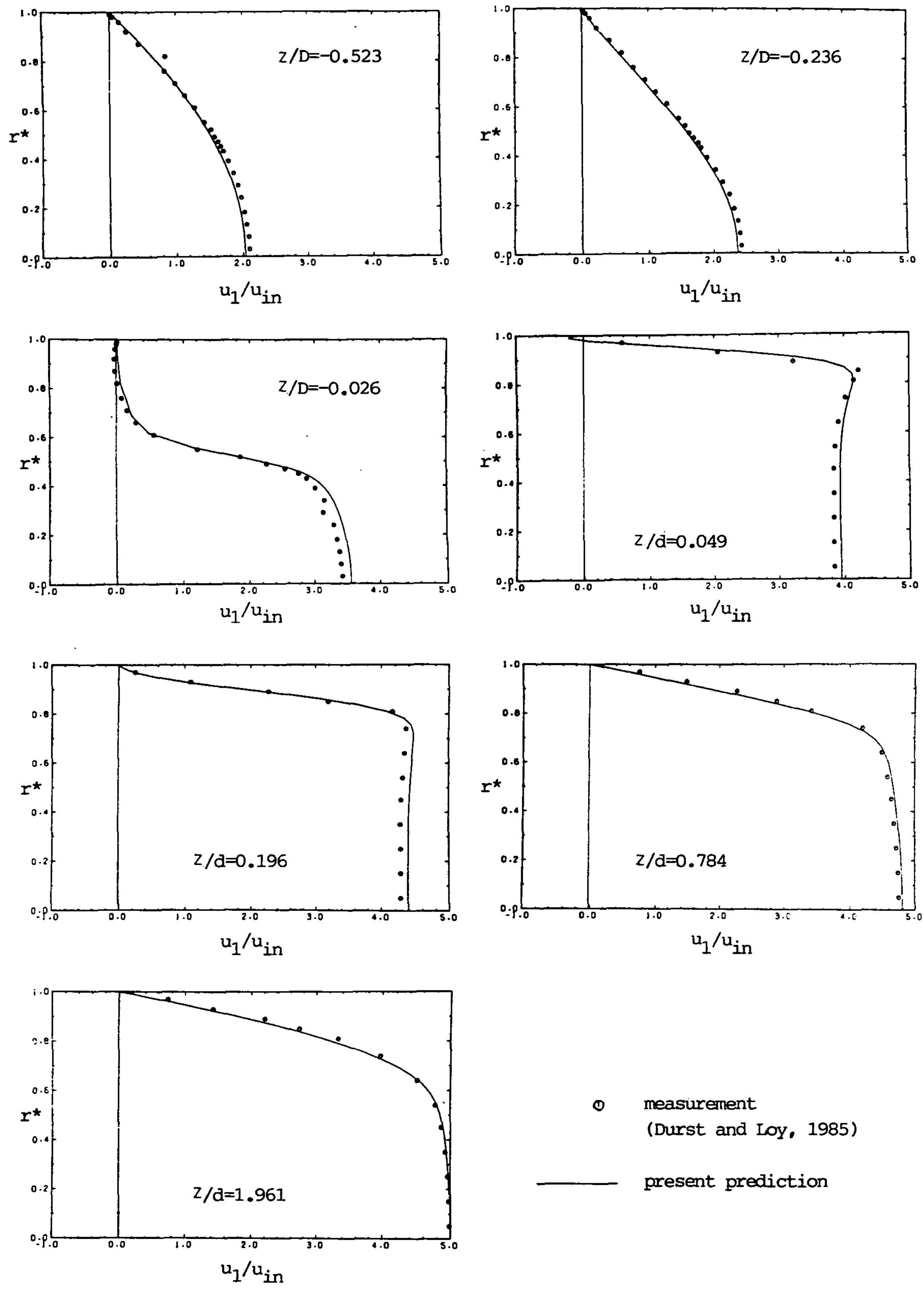


Fig. 9.5: Comparison of predicted and measured profiles of axial velocity component for flow at $Re_D = 372$.

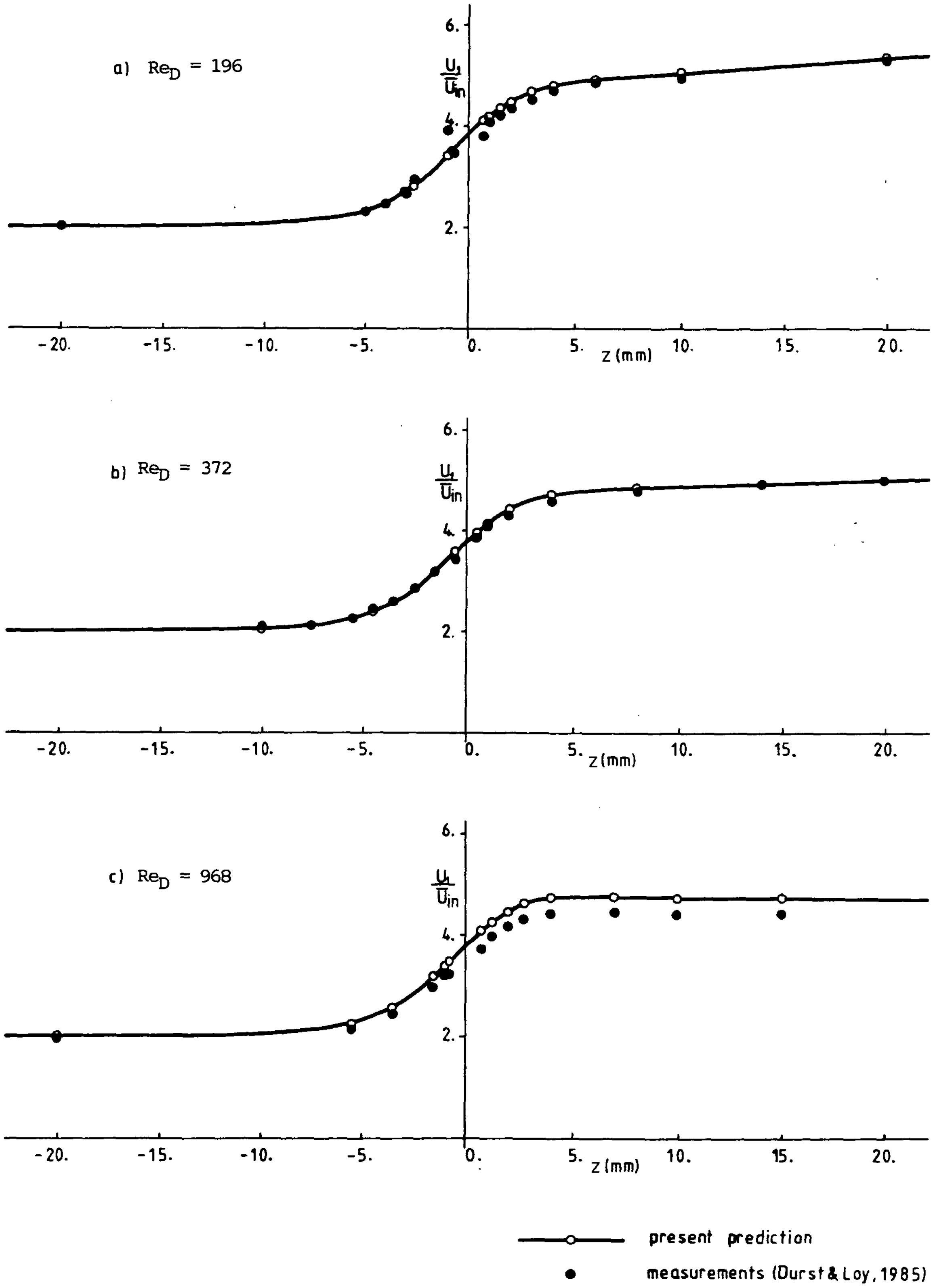


Fig. 9.6: Centerline development of axial velocity component

9.2.2 Laminar cross-flow in banks of staggered tubes

Introduction

Because of the practical inportance of flows across banks of tubes (which are extensively used in heat exchangers, boilers etc.), a large number of both experimental and numerical studies of them have been conducted in the past. An extensive survey of this work is given by Antonopoulos (1979). Due to the complexities and measurement difficulties, experimental studies were mostly limited to obtaining some mean flow characteristics of engineering interest, such as pressure drop and heat transfer coefficients, as done by eg. Aachenbach (1969, 1971), Žukauskas (1972) and Neal and Hitchcock (1967). The geometrical complexities also pose problems for numerical treatment. Some researchers (Launder and Massey, 1978; Fujii et al, 1984) used composite grids, consisting of polar-cylindrical arrangements near the tubes, and Cartesian grids in between, with some overlapping between the two. This approach, however, is cumbersome, for it requires a special computer code to be developed for this particular problem, and involves extensive interpolation in the region of grid overlap, which introduces additional inaccuracies and/or stability problems. Antonopoulos (1979) used orthogonal-curvilinear grids, which avoid these drawbacks, in his numerical method which he applied to in-line tube banks. This approach, however, also has its limitations, in that poor grid resolution near stagnation points results from the requirement that the grid lines in a particular coordinate direction cannot intersect (see, eg., Fig. 1.3), as discussed earlier in Chapter 1 . This region is very important, and adequate resolution of it is essential for good accuracy. Therefore, methods which employ non-orthogonal grids offer clear advantages for this application.

In practice, the flow in tube banks is usually turbulent, and often involves heat transfer as well. However, the study of laminar flows of this type is of vital importance for testing numerical solution methods, for it enables the assessment of numerical accuracy isolated from the inaccuracies introduced by the turbulence model. Recently a detailed experimental study of laminar flows across staggered tube banks was reported by Nowshiravani et al (1983). Their results will be used for the assessment of accuracy of the present method.

Description of problem

When a tube bank consists of many long parallel tubes, and when the flow across tubes is steady, the velocity fields after certain entrance region start repeating and the flow becomes "fully developed" (Antonopoulos, 1979). In such cases the flow can be considered as two-dimensional away from side walls, and a unit of symmetry can be identified in which the flow repeats itself, either symmetrically or anti-symmetrically. An example is shown in figure 9.7 (a). In the case studied experimentally by Nowshiravani et al (1983) the tube diameter was D=12.7 mm, the transverse pitch PT=15.9 mm, and three different longitudinal pitches PL were used: 20 mm, 25 mm and 30 mm. For all three geometries measurements were taken in three different flow regimés: one with recirculation behind the tubes (at about $Re_D \approx 140$, where Reynolds number Re_D is based on D and the mean velocity in the minimum gap between the tubes u_c), one without recirculation (at $Re_D < 15$), and one where recirculation was incipient (at about $Re_D \approx 45$). The experimental flow was intended to

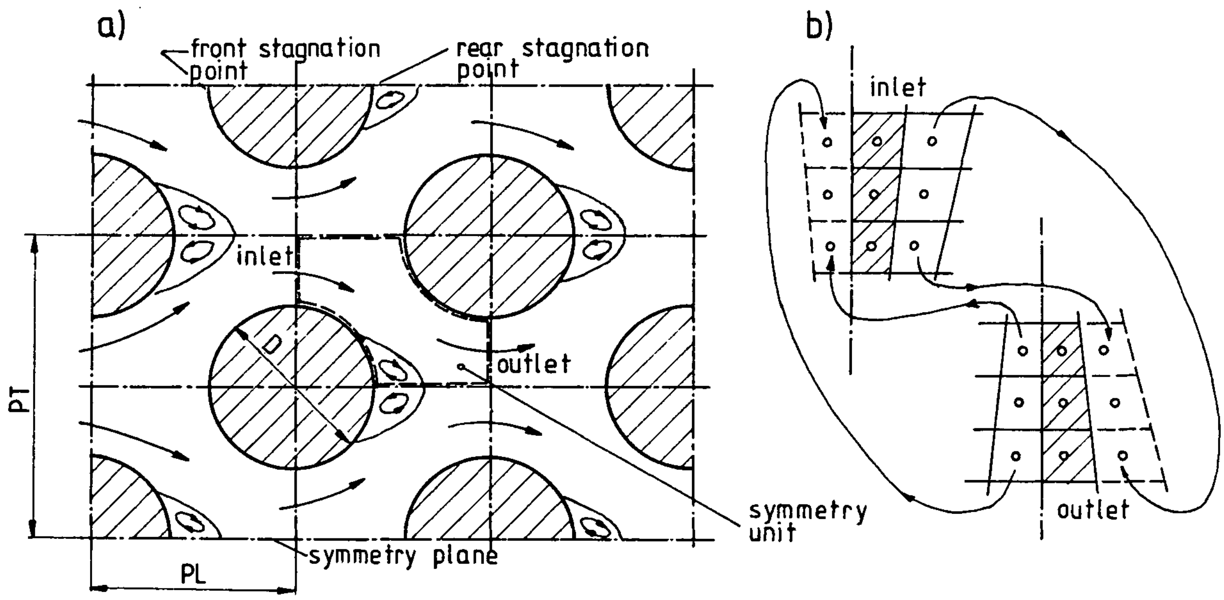


Fig. 9.7: Staggered tube bank geometry
a) definition of solution domain
b) definition of anti-symmetrical boundary conditions

be steady, fully developed and two-dimensional, but these conditions were not completely achieved: in particular some unsteadiness was observed and the velocity profiles did not display complete periodicity and two-dimensionality. The latter defect, as will be seen below, was detected from measurements made at corresponding locations on either side of the (supposed) symmetry planes.

Boundary conditions

Numerical predictions were performed for a typical symmetry unit, shown in Fig. 9.7 (a). The boundaries of the solution domain coincide with the tube walls, symmetry planes, and locations where the flow is assumed to repeat itself. The imposed boundary conditions were:

- at the wall and symmetry boundaries: the standard conditions described in section 6 of Chapter 6;
- at the inlet and outlet boundaries: antisymmetric periodicity.

The antisymmetry of inlet/outlet profiles was enforced in an explicit way. Boundary nodes at the inlet and outlet boundaries were not, as usually, placed on the boundary lines, but in centers of the imaginary preceding/following cells, as is schematically shown in Fig. 9.7 (b). To these were, after each complete iteration cycle, assigned values found in the corresponding cells within the solution domain, observing the anti-symmetry of the flow. Thus the values at nodes in the corresponding cells of the first and the last (the "extra" one) row turn out to be identical (to within convergence tolerance) when converged solution is obtained.

This explicit adjustment of the boundary values was found to slow down the rate of convergence by factors of two to three times over fixed inflow calculations, but was retained because of the ease of implementation. The only input parameter, for a given geometry, is then the mass flow rate (ie. the Reynolds number). Accuracy tests were first carried out for a single case (PL=20 mm) with different grids and differencing schemes. The same grid density was then employed for the remaining geometries (PL=25 mm and 30 mm), which entailed inserting more points to account for the larger flow domain.

Results

The accuracy tests were performed with grids of 41×18 CV and 65×30 CV (which were generated using Practice 1 of Chapter 7), and the LUDS and UDS. Plots of the calculated wall shear stress (one of the most sensitive quantities), around one half of a tube are presented in Fig.9.8. There is very little difference (~0.1%) between the two solutions obtained with the LUDS; the same was true for other flow characteristics, such as the length and width of the recirculation region. The solutions with the UDS differ slightly more (up to 2% locally), and that obtained on the finer grid is closer to the LUDS solutions in every region of the flow. This suggests that the results obtained with the LUDS - finer grid combination are (almost) grid independent. This is due in no small part to the relatively good alignment of the grid and streamlines in the greater part of solution domain, as can be seen in Fig. 9.9.

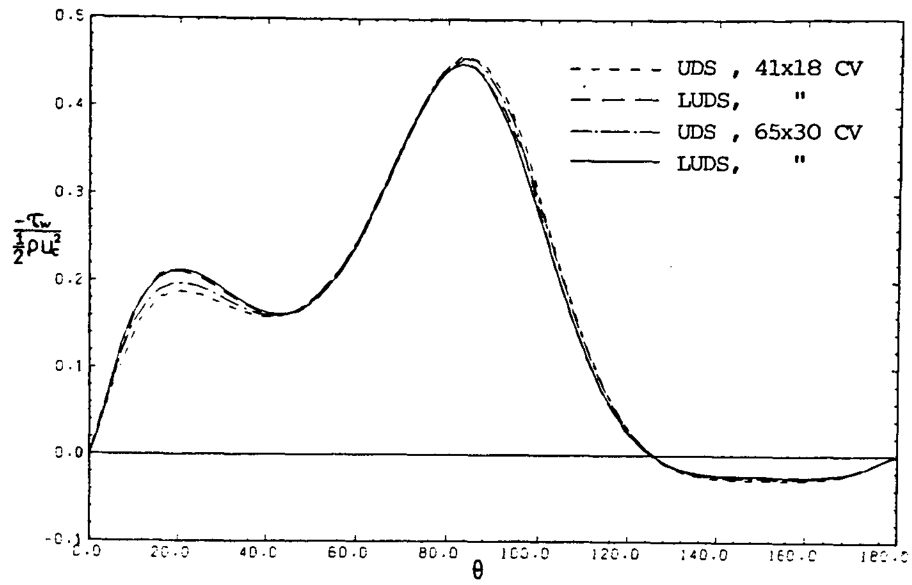


Fig. 9.8: Grid dependence test: wall shear stress around half of a tube with different schemes and grid densities, for PL=20 mm, Re_D=140

Grids of the 72 x 30 CV and 80 x 30 CV were used for PL=25 mm and PL=30 mm respectively. The solutions typically needed about 500 iterations to converge, using under-relaxation factors α_u=0.7 and α_p=0.2 , and consumed about 80 seconds on a CRAY 1 S computer. In spite of the fact that the grids used here have regions where the angle between grid lines is lower than 45°, and the aspect ratios are over 10, no stability or convergence problems were encountered.

Field plots of the velocity vectors, profiles of the Cartesian velocity components and contours of isobars and streamlines for the PL=20 mm case at Re_D=140 are shown in Fig. 9.9 (b)-(f). They show clearly where separation occurs, the size and shape of recirculation region, the high pressure at the front stagnation point and the large region of almost constant pressure within the recirculation bubble.

Comparison with experimental data

Figures 9.10 (a)-(c) show comparisons, for the flow with recirculation ($Re_D \simeq$140), of the predicted and measured profiles at four cross-sections of the streamwise velocity component u_1 , for three different longitudinal pitches. Wherever different, the measured values at loca-

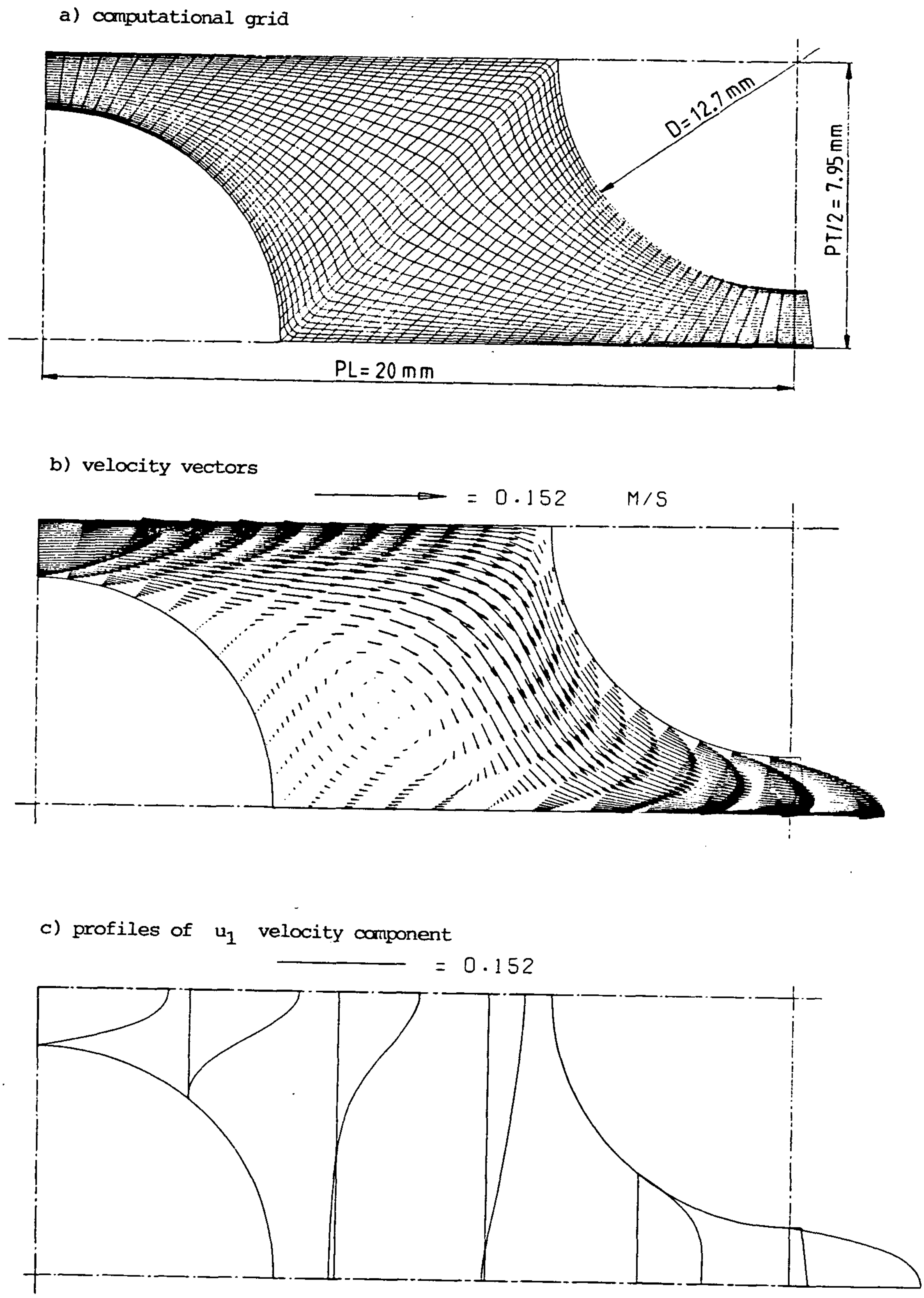
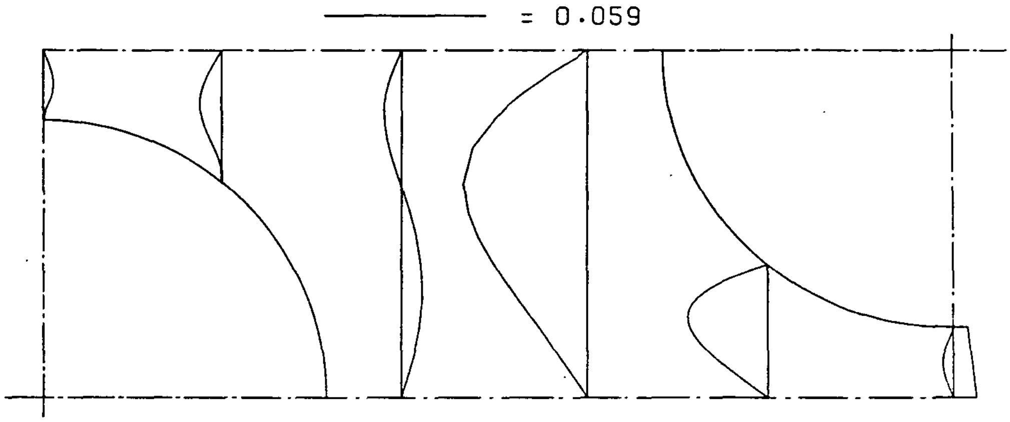
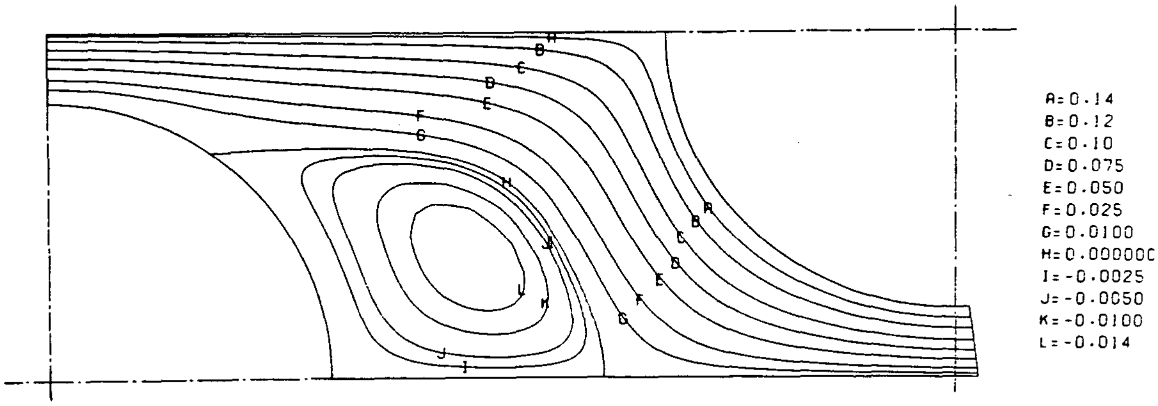


Fig. 9.9: Details of predicted flow in staggered tube bank at $Re_D \simeq 140$.

d) profiles of u_2 velocity component



e) streamlines



f) isobars ($p/\rho u_c^2$)

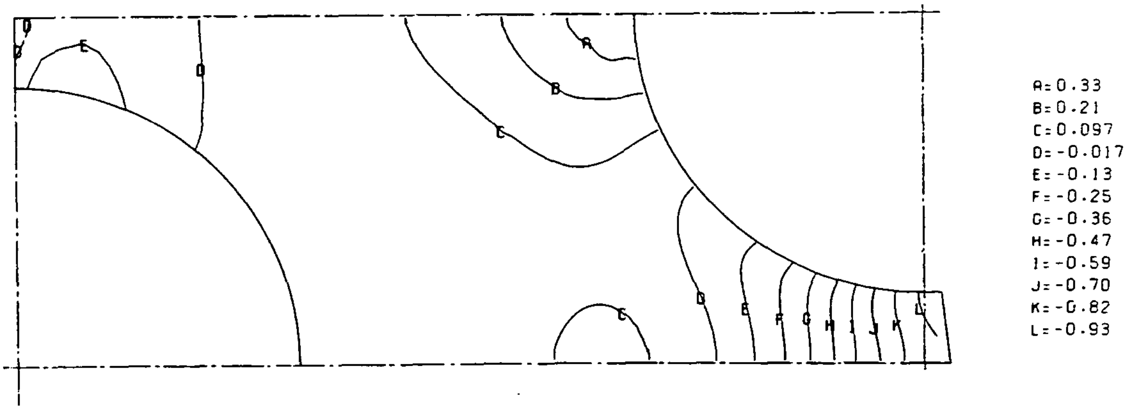


Fig. 9.9: continued

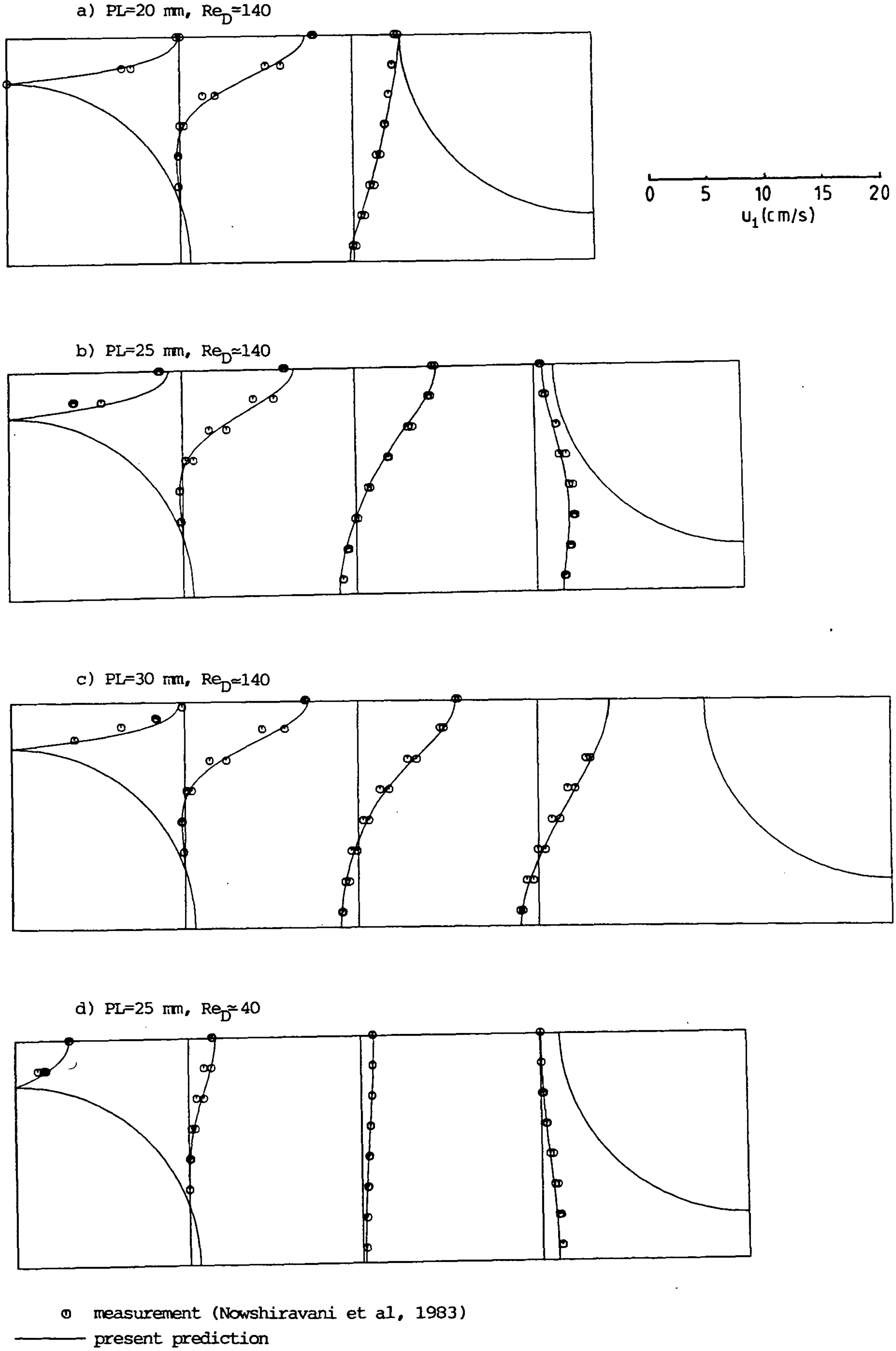


Fig. 9.10: Comparison of predicted and measured profiles of u_1 velocity component

tions on both sides of symmetry planes are shown. Although there are some discrepancies between the predictions and measurements (notably in the first two profiles for PL=25 and 30 mm), these are no larger than the discrepancies between the measured values on either side of the symmetry plane. Both the width and length of the recirculation region are well predicted for all three pitches. The minimum and maximum values of streamwise velocity component u_1 are, in general, well predicted, too. Thus both qualitative and quantitative agreement can be said to be good.

Figure 9.10 (d) shows similar comparisons for a case with PL=25 mm and $Re_D \simeq 45$: similar levels of agreement are obtained. At this Reynolds number, according to both experiment and predictions, the flow is just about to separate. The comparisons for the two other pitches at this Re_D, and for the smallest Reynolds number, were also favourable.

9.2.3 Turbulent flow across staggered tube banks

Description of problem

In this section calculations are reported of fully-developed turbulent cross-flow in tube banks. These are the counterpart of the laminar cases examined in the previous section; however, as noted there, there is a lack of detailed data for turbulent flows.

Two tube bank geometries were considered: one, designated here as TB1, with the dimensions (corresponding to the notation used in Fig.9.7(a)): D=152.4 mm, PL=177.8 mm, PT=203.2 mm , and the other, designated as TB2, with: D=38.1 mm, PL=52.65 mm and PT=60.8 mm . The TB1 case corresponds to that studied experimentally by Neal and Hitchcock (1967). They measured, using hot wire anemometry, the velocity and turbulence intensity distribution at a fixed distance from the tube surface (δ=5.8 mm) in the air flow at a Reynolds number Re_D of 140 000. The TB2 case corresponds to the geometry of tube bank being currently studied by Halim and Turner (1985) and Bruun and Khan (1985). Unfortunately, no results of these experiments were available for comparisons at the time of writing.

Computational results - TB1

The boundary conditions in this case are the same as those discussed in the previous section, with the difference that the flow is now turbulent and wall functions are employed in the treatment of wall boundaries, as explained in section 6 of Chapter 6.

Figure 9.11 (a) shows the grid used for predicting flow in TB1 case along with the basic tube bank dimensions. Based on the experience from laminar flow predictions, it was expected that this 80×40 CV grid would yield almost grid-independent results. For comparison, another grid of 48×22 CV was set up, and the results obtained on two grids using the UDS were examined. The differences between the two solutions were very small (typically less than 1%) as is shown by the wall shear stress predictions in Fig. 9.12 . They were confined to the vicinity of the peak (maximum 4% there) and a portion of the rear part of the tube. The 80x40 CV results were therefore taken as acceptably grid-independent. Calculations were also performed for laminar flows, for reasons which will become apparent later.

The rate of convergence of the numerical method was significantly affected by both the turbulence model inclusion and the explicit treatment of the periodic boundary conditions. Thus, for example, the laminar flow calculations for the TB1 geometry and 80×40 CV grid converged in 224 iterations (for $\alpha_u=0.85$ and $\alpha_p=0.15$) and used 57.8 seconds on a CRAY 1S computer. The turbulent flow calculations on the same grid, on the other hand, reached the same level of normalized residual sums ($\lambda_s=1\cdot10^{-3}$) only after 1624 iterations and used 556 seconds on the same computer (in this case the under-relaxation factors $\alpha_u=0.75$, $\alpha_p=0.2$ and $\alpha_{k\varepsilon}=0.7$ were used). One of the reasons for slower convergence in the turbulent case is the lower values of the under-relaxation factors (these were not optimized, but simply conservatively specified to be lower than in the laminar case). Another reason could be the fact that the explicit periodic boundary conditions were imposed on two more dependent variables, namely k and ε .

Figures 9.11 (b)-(d) contain the usual plots of selected features of the predicted flow field, indicating contours of constant pressure (normalized with ρu_c^2) and streamlines. The pressure contours, it should be noted, exhibit "kinks" close to the grid lines passing through the two stagnation points; those are due to the sudden change in the direction of the x^1 grid lines here, as explained earlier in Chapter 8. Although undesirable these "kinks" are not too dramatic, and they could have been reduced, if not suppressed entirely, by smoothing the discontinuity, as discussed in the previous chapter.

Contours of turbulence intensity, defined here as $\sqrt{k}/u_c$, are plotted in Fig. 9.11 (e). They show that the highest levels occur near the front stagnation point, as the oncoming flow impinges onto the tube.

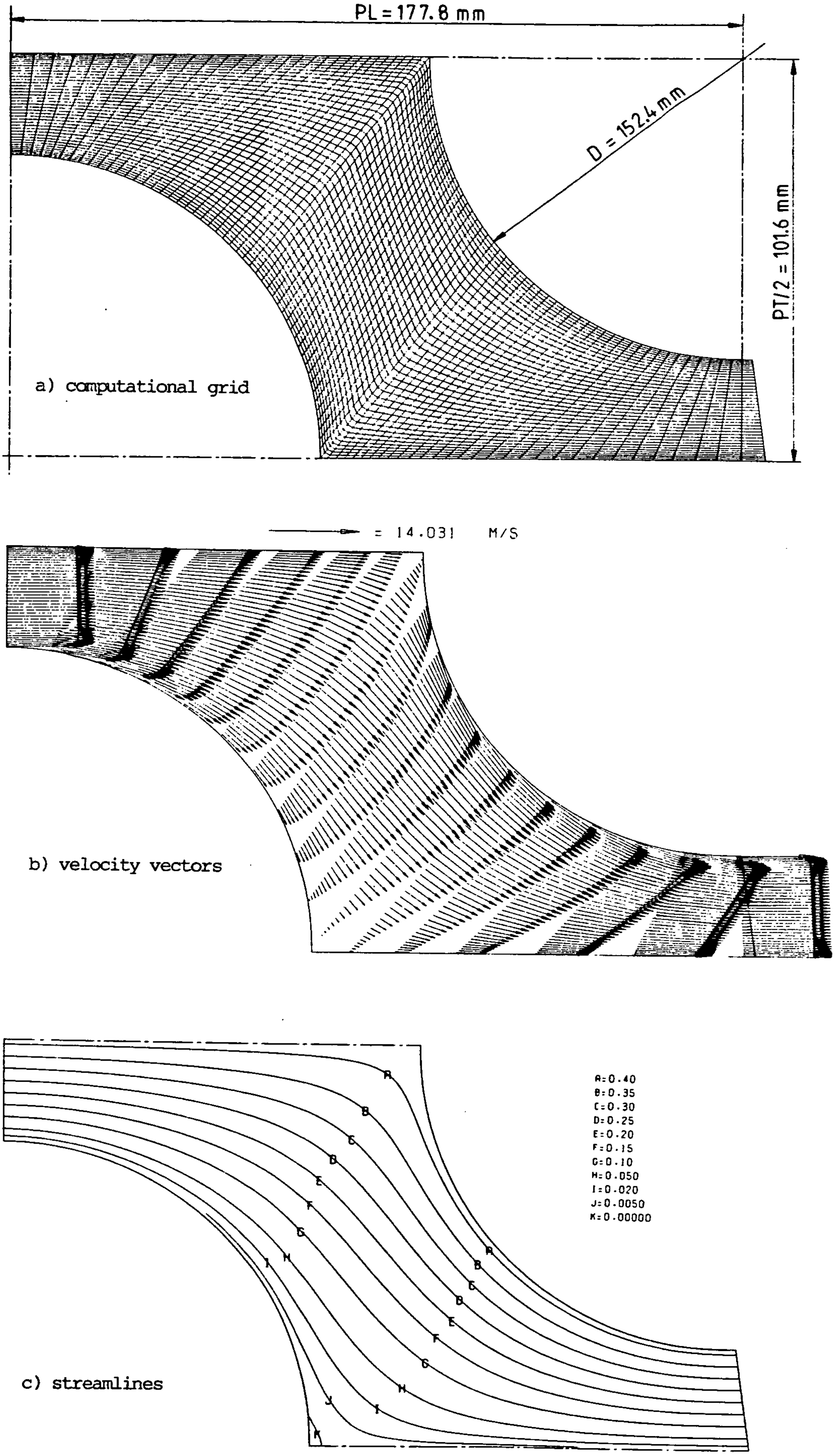


Fig. 9.11: Details of predicted turbulent flow in staggered tube bank at Re_D=140000.

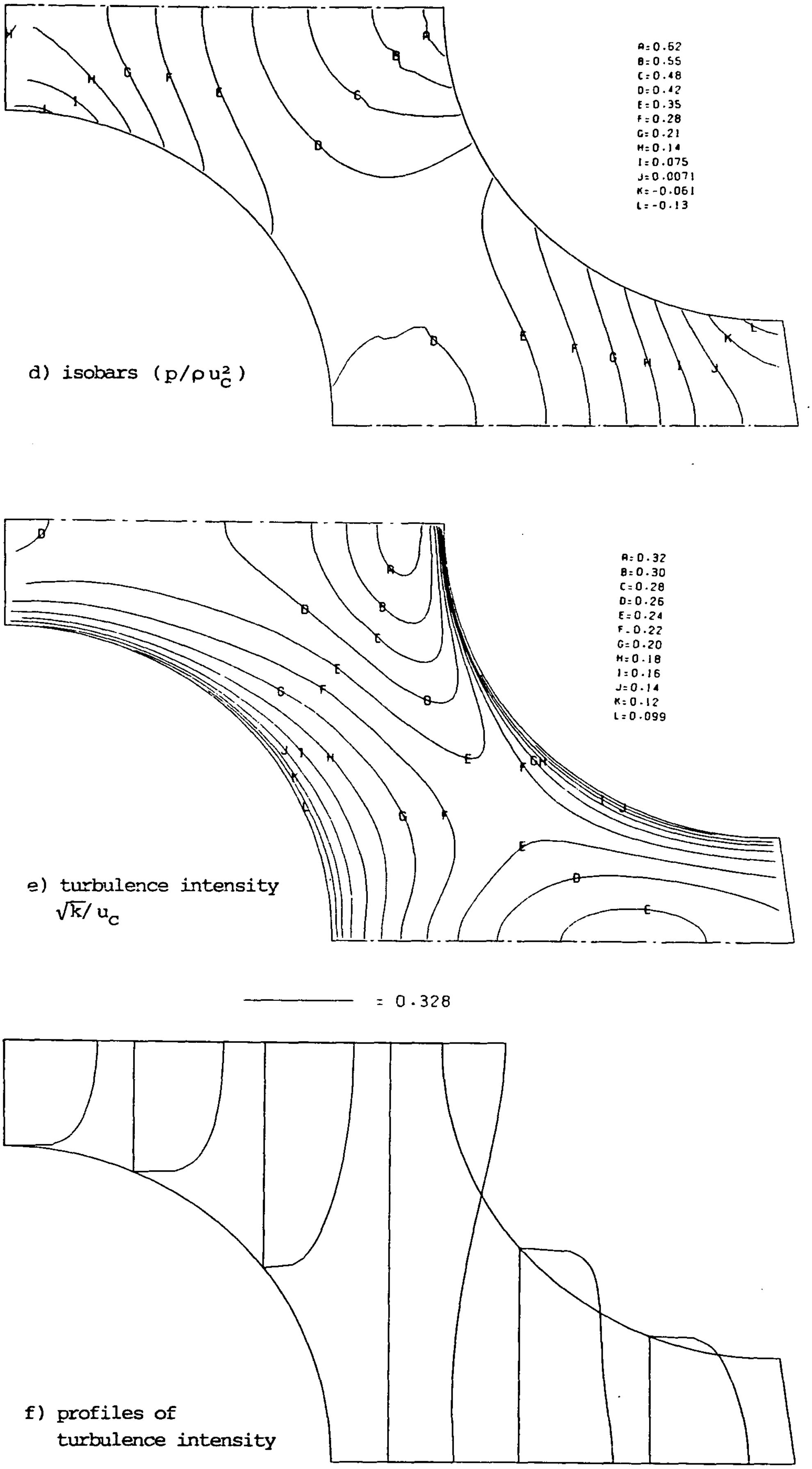
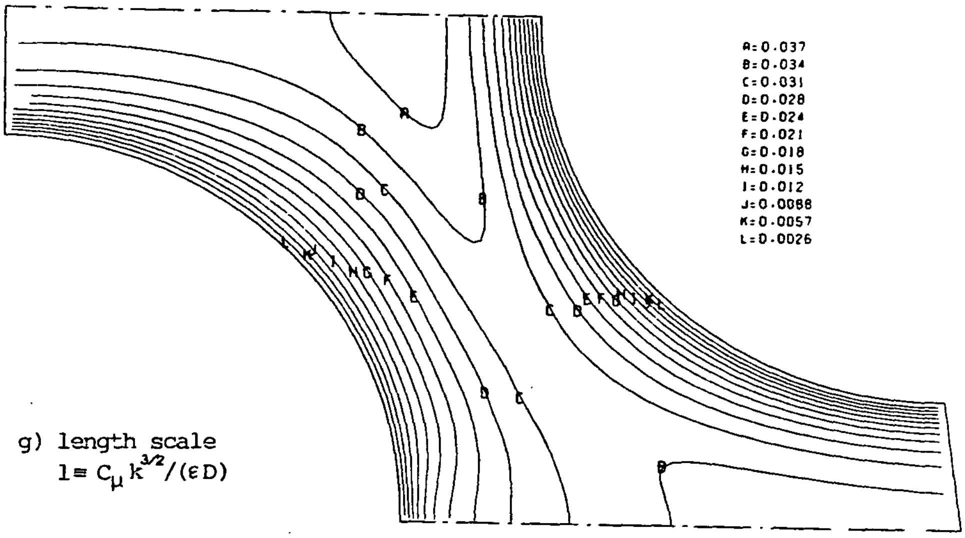
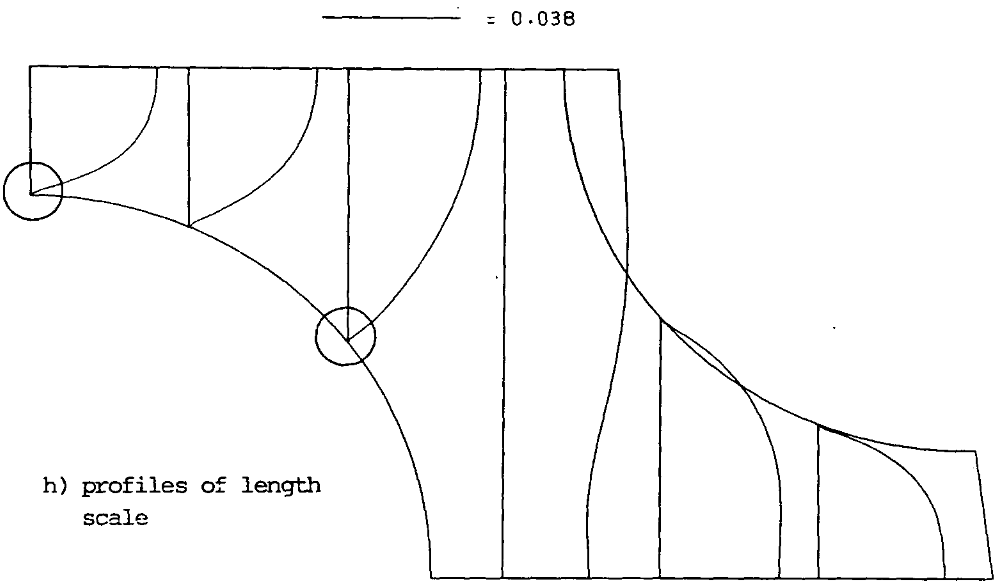


Fig. 9.11: continued



g) length scale
$l \equiv C_\mu k^{3/2}/(\varepsilon D)$



h) profiles of length scale

magnified details of length scale profiles

Fig. 9.11: continued

Another area of high turbulence is in the centre of a tube wake, close to the minimum gap between the downstream tubes. Also evident are steep gradients near the tube wall, as seen in both the contour plot and the profiles shown in Fig. 9.15 (f).

Contours of the dimensionless turbulence length scale l, defined as $l = C_\mu k^{3/2}/(\varepsilon D)$, are presented in Fig. 9.11 (g), and some profiles are plotted in Fig. 9.11 (h). They show the expected monotonic increase with distance from the wall, and reach maximum values near the front stagnation point. However, the slope of the l profiles is discontinuous at the node nearest to the wall, as is shown by the magnified inset in Fig. 9.11(h). This slope was imposed there by fixing ε through Eq.(6.29), which is, it will be recalled, strictly valid only for Couette flow (and other simple flows, eg. in pipes). Evidently this does not match the slope resulting from the transport equation for ε at nodes further away from the wall, which indicates inadequacy of the above treatment in complex flows.

From the contour plots presented here it is evident that the non-smoothness of grid lines has no visible effects on quantities other than pressure.

Comparison with experiment

Figure 9.14 shows the distributions of the velocity magnitude v (which is the quantity measured) and local turbulence intensity (here defined* as $\sqrt{\frac{2}{3}k}/v$) along a radius located 5.8 mm from the tube surface. The plots shown are from three sources: the experimental results of Neal and Hitchcock (1967), the predictions of Demirdžić (1982) and the present predictions. Although the qualitative trends are well reproduced by both calculation methods, there are significant disagreements with the experimental data. The velocity peak at about $\theta = 90°$ (where θ is the circumferential angle measured from the front stagnation point) is slightly underpredicted by both methods, while elsewhere differences of up to 15% of the peak value exist. The largest discrepancies exist in the turbulence intensity results, in particular near the two stagnation points, where calculated values are in excess of 100% . Near the rear stagnation point the high levels may be a consequence of the underprediction of the velocity there. However, near the front stagnation

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* On the assumption of isotropy, which may be in error for the near-wall locations considered, especially near stagnation points.

point the predicted and measured velocity magnitudes are in good agreement, and so the differences in the turbulence intensity must be due either to overprediction of k (and also possibly the isotropy assumption made in deducing the turbulence intensity from this quantity), or measurement errors in this region, or both.

There are also discrepancies between the two predictions, but these are much smaller and are limited to the areas near the stagnation points: these are probably due to the coarser (39 x 28 CV) grid employed by Demirdžić, and the greater sensitivity of his method to grid smoothness.

Of particular interest is the evidence about the existence and extent of the separation zone. The experimental heat transfer measurements (not shown) suggest that the separation may have occured at $\theta = 150°$, although there is no direct and incontrovertible evidence of this. The present predictions show it to occur at about $\theta = 173°$ (cf Fig. 9.12). In this connection it is noteworthy that the shape of the predicted shear stress profile (shown in Fig. 9.12) changes for the turbulent flow case on the rear part of the tube in a manner not seen in either the laminar flow predictions (also shown there) nor in the turbulent flow measurements of Aachenbach (1969). If the trend which the predicted τ_w profile has up to $\theta = 130°$ is continued as indicated by the dashed line then it would change sign at about $\theta \simeq 150°$, what is in line with the experimental observations. Also the pressure distribution around the tube is anomalous here (Fig. 9.13), for the gradient suddenly increases at about $\theta = 150°$; had the flow separated there, the profile would have followed the dashed line drawn.

One possible explanation for the unexpected behaviour of τ_w and p is inadequacy of wall functions, and perhaps the k-ε model itself, in this part of flow. Firstly, strong streamwise gradients exist in all quantities in this region, so that the Couette flow assumption (upon which the wall functions are based) is greatly in error. Secondly, the velocity becomes very small near separation, so that the high Reynolds number assumption underlying the version of the k-ε model used here may be inapproprate (Jones and Launder, 1972). In support of this, the calculated y^+ values at near-wall nodes for $\theta > 130°$ were lower than the suggested minimum value (~30) for the applicability of the wall functions and high Reynolds number k-ε model (Launder and Spalding, 1974). This is unavoidable when recirculating flows are in question.

Figure 9.12 also shows that for laminar flow at $Re_D = 508$ (which is equivalent to the mean "effective" Reynolds number in turbulent flow)

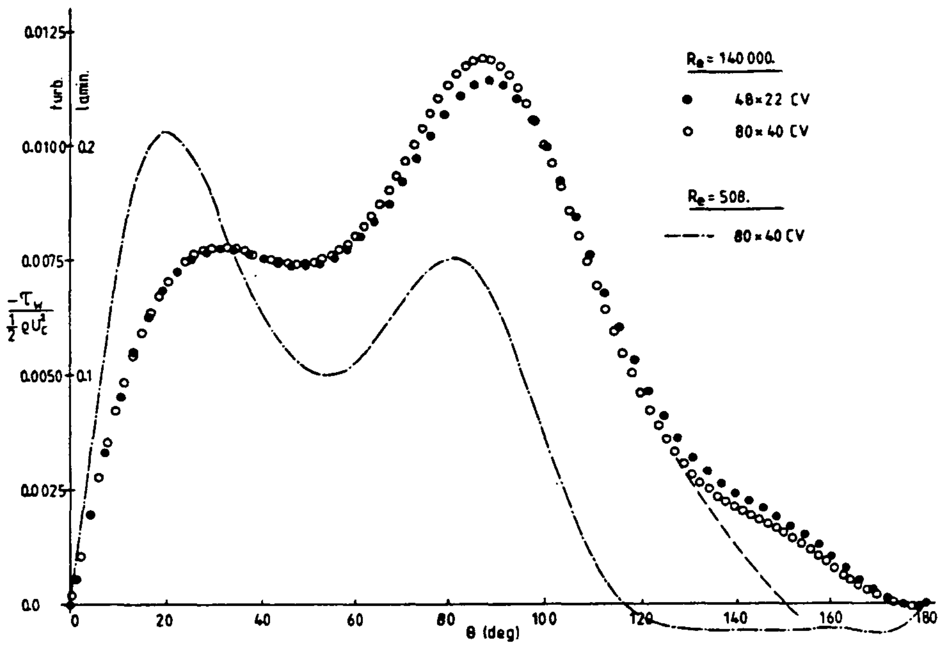


Fig. 9.12: Shear stress distribution around half a tube

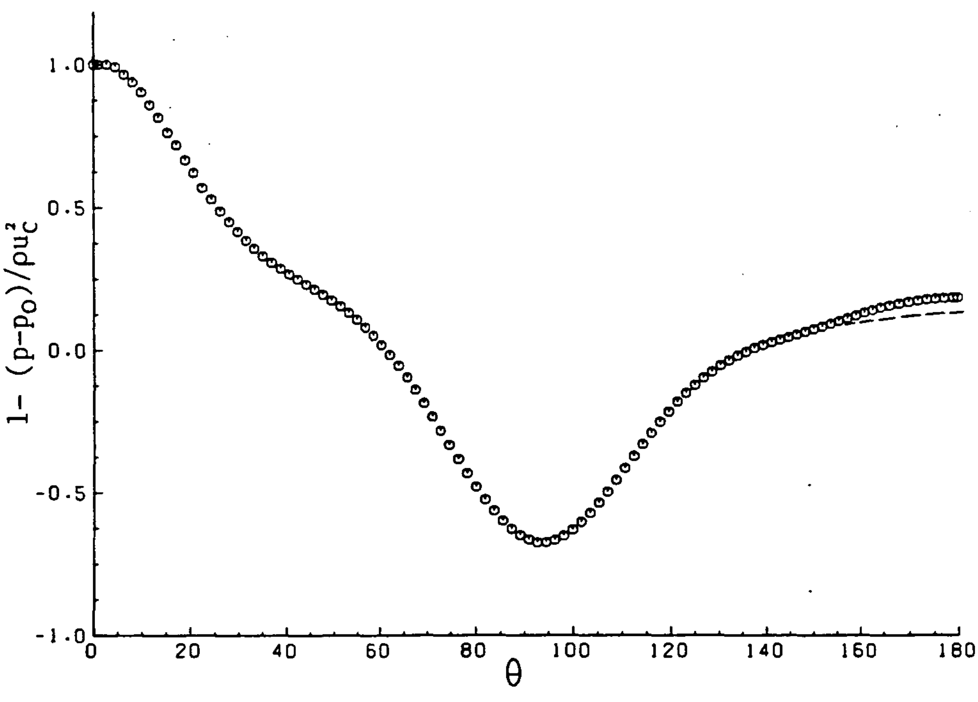


Fig. 9.13: Pressure distribution around half a tube

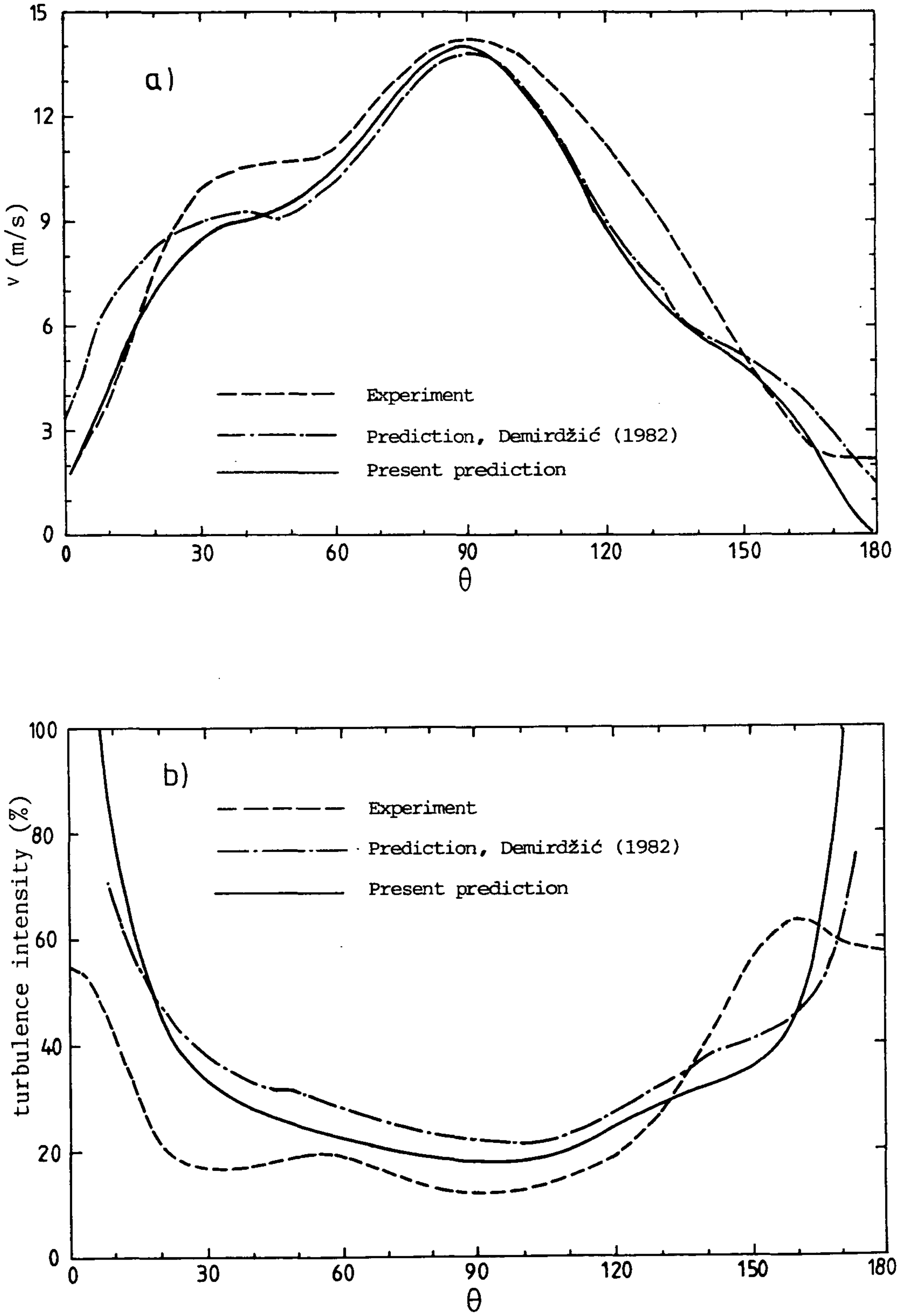
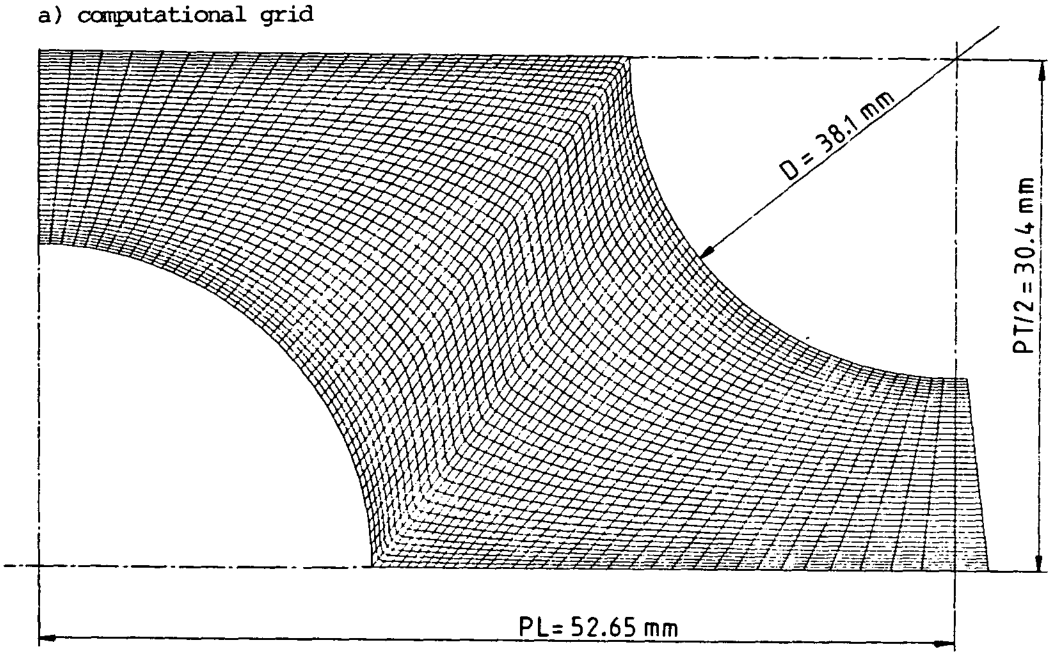


Fig. 9.14: Comparisons of predicted and measured velocity (a) and turbulence intensity (b) around half a tube, at a distance δ=5.8 mm

a) computational grid



b) velocity vectors

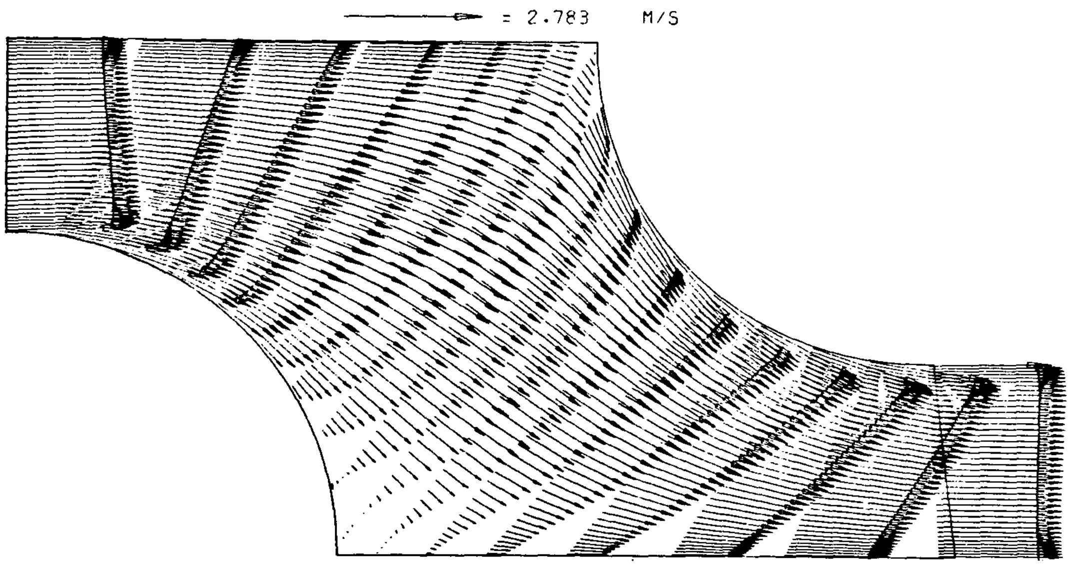
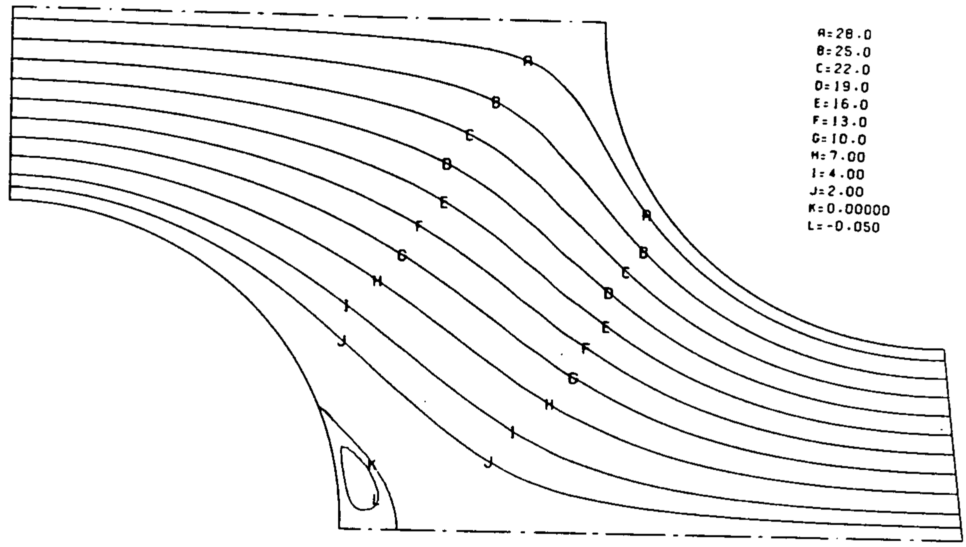


Fig. 9.15: Details of predicted flow in staggered tube bank at Re_D=100000 (TB2 case)

c) streamlines



d) turbulence intensity $\sqrt{k}/u_c$

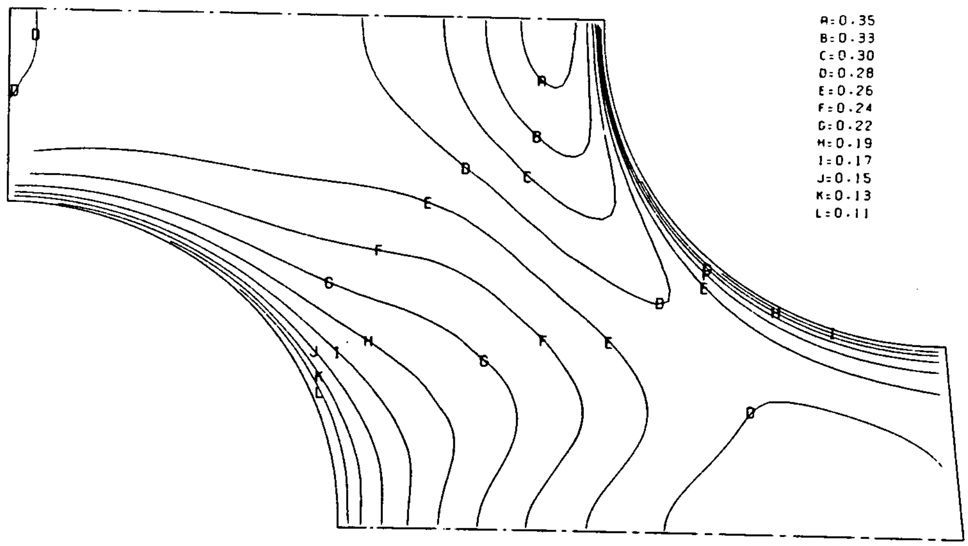


Fig. 9.15: Continued

separation occurs at about $\theta \simeq 116°$, and due to the reduced flow area the maximum shear now occurs at about $\theta \simeq 20°$; the second peak at about $\theta = 85°$ is not as large as the first one. When this is compared with the variation in Fig. 9.8 , the influence of the longitudinal pitch becomes obvious.

Results - TB2

Figure 9.15 (a) shows the grid employed in the TB2 case. Here the ratios PL/D and PT/D are both somewhat larger than for the TB1 case (1.38 and 1.6 as compared with 1.17 and 1.33 respectively). Figures 9.15 (b)-(d) show the predicted velocity vector field, streamlines and turbulence intensity ($\sqrt{k}/u_c$) contours, for Re_D=100 000 . As a consequence of the increased spacing a significantly larger recirculation zone is predicted than in the TB1 case, with the separation occurring at about $\theta = 153°$. Although larger, its size and strength may still have been underestimated for the reasons explained earlier. However, this will not be known until the experimental data become available. Otherwise, the results are similar to those of the previous case, apart from the fact that now somewhat higher turbulence levels are achieved near the front stagnation point.

9.2.4 Turbulent flow around bluff body confined in a pipe

Description of problem

The problem considered here is turbulent flow around a conical bluff body confined by a long circular pipe, as illustrated in Fig. 9.16 . Taylor (1981) and Taylor and Whitelaw (1984) presented the results of an experimental study of this flow. Their data is used here to assess the performance of the present method in predicting it.

Computational details

Figure 9.16 shows the geometry of the solution domain and its dimensions. The manner of grid generation for this problem was described earlier in section 3 of Chapter 7. The grid used here (Fig. 9.17 (a)) had 82 x 34 CV and extended over the cone and central rode holding it, the assembly being treated as a submerged body. In the absence of detailed inlet data, particularly of the turbulence dissipation rate ε , the practice adopted here was to locate the inlet boundary sufficiently far up-

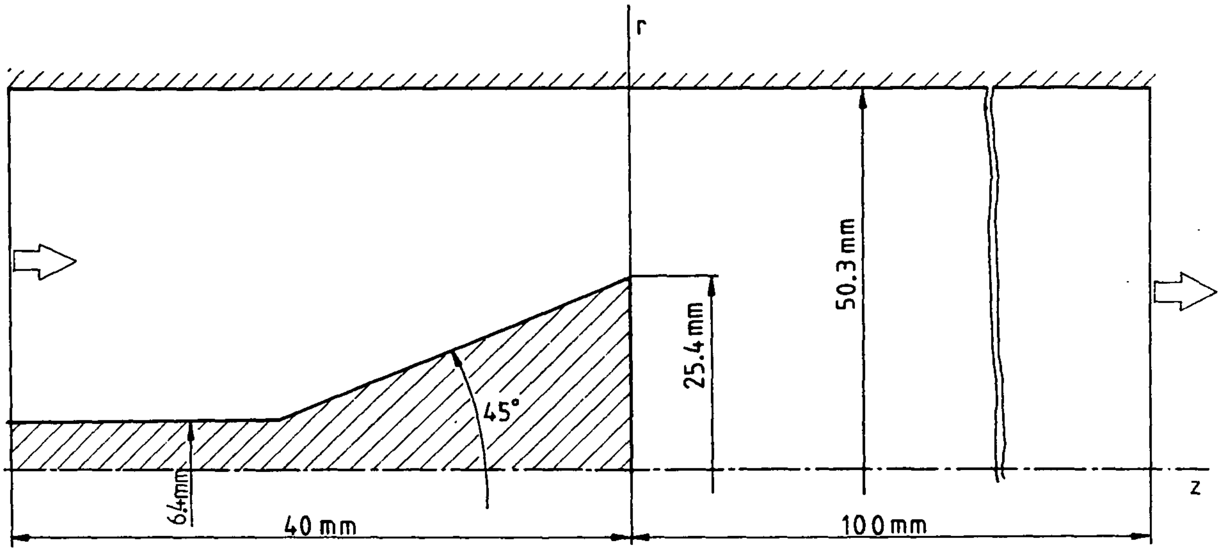


Fig. 9.16: Geometry of the solution domain for flow around bluff body confined in a pipe

stream of the cone for the assumption of fully-developed flow to be plausible. The fully-developed conditions were generated in a separate simulation of the annulus flow. It was expected that any inaccuracy in the inlet boundary conditions would not significantly influence the wake region, which is of primary interest here.

The following boundary conditions were thus applied:

- at the inlet: fully-developed annulus flow;
- at the outlet: zero axial gradients;
- at the walls and symmetry axis: standard conditions, as described in Chapter 6.

Only one grid was employed, due to a lack of time to perform refinement tests. The "grid dependnece" of the results was examined by comparing predictions obtained using the LUDS and UDS schemes: the differences were relatively small (<2%), and confined to the recirculation region. From this it was concluded that the results obtained with the LUDS are not appreciably affected by numerical diffusion, which can be attributed to the facts that reasonable alignment between grid lines and streamlines was achieved, and the grid was fine in regions of high gradients. The calculations took 240 iterations to converge, using under-relaxation factors $\alpha_u=0.7$, $\alpha_p=0.2$, and consumed 91 seconds on a CRAY 1S computer.

Results and comparison with experiments

Figures 9.17 (b)-(j) present typical features of this flow, including velocity vectors and profiles and contours of other quantities of interest. These plots are self-explanatory; noteworthy again is the fact that the length-scale profiles have irregular shapes near the pipe wall above the recirculation region indicating deficiencies in the wall functions. Further upstream and downstream the length scale increases linearly with distance from the wall, as expected. The reasons for this anomaly were discussed in the previous section.

In figures 9.18 (a)-(f) the predicted profiles of the axial velocity component at several cross-sections downstream from cone face are presented and compared with the experimental results. It is evident from these that the agreement varies from one cross-section to another, but it is undoubtly good close to the cone (ie. at $z=0.5$ mm) and poor far downstream (at $z=70$ mm). In Fig. 9.18 (g) the centerline variation of axial velocity component is presented: evidently, the predicted velocity initially decreases faster, and then increases slower, than the experiment shows. The explanation for these discrepancies will be offered shortly.

In Fig. 9.19 (a)-(d) the predicted profiles of radial velocity are shown and compared with the experimental data. The agreement is good only in the upper part of cross-section close to the cone (at $z=0.5$ mm); closer to the symmetry axis the predicted values are here significantly higher than the measured ones. Further downstream, at $z=15$, 35 and 70 mm the agreement is even poorer. The predicted radial velocity is always negative, indicating that the diameter of the recirculation bubble is decreasing in this region, and the flow is directed towards the symmetry axis. The experimental results (above the symmetry axis), on the other hand, show a somewhat different behaviour. In particular, they exhibit positive radial velocities in the outer part of pipe radius at $z=35$ and 70 mm, indicating that flow there is directed towards the pipe wall. This is difficult to appreciate considering that both the predicted and measured axial velocity components are reducing in this region and increasing near the symmetry axis, suggesting that the radial velocity component should be negative in these cross-sections.

The reasons for these discrepancies may lie in part in the fact that the measured flow was not completely symmetric. This can be concluded from axial profiles of Fig. 9.18 , where values measured along a diameter on either side of symmetry axis are shown. It is perhaps even more

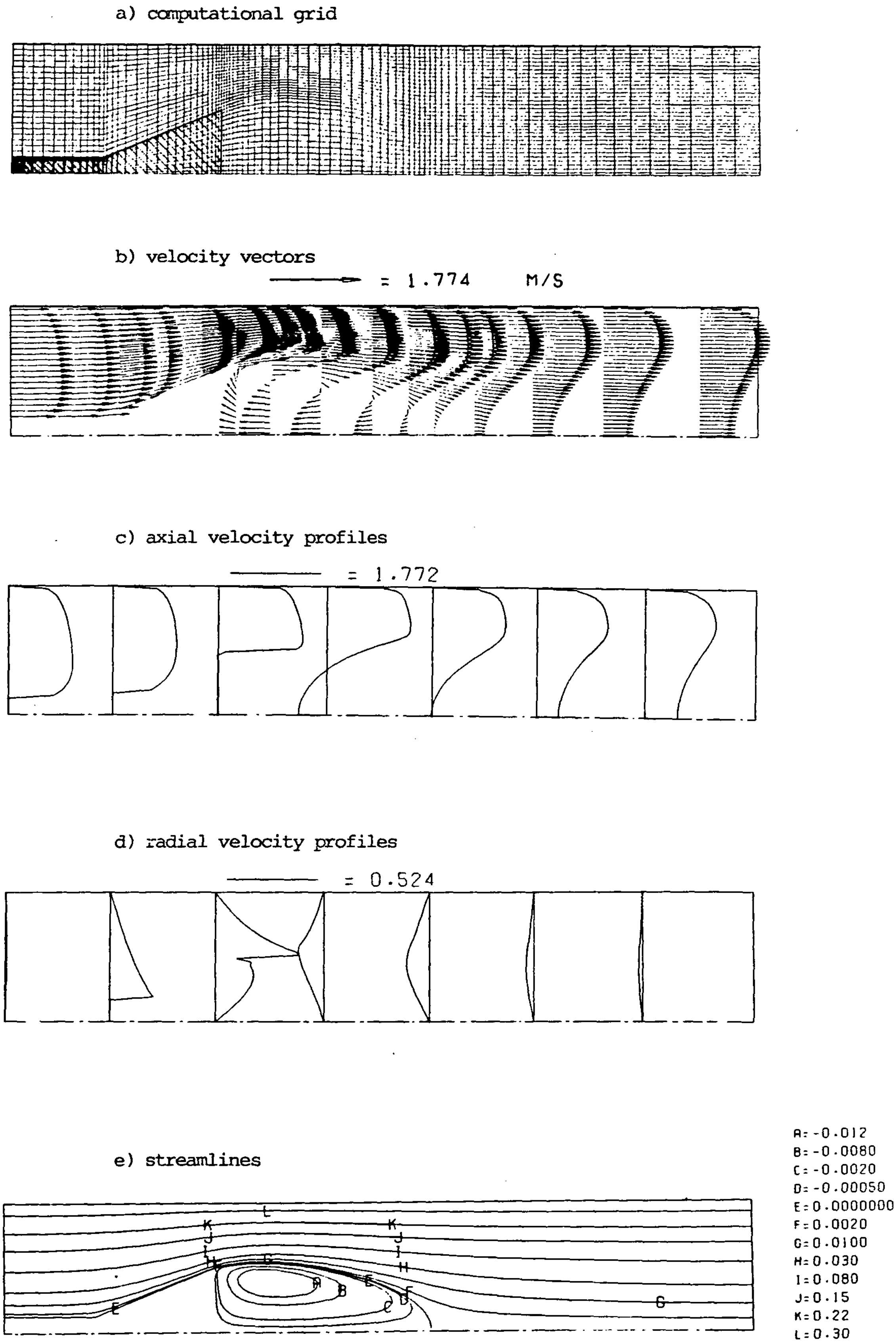
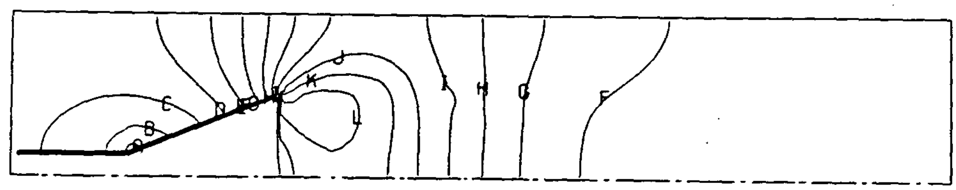
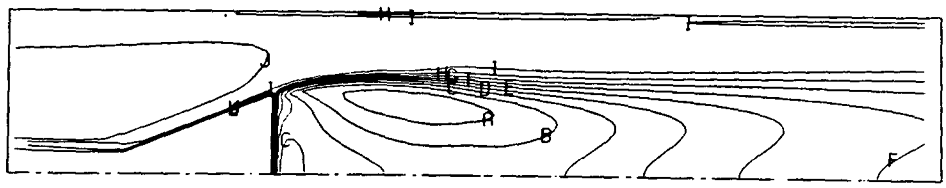


Fig. 9.17: Details of predicted turbulent flow around bluff body confined in a pipe

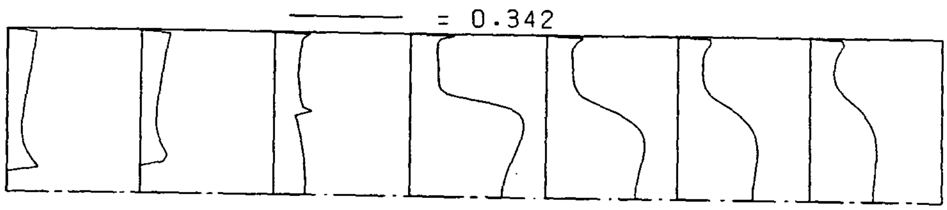
f) isobars ($p/\rho u_{in}^2$)



g) turbulence intensity ($\sqrt{k}/u_{in}$)



h) profiles of turbulence intensity



i) length scale ($C_\mu k^{3/2}/\varepsilon D$)

j) profiles of length scale

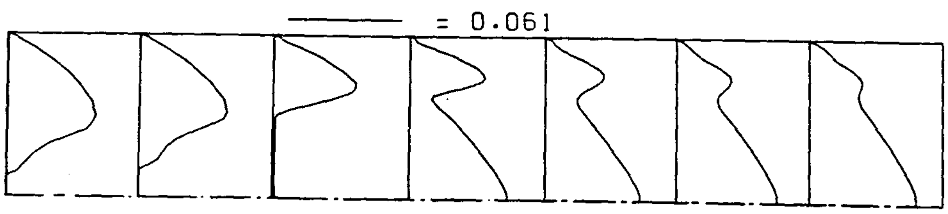


Fig. 9.17: continued

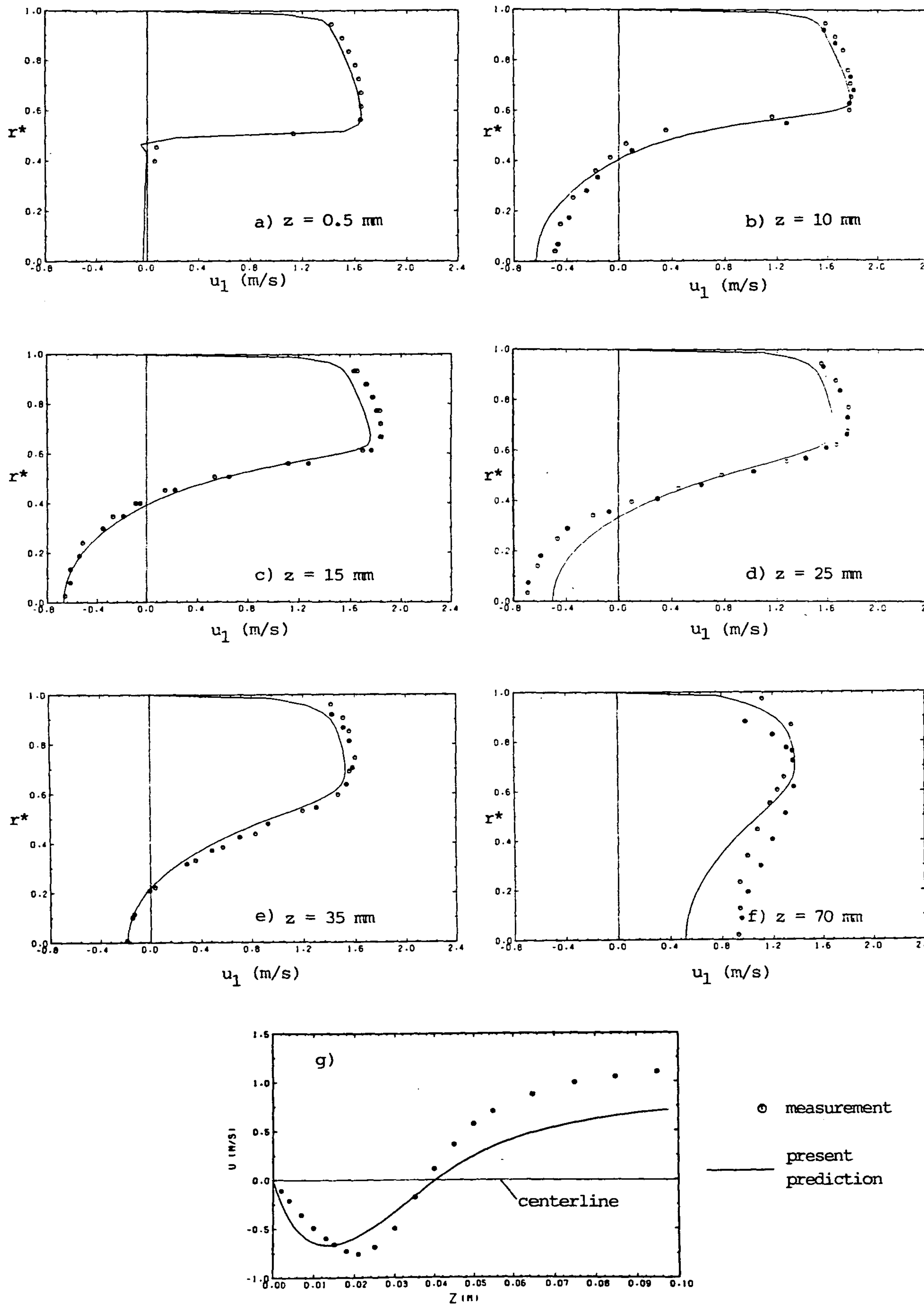
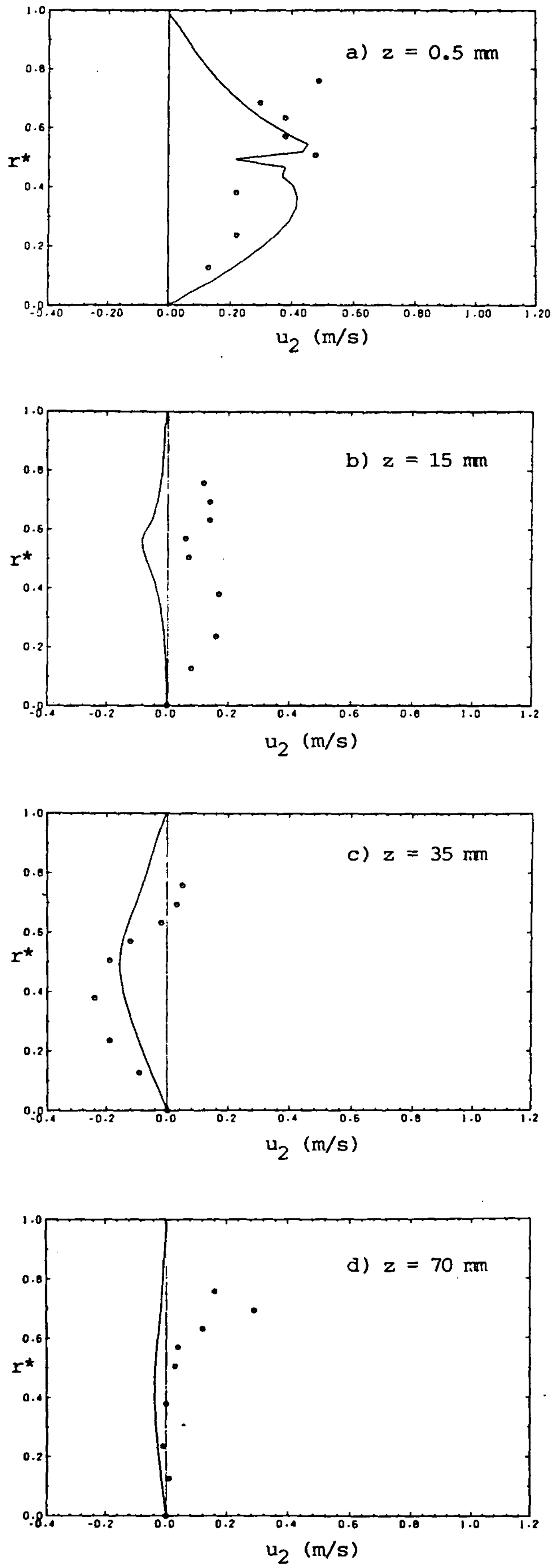
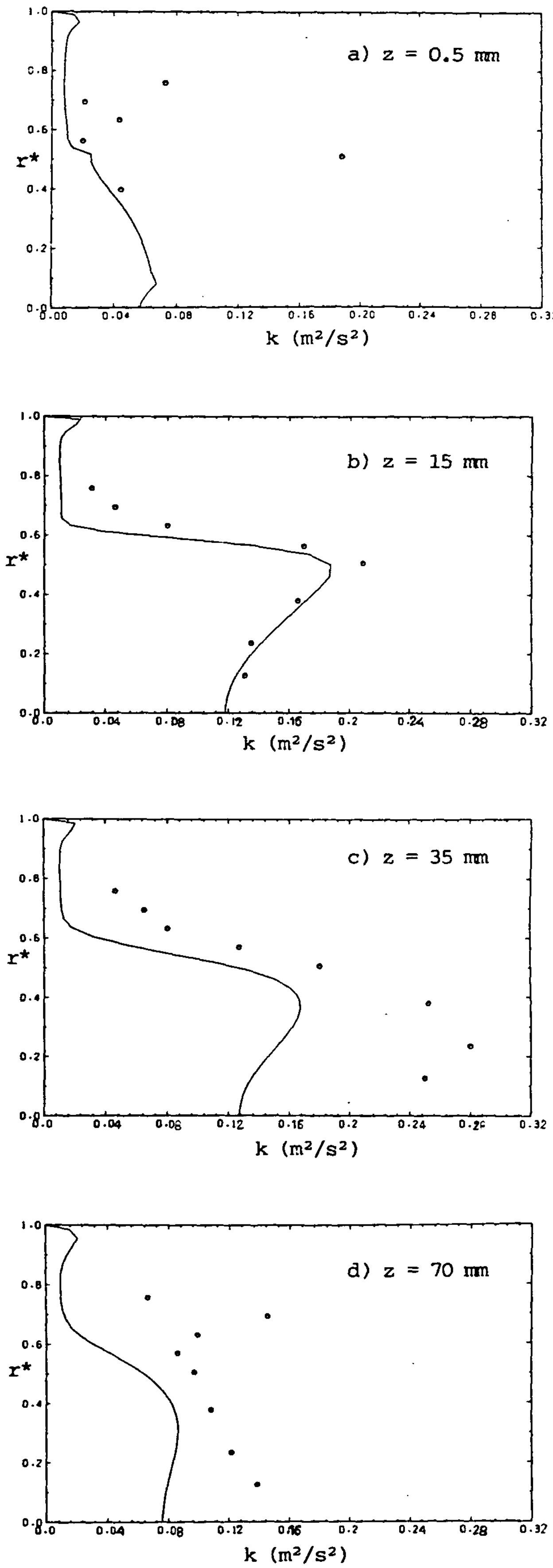


Fig. 9.18: Profiles of axial velocity component (open and full symbols represent measurements taken on opposite sides of the symmetry axis)





o measurement; ———— present prediction

Fig. 9.19: Profiles of radial velocity component

Fig. 9.20: Profiles of turbulence kinetic energy

obvious from the loci of the separation streamline, shown in Fig. 9.21 for both upper and lower part of the pipe cross-section. The experimental results show clear asymmetry, as well as indicating that in the upper half of this plane through, where the radial velocities were measured, the behaviour of the separation streamline is irregular. The shape in the lower parts is more regular, which suggests that the radial velocity profiles there (which were not measured) would have been different from those presented earlier, and qualitatively at least in closer accord with the present calculations. The latter, it should be noted, give a streamline locus which agrees reasonably wel with the "average" measured one.

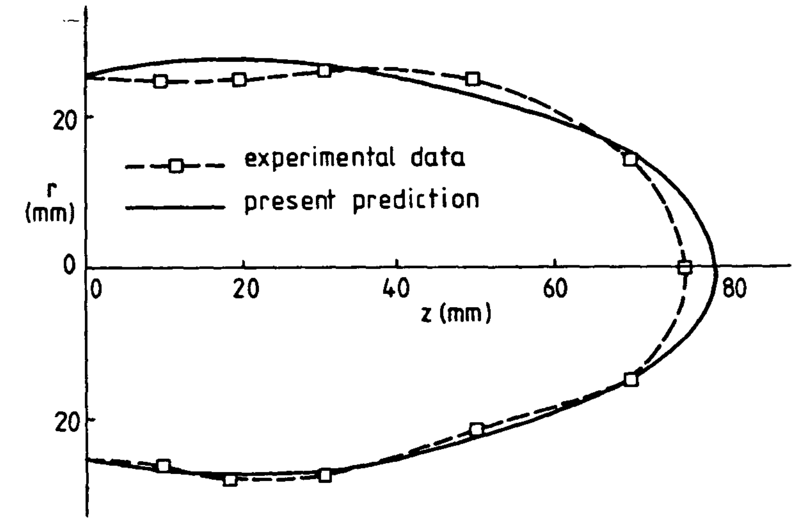


Fig. 9.21: Recirculation bubble behind the cone

In figure 9.20 profiles of turbulent kinetic energy at several cross-sections are shown and compared with experimental data* . The agreement is relatively good at z=15 mm, but deteriorates further downstream, and at z=35 and 70 mm the predicted levels are about 40% lower than the measured ones. This kind of disagreement has also been observed by McGuirk et al (1982) in their predictions of similar flow behind a disk of the same blockage ratio as in the present case with a cone. They also observed similar discrepancies in the centerline development of axial velocity to those in Fig. 9.18 (g). They claim to have reduced numerical

* Because turbulence levels in three directions were not measured at the same radial locations, some interpolation was necessary to calculate the k from them.

diffusion to insignificant levels. Such claims cannot be made here, although it is believed that it was low, for reasons mentioned earlier. The reasons for the failure to correctly predict the rate of flow recovery downstream from recirculation region lie almost entirely in the inadequacy of the $k-\varepsilon$ turbulence model, in its standard form, in such flows (McGuirk et al, 1982; Bradshaw, 1973).

The most important finding of this exercise is that the level of agreement with experimental data in the present case is comparable to that achieved by McGuik et al (1982) whose calculations were, however, confined to the region downstream from the disk, using the experimental data to define the inlet conditions. Also noteworthy is the fact that in the present case the predicted profiles at the trailing edge of a cone, which would be the inlet boundary for studies like that of McGuirk et al in which simple polar-cylindrical grids are used, were in good agreement with the experimental data there.

9.3 Applications to Three-Dimensional Flows

9.3.1 Laminar flow in a S-shaped diffuser

Definition of problem

This section describes the application of the present method to laminar flow in a S-shaped diffuser with rectangular cross-section (Fig. 9.22), which has been experimentally studied by Rojas et al (1983). The curvature of the duct results in cross-flow pressure gradients, and hence in relatively strong secondary motions. This pressure-induced secondary flow is much stronger than that induced by turbulence anisotropy, as found in straight ducts (Rojas et al, 1983). The magnitude of secondary velocity components in the case studied here can amount up to 40% of the component in the principal* flow direction.

Figure 9.22 defines the geometry of the diffuser studied by Rojas et al (1983). It consists of two sections, each comprising a 22.5° bend with the mean curvature radius of 0.28 m. The top and bottom duct walls

* Because this flow has a predominant direction, the so-called "parabolic" or "partially parabolic" solution methods (eg. those of Vanka et al, 1980; Rhie, 1983 and Rhie et al, 1984) could also be used for its prediction.

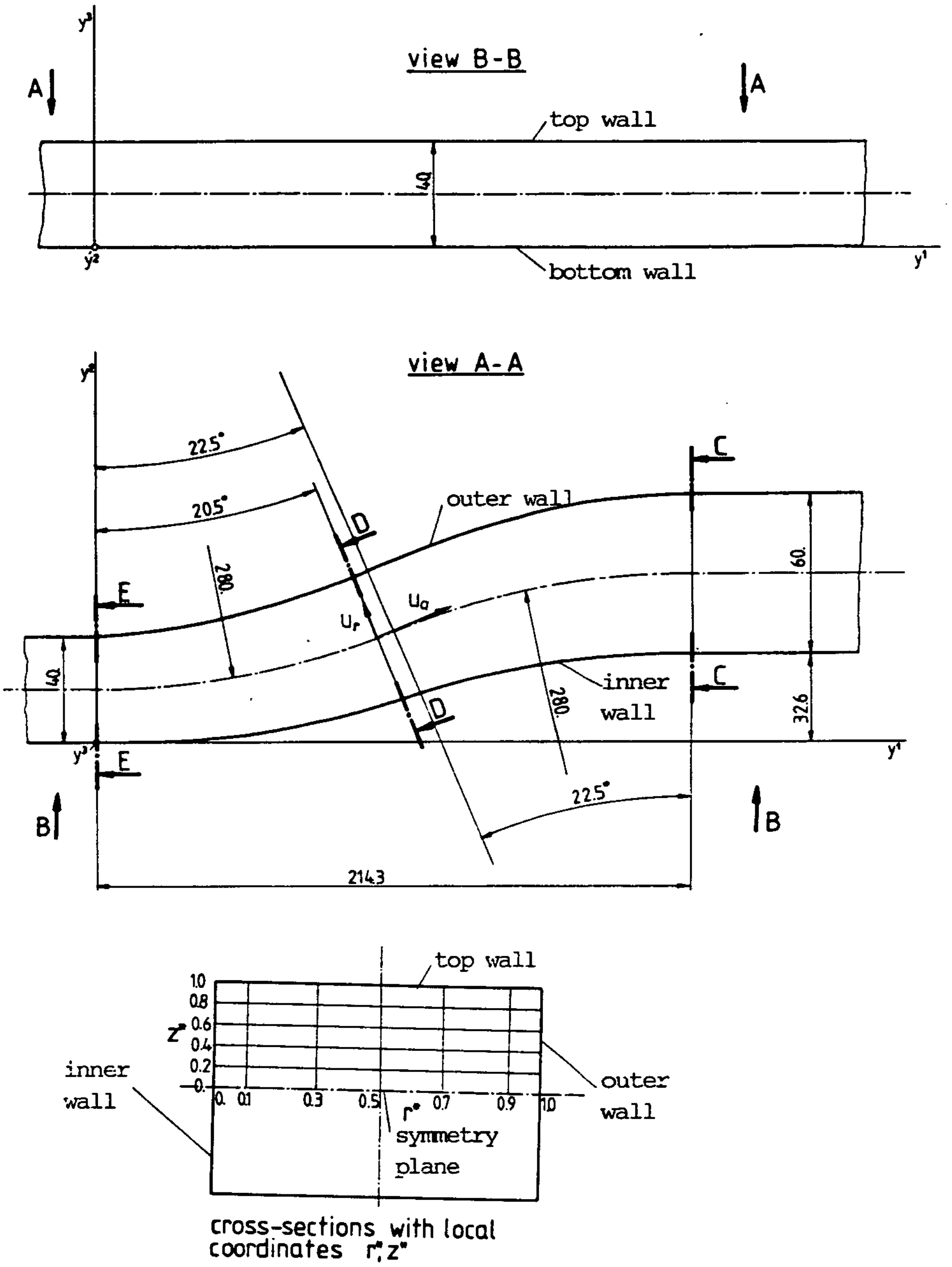


Fig. 9.22: S-diffuser geometry

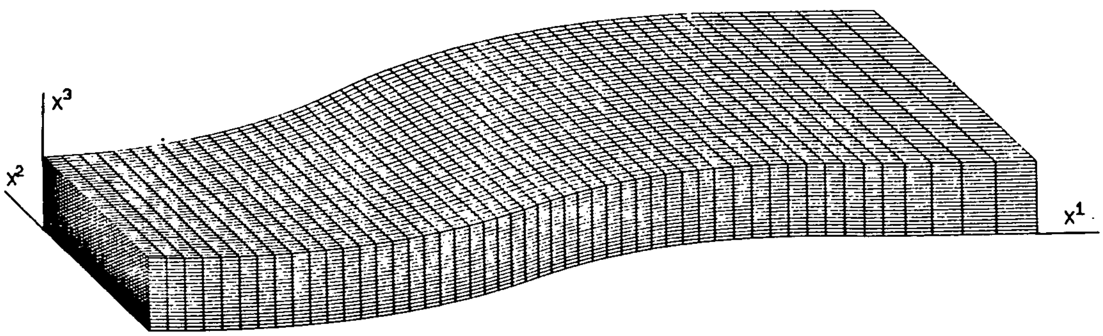


Fig. 9.23: Computational grid (50x40x20 CV)

are parallel, at a distance of 0.04 m. The side walls depart from each other linearly with downstream distance, starting at the square inlet section 0.04 m, and ending at a spacing of 0.06 m at the exit from the second bend. Thus the ratio of exit to inlet area is 1.5.

One of grids used is shown in Fig. 9.23. It was extended over an additional section of straight duct after the exit from the second bend, and only one (upper) half of the diffuser was considered, due to the existence of a horizontal symmetry plane. In the latter discussions the following terminology will be used (see Fig. 9.22):

- inner wall, denoting the side wall corresponding to the larger radius of the first bend and smaller radius of the second bend;
- outer wall, denoting the opposite side wall;
- top wall, denoting the wall parallel to the symmetry plane.

The velocity component in the direction of tangent to the mean radius of the bend curvature will be termed "streamwise", the component in radial direction - "radial", and the component in the direction normal to the symmetry plane - "spanwise".

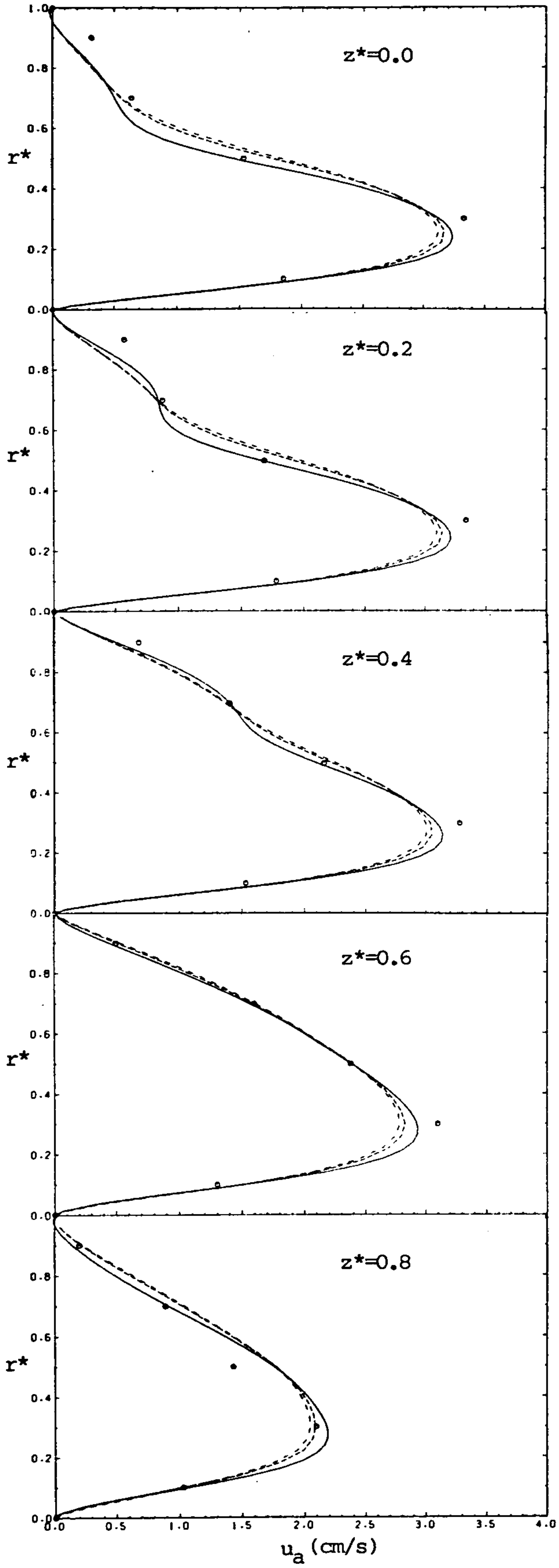
The values of the velocity components u_a (streamwise) and u_r (radial) at the inlet nodes were obtained by interpolation from experimental data, which is only given at five vertical cross-sections ($r^*=0.1$, 0.3, 0.5, 0.7 and 0.9, see Fig. 9.22), and 12 locations along each section (at $z^*=0$, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.925 and 0.95). The relative coarsness of this resolution may be one of sources of inaccuracies in the comparisons to be shown later, for in estimating the inlet grid values of these quantities, some interpolation errors were inevitable. An additional uncertainity was introduced by the fact that no experimental information about the inlet values of u_s (spanwise) velocity component was available, although it was said to be small compared to other two components (Rojas et al, 1983), and was here assumed zero. The other boundaries were treated in the usual way, as described in Chapter 6.

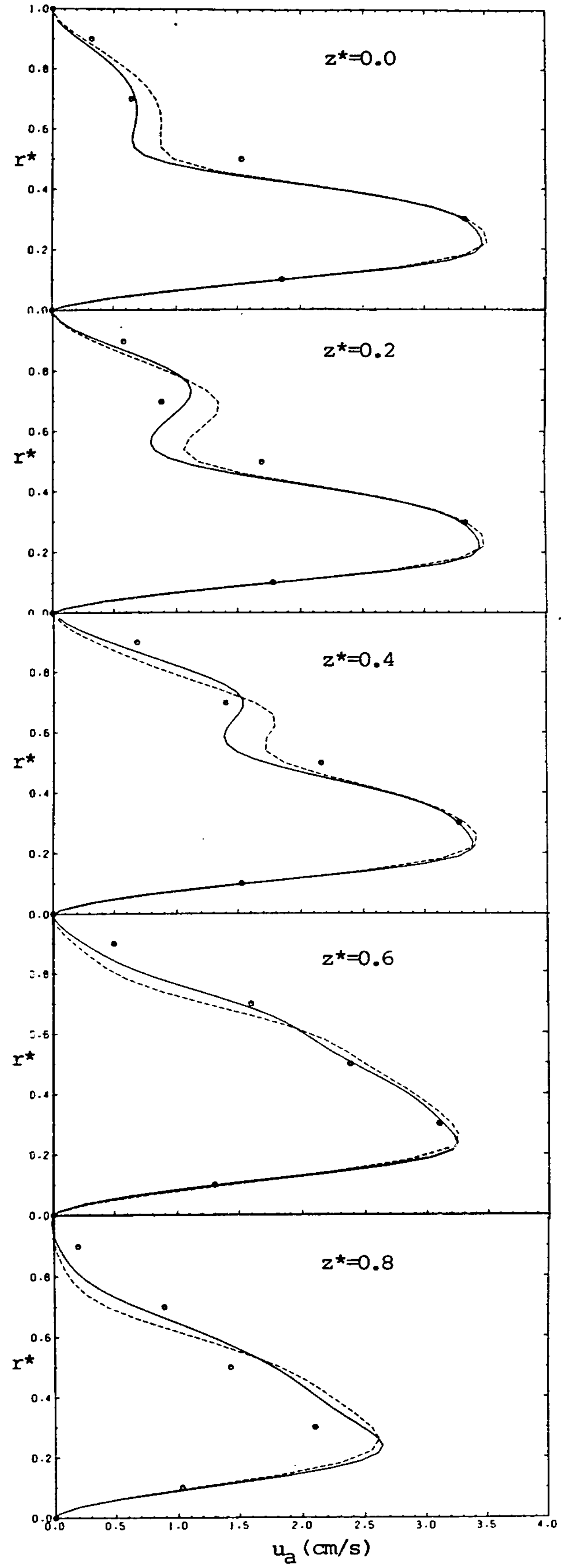
Computational results

Although a nearly orthogonal grid could easily have been generated here, it was intentionally constructed to be non-orthogonal, in order to fully test the present method. Calculations were performed for flow at a Reynolds number Re=790: this is defined by the inlet hydraulic diameter $D_H=0.04$ m and the mean inlet velocity $u_{in}=0.0198$ m/s. Three grids were used: one with $25 \times 27 \times 11$ CV in x^1, x^2 and x^3 directions respectively; one refined in the x^1-direction with $50 \times 27 \times 11$ CV , and one further refined in the x^2- and x^3-directions, with $50 \times 40 \times 20$ CV . Figure 9.24 shows a comparison of predicted (using the UDS scheme) u_a velocity profiles at the exit from the second bend in five horizontal cross-sections (cross-section C-C , at $z^*=0$, 0.2, 0.4, 0.6 and 0.8, see Fig. 9.22), for the three different grids. The differences between the results obtained on the first and the second grids are very small, suggesting that the gradients in x^1-direction are relatively well resolved, and that further grid refinement in that direction is not necessary. However, the profiles obtained on the third grid significantly differ from the forementioned ones, indicating that the numerical errors are still significant and that a grid-independent solution has not been yet reached. Further refinement of grid in x^2- and x^3-directions is evidently necessary, but was not possible due to limited computer resources.

Figure 9.25 shows the same type of comparison when the LUDS is used, for two grids: $25 \times 27 \times 11$ CV and $50 \times 40 \times 20$ CV . Again, there are substantial differences between the solutions especially for $r^*>0.5$. These results also show a second maximum in the profiles for $z^*<0.5$, which is not present in the UDS solution (although the trend with grid refinement indicates that it may form), and whose existence can not be assessed by reference to measurements, also shown, because they are too widely spaced in the radial direction. The largest differences between the two solutions occur in this region, suggesting that the second maximum may have been numerically generated or exaggerated. The reason for these suspicions is that the Peclet numbers in the x^1- and x^2-direction were high (of the order 10^2 and 10^1 respectively) in this region even with the finest grid employed, and at these levels, as shown earlier in Chapter 4, the LUDS can produce numerical oscillations. This suspicion is reinforced by the observation that the position of the second maximum shifts significantly as the grid is refined.

The obvious course of blending the UDS and LUDS, as described in section 6 of Chapter 4, was tried, but it did not result in significant changes from the LUDS solution. The reasons for this may be: (i) the criterion for boundedness, described in Chapter 4, is not adequate for the momentum equations, since the velocity components are not "transported" along streamlines like passive scalars, at least in the presence of pressure gradients; (ii) the latter gradients, which act as sources in this case, are unknown, and if incorrect can cause unrealistic extrema which may not be recognized as such by the blending procedure, which





– – – – 25x27x11 CV; ------- 50x27x11 CV; ——— 50x40x20 CV

Fig. 9.24: Grid dependence tests with the UDS

Fig. 9.25: Grid dependence tests with the LUDS

concentrates on convective terms only, and (iii) the second maximum may be "real".

It is clear therefore, that neither the UDS nor LUDS solution is grid-independent, but with refinement both are approaching the experimental results, as can be seen from figures 9.24 and 9.25 . After an overall evaluation of the comparisons, it was decided to present here the results obtained using the UDS on the finest grid; for these most closely represent the experimentally observed features of the flow, and are not suspected of any artificial effects other than numerical diffusion, the effects of which are easily percieved.

Figures 9.26 (a)-(d) show respectively the grid, velocity vectors and the isobars (normalized with ρu_{in}^2) in the symmetry plane. The velocity vectors show that, as the flow enters the second bend, it decelerates near the outer wall. Some reverse flow was predicted in this region near the symmetry plane and top wall corner, as the contours of u_a velocity components in Fig. 9.28 show : however, few measurements are available in this region to compare with. The only measurement point which falls within the predicted reverse flow region was at $r^*=0.9$, $z^*=0.95$ (1 mm from the top wall, 6 mm from the outer wall), where a zero value of u_a was reported (see Fig. 9.35). The bulk of the flow tends to choose the shortest path through the duct, so the velocity peak moves towards the inner wall.

The pressure contours in the symmetry plane are similar to those at the top wall, which are shown in Fig. 9.26 (f); however, the levels are lower at the wall. Both the maximum and the minimum pressure regions are on the inner wall: the maximum occurs where the flow impinges onto the wall, while the minimum occurs soon after the flow enters the second bend, due to its redirectioning.

Predicted vectors of secondary flow in the cross-section D-D , which is close to the transition between bends (see Fig. 9.22) are shown in Fig. 9.27 , together with both the predicted and measured contours of the velocity component normal to this plane. These results show one secondary flow vortex, with the velocities near the top wall directed towards the outer wall, and those at the symmetry plane in the opposite direction. As the flow enters the second bend, the centre of the secondary vortex created in the first bend moves towards the symmetry plane and the outer wall, while a new vortex, with opposite sense of rotation forms near the top and inner walls as a result of the change in curvature.

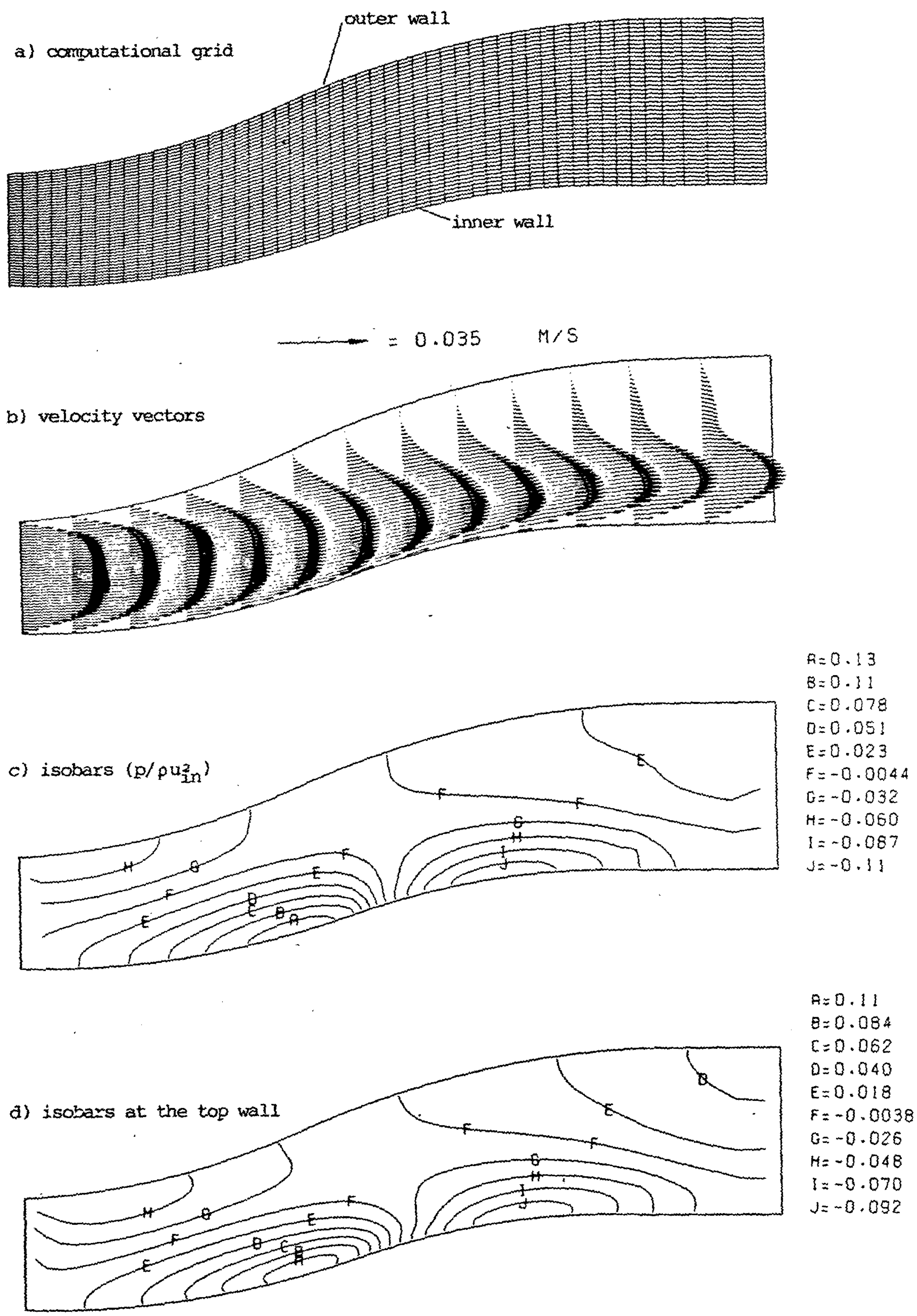


Fig. 9.26: S-diffuser: characteristics of flow in symmetry plane at Re=790

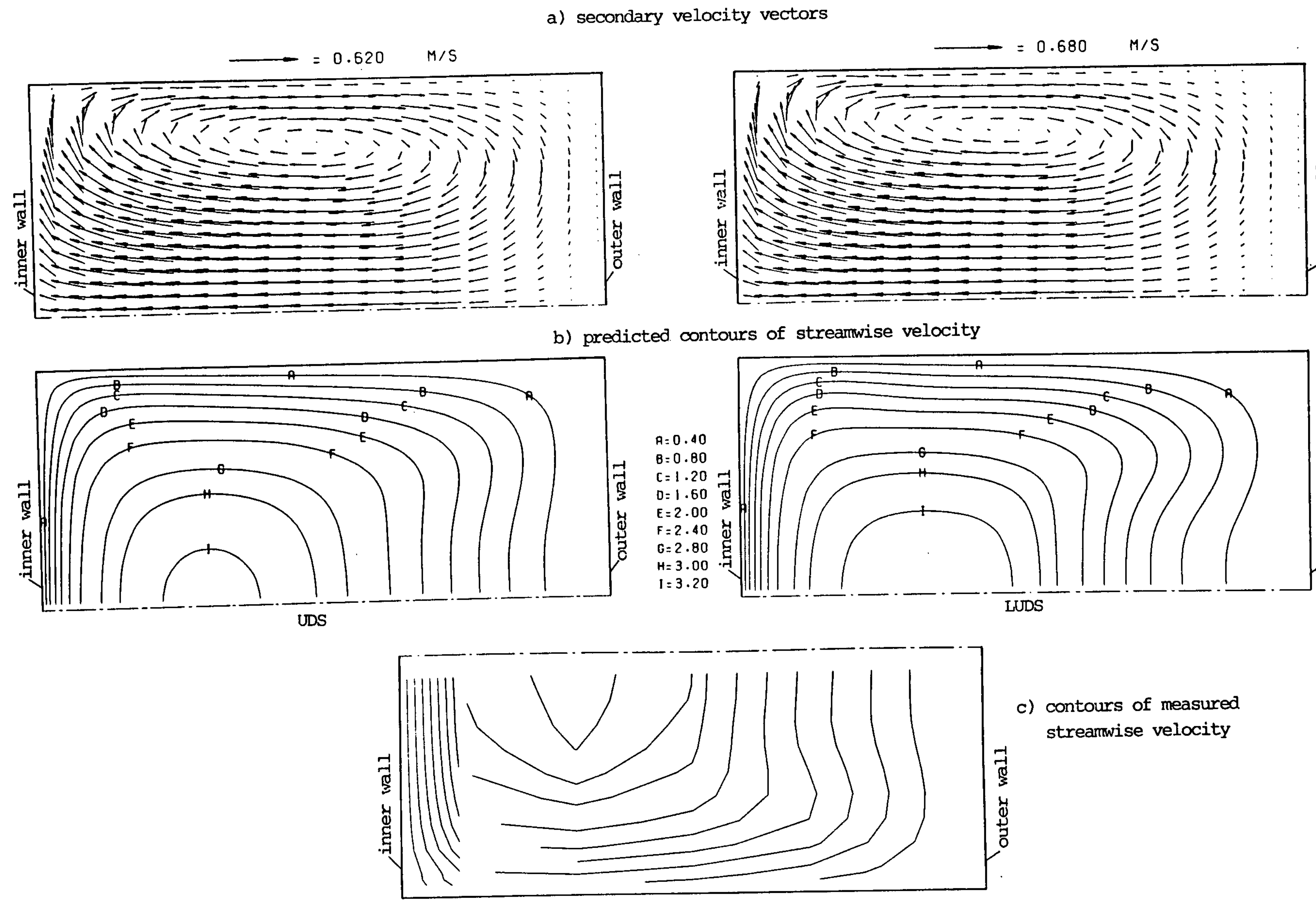


Fig. 9.27: S-diffuser: flow characteristics in cross-section D-D

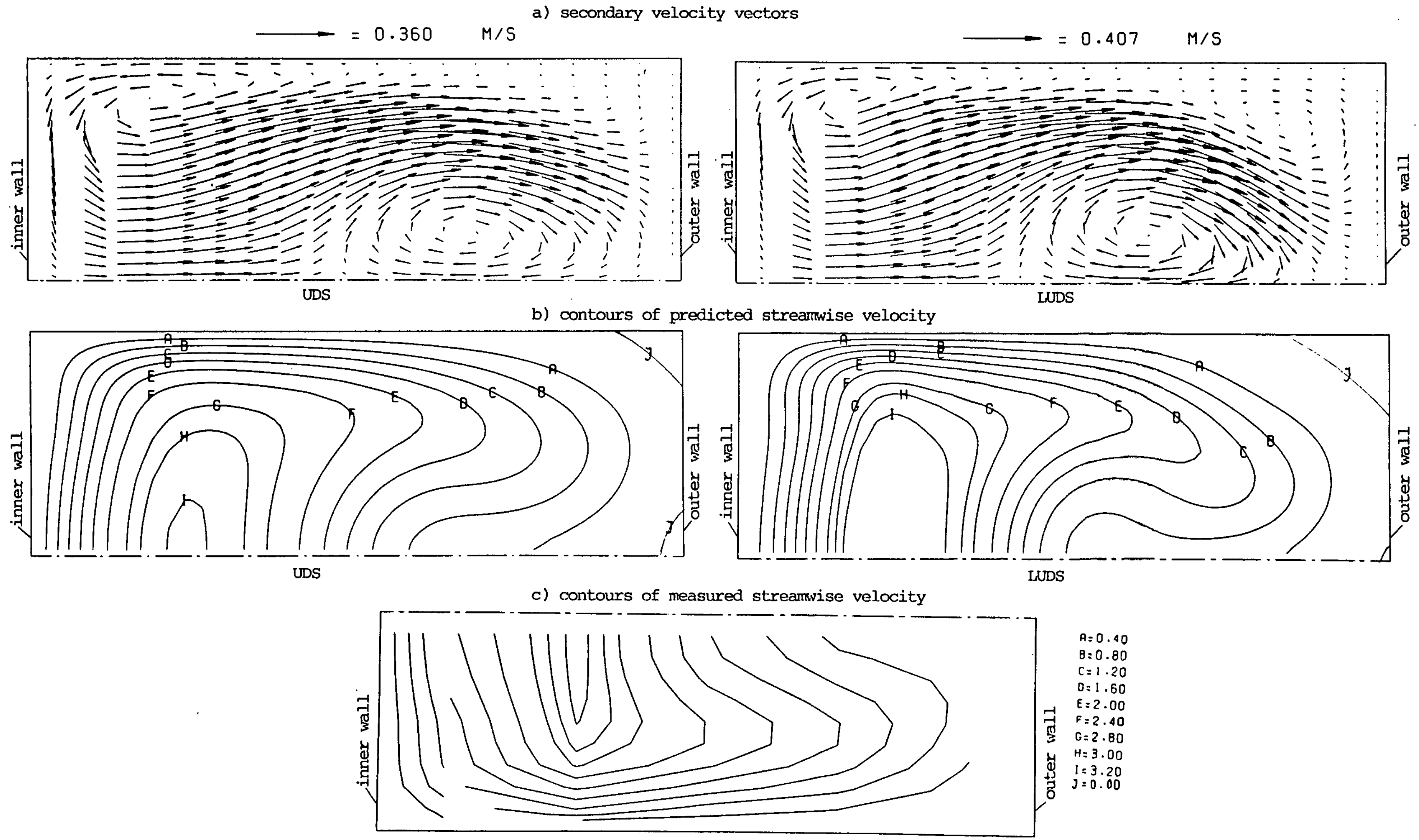


Fig. 9.28: S-diffuser: flow characteristics in cross-section C-C

The secondary flow pattern at the exit from the second bend (cross-section C-C in Fig. 9.22) is shown in Fig. 9.28 , together with the predicted and measured contours of the normal velocity. Whereas at section D-D the secondary flow was bringing fluid into the region of high velocity near the inner wall, at this cross-section it moves high momentum fluid in the opposite direction: in both cases, however, the strongest secondary flow is in the direction of increasing radius of bend curvature.

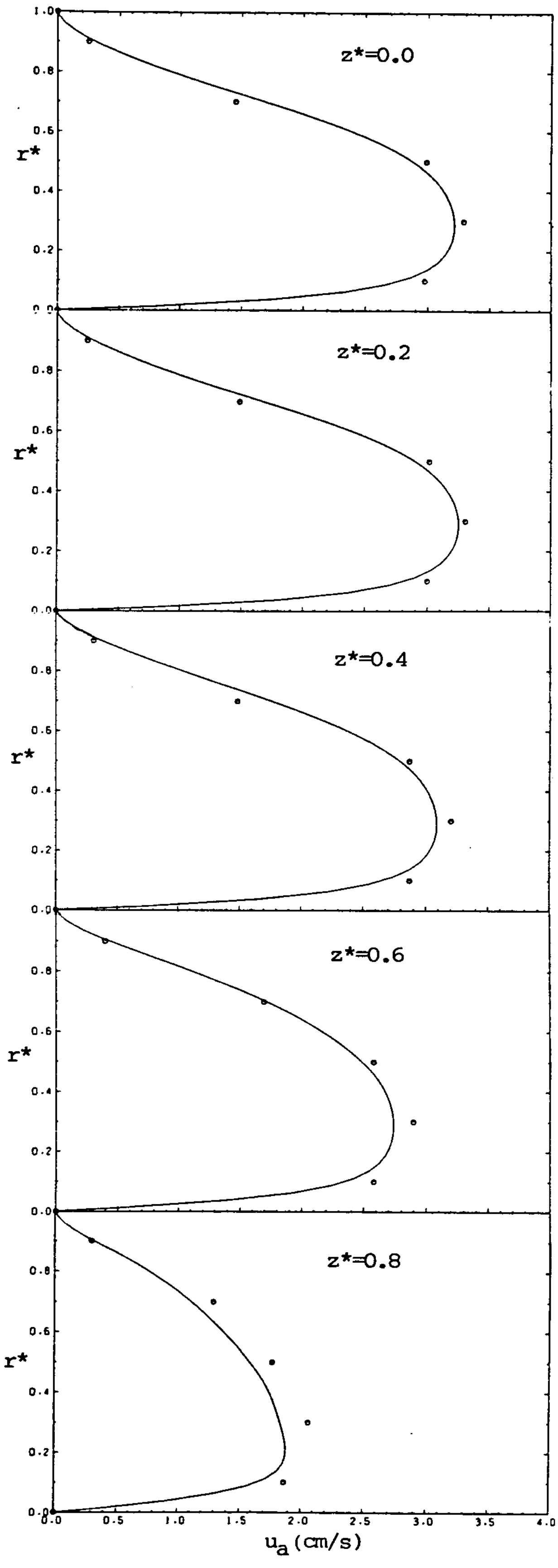
Comparisons with experiments

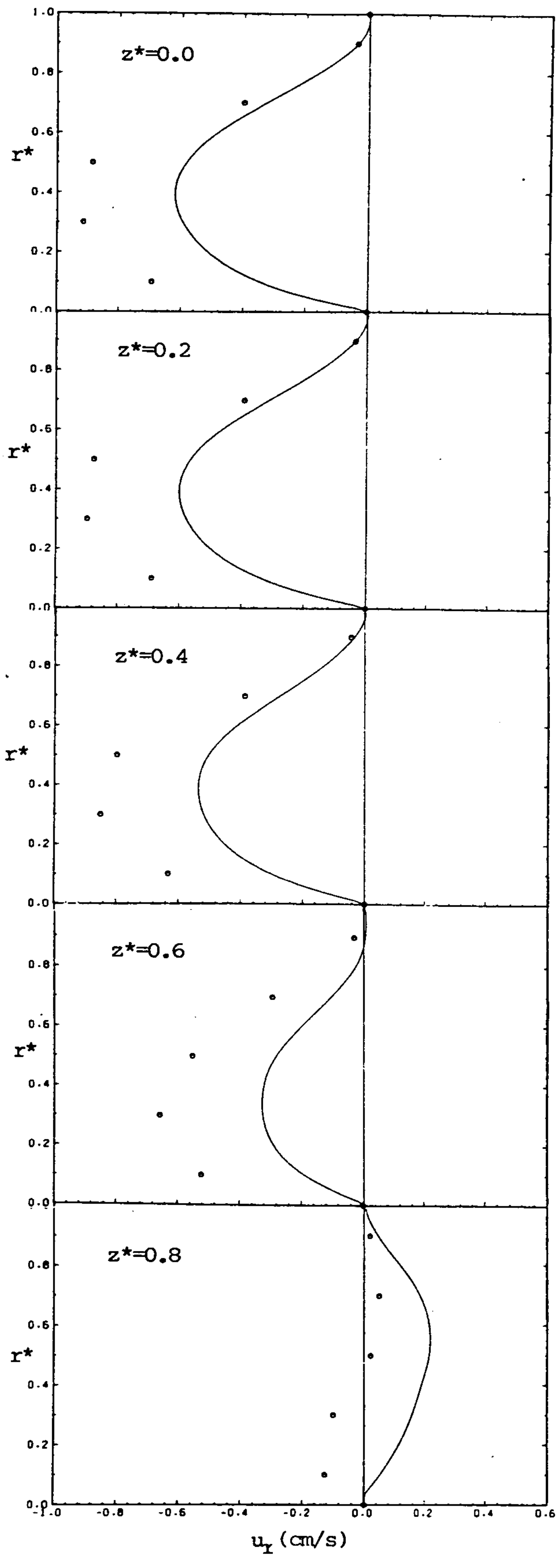
Qualitative comparisons between the predicted flow and experimental data have already been shown in figures 9.27 and 9.28 , in the form of contours of the velocity normal to the cross-sectional planes C-C and D-D . From these figures it is evident that the computated and measured results closely resemble each other (the contours constructed from the experimental data are not smooth due to the coarsness of the data).

Further quantitative comparisons are contained in figures 9.29 to 9.36 . Figure 9.29 shows predicted and measured profiles of the u_a velocity component at plane D-D, in five horizontal cross-sections at constant $z^*=0$, 0.2, 0.4, 0.6 and 0.8 , as shown in Fig. 9.22. Relatively good agreement can be seen, except for the plane closest to the top wall, where the discrepancies are somewhat larger (up to 8%). Figure 9.30 shows profiles of the radial velocity component (ie. the component in the r^* direction) lying in the same plane, at the same positions as in the previous case. Here the agreement is rather poor, with computed velocities being about 30% too low. No definitive explanation can be given for this discrepancy, which may be due to one or more of the following: (i) inlet condition uncertainities; (ii) insufficiently fine grid; (iii) errors in the interpolation of the experimental data at the comparison plane and (iv) errors in the interpolation of the predicted u_1 and u_2 velocity components while calculating the u_r at the comparison plane.

Figures 9.31 and 9.32 present the same information as the previous two figures respectively, only now in the vertical cross-sections $r^*=0.1$, 0.3, 0.5, 0.7 and 0.9 (see Fig. 9.22), running from the top wall ($z^*=1$) to the symmetry plane ($z^*=0$). Again qualitative agreement can be seen for both velocity components, while quantitatively the radial component is significantly underpredicted.

Figure 9.33 shows profiles of the u_a velocity component (here nor-

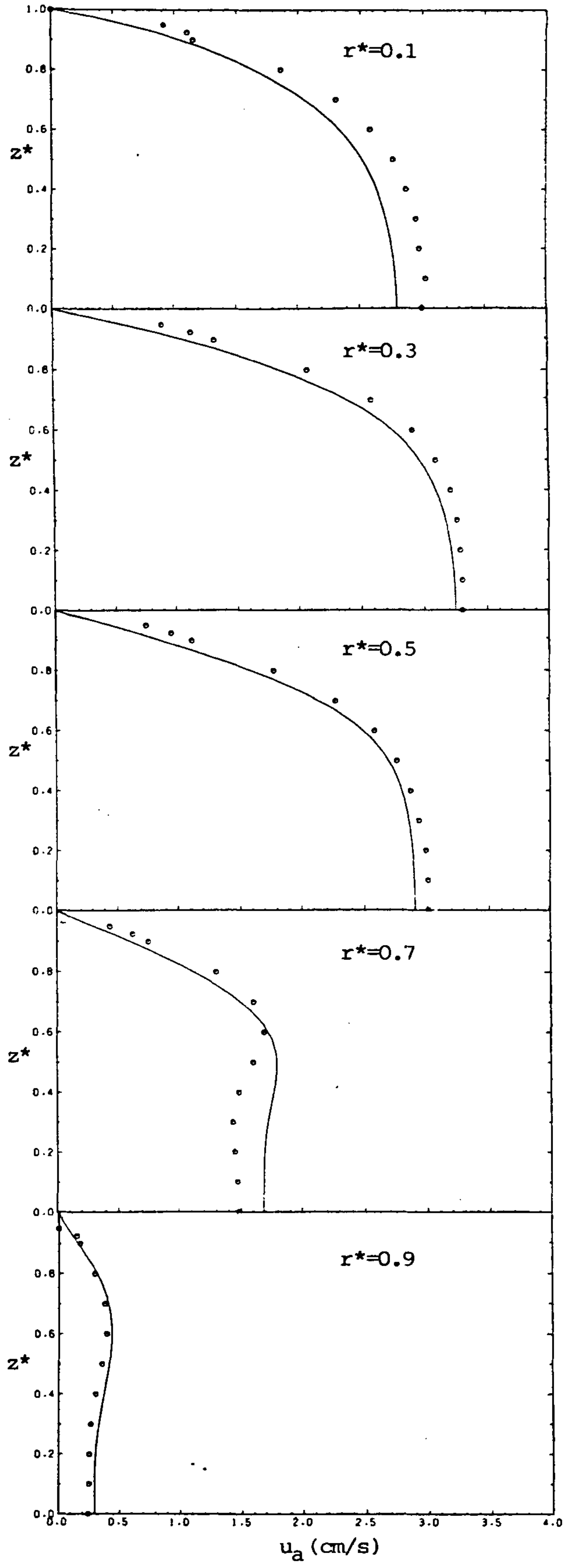


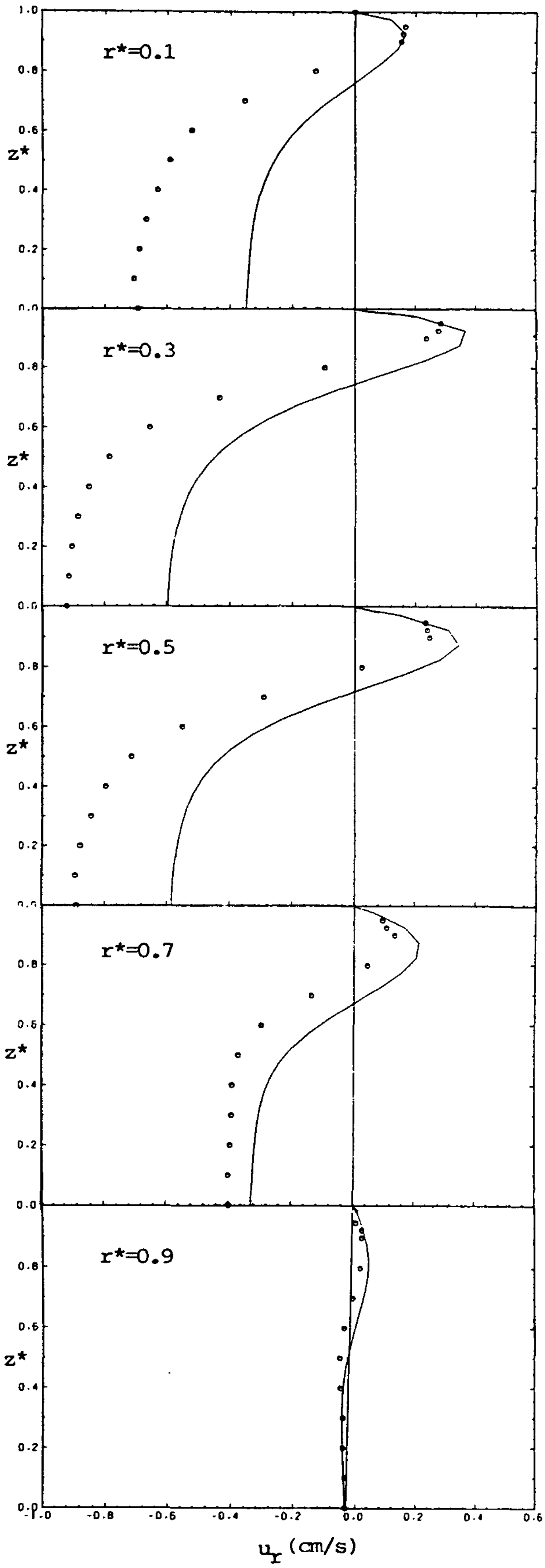


⊙ measurement; ———— present prediction

Fig. 9.29: Radial profiles of u_a in cross-section D-D

Fig. 9.30: Radial profiles of u_r in cross-section D-D

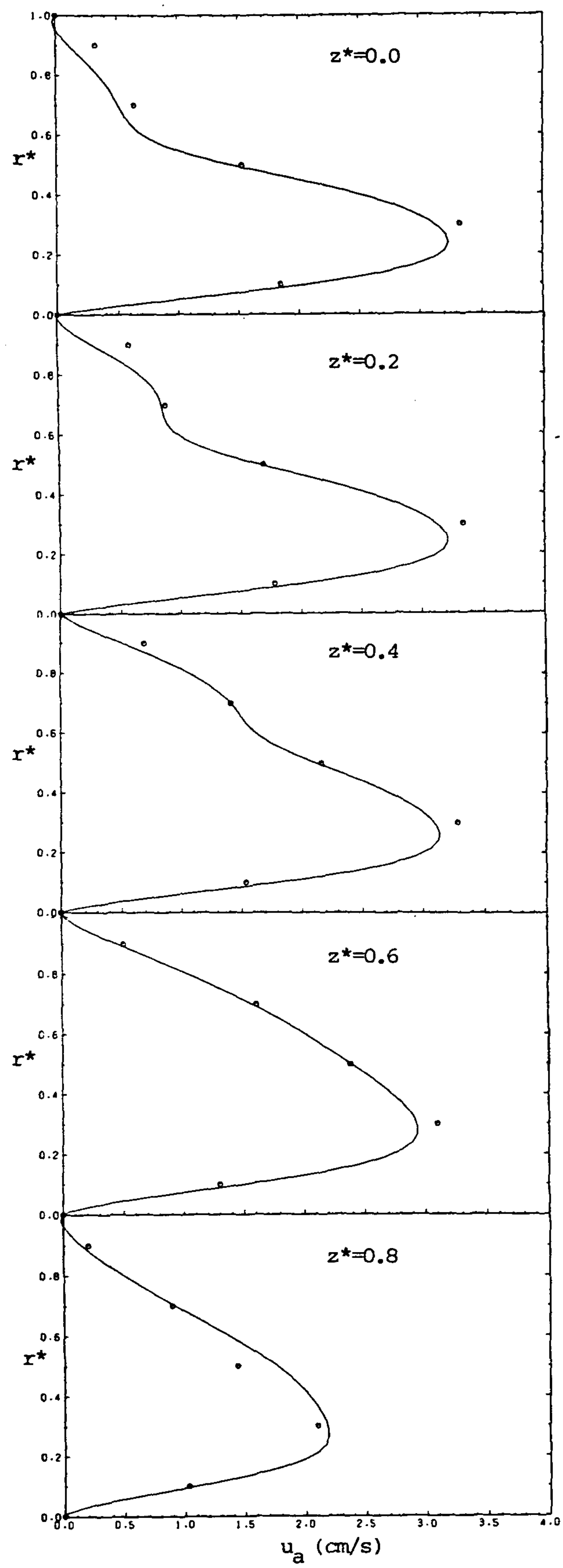


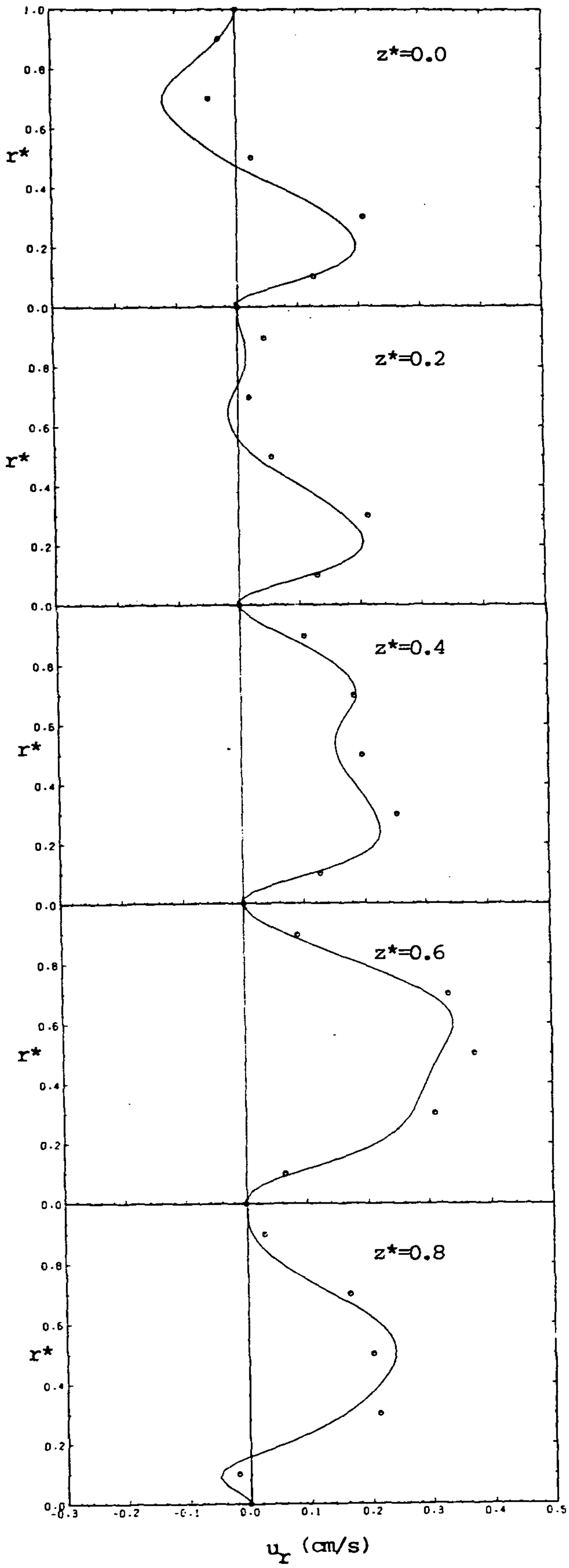


o measurement; ———— present prediction

Fig. 9.31: Spanwise profiles of u_a in cross-section D-D

Fig. 9.32: Spanwise profiles of u_r in cross-section D-D

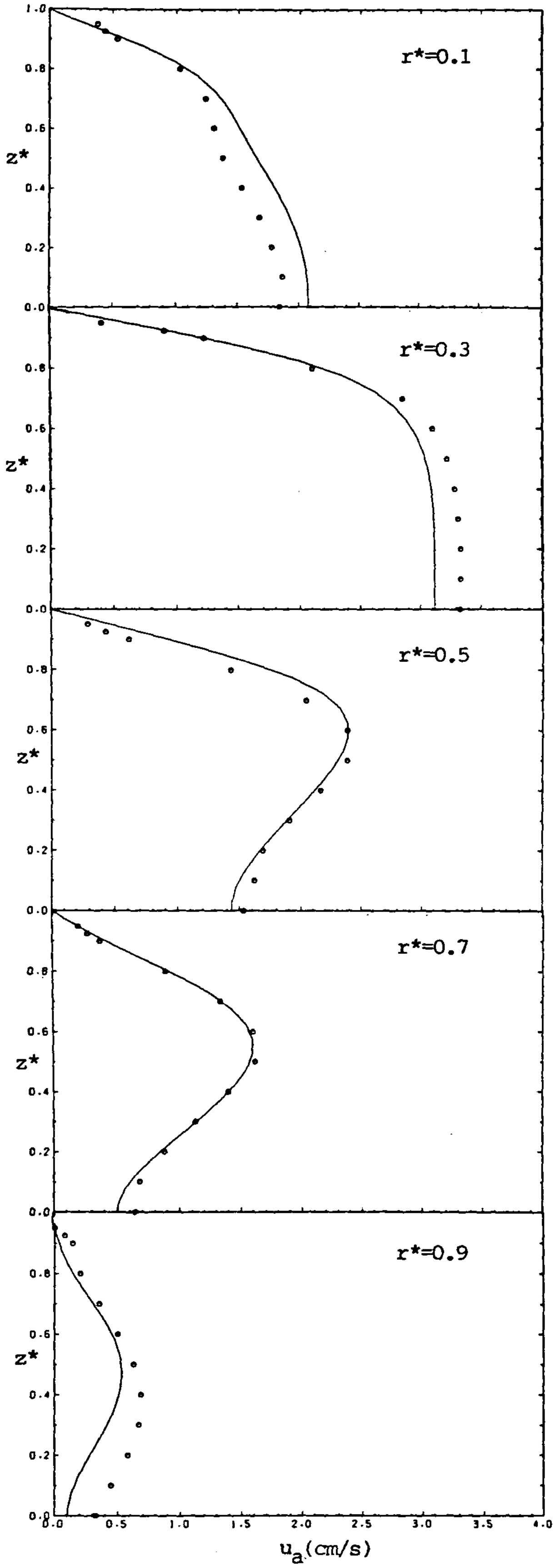


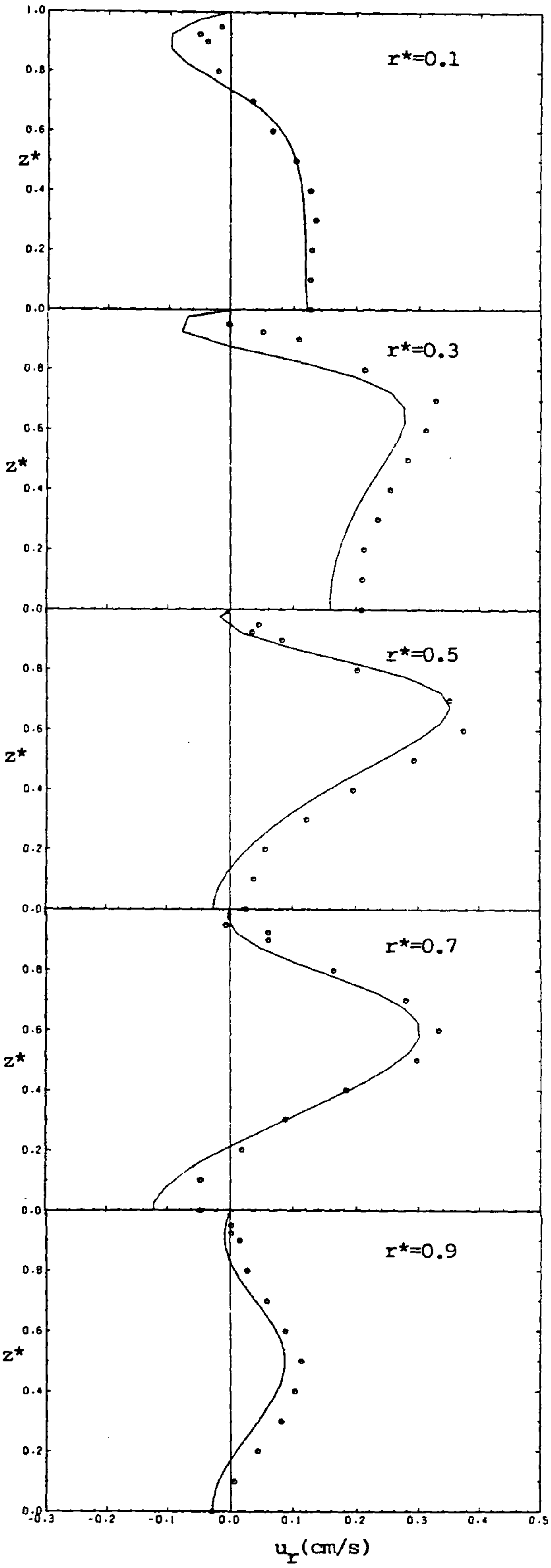


o measurement; ———— present prediction

Fig. 9.33: Radial profiles of u_a in cross-section C-C

Fig. 9.34: Radial profiles of u_r in cross-section C-C





⊙ measurement; ———— present prediction

Fig. 9.35: Spanwise profiles of u_a in cross-section C-C

Fig. 9.36: Spanwise profiles of u_r in cross-section C-C

mal to the cross-section C-C , Fig. 9.22) at the exit of the second bend in five horizontal cross-sections (z^*=0, 0.2, 0.4, 0.6 and 0.8). The qualitative agreement can be said to be very good; quantitatively, the predicted profiles do not reach the measured peaks, obviously due at least in part to numerical diffusion, which smears the profiles to some extent. This can be concluded by referring back to Fig. 9.24 , which shows the effects of grid refinement: the predicted peaks move towards the experimental points as the number of grid points is increased and in general the profiles tend to get closer to the experimental data everywhere.

Figure 9.34 shows profiles of the u_r velocity component at the same locations as in the previous figure. Here the agreement is much better than at cross-section D-D ; the predicted profiles follow the experimental points quite well.

Figures 9.35 and 9.36 present the same information as the previous two figures, only now with profiles in vertical cross-sections, at the same positions relative to side walls as in figures 9.31 and 9.32 . Discrepancies in the u_a-velocity component (Fig. 9.35) are seen in the first two profiles (6 mm and 18 mm from the inner wall) and the last one (6 mm from the outer wall), near the symmetry plane ($z^* < 0.6$). Closer to the top wall the agreement is good for all five profiles. The agreement between predicted and measured u_r velocity component in Fig. 9.36 is better seen than in Fig. 9.34 , due to the fact that there are more experimental points in the vertical direction: both qualitatively and quantitatively the agreement can be considered to be very good.

9.3.2 Turbulent flow in a S-shaped diffuser

Description of problem

In this section turbulent flow in the S-shaped diffuser, described in the previous section, is considered. The experimental results of Rojas et al (1983), obtained for Re=40 000 , are used for comparison. The same notation is used as in the laminar flow case.

Computational details

The values of the dependent variables at inlet boundary nodes were obtained by interpolation from experimental data. This again might be a source of inaccuracies due to the coarsness of experimental data matrix in the radial direction (see Fig. 9.22). This especially applies to the kinetic energy of turbulence k , of which only two components were measured (streamwise and radial). The third component was assumed here to be equal to the half of the sum of the two measured; in some regions of flow where the turbulence was nearly isotropic this assumption was not far from reality, but in other regions (eg. near walls) it could result in errors of up to 25% . Also near the side walls, where the first measurement points were located 1/10 of the duct width away, the values of k had to be estimated: they were assumed to be of the same order as those near the top wall, where closer measurements existed. The energy dissipation rate ε also had to be estimated: to this end, the length scale l was assumed to vary linearly from the wall up to 0.2 duct widths, and then stays uniform; ε was then calculated as $\varepsilon = k^{3/2}/l$. The inlet spanwise velocity component also was not measured, and in the absence of a better approximation it was set to zero, as in the laminar flow case.

Calculations were performed with the UDS on two grids: one with $25 \times 27 \times 11$ CV and one with $50 \times 40 \times 20$ CV (see Fig. 9.22). In Fig. 9.46 and 9.47 the predicted profiles of the streamwise and radial velocity components at the exit from the second bend (cross-section C-C , Fig. 9.22) are shown for the two grids. There is little difference between them, except in the slope of streamwise velocity component profiles close to $r^*=1$. Some of this discrepancy could be attributed to the fact that the interpolation of the inlet data may have produced differences in the inlet conditions, especially for the coarser grid. It is believed that the numerical errors were not significant in the solution on the finest grid.

It should be noted that the calculations in this case converged rather slowly, due to the inability of the Stone method to solve the pressure-correction equations to the necessary tolerance ($\lambda_p = 0.3$; see section 7 of Chapter 6) within a reasonable number of inner iterations (the maximum was set at 20). This happened because the aspect ratios $\frac{\delta x^{(1)}}{\delta x^{(2)}}$ and $\frac{\delta x^{(1)}}{\delta x^{(3)}}$ were rather high (> 10), which affects the convergence of the Stone solver used, as discussed in Chapter 7. For this reason ~ 500 outer iterations were required to reach a satisfactory solution (with $\max \lambda_s < 2 \cdot 10^{-3}$ in this case), with $\alpha_u = 0.75$, $\alpha_p = 0.2$, $\alpha_{k,\varepsilon} = 0.7$, and over 6 hours of computational time on an AMDAHL V8 computer. Clearly a better solver would be advantageous in those circumstances.

Results and comparisons with experiment

In Fig. 9.37 the characteristics of flow in the symmetry plane are shown. The differences from, and the similarities with, the corresponding figures for the laminar flow case (cf Fig. 9.26) are obvious: here the boundary layers are thinner, the high velocity core is wider, the low velocity region is smaller and the pressure contours are similar. The turbulence intensity contours and profiles reveal low turbulence in the velocity core region and high levels near the side walls. On entry into the second bend the turbulence intensity peak moves away from the outer bend wall towards the centerline, for reasons which will become obvious when the detailed velocity profiles are examined.

In Fig. 9.38 the flow characteristics in cross-section D-D are shown, including contours of the measured streamwise velocity component and turbulence intensity. Qualitative agreement between the measured and the predicted results can be observed. In Fig. 9.39 the same information is given for cross-section C-C . Again qualitative agreement between measured and predicted contours can be seen, although the streamwise velocity contours exhibit somewhat different trends near the corner $r^*=1$, $z^*=1$. The secondary flow vectors are similar to those for the laminar case, only now a stronger motion exists near the forementioned corner.

Figures 9.40 and 9.41 present spanwise profiles of the streamwise (u_a) and radial (u_r) velocity components in the cross-section D-D at five radial locations (see Fig. 9.22), normalized with the inlet velocity (~1 m/s) and compared with the experimental data. In contrast to the laminar case, here the predicted radial velocities are in good agreement with the experimental data at all locations. The streamwise velocities also agree reasonably well, except for the $r^*=0.1$ and $r^*=0.9$ profiles (which are nearest to the side walls), where close to the top wall (ie. the duct corner) local disagreements of up to 10% are evident. In Fig. 9.42 and 9.43 radial profiles of the same velocity components in five horizontal cross-sections are presented. The agreement with experimental data is good, with typical differences of about 3 - 5% everywhere apart from the $z^*=0.8$ plane.

Figures 9.44 and 9.45 present spanwise velocity profiles in the cross-section C-C , compared with the experimental results. Qualitative agreement is again obtained, but for $r^*=0.9$, and to a lesser extent close to the top at $r^*=0.1$, significant discrepancies are evident. Figures 9.46 and 9.47 show radial profiles of the same velocity components in the same cross-section, at five spanwise locations. The streamwise velo-

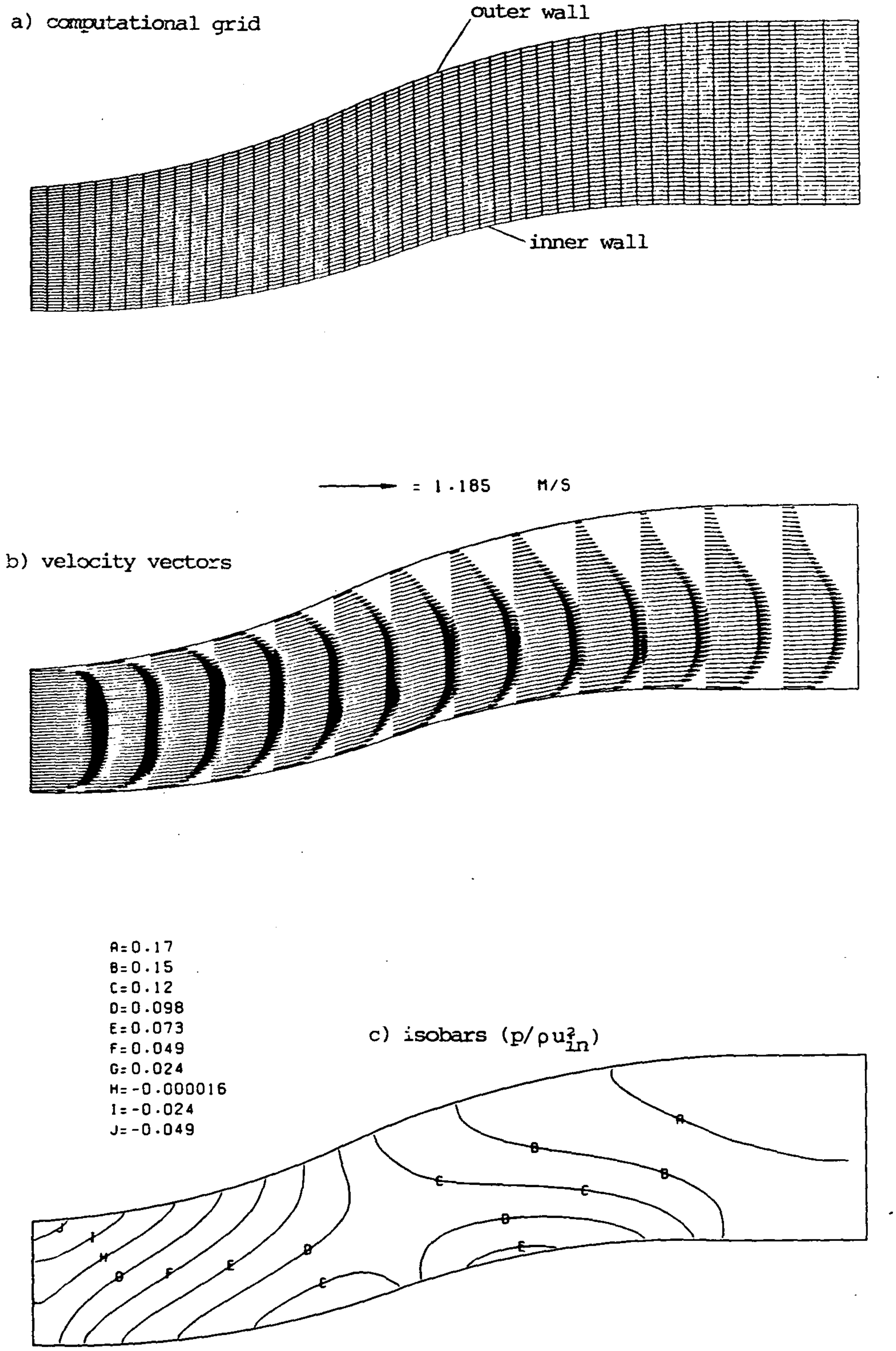
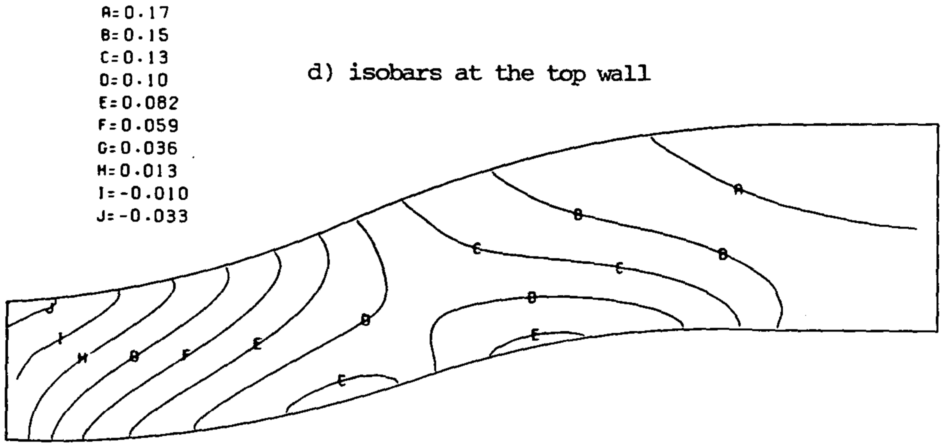
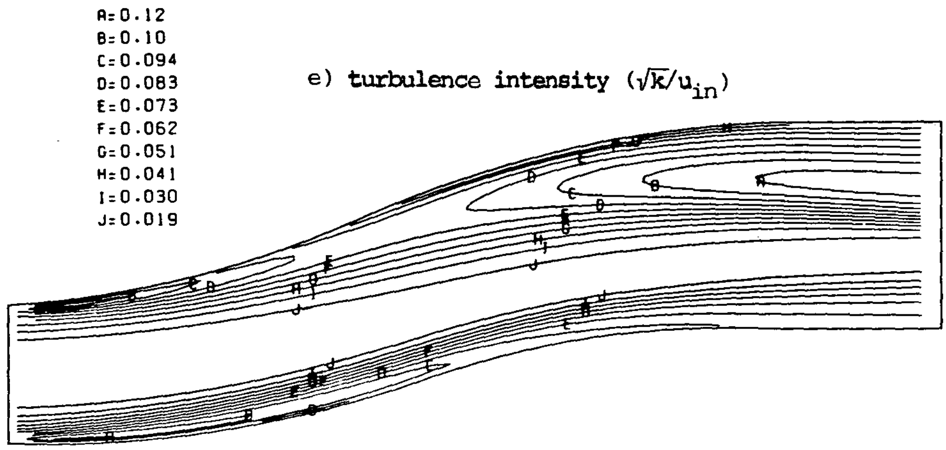


Fig. 9.37: Characteristics of flow in symmetry plane





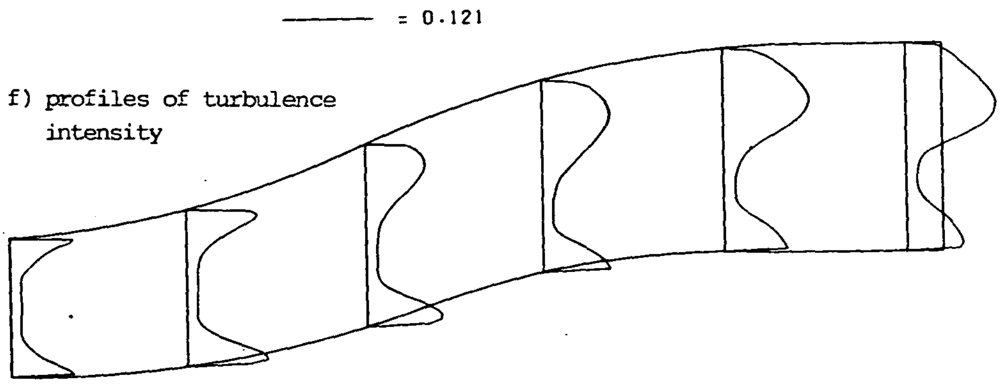


Fig. 9.37: continued

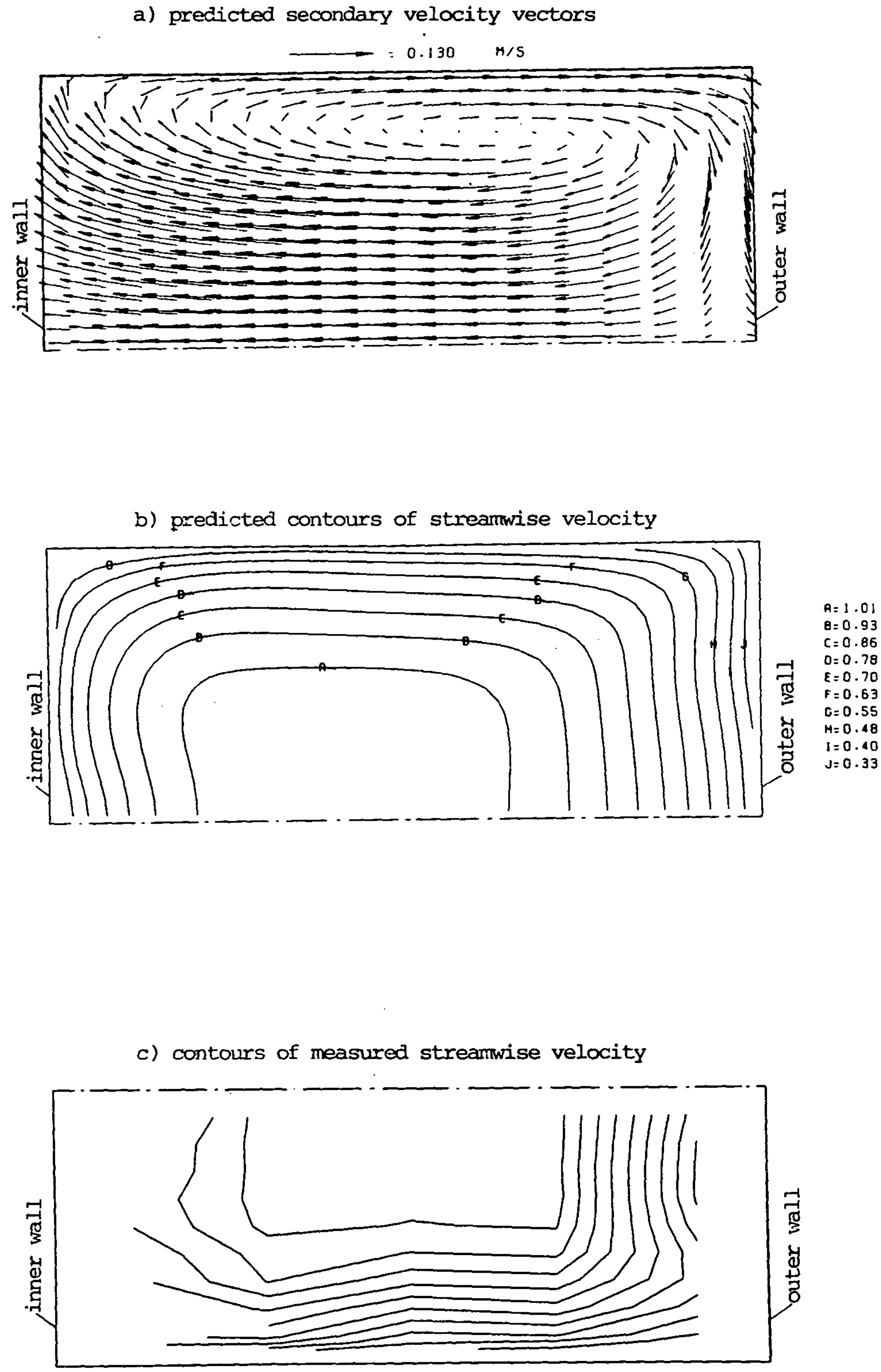
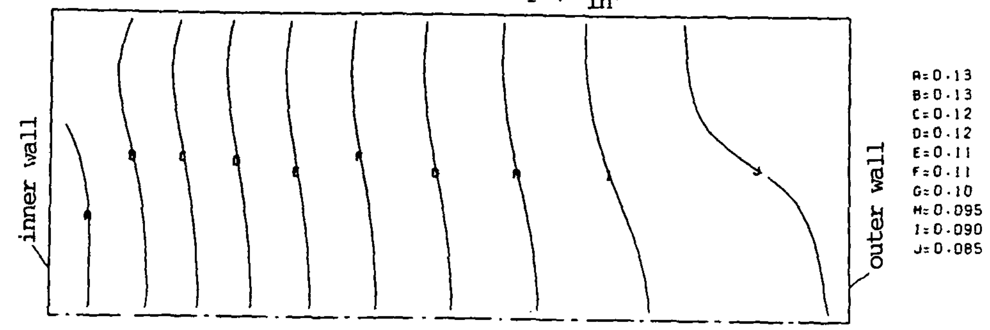
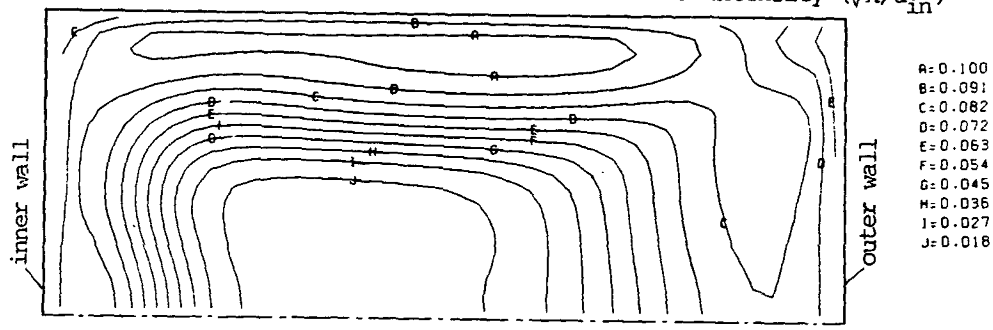


Fig. 9.38: Turbulent flow characteristics in cross-section D-D

d) predicted isobars ($p/\rho u_{in}^2$)



e) predicted contours of turbulence intensity ($\sqrt{k}/u_{in}$)



f) contours of measured turbulence intensity

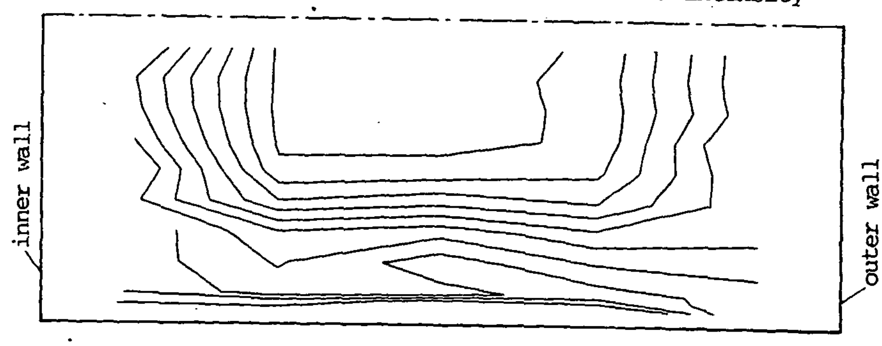
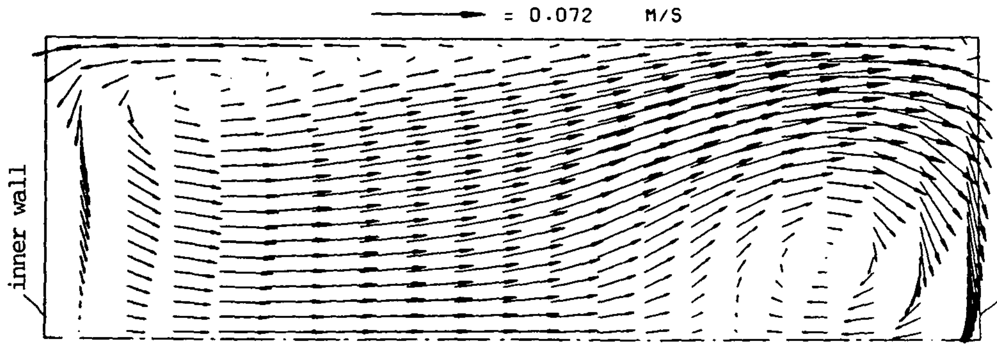
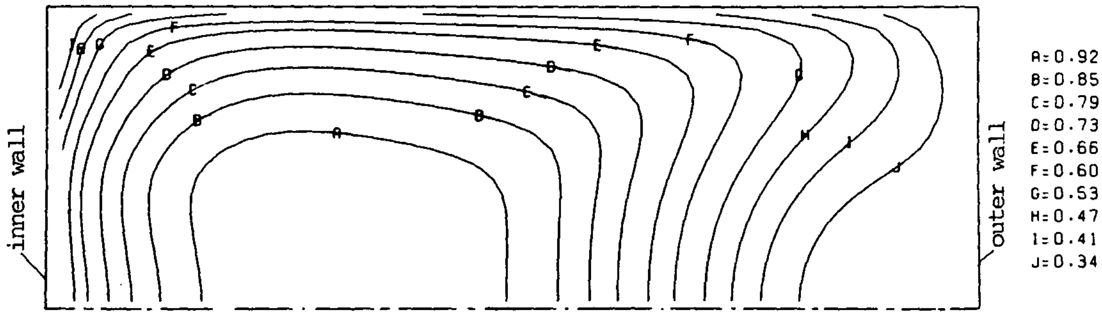


Fig. 9.38: continued

a) predicted secondary velocity vectors



b) contours of predicted streamwise velocity



c) contours of measured streamwise velocity

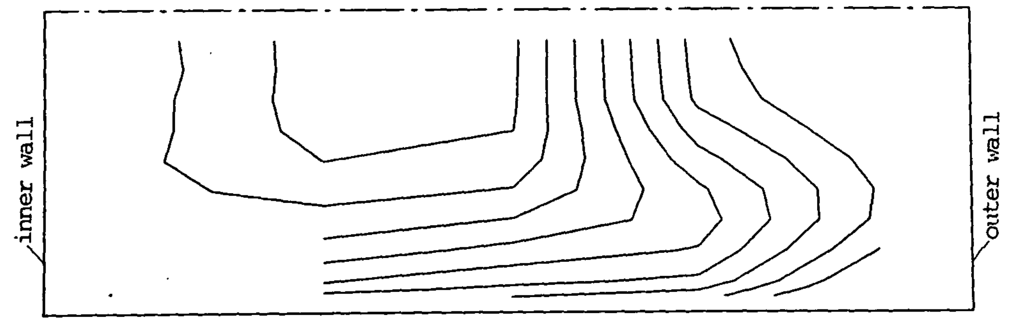
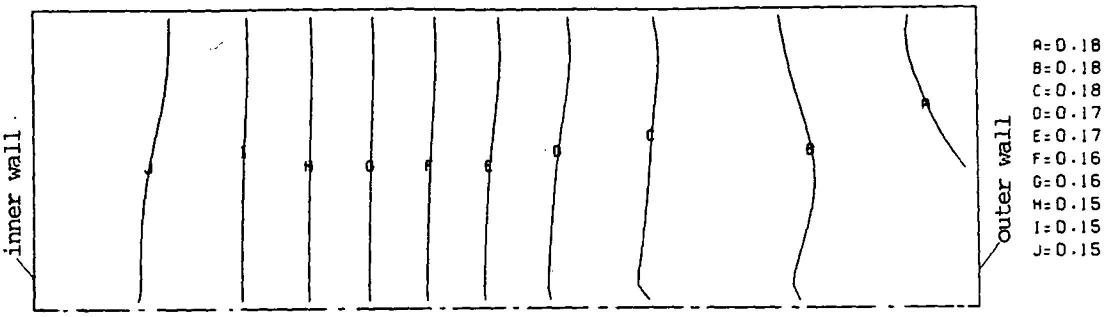
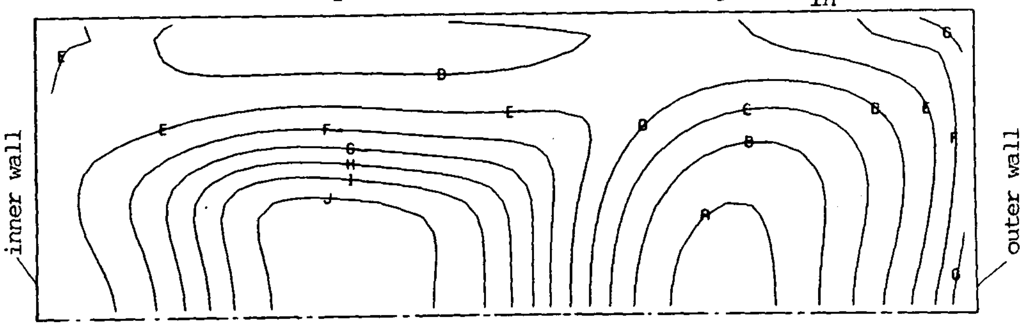


Fig. 9.39: Turbulent flow characteristics in cross-section C-C

d) predicted isobars ($p/\rho u_{in}^2$)



d) contours of predicted turbulence intensity ($\sqrt{k}/u_{in}$)



e) contours of measured turbulence intensity

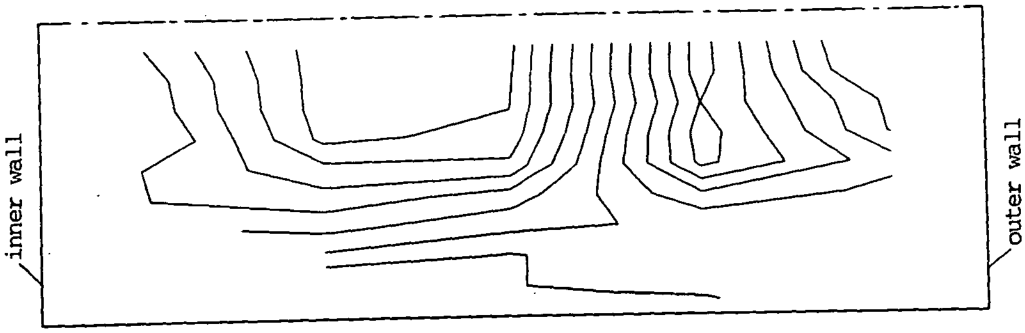
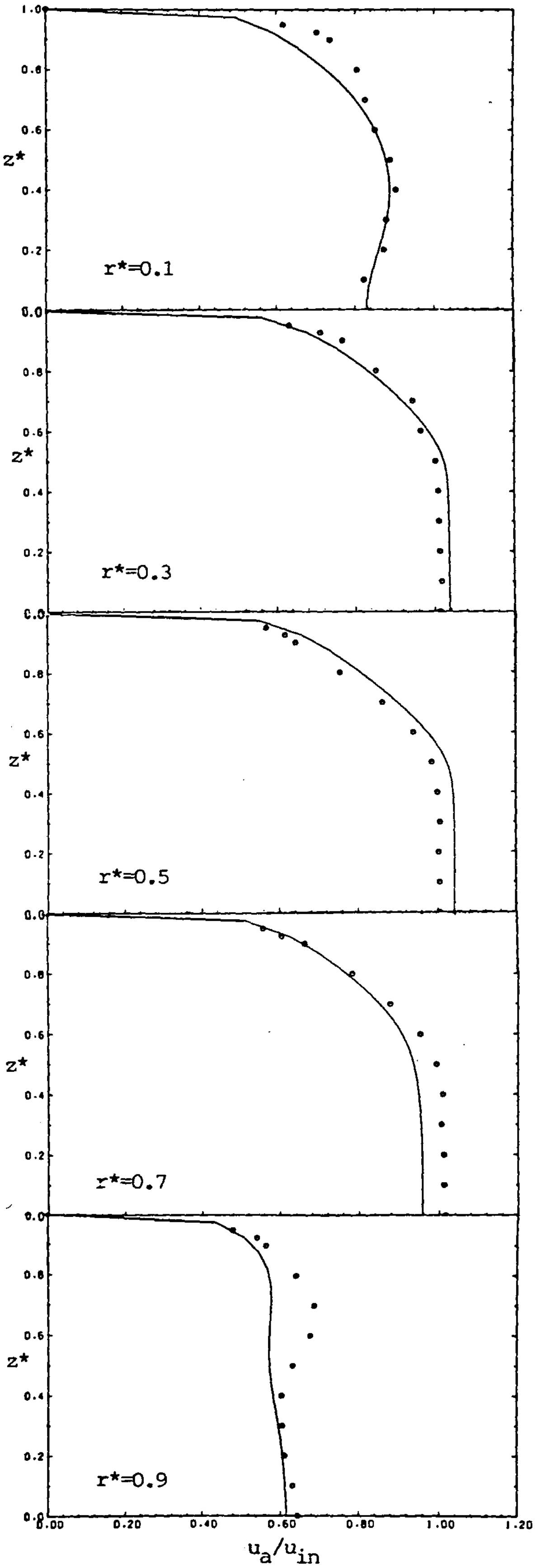
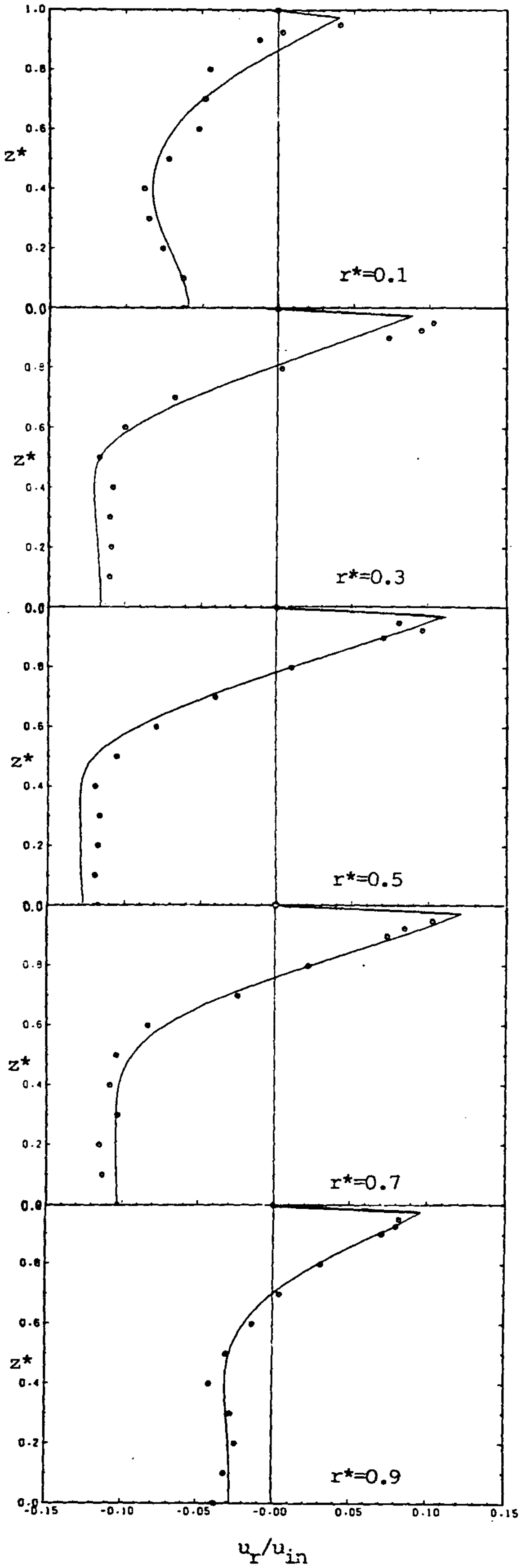


Fig. 9.39: continued

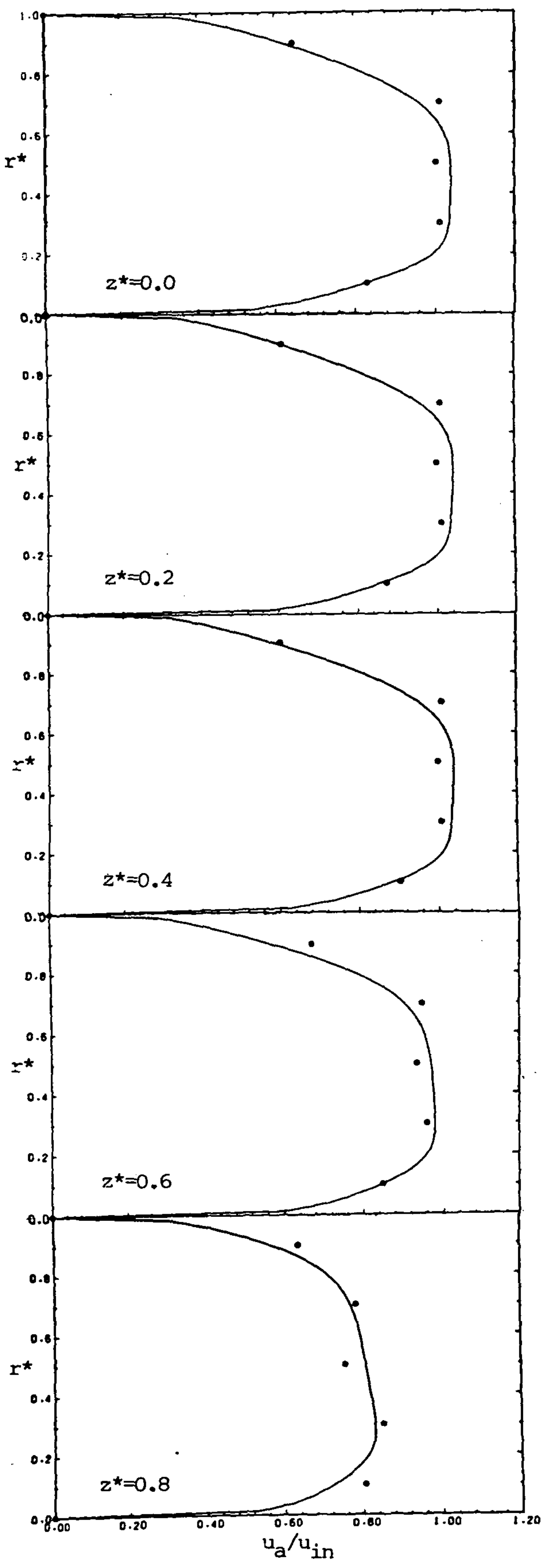


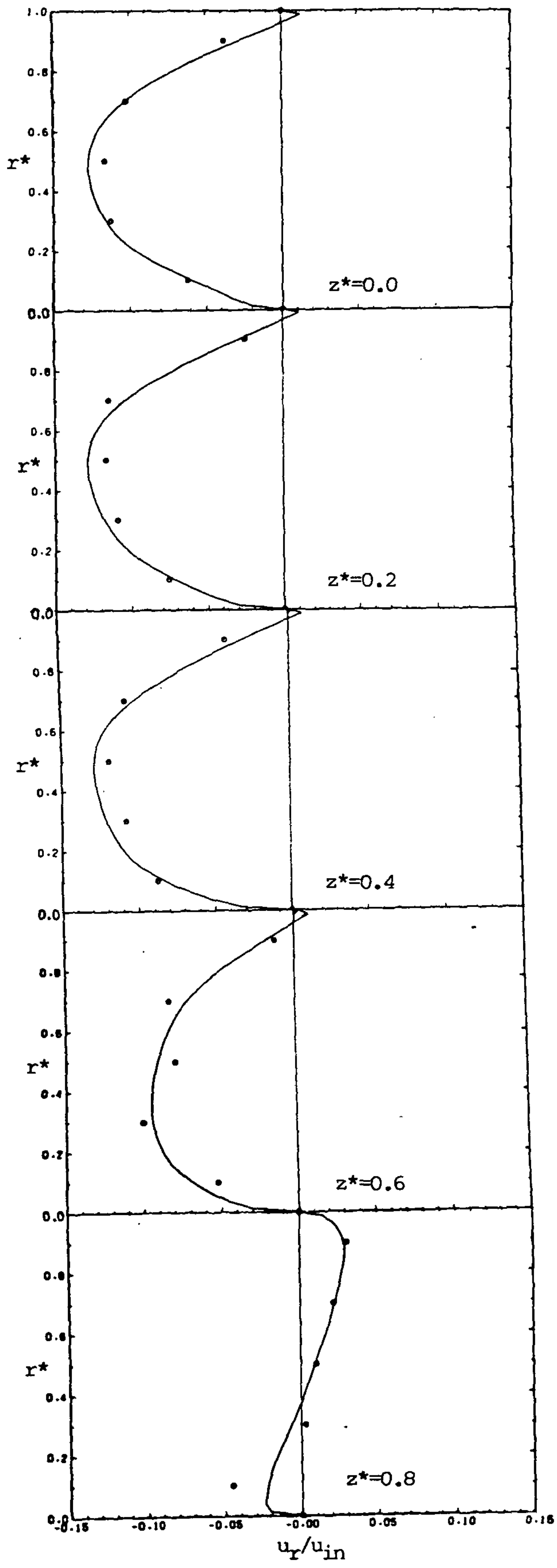


o measurement; ———— present prediction

Fig. 9.40: Spanwise profiles of u_a in cross-section D-D

Fig. 9.41: Spanwise profiles of u_r in cross-section D-D

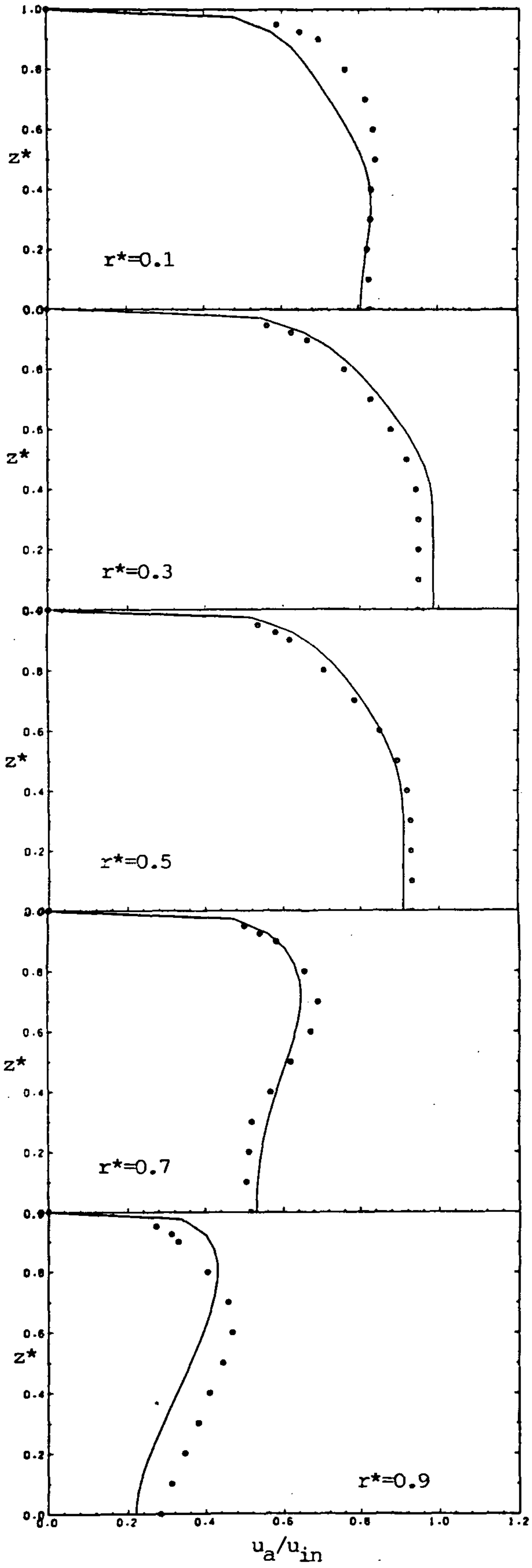


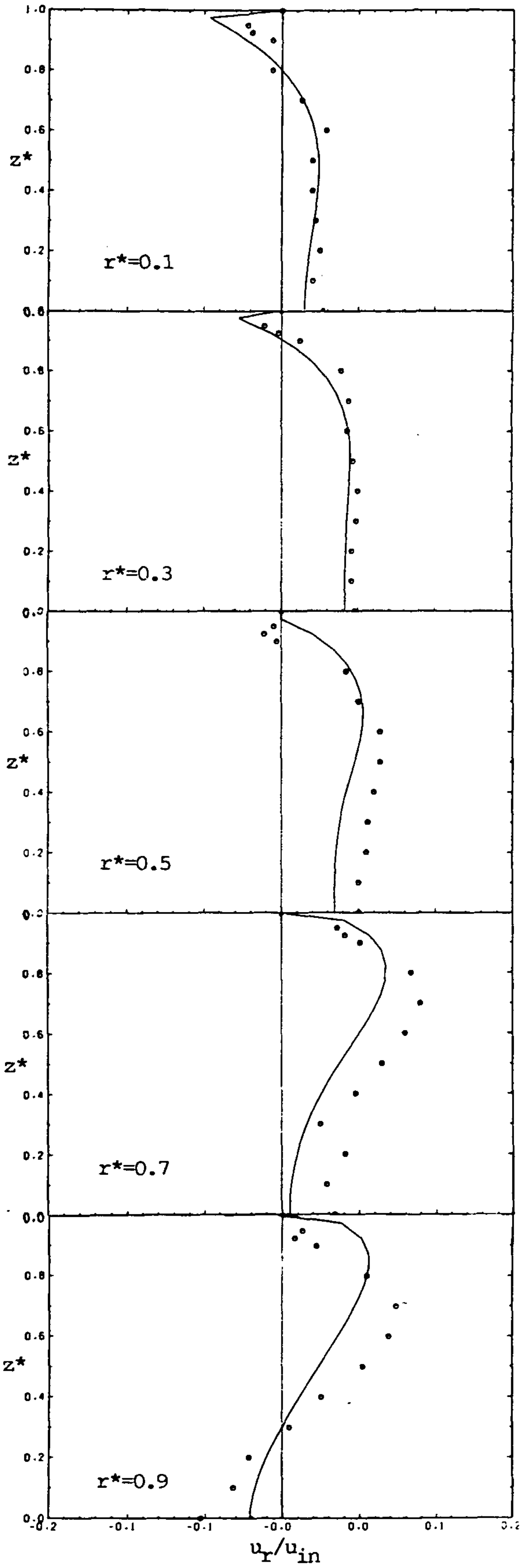


o measurement; ——— present prediction

Fig. 9.42: Radial profiles of u_a in cross-section D-D

Fig. 9.43: Radial profiles of u_r in cross-section D-D

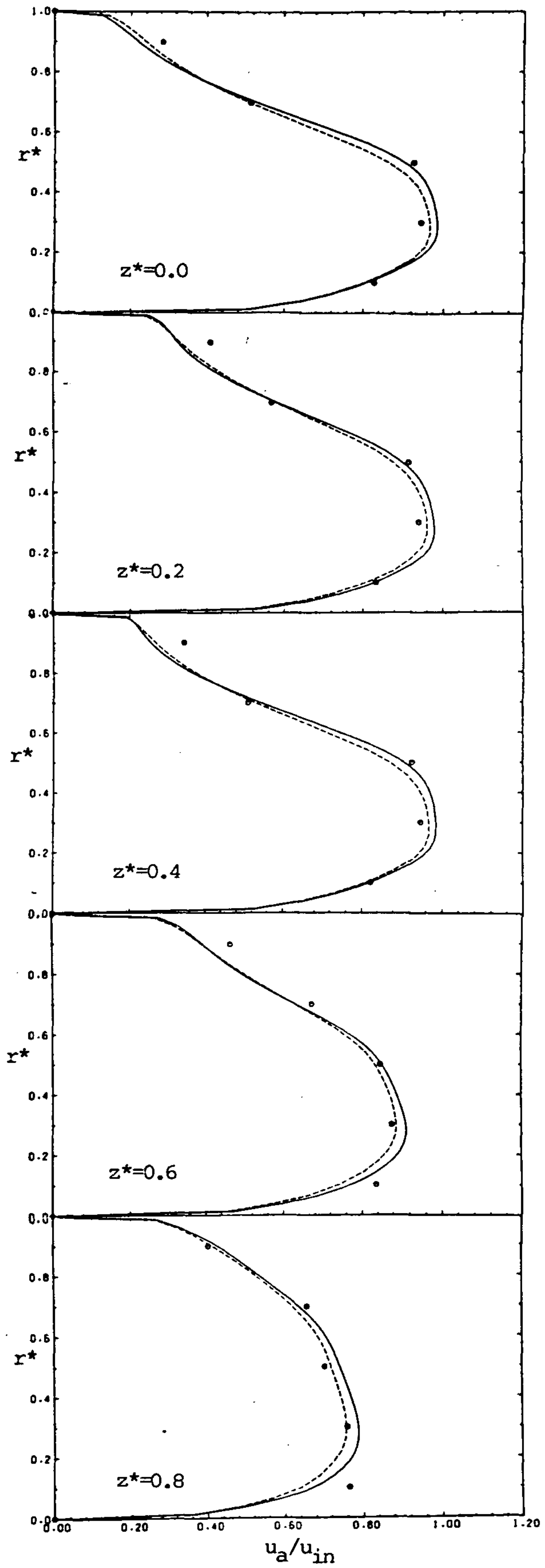


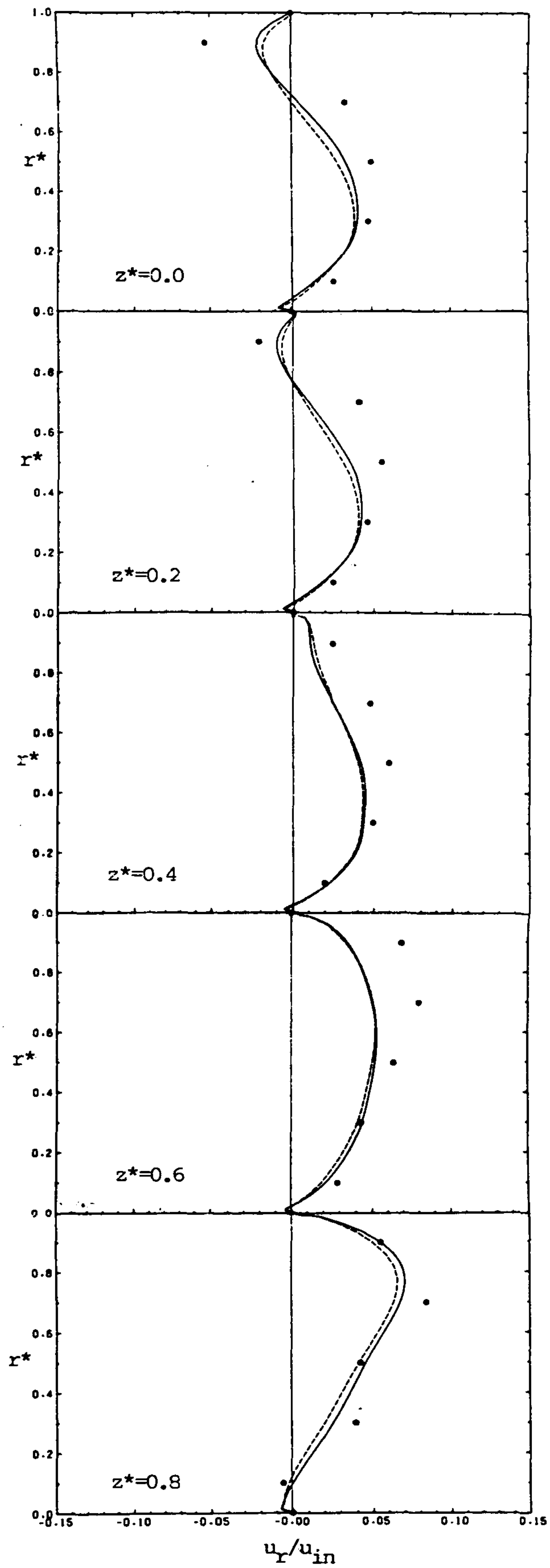


⊙ measurement; ———— present prediction

Fig. 9.44: Spanwise profiles of u_a in cross-section C-C

Fig. 9.45: Spanwise profiles of u_r in cross-section C-C

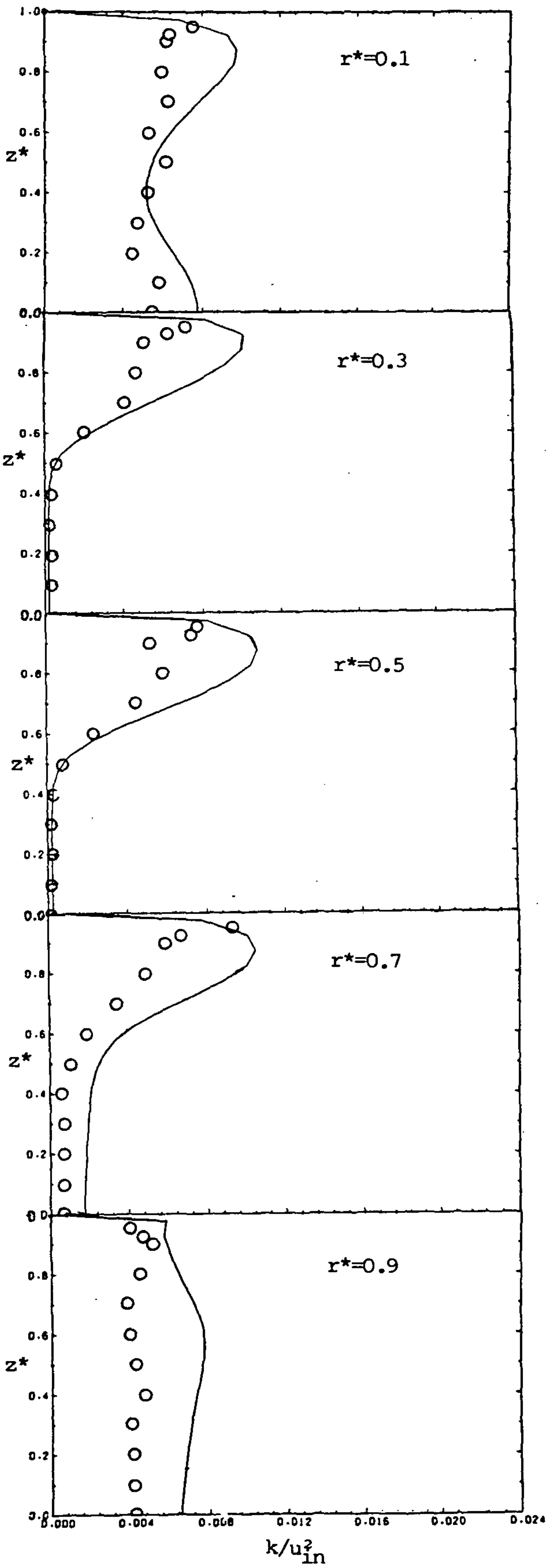


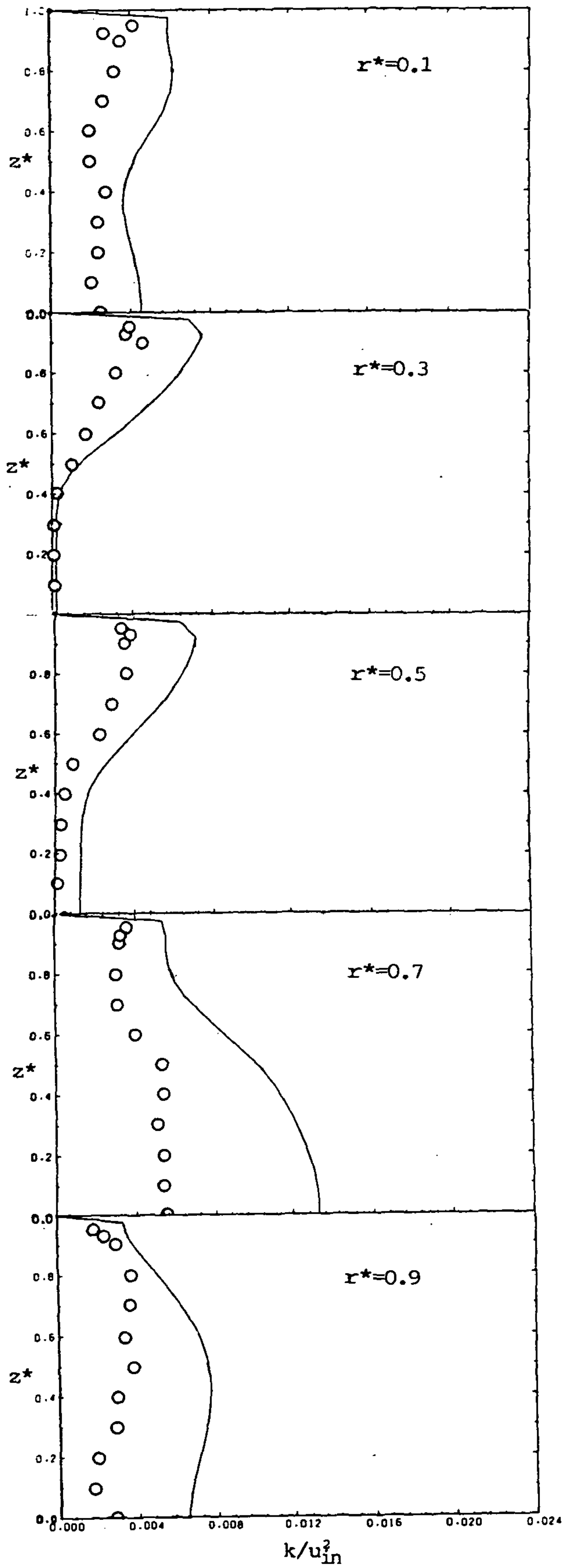


o measurement; predictions: -------25x27x11 CV; ——— 50x40x20 CV

Fig. 9.46: Radial profiles of u_a in cross-section C-C

Fig. 9.47: Radial profiles of u_r in cross-section C-C

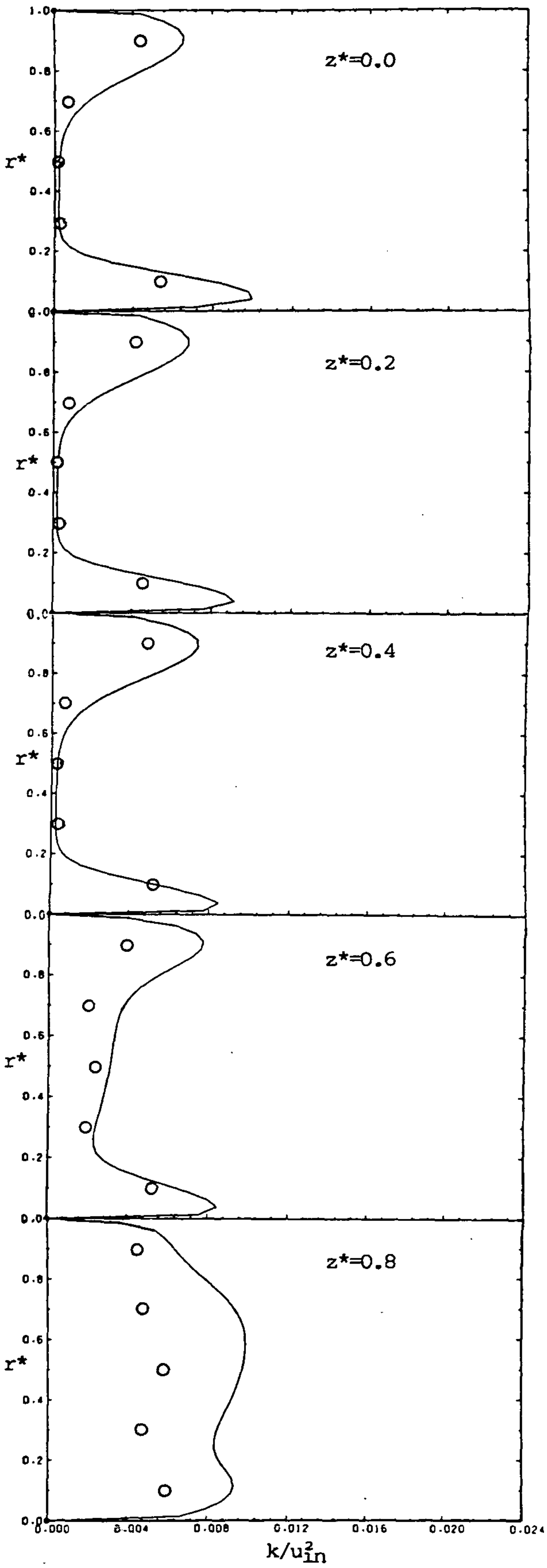


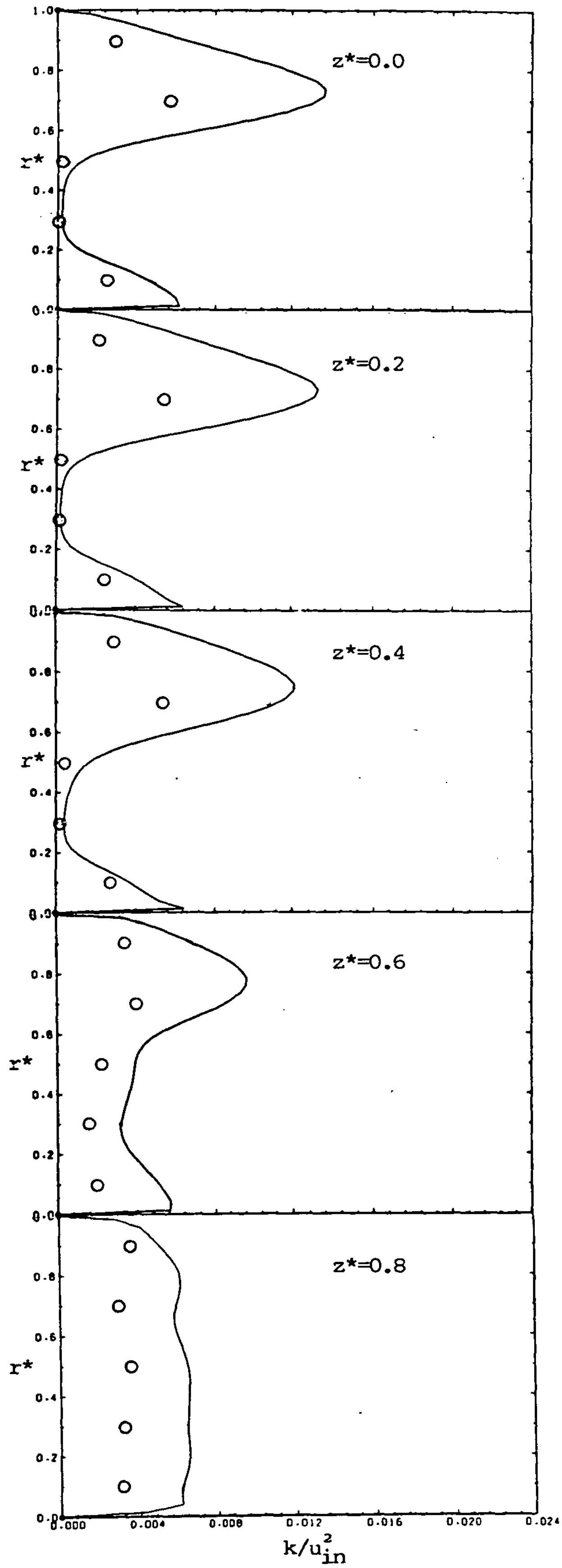


o measurement; ———— present prediction

Fig.9.48: Spanwise profiles of k in cross-section D-D

Fig. 9.49: Spanwise profiles of k in cross-section C-C





o measurement; ———— present prediction

Fig. 9.50: Radial profiles of k in cross-section D-D

Fig. 9.51: Radial profiles of k in cross-section C-C

city is under-predicted for $r^* > 0.8$, and radial more significantly for $r^* > 0.5$, except at $z^* = 0.8$.

Figures 9.48 and 9.49 show spanwise, and Fig. 9.50 and 9.51 radial, profiles of turbulence intensity k/u_{in}^2 , compared with the experimental results* . Although the predicted profiles, in general, have the same shape and relation between themselves as the experimental ones, they show significantly higher levels of turbulence, more so in the cross-section C-C where the predicted values are about 50% higher than the measured ones.

Discussion

There are several possible reasons for the discrepancies between predicted and measured quantities in this case. The main ones are discussed below.

1) Numerical diffusion may have contributed to observed discrepancies, although its role is believed to be a minor one, as discussed earlier.
2) Uncertainties in specifying inlet boundary conditions may have caused the numerical flow to differ from the experimental one.
3) Inadequacies of the k-ε turbulence model and the treatment of wall boundaries in some regions of flow are also believed to be sources of errors. In particular, the curvature of the streamline, the rotation of the velocity vector near the wall and the non-existence of equilibrium of production and dissipation of turbulence are all present to some extent and are in conflict with certain of the assumptions inherent in the modelling. It is also well-known that the k-ε model is not capable of predicting turbulence induced secondary flow in corners (see Launder and Ying, 1973) , where, it will be recalled, the largest discrepancies exist in the present comparisons (by contrast, in the laminar case agreement was always good near the top wall).
4) Asymmetry of the measured flow could also be responsible for some discrepancies. This can be concluded from the reported spanwise velocity component measurements, which show unexpected non-zero values

* Again, because the spanwise component was not measured, it was assumed to be equal to the half of the sum of the other two. If, on the other hand, it was equated to the sum, it would increase the "measured" k by 25% .

in the symmetry plane of up to 4% of the mean streamwise velocity. Also the measured spanwise component was everywhere directed towards the symmetry plane, while the predicted secondary velocity vectors shown in Fig. 9.38 and 9.39 (which appear to be at least qualitatively correct) suggest that values of both signes should be present.

Nonetheless, the results of the present prediction do account for most features of the flow considered, and represent - bearing in mind all the limitations - proof that the present method can be used to reasonably predict complex three-dimensional turbulent flow.

9.3.3 Turbulent flow in a C-shaped diffuser

Description of problem

In this section turbulent flow in a C-shaped diffuser is considered. The solution domain is shown in Fig. 9.52 along with the basic dimensions. The duct consists of two parallel walls at a spacing of 40 mm (labelled top and bottom wall), and two side walls (labelled inner and outer wall) which form a 45° bend with a mean radius of curvature of 280 mm. The cross-sectional area of the bend increases linearly from the square 40 mm x 40 mm at inlet to 60 mm x 40 mm at the exit. An additional length of a straight duct after exit from the bend is also included in the calculations. The flow is turbulent and Reynolds number is set at 40 000 (defined with respect to inlet mean velocity u_{in} and hydraulic diameter D_H=40 mm) to conform with the experimental study of Rojas et al (1984). Their results will be compared with the present predictions.

Due to existence of a horizontal symmetry plane between top and bottom walls (see Fig. 9.52), only half the duct was considered. A computational grid of 24 x 40 x 20 CV was constructed using Practice 1 of Chapter 7 in horizontal sections (see Fig. 9.53 and 9.54 (a)). No grid dependence tests were performed, due to lack of time and limited computing resources. Instead, the results of the grid tests in the S-diffuser calculations presented earlier were used as guidance. In the presentation which follows, the velocity component normal to radial cross-sections is termed "streamwise" (u_a), the velocity in the radial direction is called "radial" (u_r) and the velocity component normal to the symmetry plane is called "spanwise" (u_s). The boundary conditions applied were the same as in the turbulent S-diffuser case: in particular the same inlet values were used, since the experimental conditions there

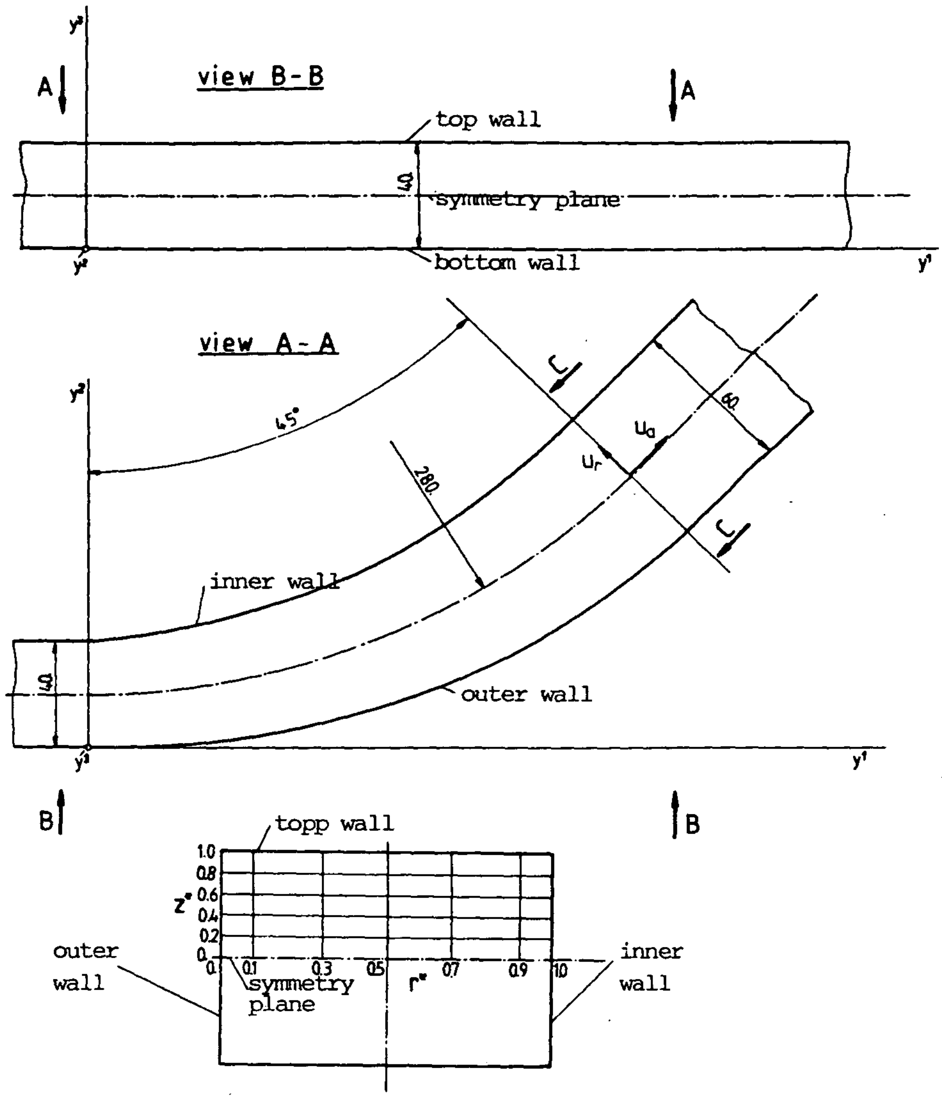


Fig. 9.52: C-diffuser geometry

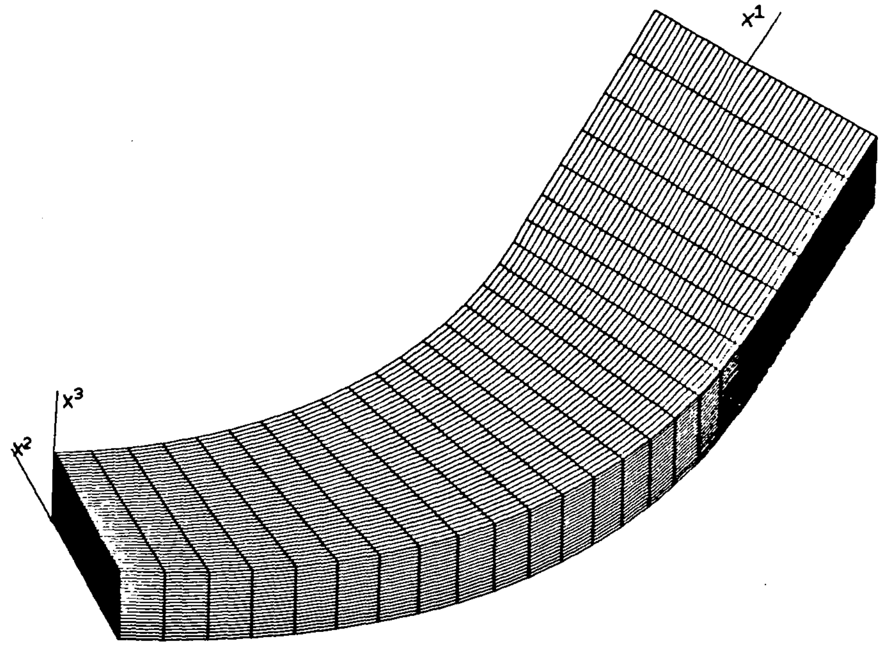


Fig. 9.53: Computational grid, 24x40x20 CV

were identical for the two diffusers. The comments about slow convergence made for the S-diffuser calculations also apply here , for the same reasons. A total of 350 iterations was needed to reach a converged solution ($\lambda_s \simeq 2\cdot 10^{-3}$ in this case), with α_u=0.75 , α_p=0.2 and α_k =0.8 . This required 42 minutes of computing time on a CRAY 1S computer and 800 000 words of memory.

Results and comparison with experiment

In Fig. 9.54 some predicted flow characteristics in the symmetry plane are presented. Contrary to the S-shaped diffuser case, here the core of uniform velocity disappears before the exit from the bend, and a pronounced peak forms near the outer wall (at about r*=0.2 , where r* is the normalized radial distance, measured from the outer wall). The pressure contours (normalized with ρu_{in}^2) are very similar in the symmetry plane and at the top wall. The radial pressure gradient is everywhere directed towards the inner wall, which results in a single secondary flow vortex, as will be seen later. The difference in pressure level on the two curved walls, as is evident from isobar contours, is about $0.1\rho u_{in}^2$, and the recovery at the exit from the bend is about $0.2\rho u_{in}^2$. These observations accord with those of Rojas et al (1984).

As the flow goes through the bend, the peak in turbulence intensity profiles near inner wall moves towards the centreline, due to the existence of steep velocity gradients there. The turbulence levels are, in general, somewhat lower than in the S-shaped diffuser case.

Figure 9.55 shows some flow characteristics in cross-section C-C (see Fig. 9.52). The secondary flow vectors show one vortex centered at about r*=0.5 , z*=0.7 : here z* is the normalized spanwise coordinate, measured from the symmetry plane. The maximum secondary velocity is about 15% of the u_{in} . The predicted contours of streamwise velocity component are similar to those in the S-shaped diffuser, and represent a typical pattern found in curved-duct flows (Rojas et al, 1984). The measured contours are also shown, and the agreement is reasonable.

Also shown are the predicted and measured turbulence intensities ($\sqrt{k}/u_{in}$), in Fig. 9.55 (c). These too have features in common, although the agreement is not obvious everywhere, due partly to the coarsness of the data.

Figures 9.56 and 9.57 present comparisons between the predicted and measured spanwise profiles of the streamwise and radial velocity components in cross-section C-C , at five locations of constant r* .

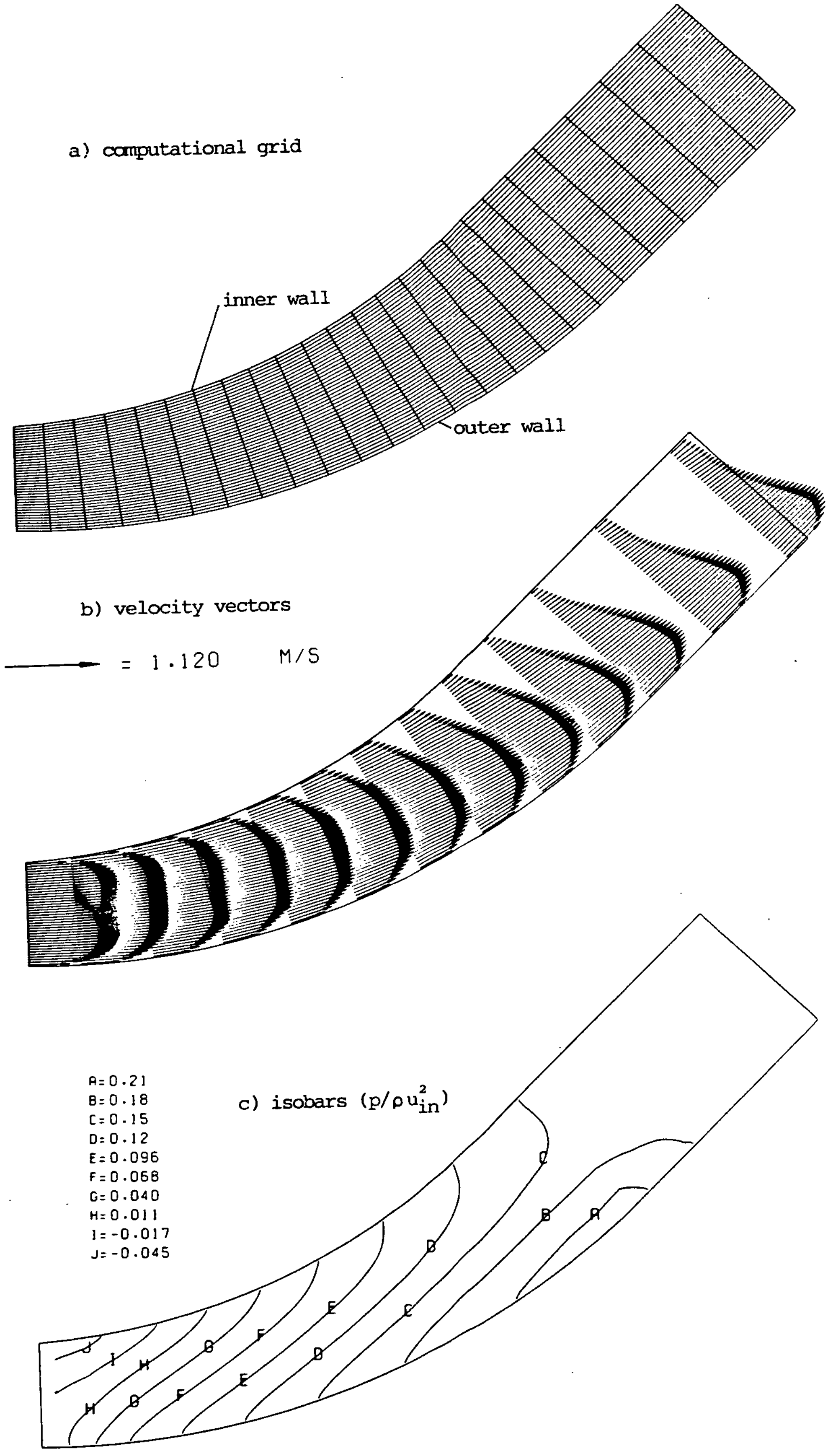


Fig. 9.54: Turbulent flow characteristics in symmetry plane

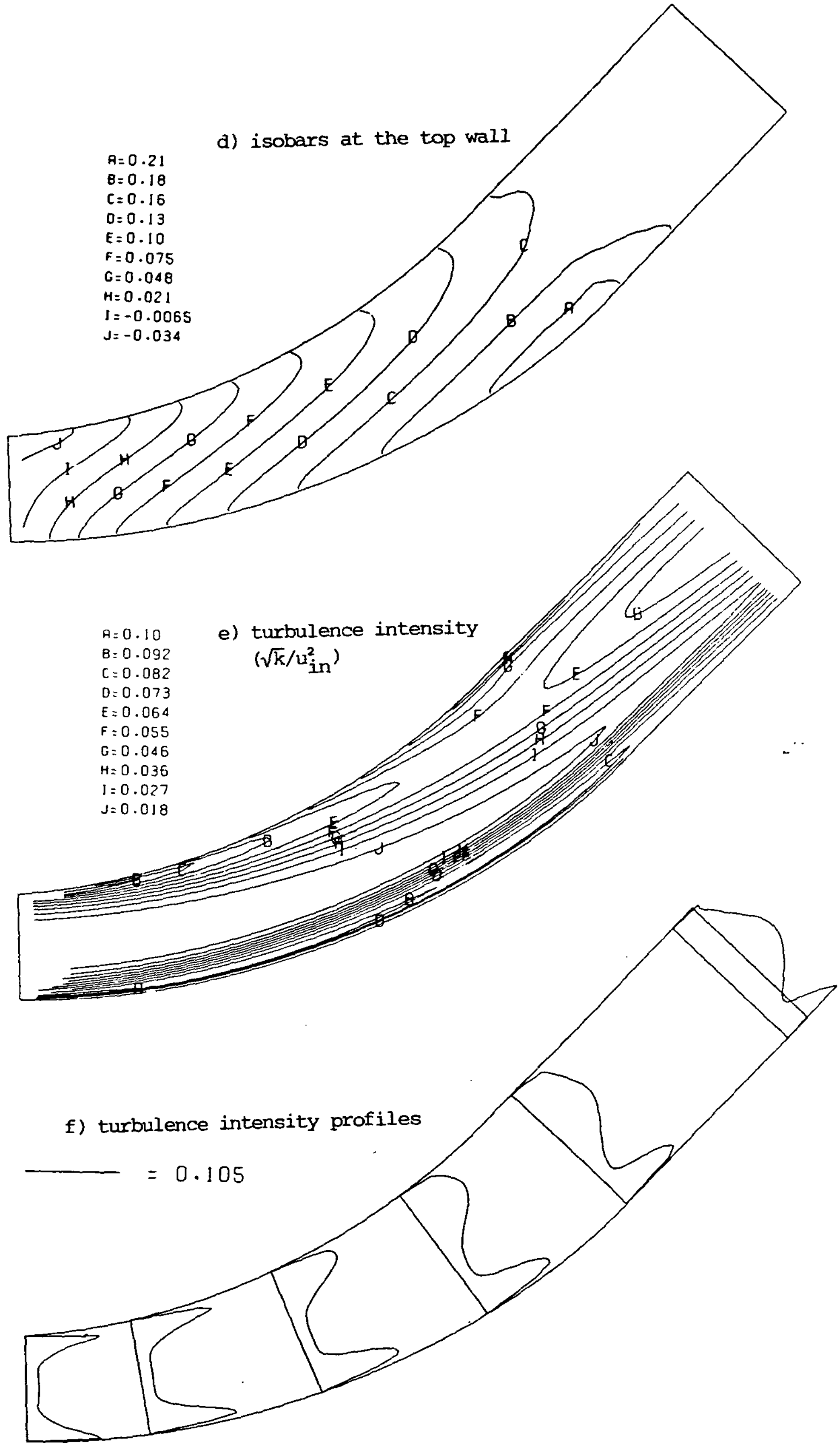


Fig. 9.54: continued

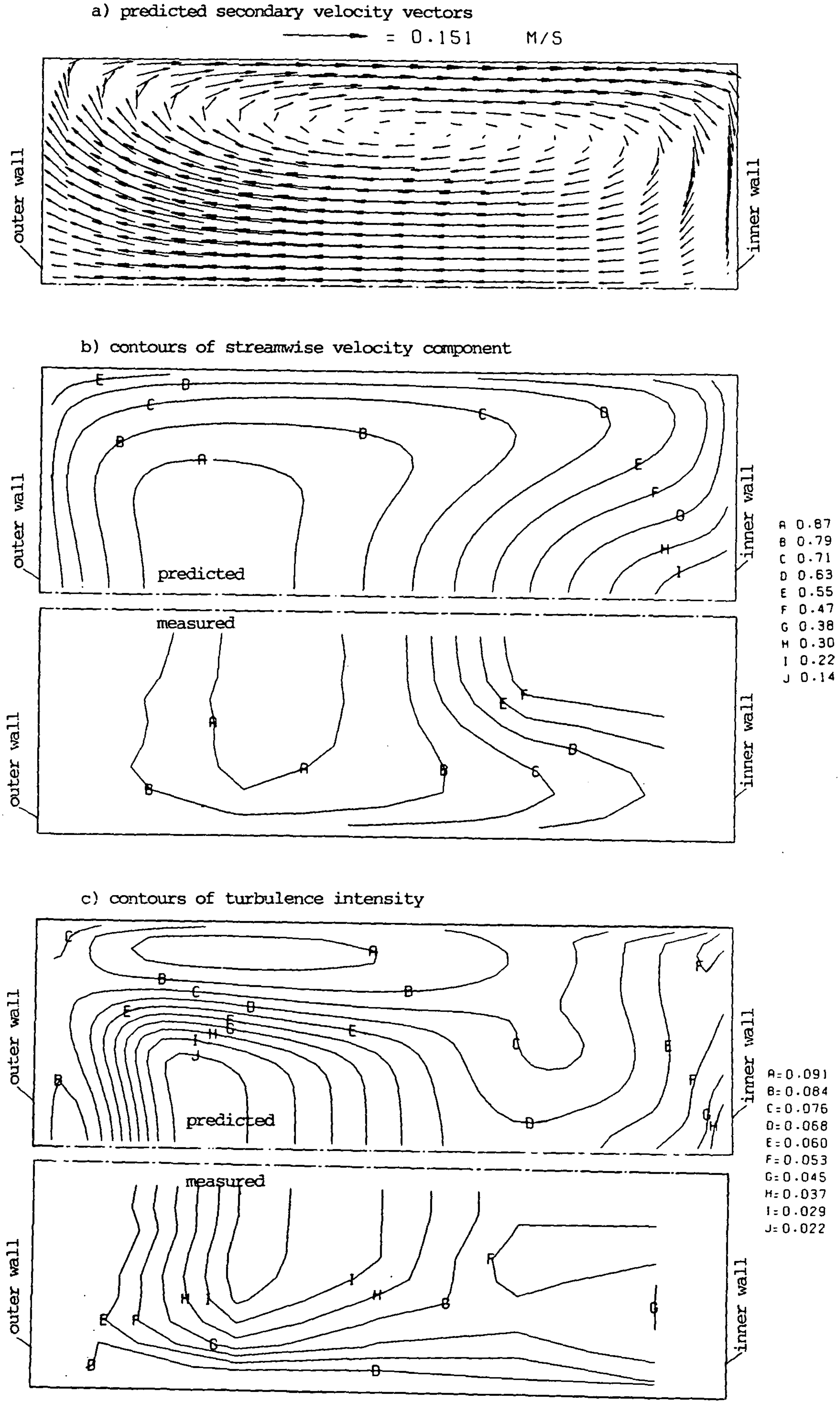


Fig. 9.55: Turbulent flow characteristics in cross-section C-C

Although showing qualitatively the same trends, the predicted profiles differ from the measured ones in some regions by up to 10% u_{in} . The radial velocities are in relatively good agreement at $r^* > 0.5$ (except for $r^* \sim 0.7$ and $z^* < 0.4$), but differ significantly near the outer wall, notably the $r^*=0.1$ and 0.3 profiles. In particular, at $r^*=0.1$ and $z^* > 0.8$ the predicted radial velocities are directed towards the inner wall (see also Fig. 9.55 (a)), while the measured ones are in the opposite direction.

Radial profiles of the same two velocity components are shown in Fig. 9.58 and 9.59 , for five constant z^* locations. The qualitative agreement is obvious, except for the radial velocity component at $z^*=0.8$ and $r^* < 0.5$. However, the gradients of the streamwise velocity component for $r^* > 0.5$ are underpredicted, as well as the peak in $z^*=0.8$ profile (for $\sim$10%).

Figures 9.60 and 9.61 show the predicted spanwise and radial profiles of turbulence intensity in cross-section C–C , compared with values calculated from the experimental data* . Again qualitative agreement can be seen, except for the radial profile at $z^*=0.8$ and $r^* < 0.5$, where disagreement was also observed in the radial velocity component (cf Fig. 9.59). In the symmetry plane ($z^*=0$), as well as in the two nearest parallel planes ($z^*=0.2$ and $z^*=0.4$) even quantitative agreement could be considered satisfactory, bearing in mind the uncertainity in presentation of experimental data and all the other limitations. Elsewhere, in general, the predicted turbulence levels are higher than the measured ones, by up tp 50% (eg. on $r^*=0.3$ and $r^*=0.5$ profiles).

The reasons for the observed discrepancies between the predicted and measured values are believed to be the same as in the previous case, with perhaps somewhat larger contribution from numerical inaccuracies, due to coarser grid used here.

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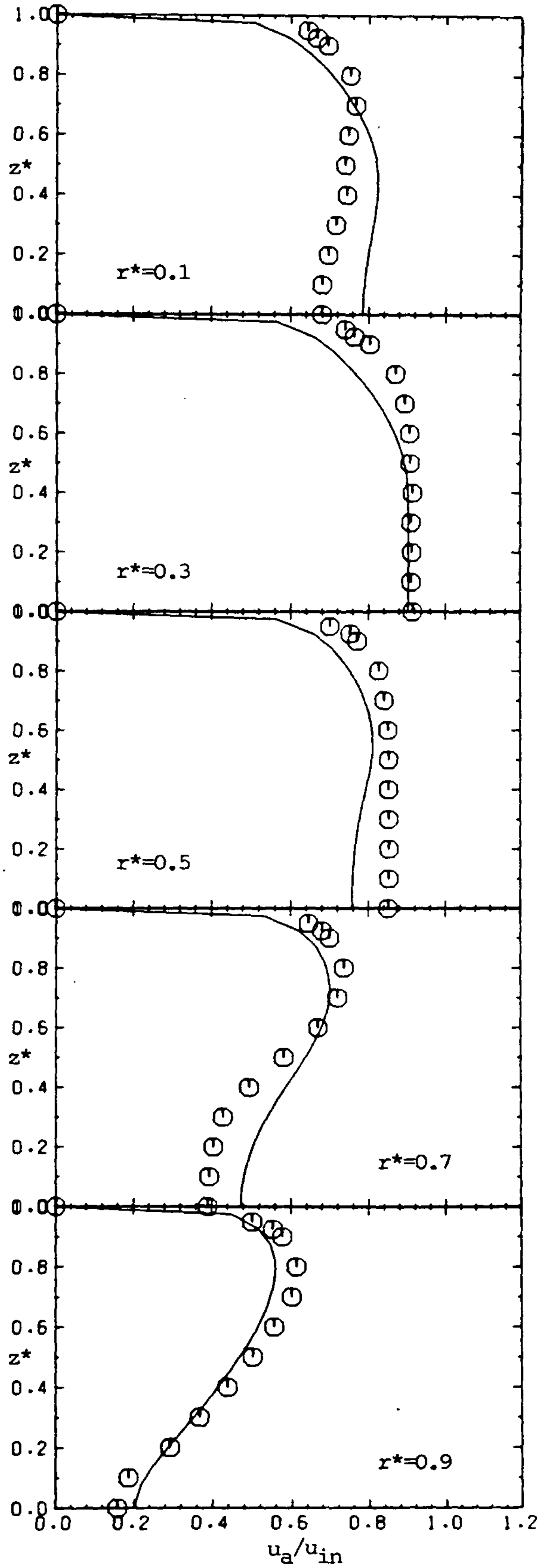
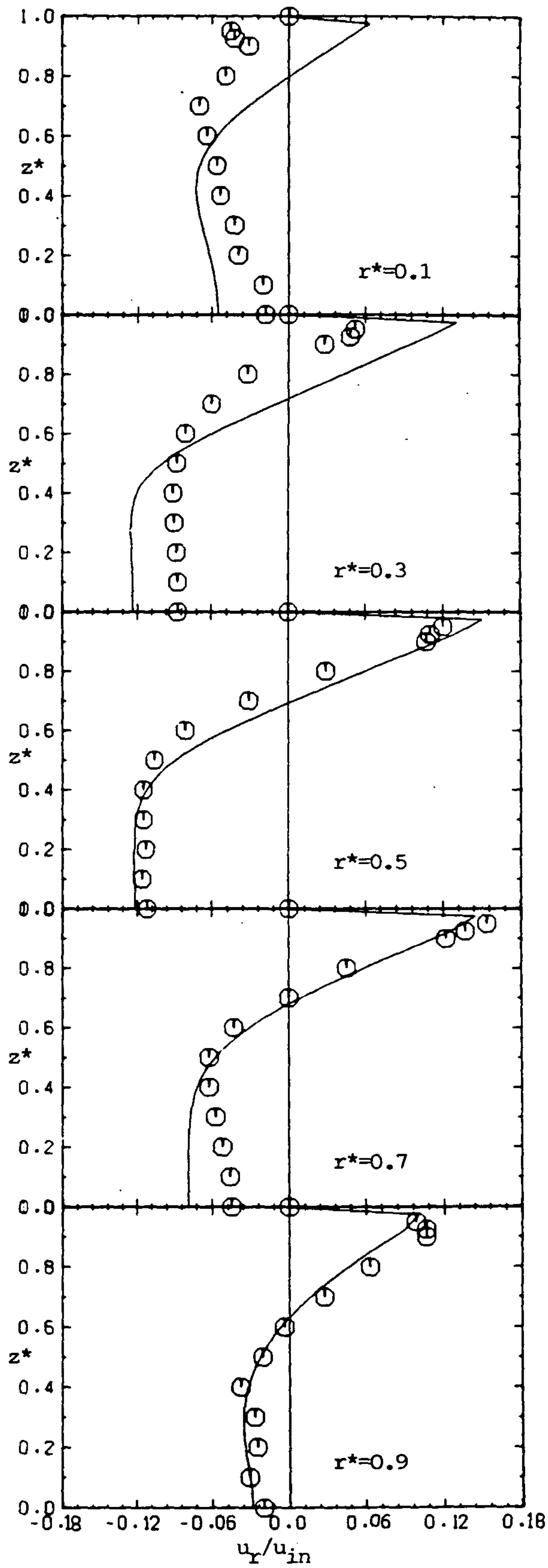
* These were determined in the same way as in the S-diffuser case, and are subject to the same uncertainities.

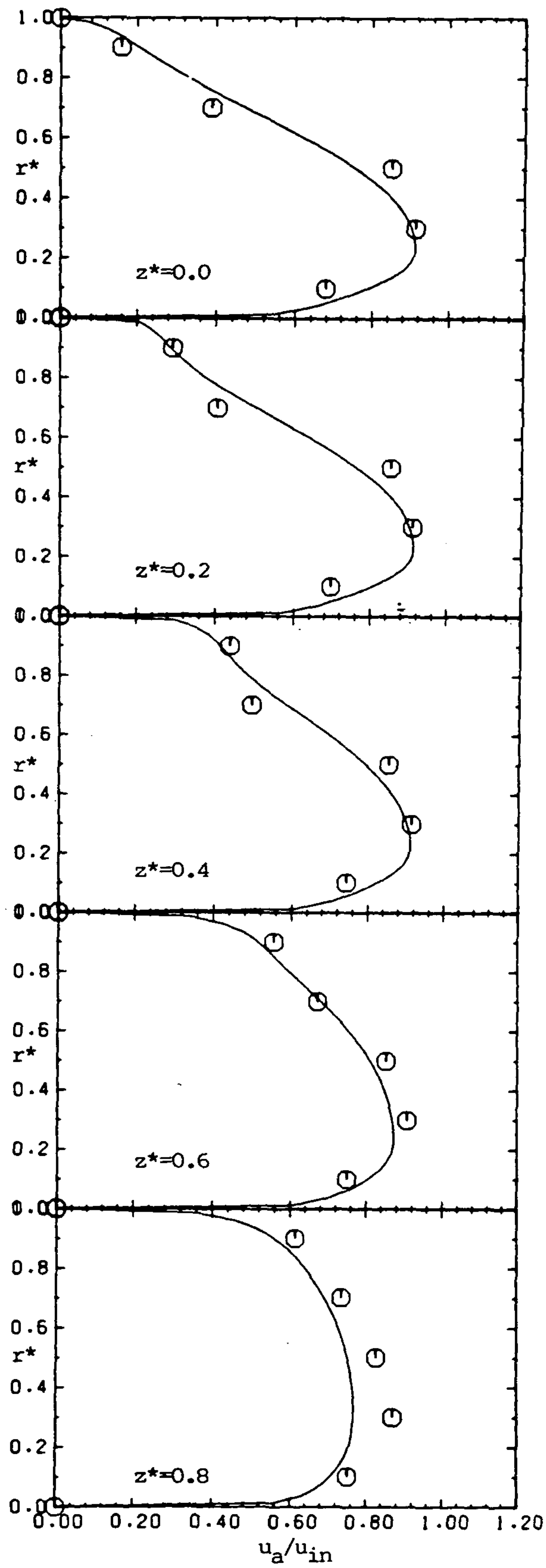


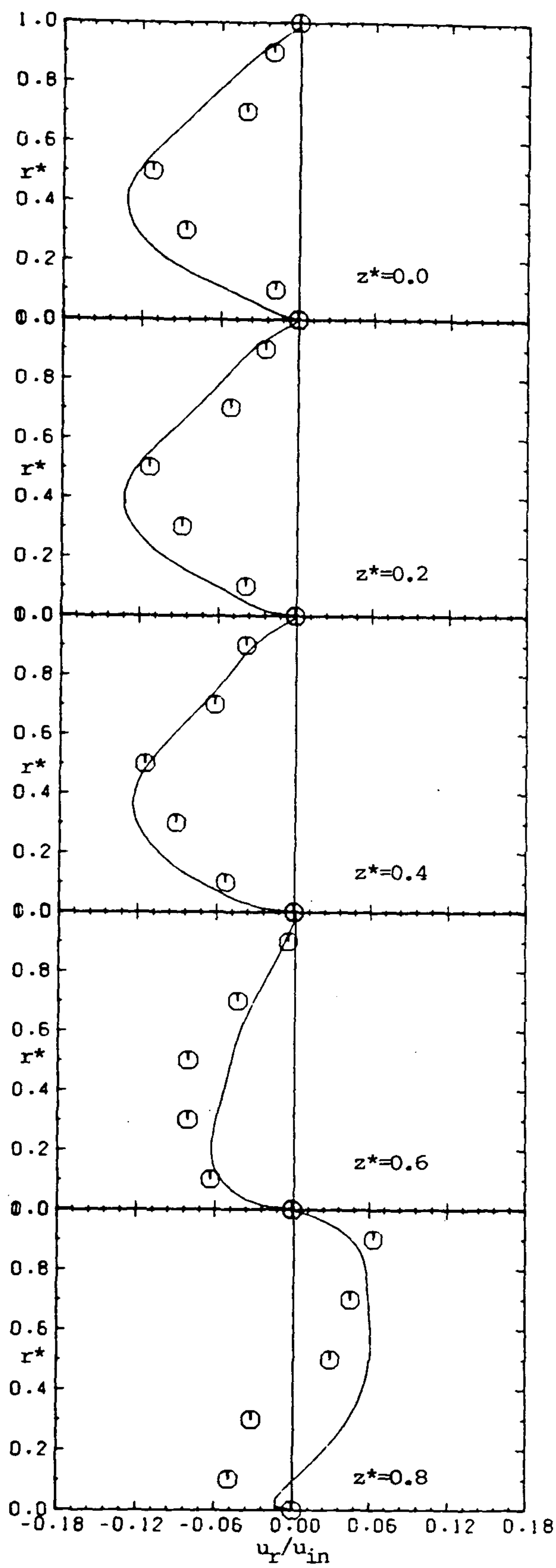


measurement; ——— present prediction

Fig. 9.56: Spanwise profiles of u_a in cross-section C-C

Fig. 9.57: Spanwise profiles of u_r in cross-section C-C

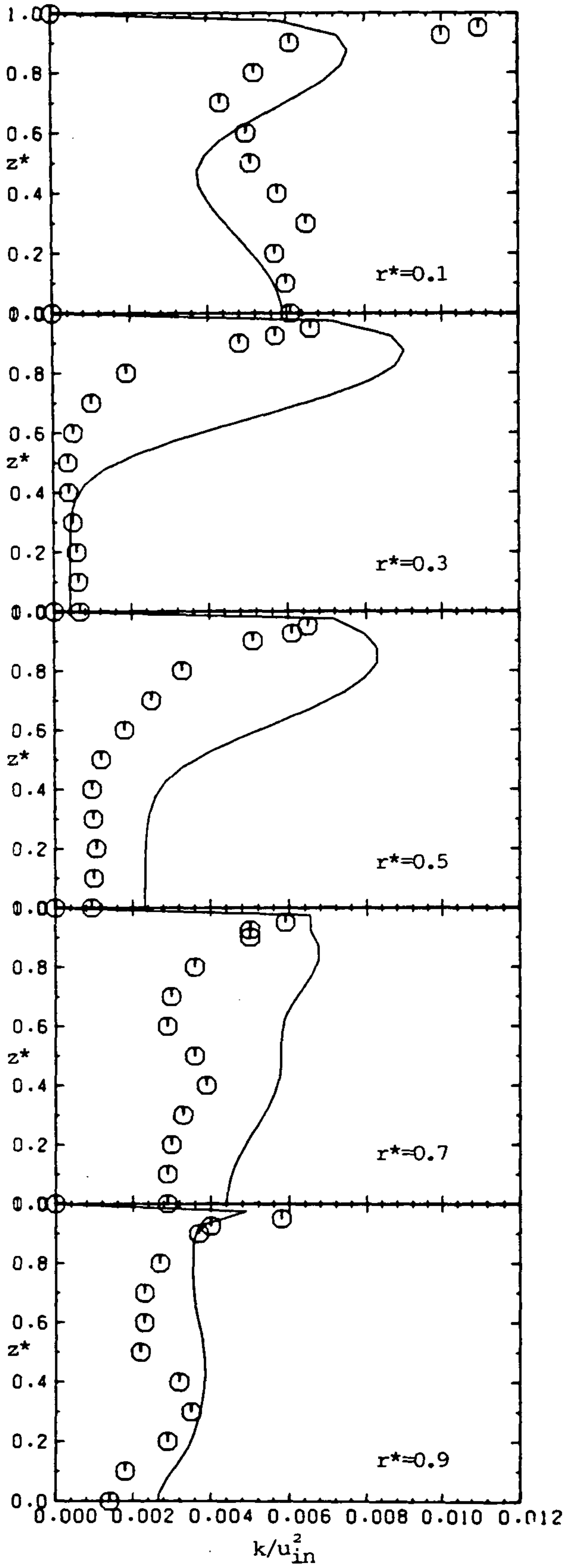


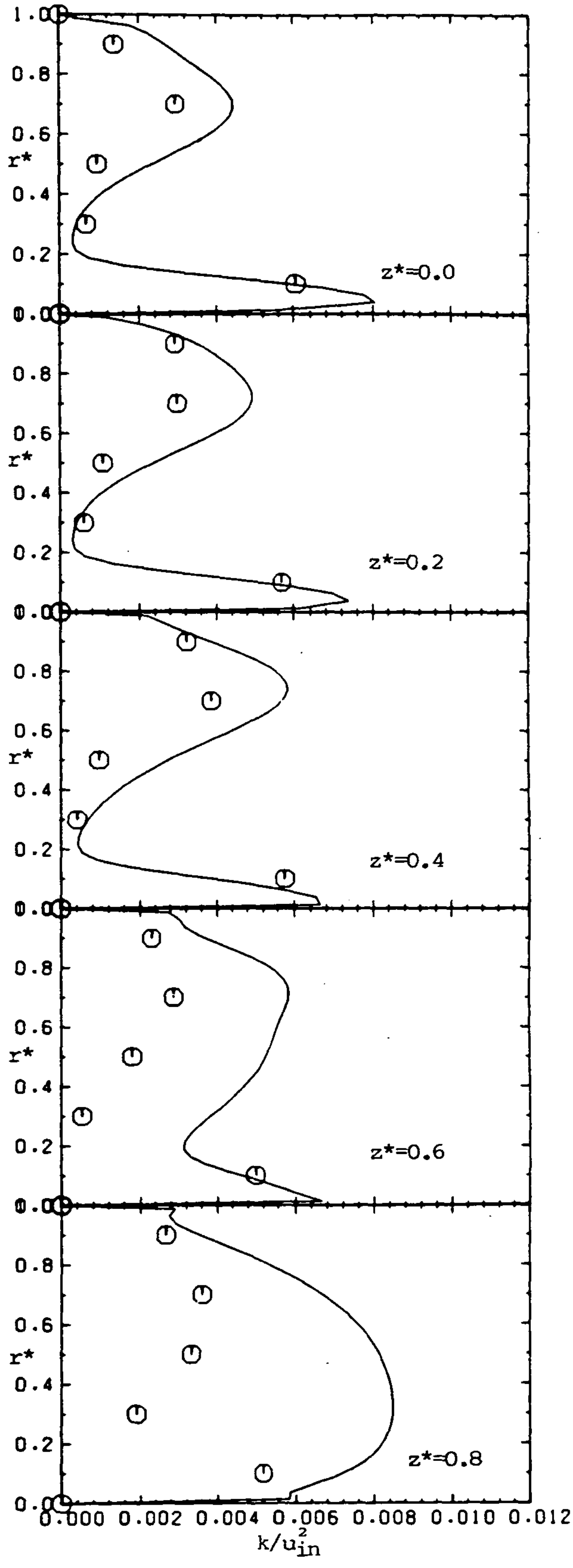


measurement; ——— present prediction

Fig. 9.58: Radial profiles of u_a in cross-section C-C

Fig. 9.59: Radial profiles of u_r in cross-section C-C





measurement; ——— present prediction

Fig. 9.60: Spanwise profiles of k in cross-section C-C

Fig. 9.61: Radial profiles of k in cross-section C-C

9.4 Demonstration of Possible Applications of the Method

9.4.1 Flow across in-line tube banks

In-line banks of tubes are also often found in engineering practice (eg. in heat exchangers). In the fully developed regime a unit of symmetry can be identified, as shown in Fig. 9.62 (a) , where a 29×13 CV grid was constructed over it. At inlet and outlet periodic boundary conditions apply, while the boundaries elsewhere are made up of tube walls and symmetry planes.

Figures 9.62 (b)-(f) show typical features of the predicted flow at $Re_D = 50\,000$, where the Reynolds number Re_D is defined with respect to mean inlet velocity and tube diameter. The plot of streamlines (c) indicates the existence of two vortices, which would probably detach, if the longitudinal pitch was increased. The isobars (d) show a high pressure region where the flow impinges onto the tube, a rapid decrease over the front, and a second increase over the rear part of the tube until separation occurs, and finally a large zone of almost uniform pressure between the tubes. A region of high turbulence (and presumably high heat transfer) is also seen in the impingement area, while the maximum length scales are, as expected, near symmetry plane between the tubes.

9.4.2 Flow around submerged body

In the second demonstration case, flow around a submerged body resembling a car shape confined between two parallel walls (a "wind tunnel") was considered. The geometry and the grid used (48×21 CV) are shown in Fig 9.63 (a) ; the manner of grid generation for this problem was described in Chapter 7. Around 60 iterations were required to obtain a converged solution, using under-relaxation factors $\alpha_u = 0.75$, $\alpha_p = 0.2$ and $\alpha_{k,\varepsilon} = 0.75$.

Figures 9.63 (b)-(j) show the predicted features of this complex flow. These are self-explanatory and need no further comment. Again, the expected features of the flow, such as the recirculation behind the body, pressure recovery and high turbulence in the wake are correctly predicted. Also evident is the irregularity of the length scale profiles near duct walls in the vicinity of the body and its wake, as noted earlier.

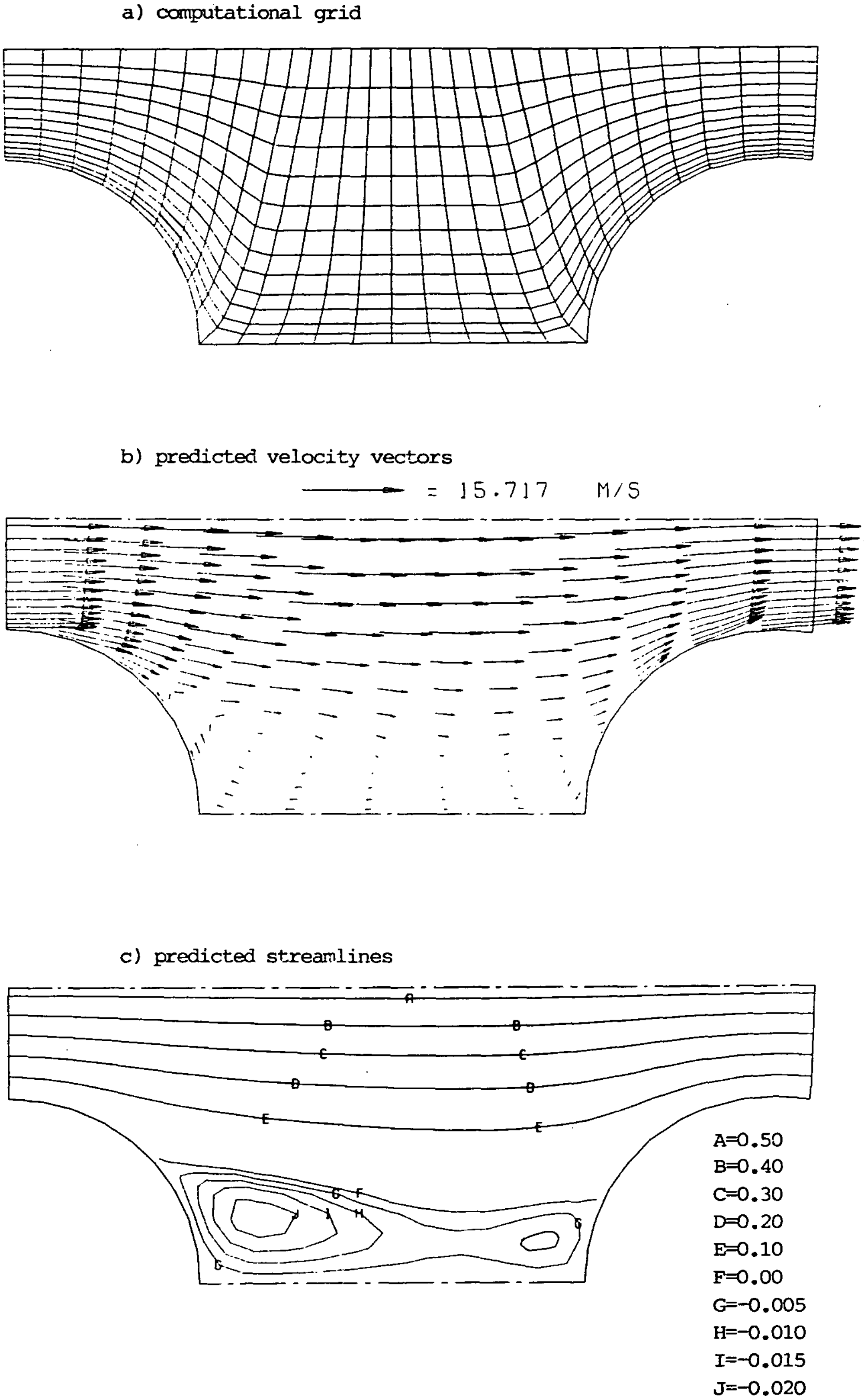
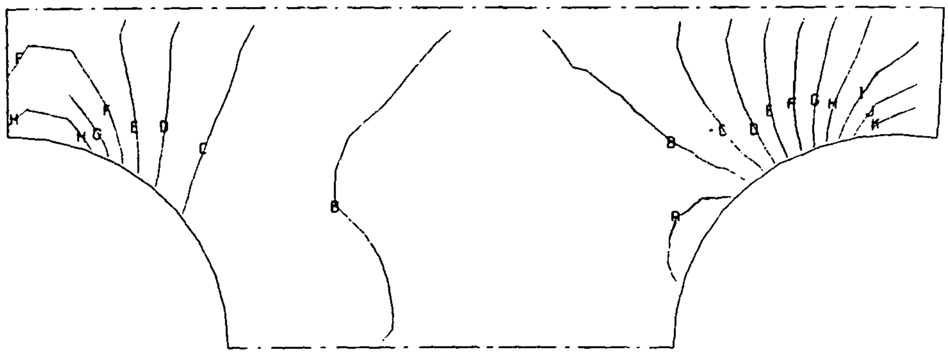


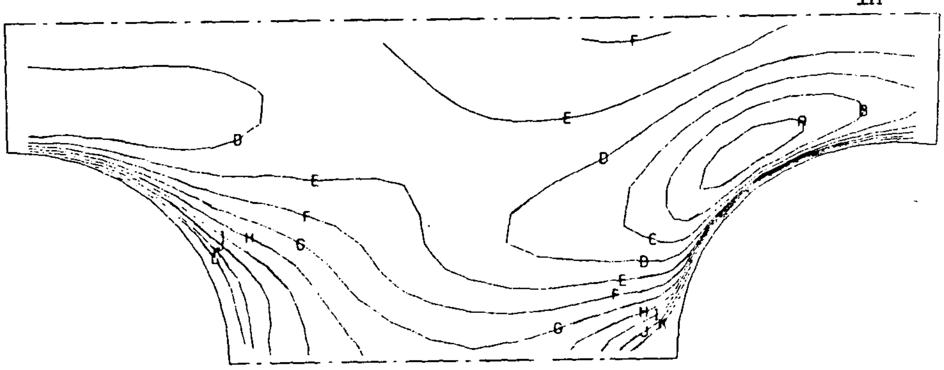
Fig. 9.62: Predicted features of turbulent flow across in-line tube bank at Re_D=50000

d) predicted isobars ($p/\rho u_{in}^2$)



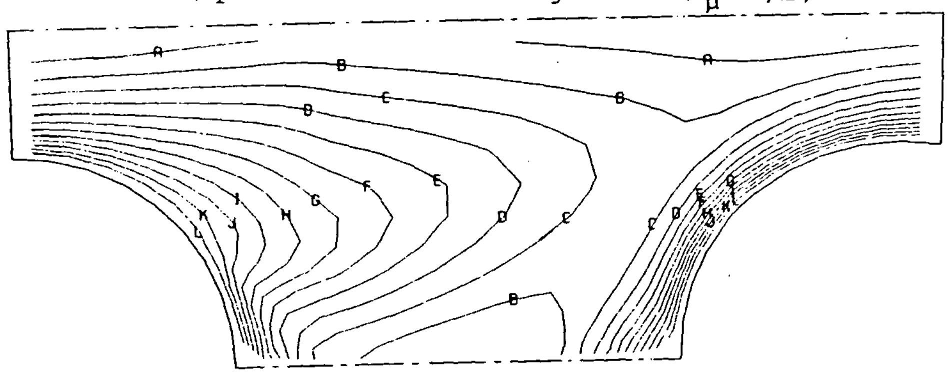
A=0.44
B=0.39
C=0.34
D=0.29
E=0.24
F=0.19
G=0.14
H=0.091
I=0.040
J=-0.01
K=-0.06
L=-0.11

e) predicted contours of turbulence intensity ($\sqrt{k}/u_{in}$)



A=0.23
B=0.22
C=0.21
D=0.20
E=0.19
F=0.17
G=0.16
H=0.15
I=0.14
J=0.13
K=0.11
L=0.10

f) predicted contours of length scale ($C_\mu k^{3/2}/\varepsilon D$)



A=0.056
B=0.051
C=0.047
D=0.042
E=0.038
F=0.033
G=0.029
H=0.025
I=0.020
J=0.016
K=0.011
L=0.006

Fig. 9.62: continued

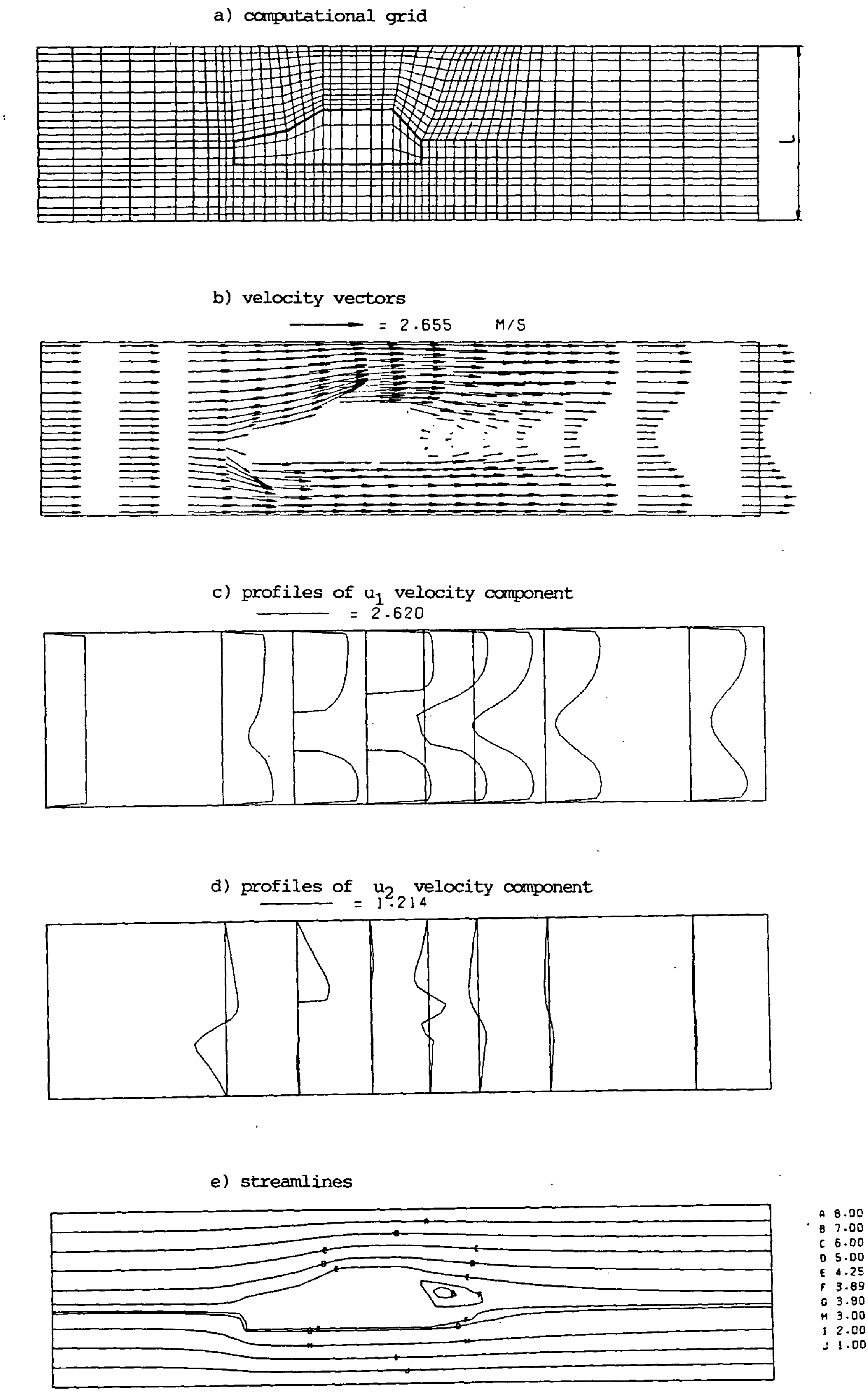
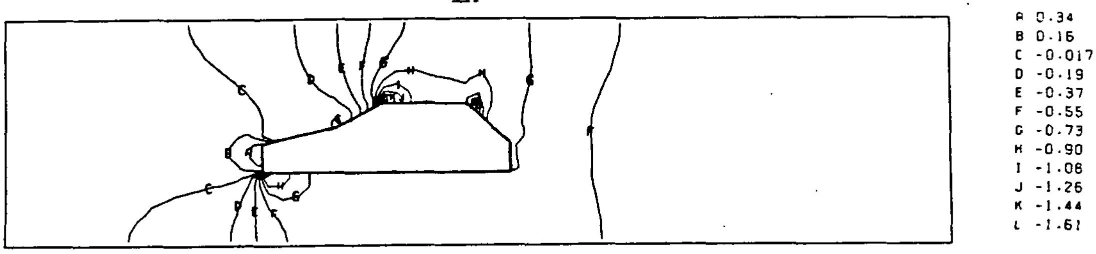
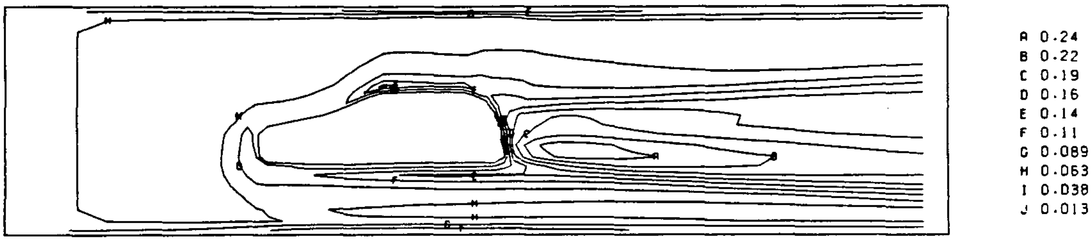


Fig. 9.63: Details of predicted turbulent flow around submerged body

f) isobars ($p/\rho u_{in}^2$)

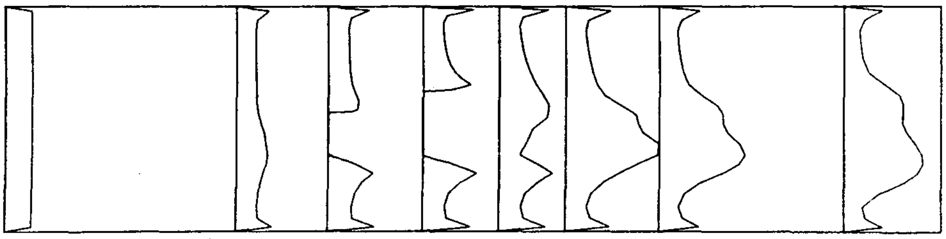


g) turbulence intensity ($\sqrt{k}/u_{in}$)

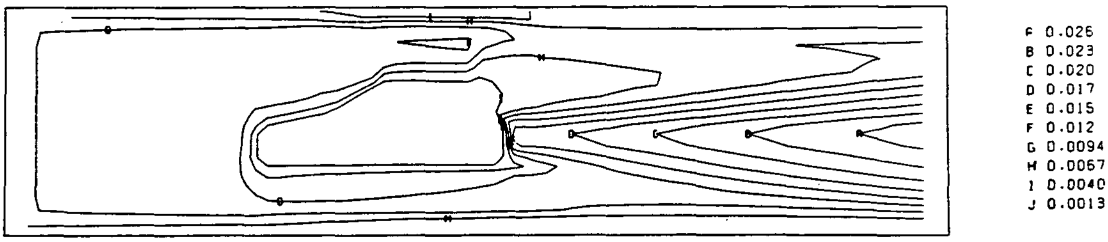


h) profiles of turbulence intensity

——— = 0.253



i) length scale ($C_\mu k^{3/2}/\varepsilon L$)



j) length scale profiles

——— = 0.027

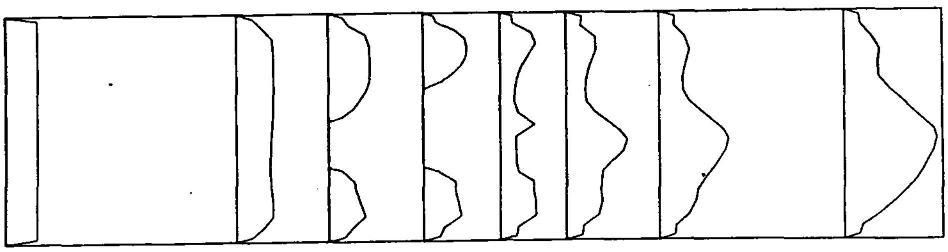


Fig. 9.63: continued

9.4.3 Flow in a complex duct

The example in Fig. 9.64 (a)-(f) shows predictions of turbulent flow in a complex duct. The grid, consisting of 100×12 CV , is shown in Fig. 9.64 (a); two curved boundaries have several sharp corners and turnings through $90°$ or $180°$. There are regions of high grid non-orthogonality (up to $30°$), high cell aspect ratios and also grid non-smoothness. Around 100 iterations were needed to obtain a converged solution, using $\alpha_u=0.7$, $\alpha_p=0.2$ and $\alpha_{k\varepsilon}=0.7$. Figures 9.64 (b)-(f) show some typical flow characteristics. There are three major recirculation regions, and a very high turbulence levels exist where the flow impinges at right angle onto the duct wall near exit.

The same two-dimensional flow was predicted using the computer program for three-dimensional flows, by taking the third dimension to be two orders of magnitude larger than those in cross-sectional plane. The number of iterations and the results were the same to within convergence tolerance.

Three-dimensional flow in a duct whose cross-section is represented by Fig. 9.64 (a), with two other walls being parallel and at a distance 1.25 L was also performed. Due to existence of symmetry plane between the parallel walls, only half the duct was considered. The grid had $100 \times 12 \times 10$ CV , and was the same in each cross-section parallel to symmetry plane. The number of iterations needed to obtain converged solution was about the same as for the two-dimensional case, using the same set of under-relaxation parameters.

Interesting secondary flow patterns at several cross-sections marked in Fig. 9.64 (a) are shown in Fig. 9.65 (a)-(f). The three-dimensionality effects are clearly seen, as well as the edge of recirculation zone in plots (c) and (f). In the symmetry plane the flow was similar to that obtained for two-dimensional case, because of relatively large distance between the parallel walls.

9.5 Closure

In this section results of applications of the present method to some complex laminar and turbulent flows were presented and compared with available experimental data. Based on the evidence from these applications, the following conclusions can be drawn:

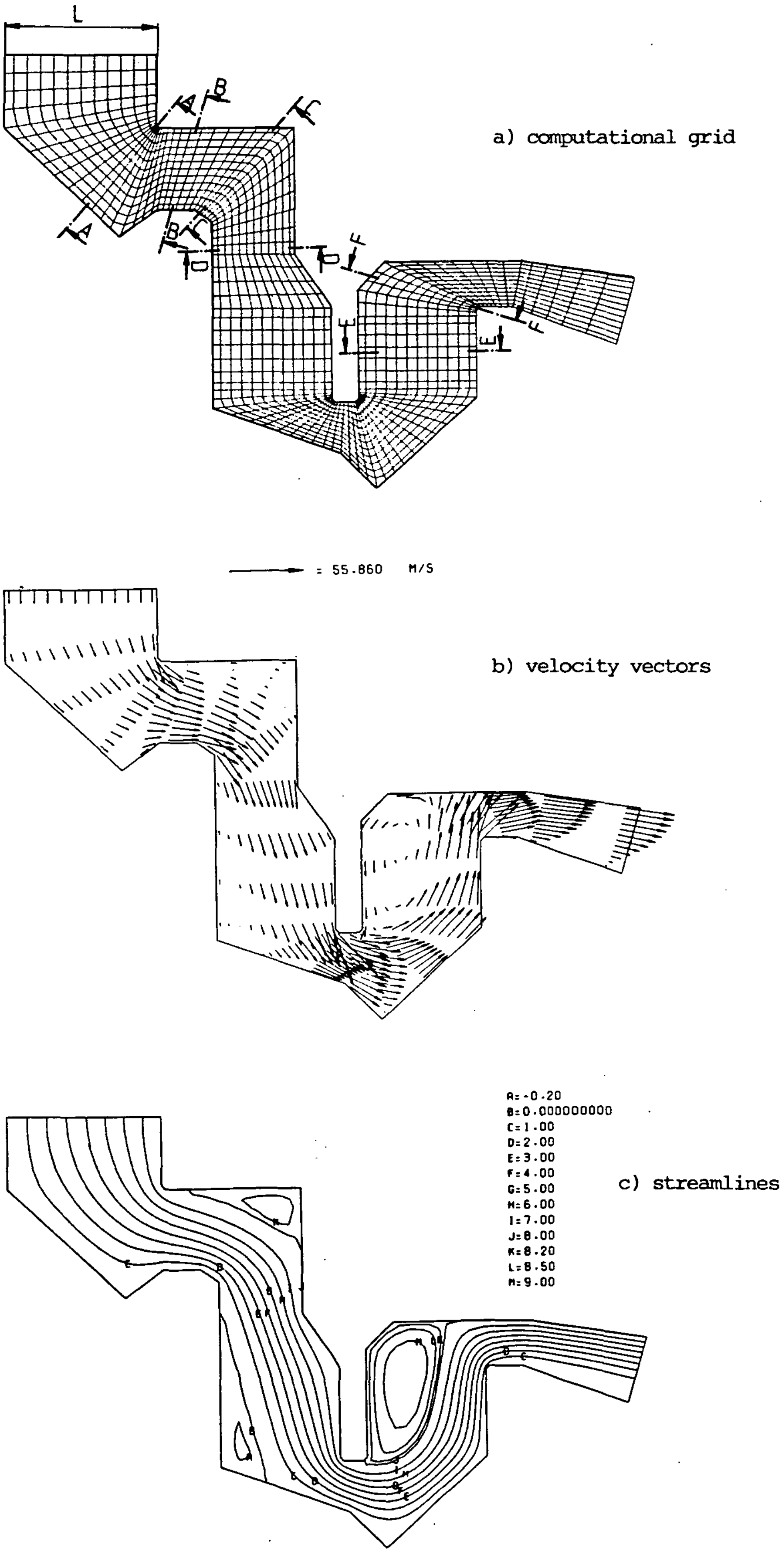
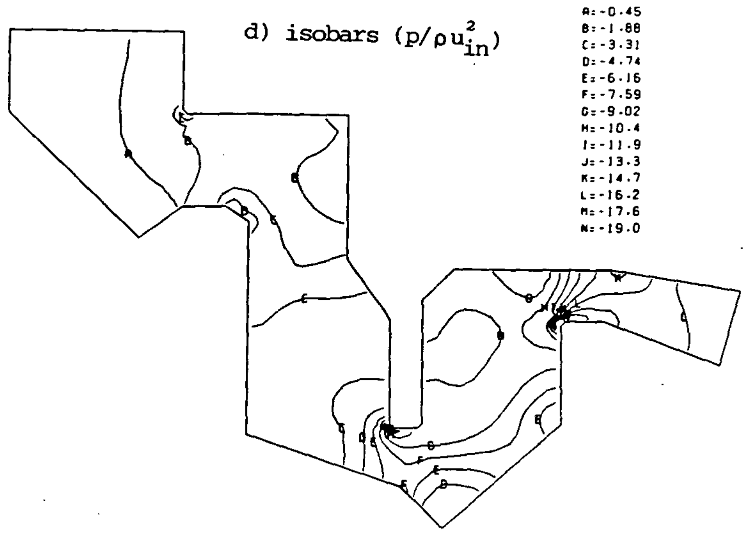
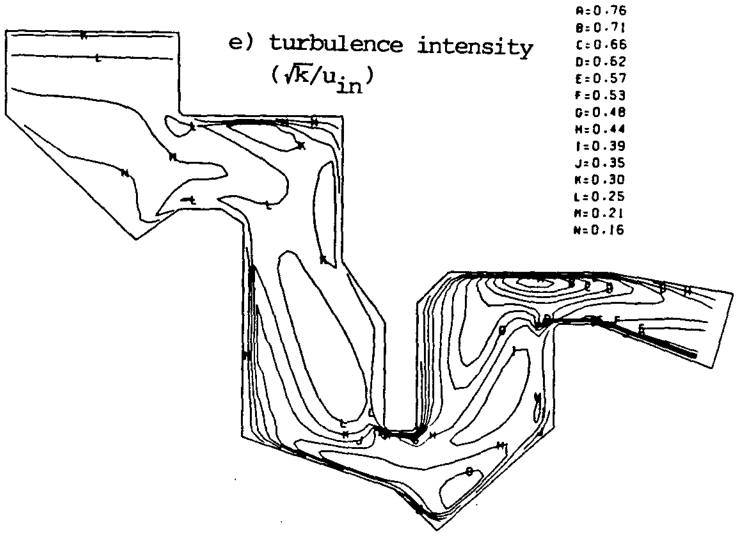


Fig. 9.64: Predicted features of two-dimensional turbulent flow in a complex duct





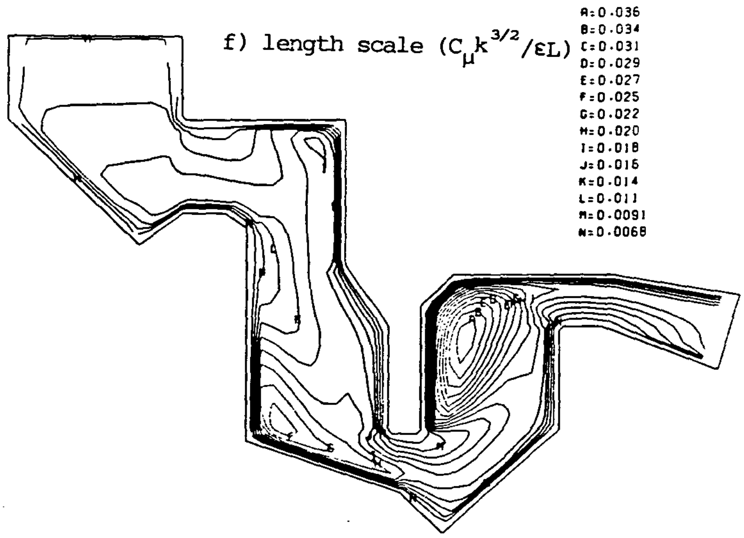


Fig. 9.64: continued

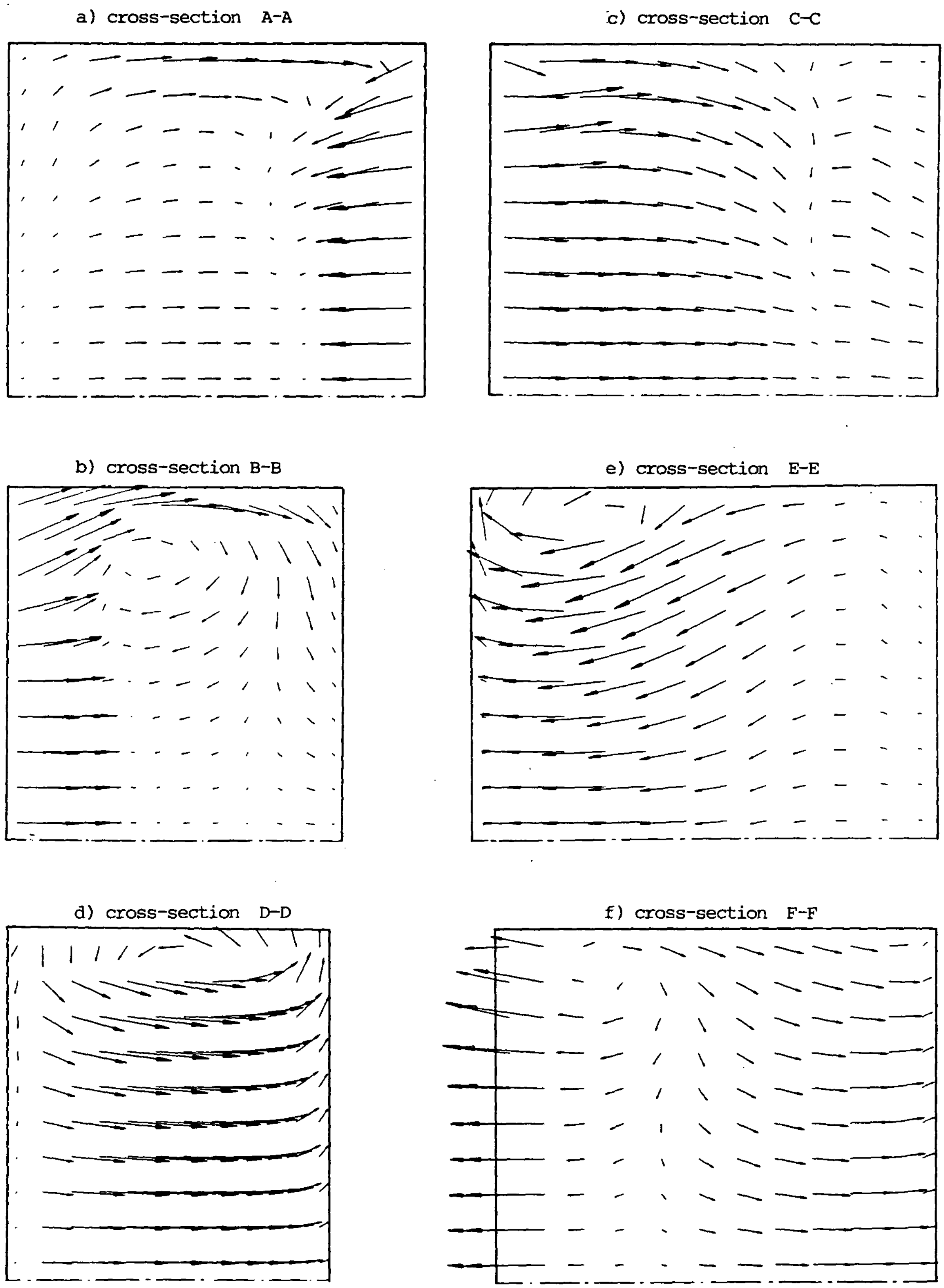


Fig. 9.65: Predicted secondary velocity vectors in some cross-sections of a complex three-dimensional duct

1) The present method is capable of predicting both two- and three-dimensional fluid flow in domains of arbitrary geometry. No major problems of instability or any other kind were encountered, in spite of the fact that grids with regions of high non-orthogonality, high aspect ratios and non-smooth grid lines were employed.
2) Laminar flow can be predicted with good accuracy on non-orthogonal grids, provided that the grid is fine enough for adequate resolution of steep gradients. In all two-dimensional cases considered here almost grid independent solutions were achieved with a reasonable number of nodes. Computer limitations prevented similar results being obtained in the three-dimensional cases studied.
3) Predictions of turbulent flows showed qualitative agreement with experimental data and reproduced reasonably well the expected flow behaviour and mean flow parameters of engineering interest. However, quantitative comparisons showed some appreciable differences between the predictions and experimental data. These were partly due to numerical errors, but mostly to inadequacies of the $k-\varepsilon$ turbulence model and associated wall functions when complex recirculating flows are in question.
4) The accuracy of numerical predictions could be further improved by: (i) using computational grids of higher spatial resolution, (ii) using higher-order interpolation practices (differencing schemes), (iii) using multi-layer wall functions (eg. those of Amano, 1984), and (iv) using higher-order turbulence closure models (eg. Reynolds stress model of Hanjalić and Launder, 1972, or algebraic stress model of Rodi, 1976).
5) In order to further evaluate the accuracy of methods of the present kind, and especially in order to improve turbulence modelling, more detailed, reliable and complete experimental data for complex flows are needed.

CHAPTER 10

SUMMARY AND CONCLUSIONS

10.1 Achievements and Findings

In the course of work presented in this thesis a detailed study of the problem of numerical calculation of fluid flow and associated phenomena in domains of complex geometry has been carried out. As a result a complete finite volume method for solving this problem was developed, assessed and applied. Here the most important findings and major contributions of the present study will be summarized.

Differential Equations

Along the lines of the earlier study of Demirdžić (1982), various forms of the differential transport equations in general coordinates were analysed. The strong conservation form of these equations was found favourable from the numerical solution point of view, and the reasons for this were identified, the main one being the absence of curvature terms. The form of equations in which vectors and tensors are expressed in terms of their Cartesian components was selected, for its simplicity and suitability for iterative solution methods.

Discretisation

The transport equations in selected form were discretised using the finite volume approach. This was performed in physical space, in a way which minimizes the required information about the computational grid, ensures conservation of space and minimizes the sensivity of the method to certain grid properties. The result is that the only grid information needed is the Cartesian coordinates of control volume (cell) vertices, which are then assumed to be joined by straight lines. Thus the method is not restricted to particular type of grids: the forementioned information can be generated by any suitable method, including hand sketching.

Various interpolation practices (differencing schemes) for the evaluation of convection terms were examined for accuracy, stability, economy and suitablity for three-dimensional calculations on non-orthogonal grids. All the higher-order schemes studied, which are believed to offer higher accuracy than the basic upwind scheme, were found to suffer, to varying extents, from stability and unboundedness problems under certain circumstances due to the negative coefficients which they generate in the discretised equations. To overcome this problem a new method of blending of the higher-order schemes with less accurate, but unconditionally stable and bounded lower-order ones was developed. It assures, under some conditions, that the solution is properly bounded while retaining the maximum possible contribution from the higher-order scheme.

Attention was also paid to the cross-derivative diffusion terms and their importance. Two interpolation practices for their calculation were evaluated: both can generate negative coefficients and therefore unbounded solutions, but one can be made to be bounded under certain conditions. In the present method these terms were treated explicitly, ie. as pseudo sources, calculated from values obtained in the previous iteration, for reasons of economy and storage reduction.

Solution

Several possible variable arrangements on the computational grid, suggested and used by various research groups, were evaluated from the point of view of their suitability for use with general grids and selected form of transport equations. All the arrangements studied were found to suffer from possible decoupling problems. The non-staggered arrangement was found the most attractive, as it requires minimum computational effort for the assembly of coefficients of discretised equations. The decoupling problems with this arrangement were overcome by means of a special interpolation practice for the calculation of mass fluxes, inspired by works of Rhie (1981) and Rhie and Chow (1982). This formulation also enables the use of several efficient pressure calculation algorithms, and four of these were suitably adapted and evaluated for applications on non-orthogonal grids. The selected algorithm, which is a variant of the SIMPLE method, was further modified in two important ways: firstly a useful simplification was made to the pressure-correction equations which reduced the effort required to solve them, and secondly an analysis was performed of the effects of the under-relaxation parameters for velocities and pressures on the rate of convergence of the solution process which led to an optimum relation between the two. This relation enabled the velocity under-relaxation factors to be increased, which resulted in reduction of the number of iterations, needed to obtain converged solution, by up to five times as compared with the use of "conventional" set of under-relaxation parameters found in literature (eg. in Patankar, 1980, 1981). However, when the grid becomes severely non-orthogonal the under-relaxation parameter for the pressures needs be reduced below the value suggested by the forementioned relation, which is due to the simplification of the pressure-correction equation.

Testing and applications

The present method was initially assessed by applying it to some flows for which "exact" solutions were known, with emphasis on exploring the effects of grid non-orthogonality and non-smoothness. It was found that the former, alone, does not introduce any additional inaccuracies: only the rate of convergence is reduced, which is due to the explicit treatment of cross-derivatives. Non-smoothness of grid lines (the definition of smoothness is given in Chapter 7), though, does affect accuracy, mostly through the errors introduced by linearly interpolating pressures along grid lines. However, the effects are of limited extent, obvious in the pressure field only, and usually can be reduced to acceptable levels by some simple grid smoothing techniques described in Chapters 7 and 8 .

The method was also applied to several complex two- and three-dimensional laminar and turbulent flows. These included laminar flow in a circular pipe with sudden contraction, laminar and turbulent flow across staggered tube banks, turbulent flow around an axisymmetric bluff body confined in a circular pipe and laminar and turbulent flow in S-shaped and C-shaped three-dimensional diffusers. In cases of laminar flows the agreement between the predictions and the experimental results was found to be good whenever grid-independent solution was achieved. For turbulent flows, however, the agreement was somewhat poorer. This is, apart from numerical errors due to insufficiently fine grids, mostly due to the limitations of the k-ε turbulence model and wall functions employed here.

To highlight the possible applications of the method, results of calculations were also presented for some complex practical flows for which no experimental data exist.

10.2 Directions for Future Research

In this section some proposals are made for improving and extending the present method. They are listed below.

1) The generation of grids for general three-dimensional problems needs further study with emphasis on surface description, particularly when

submerged bodies are present within the solution domain, as is often the case in practical situations (eg. in the flow through inlet port/valve assembly of reciprocating engines). The use of transfinite mappings like those employed in the present study, but of a more general character (Gordon and Hall, 1973) together with appropriate surface representation seems to be a promising path.

2) Indirect addressing of grid nodes could help in optimizing grid arrangements and saving on memory requirements, especially in cases where cut-outs or submerged bodies within the solution domain exist. This would, however, destroy the diagonal structure of the coefficient matrix, and therefore require a different linear equation solver from the one used here.

3) Further study of differencing schemes for both convection and diffusion terms is needed, with the aim of increasing accuracy without affecting stability and boundedness. This is especially important for three-dimensional applications, where improvement of accuracy by grid refinement is not always possible, due to excessive storage and run time requirements. Also of importance is the study of higher-order interpolation practices for calculation of pressure terms, as the present grid smoothness limitation is believed to arise from inadequacy of the linear interpolation practice presently employed. It should, however, be noted that although the higher-order interpolation practices may improve accuracy, they will also affect structure of the coefficient matrix in ways which may possibly produce more negative coefficients and increase the cost of the computations. All these effects need careful evaluation.

4) Treatment of cross-derivative diffusion terms and contributions from higher-order interpolations to the coefficients of discretised equations also require further evaluation. The explicit treatment employed in the present study causes deterioration in convergence in cases of high non-orthogonality, which is sometimes unavoidable. Implicit treatment would increase the storage requirements (by up to two or three times) and require different solvers but may result in faster convergence (partly, perhaps, through ensuring that the optimum relation between the under-relaxation factors remains applicable).

5) Linear equation solvers need be further studied, with the aim of (i) achieving faster convergence in case of non-diagonally-dominant matrices, (ii) avoiding the necessity to employ empirical parameters and (iii) reducing sensivity of the convergence rate to the

properties of grid and equations being solved (eg. high aspect ratio, Neuman's boundary conditions etc.). The need also arises from the other related problems mentioned earlier.

6) Multigrid/multilevel techniques, as used eg. by Phillips and Schmidt (1984) and Ghia et al (1982), which have proven to give substantial improvements in accuracy and convergence rates on regular (Cartesian) grids need be evaluated for the non-orthogonal ones. Local grid refinement in particular may help in improving accuracy without excessive increase in computational effort and storage requirements.

7) The time-marching technique of calculating steady flows should also be considered, especially in conjunction with the flux blending technique developed here, some limitations of which may thus be avoided (eg. in particular the "irreversibility" of the blending factor decrease).

8) Turbulence modelling requires urgent attention. The k-ε model and associated wall function, still most widely used and also employed here, have proven to be inadequate in their present form in many complex flows. Improvements in this area are necessary in order to fully exploit the possibility of predicting flows in complex geometries offered by the solution methods of the present kind.

9) The present method can be extended to enable calculations of flows which possess one or more of the following additional complexities: unsteadiness, compressibility and moving boundaries. It may also be extended to include chemical reactions, heat and mass transfer and multiple phases. Some of these extensions are already in progress.

10) The problem of post-processing and presentation of results of three-dimensional flows also deserves attention. Particular difficulties exist in presenting velocity fields, more so if unsteady flows are considered, where both spatial and temporal presentation is difficult. New graphics and video technology make several routes possible, as discussed by Brandstätter et al (1985), among others.

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APPENDIX

MODIFICATION OF THE DISCRETISED EQUATIONS FOR TWO-DIMENSIONAL AXISYMMETRIC PROBLEMS

Axisymmetric flows are three-dimensional with respect to Cartesian velocity components, but only two-dimensional with respect to polar-cylindrical ones, where the derivatives in the circumferential direction are zero. Therefore, using the latter components greatly simplifies the problem of obtaining solutions in such cases.

The transport equations in polar-cylindrical coordinates (which can be found in many textbooks, eg. Hershley, 1973) are in the axisymmetric case similar to the two-dimensional Cartesian ones. In case of general, non-orthogonal grids, the coordinate transformation is performed only in (y^1, y^2) plane (with y^1 denoting the axial, and y^2 the radial coordinate direction). The appropriate forms of the transport equations can easily be obtained from the standard polar-cylindrical forms, using the coordinate transformation

$$y^i = y^i(x^1, x^2) \quad , \quad i = 1,2 \tag{A.1}$$

or as a degenerate case of the general three-dimensional forms as given by eg. Demirdžić (1982). They can then be cast into the same form as arises for planar two-dimensional case, with u_1 and u_2 now denoling the velocities in the axial and radial directions respectively. The two versions of the general (scalar) equation then read:

planar:

$$\frac{\partial}{\partial x^1}\left[U_1\phi - \frac{\Gamma_\phi}{J}\left(\frac{\partial \phi}{\partial x^1}B_1^1 + \frac{\partial \phi}{\partial x^2}B_2^1\right)\right] + \frac{\partial}{\partial x^2}\left[U_2\phi - \frac{\Gamma_\phi}{J}\left(\frac{\partial \phi}{\partial x^1}B_1^2 + \frac{\partial \phi}{\partial x^2}B_2^2\right)\right] = Js_\phi \tag{A.2}$$

axisymmetric:

$$\frac{\partial}{\partial x^1}\left[U_1 r\phi - \frac{\Gamma_\phi}{J}r\left(\frac{\partial \phi}{\partial x^1}B_1^1 + \frac{\partial \phi}{\partial x^2}B_2^1\right)\right] + \frac{\partial}{\partial x^2}\left[U_2 r\phi - \frac{\Gamma_\phi}{J}r\left(\frac{\partial \phi}{\partial x^1}B_1^2 + \frac{\partial \phi}{\partial x^2}B_2^2\right)\right] = Js_\phi r \tag{A.3}$$

The (A.2) form is derived from Eq. (3.11) by setting the derivatives with respect to the x^3 coordinate equal to zero. The difference between the above two equations is only in the appearance of the radius from the symmetry axis $(r=y^2)$ in the latter. The integrated and discretised forms are also of similar forms; only the mass fluxes F_i , coefficients D^i_j and cell volumes δV need be calculated differently for the axisymmetric case, thus:

$$F_{ic} = \rho r_c(u_1 b^i_1 + u_2 b^i_2)_c \tag{A.4}$$

$$\begin{aligned} D^1_1 &= r(b^1_1 b^1_1 + b^1_2 n^1_2) \\ D^1_2 = D^2_1 &= r(b^2_1 b^1_1 + b^2_2 b^1_2) \\ D^2_2 &= r(b^2_1 b^2_1 + b^2_2 b^2_2) \end{aligned} \tag{A.5}$$

and

$$\delta V = \int_{\delta V} Jr \, dx^1 \, dx^2 \tag{A.6}$$

The coefficients b^i_j are calculated according to Eqs. (3.24), by taking account of the fact that the derivatives with respect to x^3 in the corresponding β's are equal to zero for y^1 and y^2 , and to unity for y^3 . The radii r (equal, as noted earlier, to the y^2 coordinate distances), are stored at the same locations as the y^2 and for other locations are interpolated in the same way. The calculation of cell volumes was described earlier in Chapter 7. Also due to the transformation being performed in one plane only, the δV in equations like (3.35) need to be substituded by δA , which is the cell area in the (y^1,y^2) plane and is calculated according to Eq. (7.12). Thus the final expressions for convection and diffusion fluxes, analogous to (3.31) and (3.35), become, for eg. the "e" cell face:

$$I^C_e \simeq F_{1e}\,\phi_e \tag{A.7}$$

$$I^D_e \simeq I^{DN}_e + I^{DC}_e = \left[\frac{\Gamma_\phi}{\delta A} D^1_1\right]_e (\phi_E - \phi_P) + \left[\frac{\Gamma_\phi}{\delta A} D^1_2 (\phi_n - \phi_s)\right]_e \tag{A.8}$$

The momentum equations for the axisymmetric case can also be cast into the global form of (A.3). The extra stress terms are obvious from the two-dimensional forms of Eqs. (3.17) and (3.18). The pressure terms also contain the radius r , and so the S_P term of Eq. (6.4) becomes in this case:

$$S_P = -r_P[(p_e - p_w)b^1_i + (p_n - p_s)b^2_i]_P \tag{A.9}$$

An additional term exists in the u_2 (radial) momentum equation; it arises from τ_{33}, is non-conservative and is therefore integrated as a source, ie.:

$$-\int_{\delta V} 2\mu \frac{u_2}{r^2} Jr\, dx^1\, dx^2 \simeq -\left(2\mu \frac{\delta V}{r^2}\right)_P u_{2P} \tag{A.10}$$

This term is introduced through the linearized source term S" .

Finally, an additional term exists in the expression for the production of turbulent kinetic energy G (see eg. Amano, 1984), which is calculated as:

$$G_P^{add} = 2\left[\mu_t\left(\frac{u_2}{r}\right)^2\right]_P \tag{A.11}$$

This is added to other terms calculated from Eq. (3.39), in which all terms involving superscript "3" are zero, and δV is replaced by δA, as was earlier explained.